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An application of hydraulic tomography to a deep coal mine: combining traditional pumping tests with water inrush incidents

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Journal of Hydrology (2018); DOI: 10.1016/j.jhydrol.2018.09.058

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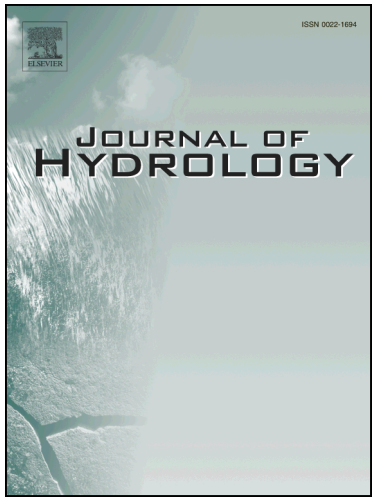

Research papers

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PII: S0022-1694(18)30752-2
DOI: https://doi.org/10.1016/j.jhydrol.2018.09.058
Reference: HYDROL 23155

To appear in: *Journal of Hydrology*

Received Date: 10 May 2018
Revised Date: 27 August 2018
Accepted Date: 25 September 2018

Please cite this article as: Mao, D., Liu, Z., Wang, W., Li, S., Gao, Y., Xu, Z., Zhang, C., An application of hydraulic tomography to a deep coal mine: combining traditional pumping tests with water inrush incidents, *Journal of Hydrology* (2018), doi: https://doi.org/10.1016/j.jhydrol.2018.09.058

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An application of hydraulic tomography to a deep coal mine: combining traditional pumping tests with water inrush incidents

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Submitted to *Journal of Hydrology*

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Abstract

Water inrush incidents threaten the safety of coal mining. Understanding of hydrogeologic parameter distributions is critical for preventing water-related hazards in coal mines. During the deep mining (> 1000 m) under the North China Plain, water from water-bearing strata discharges into coal seams through geologic conduits (i.e. water inrush) due to the fractured zone under the floor of working faces . In this study, a water inrush incident was exploited as an active stimulus. A 3D groundwater flow model was built for the eighth member of the Middle Ordovician system in Xingdong coal mine. Using this model and an inverse approach, we first checked if the data from the incident and an independent pumping test carry non-redundant information about the heterogeneity of the mine. Afterward, we combined these datasets to conduct a large-scale (approximately 10 km) hydraulic tomography (HT) analysis. The estimated hydraulic conductivity distribution from the HT analysis is found consistent with the distribution of known geologic faults. That is, a cluster of faults is characterized as a high-conductivity zone. A high conductivity zone is identified at locations close to the water inrush location, which is the high cement consumption zone during the grouting project. Finally, results of this study promote exploiting the water inrush events as a HT survey for mapping geologic structures over a large area.

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1. Introduction

China continues to rely on coal for about 69.6 % of its energy (Yearbook China Statistical, 2017). Therefore, coal production is of vital importance for China's economy and development. The increasing coal extraction depth results in significant increase of water pressure and groundwater inrush incidents from underlying aquifers, which could lead to disastrous consequences (Vutukuri and Singh 1995; Zhou 1997; Wu et al. 2004; Zhang 2005; Klose 2007). Among the reported incidents in China, more than 90% are related to limestone aquifers, especially the Ordovician limestone aquifers (Shi and Singh 2001), a featured aquifer under coal seams in the North China Plain (Wu and Wang, 2006).

For the past decades, engineering measures, including depressurization, grouting, backfill mining, etc., have been used to prevent these inrush incidents in coal mines and tunneling projects (Gao and Shi, 1999; Sui et al, 2011. In principle, these measure involve pumping water out from the high pressure confining aquifers to reduce the pressure or the sealing water conducting channels. Nevertheless, a detailed characterization of hydraulic parameters is crucial for the successful applications of these engineering measures as well as for the safe mining.

Due to the presence of fractures, faults, and karst conduits, limestone aquifers are highly heterogeneous and hydrological characterization of limestone aquifers are challenging (Teutsch, 1993; Scanlon et al., 2003; Illman and Hughson, 2005). In order to face the challenge, Audouin and Bodin (2008) employed cross-well slug interference tests to identify the channels and fracture flow paths of a karstic limestone aquifer. Robert et al. (2012) applied electrical resistivity tomography to detect preferential flow paths. These techniques work well for delineations in small scales. Based on the recent advances in aquifer characterization technologies, Illman (2006)

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proposed the application of hydraulic tomography (HT) to characterize aquifers with preferential flow channels.

HT (Yeh and Liu, 2000; Zhu and Yeh, 2005; Mao et al., 2016) is a recently developed and matured technology for characterizing aquifer at high resolutions. It employs a series of interference tests to stimulate aquifers by pumping water from or injecting water into a well while monitoring the aquifer's responses in other wells. By stressing the aquifers at different locations, non-redundant information about the distribution of aquifer parameters (Oliver, 1993; Mao et al., 2013a) can be obtained. Over the past decades, this technology has proven to be robust and flexible in dealing with different types of applications. For examples, Sharmeen et al. (2012) proved its effectiveness in delineating micro-fractures in a laboratory experiment; Lee and Kitanidis (2014) applied HT for joint inversion of hydraulic head and tracer data; Illman et al. (2009) and Zha et al. (2017) successfully applied HT to map fractures and faults over kilometer-scale granite geologic formations. For increasing the resolution of the characterization of fracture distribution, Zha et al. (2014) included flux data in the HT survey; Zhao and Illman (2017, 2018) and Zha et al. (2017) included prior geologic information in HT analysis; Soueid Ahmed et al., 2015) applied an image-guided method to include structure information.

HT has also been verified at different scales. For examples, Zhao et al., (2017) used it to delineate the heterogeneity of unconfined aquifers in a laboratory sandbox (2.4 m × 1.2 m). In a small-scale field test described in Brauchler et al. (2011), the hydraulic parameters of a field site (5 m × 5 m × 2 m) were estimated. Straface et al. (2007) employed HT to obtain the transmissivity and storativity for a confined aquifer (100 m × 100 m). Illman et al. (2009) explored HT to characterize a large-scale granite fractured aquifer successfully (884 m × 392 m × 1054 m). Yeh et al. (2009) used synthetic examples to promote river stage tomography with river stage acting as

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a driving force for characterizing a groundwater basin (130 km × 130 km). Wang et al. (2017) utilized this river stage tomography concept to obtain the diffusivity of an 1800 km^2 basin but was unable to retrieve hydraulic conductivity and specific storage separately due to the ill-defined problems (Mao et al., 2013). The lack of large-scale stimuli constrains the usage of HT. On the other hand, integration of different types of information may advance the development of tomographic surveys research. As advocated by Yeh et al. (2008), taking advantage of the natural stimuli for conducting basin scale hydraulic tomography tests is the future of the subsurface science.

The purpose of this study was to map the distribution of hydrogeologic properties of the eighth member of the deep Ordovician limestone aquifers in Xingdong coal mine. Crucial hydrogeologic information was acquired for safe mining and future grouting projects. For this purpose, we innovatively exploited data collected during water inrush incidents in the coal mine and combined it with one traditional pumping test to perform a large scale HT test (approximately 10 km) to reveal the geologic structure of the coal mine.

2. Site description

2.1 Site location

Xingdong coal mine is located 4 km northeast of Xingtai city (Fig. 1). The ground elevation is 55 m~65 m above the sea level. Geographically, it belongs to the North China Plain, which is one of the main coal production regions in China. Approximately 70% of all the identified coal resources in China are deposited under the North China Plain. Xingdong coal mine has an administration range of 3.4 km × 3.8 km, 14.5 km^2 in total area. The coal mine was planned and undergone geologic investigations in 1998 and started mining operation in November 2001 with

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an estimated annual coal production 600,000 tons. The current mining area is bounded by faults *F*1, *F*2, *F*19 and *F*30 in four directions as shown in Fig. 1. The coalfield north of fault *F*19 is reserved for future exploration.

2.2 Geology background

The stratigraphy of the mine consists of Ordovician (O), Carboniferous (O), Permian (P), Triassic (T), and Quaternary (Q) systems sequentially. Cretaceous strata (K) only exist in the southeast corner of the coalfield. Two cross-sections of the mine area are shown in Fig.2, indicating that the inclined strata strike NW with a low dip angle 10°～15°, dipping NE. The depth of 2# coal seam floors extends from -420 m in the west to -980 m in the east part of the coalfield, resulting in a higher confining pressure in the Ordovician strata in the east part.

Middle Ordovician strata are the main water-yielding layers, possessing threats to deep mining activities. The strata have an average thickness of 545 m consisting of three formations, XiaMajiagou formation (O_{2x}), ShangMajiagou formation (O_{2s}), and Fengfeng formation (O_{2f}) from bottom to top in sequence. Fengfeng formation with a thickness of 150 m is the research target of this study, which has the shortest distance to the 2# coal seam. This formation can be further divided into two members, the seventh member (O_{2f}^{1}) with an average thickness of 50 m (varying from 40 m to 69 m) and the eighth member (O_{2f}^{2}) with an average thickness of 100 m (varying from 70 m to 130 m). O_{2f}^{1} is mainly comprised of clayey breccia and dolomitic breccia and is considered as an aquitard hydrogeologically. While O_{2f}^{2} is a relative pure limestone unit, consisting of thick dark gray limestone and thin brecciated limestone. This member of Middle Ordovician strata is the main water-bearing aquifer that threats the safety of the $2^{\#}$ coal seam. Borehole drilling reveals that the water level in O_{2f}^{2} is only dozens of meters below the land surface. For example, before the pumping test conducted in well D1, the water level in well D3 is 25.9 m

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about sea level. This leads to a confining pressure of 13.35 MPa (an equivalent of 1362 m water column) calculated from the top of the O_{2f}^2 layer.

The Permian and Carboniferous strata are the main coal-bearing formations. There are 15 layers of coal seams in total, but only six layers are suitable for mining. At present, 2# coal seam is the production seam with a thickness of 4.0 m to 4.6 m. The floor of 2# coal seam is about 85 m to 263 m (an average thickness of 157 m) above the water-bearing Ordovician Majiagou Formation (including members O_{2x} and O_{2s}). In between the Majiagou Formation and the 2# coal seam, there are several layers of aquitards and less developed aquifers acting as a buffer zone preventing water inrushes from the Middle Ordovician system. Comparing with the Ordovician strata, the contribution of these less developed aquifers to water inrush incidents were negligible.

Even though there is a buffer zone between the 2# coal seam and the Middle Ordovician system, the presence of localized flow channels (such as fractures or high permeability zone) could result in a direct discharge of Ordovician water to working faces. Out of the four possible flow channels summarized by Wu and Wang (2006), faults or fracture zones are considered as potential hydraulic paths in Xingdong coalfield. In Xingdong coalfield, there are 243 faults revealed from extensive geologic surveys, borehole drillings, and geophysical surveys. All the faults are normal faults with a high dip angle 50°~65°, see Fig.1 and 2 for an illustration of the major faults. 55% of the faults have drop distances less than 5 m and 5~20m for 20% faults, while the drop distance of the rest faults is more than 20 m. The faults strike direction in this area is generally NNE~NE, dipping SE.

2.3 Monitoring boreholes

Observation wells were drilled to monitor the water level variations in the O_{2f}^2 layer. In this study, seven observation wells were employed, i.e. D1, D3, D4, D5, D6, D9, and D11,

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covering the boundaries of the current mining area (Fig. 1). They were installed as long-term water level monitoring wells for mining safety. No observation wells were installed inside mining area due to safety consideration because of their potential as conduits between aquifers and coal seams. These seven wells all fully penetrated into the eighth member of the Fengfeng formation O_{2f}^2, and partially penetrate into the seventh member O_{2f}^1 (Table 1). All the wells were cased from the land surface to the top of O_{2f}^2 layer, and the rest was left as unscreened open boreholes. Their fluctuations represent the depth-averaged head response in the eighth member O_{2f}^2 layer.

3. Water inrush incidents

3.1 Water inrush

Water inrush incidents in China have recorded a maximum water inflow of 2,053 m^3/ min (Zhang et al, 1997). As discussed by Zhang (2005), several factors control and influence the occurrence of water inrushes including strata pressures, geologic structures, mining sizes, and water pressures. Previous geologic surveys indicated that water pressures and geologic structures are the most important factors affecting the water inrushes in Xingdong coal mine.

The Ordovician limestone is a confined aquifer with a high water pressure. While there are strata between coal seams and the aquifer to prevent water from lower aquifers, a fractured damage zone may occur below the floor of the coal seam, extending dozens of meters due to mining-induced strata failures during mining (Zhang, 2005). As a result, a large amount of groundwater could discharge into the working face when the water pressure is greater than the strength of the coal seam floor. If there are geologic structures, such as water conducting faults, close to the damaged zone, the high-pressure groundwater can break through the coal seam floors more easily, shortening the distance between the aquifer and the coal seam, see Fig. 3 for an illustration.

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Therefore, the water inrush incident on the mining floor with conductive channels presents a significant threat to the mining safety.

3.2 Grouting project design

Grouting is an effective way to cut off the flow paths to coal mines (Li and Zhou, 2006). Grouting materials are injected into the target geologic stratum, filling the fractures and pore spaces, and replacing water in voids with the grout material. After the water inrush incident at location W1, a grouting project targeting O_{2f}^{2} strata was initiated to avoid further hazards. One vertical borehole was drilled as the main operation borehole. A series of parallel directional boreholes were then drilled spreading into the Ordovician limestone at depths between 70 m and 90 m below the top of O_{2f}^{2} strata. The distance between every two parallel boreholes was set to be 70 m based on experience which will ensure a continuous curtain. Different colors of the parallel boreholes represent different stages of the grouting projects (Fig. 4a). Cement and water with a mass ratio 1:3 was mixed for injection. The cement slurry was injected while drilling and the weight of cement consumption was recorded wherever the slurry could be injected. A total of 17,105 tons of cement were consumed during this project. The cement consumption recording locations, as well as major faults, are shown in Fig. 4b. The solid frame in Fib. 4b outlines the influence area of this grouting project, approximately 1.1 km ×2.1 km.

4. Hydraulic tomography

Most faults in this mine have high angles and therefore they could be treated as vertically conductive channels. The water inrush incident is therefore merely as a pumping test where the pumping well is fully screened through the O_{2f}^{2} layer. With such conceptualization, water inrush incidents could be jointly analyzed with traditional pumping tests conducted in the O_{2f}^{2} layer to

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perform a tomographic survey. Consequently, we combine one pumping test and one water inrush incident to characterize the coal mine. The hydraulic tomography algorithm Simultaneous Successive Linear Estimator (SimSLE) (Xiang et al., 2009; Mao et al., 2013b) was adopted to interpret the head responses from these two datasets. The groundwater flow model was implemented with finite element code VSAFT3 (Srivastava and Yeh, 1992).

4.1 Inversion algorithm description

Here, we briefly discuss the main elements of SimSLE. During SimSLE estimation process, the head responses from all the interference tests are incorporated into the algorithm simultaneously. There is no need to evaluate the effect of data incorporating sequence on the estimated parameters. SimSLE starts with cokriging interpolation to obtain an initial estimate of hydraulic parameter distribution. By comparing the difference between simulated and observed head responses, the parameters are updated successively by the successive linear estimator (Yeh et al., 1996), which considers the covariance of the hydraulic parameters and cross-covariance between the head and the parameter. During the update, the covariance function of the parameters is also revised to reflect the gain of additional information about the parameter in each iteration. Therefore, at the end of the estimation, the diagonal elements of the updated covariances (i.e., residual variances) can be employed to represent the uncertainty of estimated parameters.

4.2 Drawdown data

One pumping test conducted in well D5 and one water inrush incident occurred at location W1 were included in the hydraulic tomography test. The depth-averaged head responses in the eighth member of Fengfeng formation (O_2f^2) were recorded during both the pumping test and the water inrush incident. The pumping test began on 18 January 2014 and lasted until 25 January 2014. The pumping rate started at 800 m^3/day and decreased to 600 m^3/day after two days till the

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end of the test. Hydraulic heads were monitored every 30 minutes in five wells, D1, D4, D6, D9, and D11. Pumping rate was measured every 30 minutes as well.

The water inrush incident at location W1 occurred on 1 March 2015 during the coal mining at working face 2222 of the 2# coal seam. Inrush location W1 is close to fault F22 (Fig. 1). With an initial rate of 10 m^3/hour, it quickly increased to 100 m^3/hour in a few hours. As one of the emergency responses, the discharge was pumped out from the working face to avoid further damages. The inrush rate was measured based on this emergency pumping rate. This incident lasted for about one year with a gradual decrease of discharge rate due to grouting. For this study, the data from the first seven days were utilized, considering that the dataset from such a short duration will be less influenced by the boundaries. During the incident, the hydraulic head response in six wells (D1, D3, D4, D5, D9, and D11) and the discharge rate were monitored every eight hours.

The hourly atmospheric pressure was recorded in the nearest weather station located in Ren county, 12 km east of Xingdong coal mine, and its influence was removed from recorded hydraulic head data. For hydraulic tomography inversion, both the head response from the pumping test and the water inrush incident, and discharge rate were averaged every 0.5 days, see Fig. 5. Due to the large drop of hydraulic head in well D9 during the water inrush incident, it was plotted separately in Fig. 5c. In total, there are 70 drawdown records from the five observation wells in the pumping test and 84 drawdown data from the six observation wells in the water inrush incident. Because the number of observation wells is limited, all drawdown data were used for inversion and there were no data reserved for validation.

4.3 Numerical model setup

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With the depth-averaged hydraulic head responses in the eighth member O_2f^2 of Fengfeng formation, a 3D groundwater flow model was built. It had dimensions 15 km × 20 km in *x* and *y* directions, with 100 m in the *z* direction which was determined by the thickness of the O_2f^2 layer. This simulation domain encompasses both coal mine and grouting project influence area (Fig. 6). The right boundary was only 500 m to the right of the well D9, which was determined by fault *F*2. The impact of this impermeable boundary is discussed in section 6.1. Considering only seven observation wells and two discharge events are available, we expect that the HT can only reveal large-scale hydrogeologic features. Therefore, relative large elements were used in this analysis. The numerical domain was discretized into 7,500 elements with element dimensions 200 m× 200 m× 100 m. The top, bottom, and left boundaries were assigned with constant total head 1000 m, and the right boundary was zero flux due to the impermeable nature of the *F*2 fault. An initial total head of 1000 m was used.

As for the input parameters required by SimSLE, 1000 m and 1000 m were used for the correlation scales in *x* and *y* directions. The variance of ln*K* and lnS_s is set as 0.3. The initial mean of *K* and S_s were obtained by running an estimation with the forward model VSAFT3 and PEST (Doherty, 2010). The initial mean of *K* is 1.5 m/day and the uncertainty is indicated by 95% confidence interval, which has an upper limit of 1.8 m/day, and a lower limit of 1.25 m/day. For S_s, the estimated mean is 5×10^{-5} /m with an upper limit of 6.1×10^{-5} /m and a lower limit 4.1×10^{-5} /m. Due to the limited heterogeneity information about the deep O_2f^2 layer, different values of correlation scales and variance magnitudes were tested in inversion. It is found that they did not change significantly the estimated hydraulic conductivity and specific storage distributions. This corroborates the conclusion from Yeh and Liu (2000) that these input parameters have minor impacts on the pattern of the estimated parameters when sufficient hydraulic head data are

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available. The pattern of the estimated parameters will ultimately be conditioned by the head responses from the seven observation wells, and therefore only the inversion results from the above-mentioned input parameters were presented.

5. Results

With two datasets from the pumping test in well D5 and the water inrush incident at location W1, three inversion scenarios were attempted. The first two scenarios invert the two datasets independently, and the third scenario uses the two datasets as a hydraulic tomography. By running two datasets independently, we investigated the feasibility of treating water inrush incident as a pumping test. For all the three scenarios, only part of the numerical domain was illustrated because the estimated parameter values at the rest of the domain were close to their initial uniform values.

With head responses only from the pumping test in well D5, a less conductive region is unveiled adjacent to the six wells, which is surrounded by a conductive structure. The conductive structure forms a "C" shape on the west of the well D11 (Fig. 7). On the other hand, there are no major patterns in the estimated specific storage distribution, except a high storage region close to the pumping well D5.

As shown by the middle column of Fig. 7, the water inrush incident reveals a less conductive region to the north of inrush location W1 and a conductive region to the south of location W1. However, there is not noticeable heterogeneity present in the west of well D5. With the largest drawdown in well D9 among all wells during the incident, this head response yielded a relative conductive channel between well D9 and inrush location W1, and a low storage specific region around well D9. There is also a high storage specific region around well D3.

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While these two tests revealed different heterogeneity distributions, similarities between them do exist. The conductive region south of water inrush location W1 is an extension of the conductive part of "C" shape from the inversion of the pumping test data in well D5. More details are found using the data from the inrush incident about the low conductive region surrounded by the "C" shape. The similarities and differences shared by these two tests ultimately suggest that the data from the two events carry non-redundant information about the heterogeneity in this area. Thus, it is useful to combine the pumping test with the water inrush incident as a hydraulic tomography survey. The inversion results are shown in the right column of Fig. 7. The features identified by the two independently inversions now are assimilated. For example, the conductive region identified from the water inrush incident now connects with the "C" shape structure discovered by the pumping test. The less conductive region merges with each other as well. However, the result of HT analysis indicates that there are no major structures in the estimated specific storage field.

Fig. 8 shows the residual covariance distributions corresponding to these three different scenarios, which indicate the uncertainty of the estimates. A low uncertainty area implies that great hydraulic parameter information gained from the tests and our confidence is high about the estimated parameters in these areas. When combining these two tests, we see a merge of low residual covariance areas from these two tests (the last column of Fig. 8). The low residual covariance areas expand to the outskirt of the seven well cluster, and away from the cluster, the residual variance remains as the initial variance 0.3. In other words, the estimated parameter distributions from this inversion are likely representatively of the true field inside the well cluster.

The comparison between observed and simulated drawdowns are also shown in Fig. 9. In general, most of the 11 datasets agree with the observed drawdowns reasonably, especially the five

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datasets during the pumping test in well D5. For the water inrush incident, the head response in observation well D9 is well reproduced based on the estimated parameter distributions. However, at the late period of the observation wells D1, D3, D5, and D11, simulated drawdowns are higher than the observed data. This hints that after the initial discharge of water from the underlying aquifer, there may be a slight change of water inrush channels.

6. Assessment of inversion results

6.1 Effect of boundary conditions

A larger numerical domain 15 km × 20 km than Xingdong coal mine was set to minimize the impact of boundary conditions during both forward and inverse simulations. A no-flux boundary was set on the right side, which is only 500 m away from well D9. The no-flux condition was determined by the known fault *F2*. However, fault *F2* is not strictly vertically distributed, and connected with fault *F*30 at the bottom which goes left at the location close to well D3. In order to evaluate the effect of the right boundary, the no-flux boundary assignment was then removed. The right boundary was extended for another 5 km and was set as constant head type.

Fig. 10 is the estimation results under the extended domain and the dashed lines indicate the locations of the original no-flux boundary. Comparing with the results in the last column of Fig. 7, there are no significant changes for regions close to the dashed lines. The pattern of the estimated parameters is mainly dictated by the observed head responses. The low conductivity region in the expanded domain with a constant head assignment supports the impermeable nature of fault *F*2. This is consistent with the finding by Sun et al. (2013) that the estimated hydraulic conductivity could indicate the information about boundary types if the boundary condition is wrongly assigned.

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6.2 Grouting records verification

Figs. 11a and 11b are the estimated hydraulic conductivity and specific storage distributions around the areas influenced by the grouting project. Fig. 11c is the contour map of the amount of cement consumption interpolated from all the recording locations, and the black dots in Fig. 11c are the recording locations. The solid frames in Fig. 11 represents the scope of the grouting project.

During the grouting project, with the directional boreholes going 70 ~ 90 m below the surface of O_{2f}^2 strata, the cement goes into the pores of the aquifer. The amount of injected cement indicates the volume of pore space that the boreholes encounter, and to some extent, it reflects the conductivity of the aquifer (Fig. 11b) by comparing the contour patterns, not the specific storage in the current project. When comparing Fig. 11a and 11c, we observe a clear feature: more cement was consumed in the bottom right, and less cement is used in the top part and a relative increase of cement consumption right below this top part. The pattern is similar to the distribution of hydraulic conductivity. However, there are more variations in cement consumption than the estimated hydraulic conductivity. This is because of the scale differences between the grouting project and the hydraulic tomography test. We do not expect a high similarity between the depth-averaged hydraulic conductivity and cement consumption along one horizontal level. The purpose of this comparison is to show the amount of cement is related to the pattern of hydraulic conductivity. In turn, more attention is needed on the high conductive zone located at the bottom part of Fig. 11 for future grouting project.

6.3 Fault distributions verification

In this section, the estimated parameter distributions are compared with fault information collected at Xingdong coal mine (Miao et al., 2014). As suggested by Zha et al. (2016), HT is

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another way to map faults and is complementary to geological surveys. The estimated parameters overlay with the major faults in the coalfield (Fig. 12). As shown in the figure, the conductive zone located south of the water inrush location W1 agrees with a cluster of faults. Another conductive zone is located to the left of fault *F*1 which was identified as a conductive fault during the geologic survey. On the other hand, *F*19 was thought of as a conductive fault. However, the inverted hydraulic conductivity shows that it is less conductive compared with the above two conductive zones. It may be because that damage zone of the fault is not significant, or the fracture zones are filled with fault gouges and consequently are not as conductive as one expected.

Non-conductive fault *F*2 has already been well determined by the geologic survey. The inverted low conductivity located close to *F*2 and across *F*2 shows the impermeable nature of this fault is well included in the hydraulic head responses. *F*30 was treated as a non-permeable boundary in the geologic survey. However, it is blurred with the estimated conductive zone. This can be explained by the inadequate observation wells in this region. More observation wells are needed to further condition the hydraulic properties of fault *F*30.

For specific storage, the low storage region coincides with fault *F*2. This low storage region coexists with the less conductive region. The two high storage regions only concentrate around the pumping well D5 and the water inrush location W1 and these two regions have a high conductivity value. This pattern is the opposite of the finding by Illman et al. (2009) in Japan Mizunami site.

7. Discussion and conclusions

By combining the traditional pumping test and the water inrush incident in Xingdong, the parameters estimated from the hydraulic tomography test is in general consistent with the local fault distributions and the cement consumption data. The assumption of treating water inrush

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incidents as pumping tests was verified by comparing the pattern of the estimated parameters with the result from the traditional pumping test. These findings promote the usage of this type of stimulus in a large-scale hydraulic tomography test as advocated by Yeh et al. (2008).

The verifications providing us with further confidence in the validity of our results. The results from the aforementioned approaches of evaluating estimated results are encouraging. More rigorous and quantitative means to evaluate the estimated parameter distributions are necessary, such as to test the predictability of head responses using the estimated parameter fields under different flow scenarios (Liu et al., 2007). In the current application, only two tests were employed in the analysis, this yield low-resolution map about major geologic information but with more tests, one should be able to obtain higher resolution information (Illman et al., 2009, Zha et al., 2017).

The target aquifer of O_{2f}^2 strata is not perfectly flat and its continuation is interrupted by geologic faults (Fig. 2). Our assumption of a perfect flat aquifer with 100 m in thickness likely yields some incorrect estimates. Secondly, treating the seventh member O_{2f}^1 strata as an aquitard was based on a limited number of boreholes. In addition, there may be a connection between eighth member O_{2f}^2 aquifer and deeper members in the Middle Ordovician strata, and these were not considered in the current model configuration. Our assumptions are necessary to conduct a quantitative analysis. Some major conclusions can be drawn from this research,

1) To the best of our knowledge, it is the first HT test to combine a traditional pumping test and a water inrush incident. The estimated results are encouraging. The large-scale hydraulic tomography test (approximately 10 km) propels us to advance hydraulic tomography technology into more innovative applications.
2) The water inrush incident on the coal seam floor is treated as a pumping test. Water discharge into the coal seam is assumed as a pumping test from the underlying aquifer O_{2f}^2

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layer through conductive channels. This assumption is verified by the pattern of the inverted parameters from an independent pumping test conducted in a different location of the same aquifer.

3) The hydraulic conductivity distribution is in general consistent with the cement consumption in the grouting project. The conductive zone is related to the high cement consumption zone at the bottom of the grouting project, and the less conductive zone is well linked to a low consumption zone. The estimated hydraulic parameters can be furthered employed to guide the design of future grouting projects.
4) The inversion results from the HT test agree with major faults distribution in this coal field. The high conductive zone overlays with a fault cluster in the south of the well cluster. The identified impermeable fault *F*2 is well characterized by a low conductive and small storage zone.

Acknowledgments

The first author would like to thank the support from Qilu Youth Scholar program of Shandong University and research support from Xi'an Research Institute Company of China Coal Technology and Engineering Group is also appreciated. The second author acknowledges the support from the National Key Research and Development Program of China, grant No. 2017YFC0804102. The third author thanks the support of from NSFC, grant No. U1603243.

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Tables

Well	*x* (m)	*y* (m)	Top of O_2f (m)	Bottom of O_2f (m)	Thickness of O_2f (m)
D1	11100	8100	-745	-846	101
D3	14100	9500	-1336	-1458	122
D4	13100	11700	-1000	-1139	139
D5	11700	10500	-828	-961	133
D6	13300	10900	-1135	-1231	96
D9	14500	10700	-1551	-1667	116
D11	10300	9700	-516	-624	108
W1	13900	10100	-	-	-

Table1: Well coordinates and Fengfeng formation O_2f thickness encountered by the seven observation wells information. Here the thickness of O_2f includes part of seventh member O_{2f}^1 and all eighth member O_{2f}^2 in each location.

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Figures

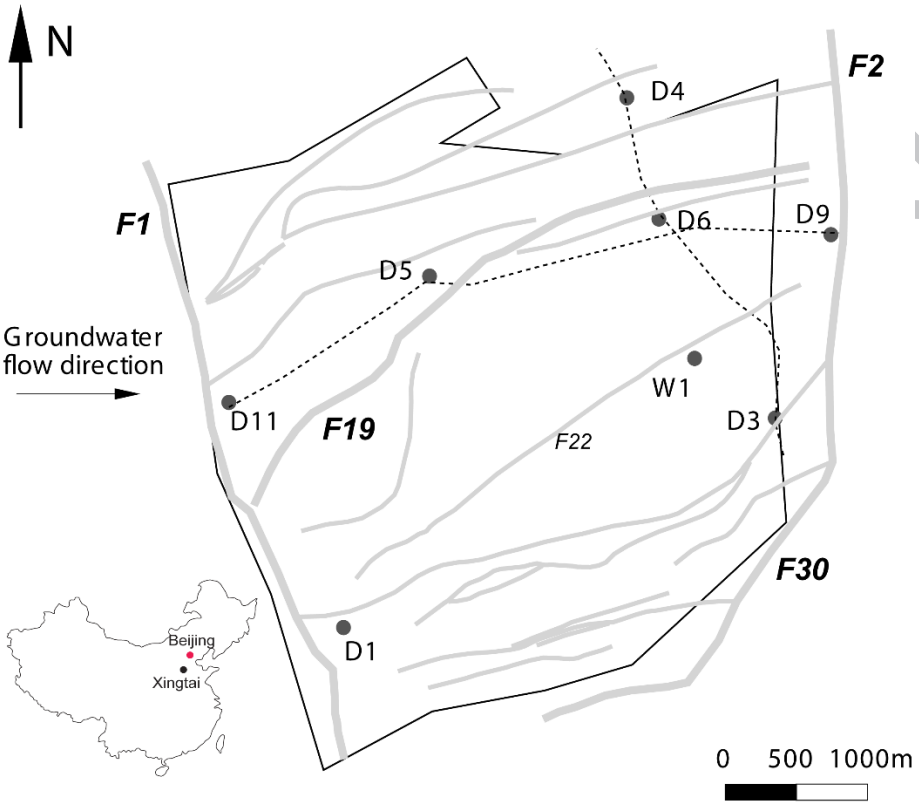

Figure 1: Location of the Xingdong coal mine in China and its geology background. The dark grey lines indicate major faults discovered in this region. The black frame shows the administration range of the coalfield. Four faults *F*1, *F*2, *F*19 and *F*30 mark the current extraction range, and they were represented with thicker lines. One pumping test conducted in well D5 and one water inrush accident occurred at location W1. *F*22 is the closest fault to the water inrush location (inclination angle 65˚). Drawdown data were taken from seven wells (D1, D3, D4, D5, D6, D9, and D11) during these two tests. Regional groundwater flows from west to east. Horizontal dashed line starting from D11 represents A-A' cross-session. B-B' vertical cross-section starts from D4.

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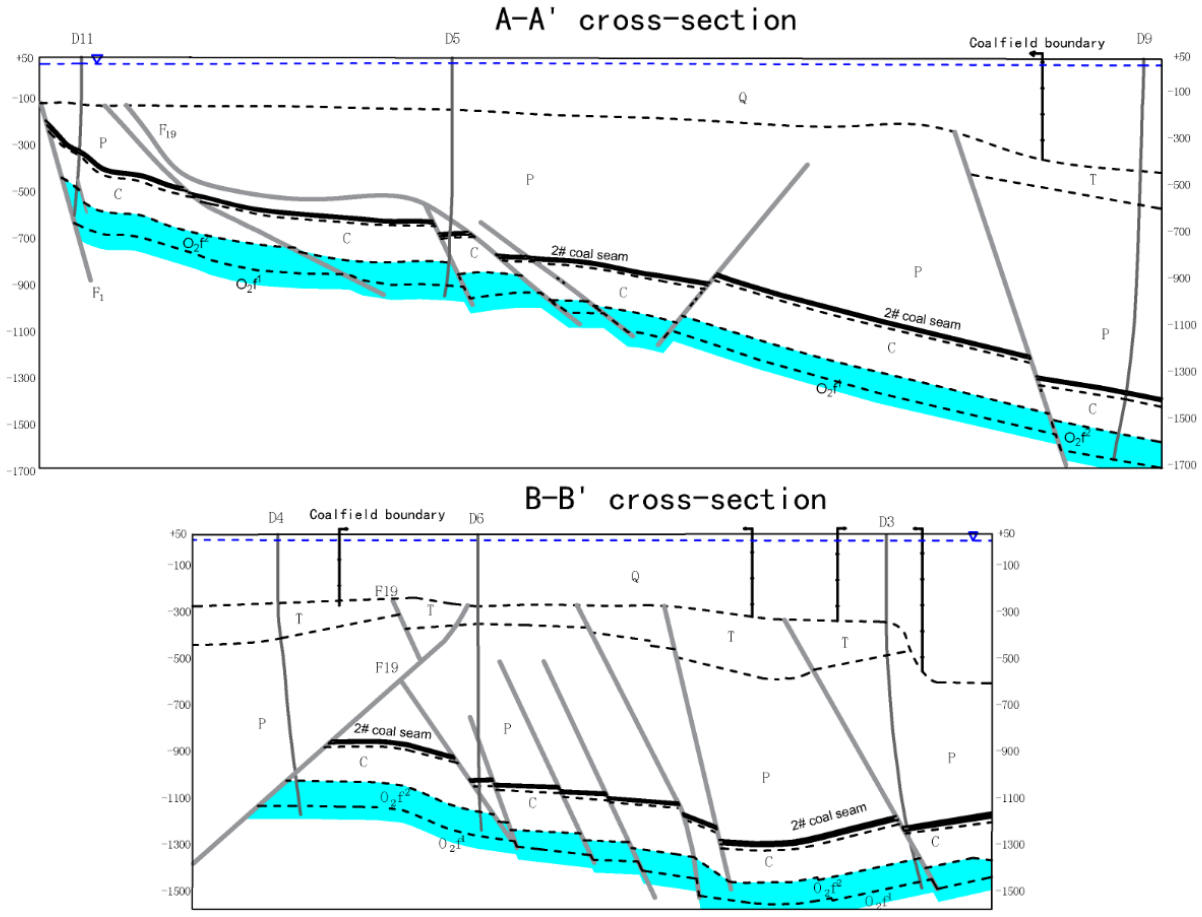

Figure 2: Geological cross-section of Xingtai coal mine (A-A’ and B-B’). Depth unit is meter. The stratigraphy consists of Ordovician (O), Carboniferous (C), Permian (P), Triassic (T), and Quaternary (Q) systems. The eighth member of Middle Ordovician strata O_2f^2 is the main aquifer threating the safety of coal mining at 2# coal seam, and the seven member O_2f^2 is treated as an aquitard.

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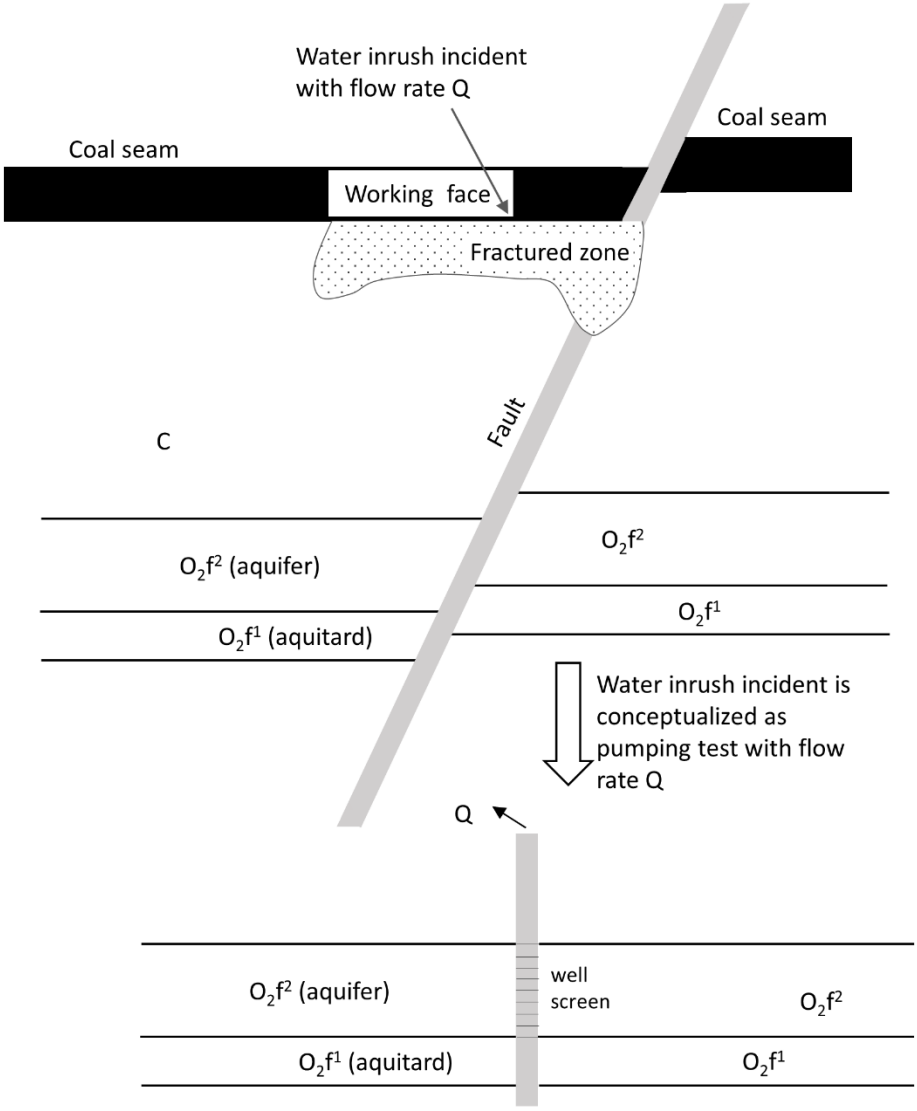

Figure 3: Conceptualization of water inrush incident as a pumping test. Water inrush incident is connected with the eight member of Middle Ordovician strata O_2f^2 aquifer through fault during the creation of fractured damage zone while coal mining in the work face. The seventh member O_2f^1 is treated as an aquitard.

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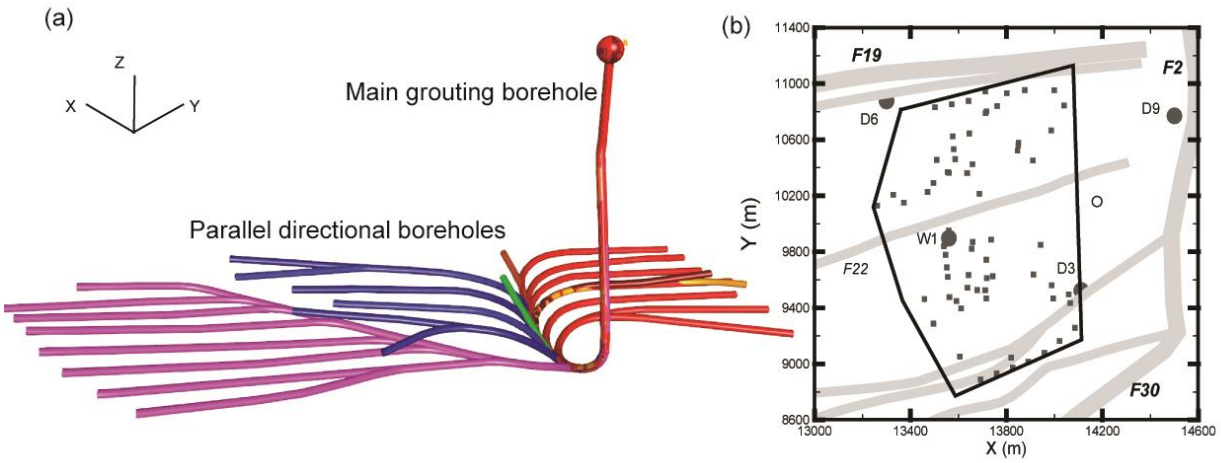

Figure 4: Illustration of grouting projection in aquifer O_2f^2. (a) Grouting projection design is shown in 3D format. Different colors represent the different stages of the project. Parallel directional boreholes were drilled for grout injection. (b) 2D map shows grout consumption recording locations and grouting influence boundary (solid black frame). The black circle marks the main grouting borehole. The gray lines are the major faults in the influence domain of the grouting project.

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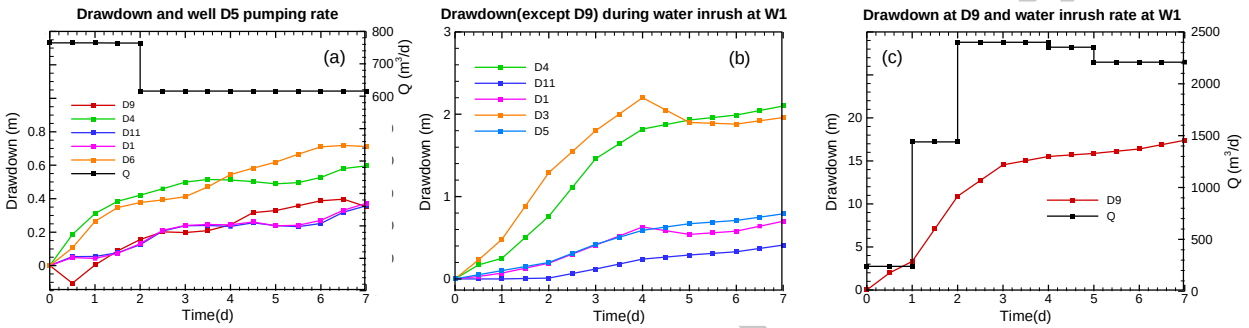

Figure 5: Drawdown and pumping rate information. (a) It shows the drawdown and pumping rate information during the whole seven days test period. (b) It shows the drawdown data in six observation wells during water inrush accident at W1. (c) Drawdown at well D9 was separated with the rest observation wells due to its large magnitude. The flow rate from this incident is also included.

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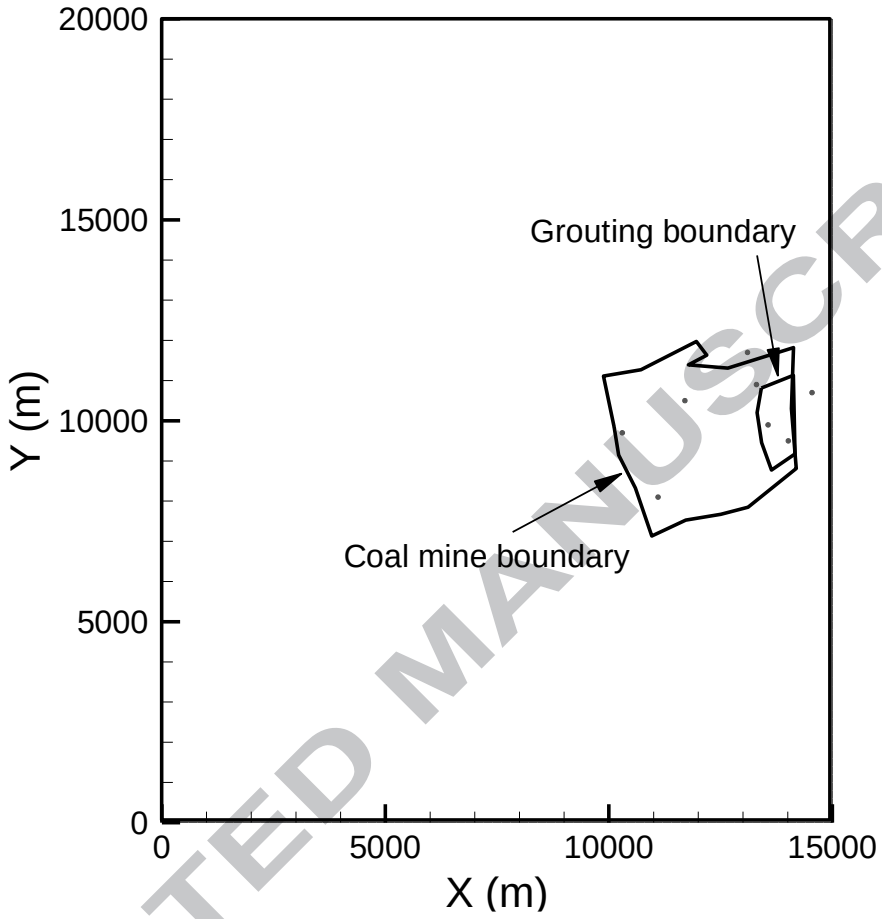

Figure 6: Illustration of different domains. The simulation domain is 15 km × 20 km. The coal mine is located on the right side of the domain. The grouting region is close to the right boundary of the coal mine. The black dots represent observation wells and water inrush locations.

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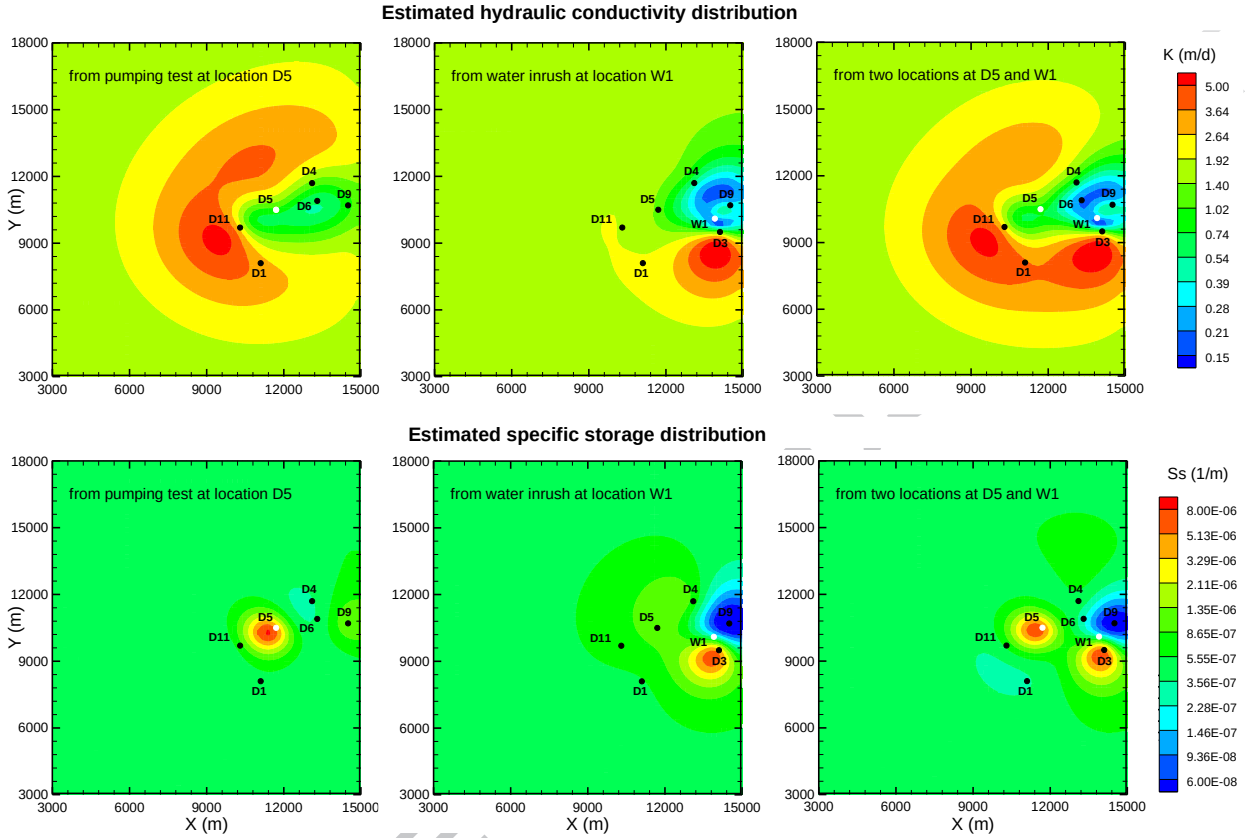

Figure 7: Estimated hydraulic conductivity and specific storage distributions from three different scenarios. Drawdown data from five wells (D1, D4, D6, D9, and D11) were recorded during pumping test at D5 and drawdown data from six wells (D1, D3, D4, D5, D9, and D11) were recorded during water inrush incident at W1. The first column shows the result with data only from pumping test at D5. The middle column is the inversion result with data only from water inrush incident at W1. The last column is the hydraulic tomography result by incorporating data from both the pumping test and the water inrush incident.

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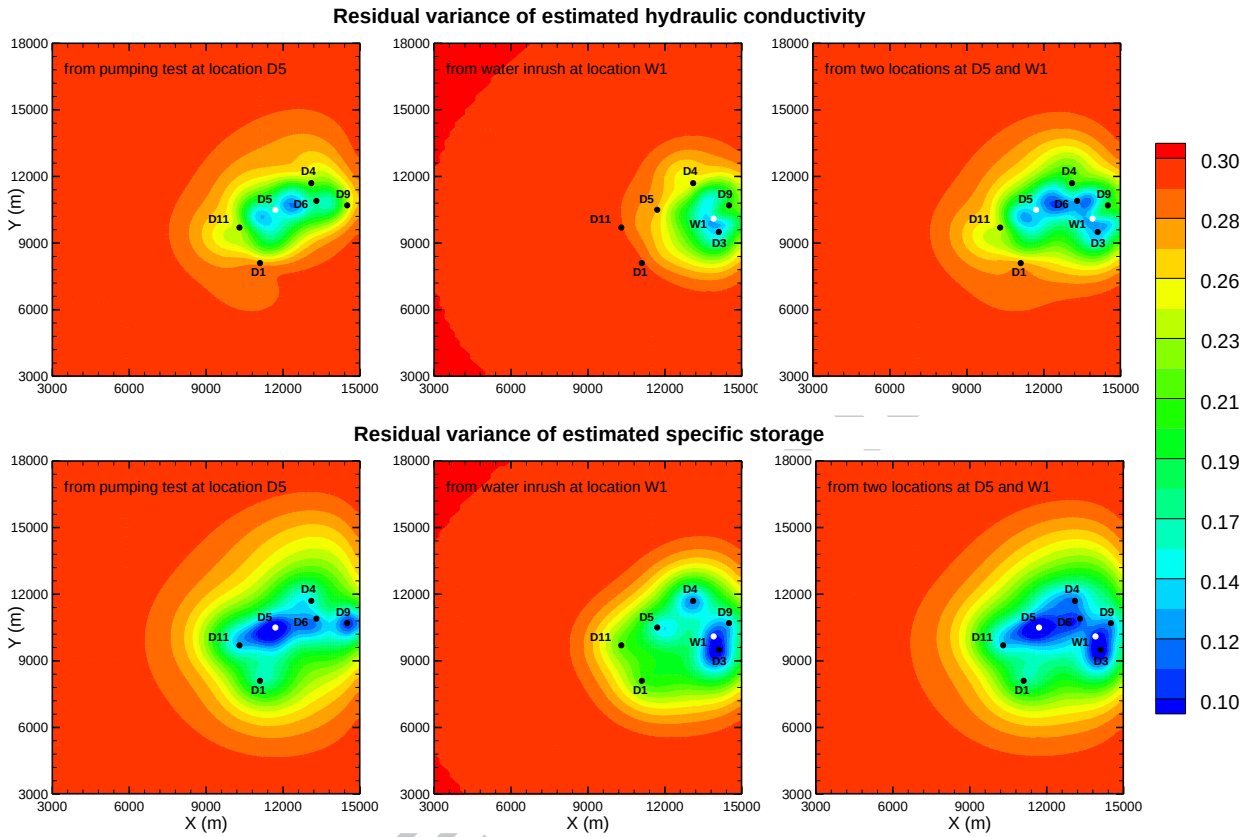

Figure 8: Comparison of residual covariance among three different inversion scenarios. The background red legend equals to the initial value 0.3. See Fig. 7 for legend explanation.

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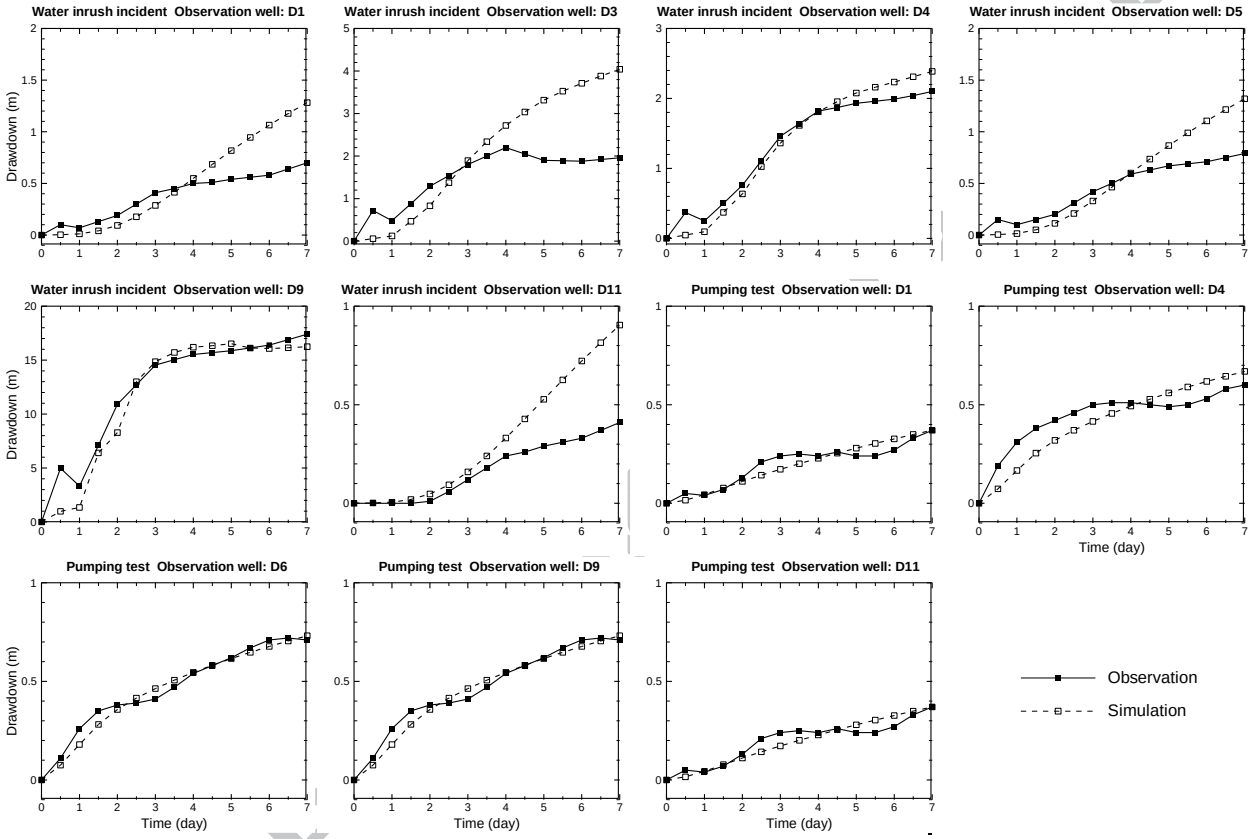

Figure 9: Comparison between observed and simulated drawdowns during hydraulic tomography test with data from both pumping test in well D5 and water inrush incident at location W1.

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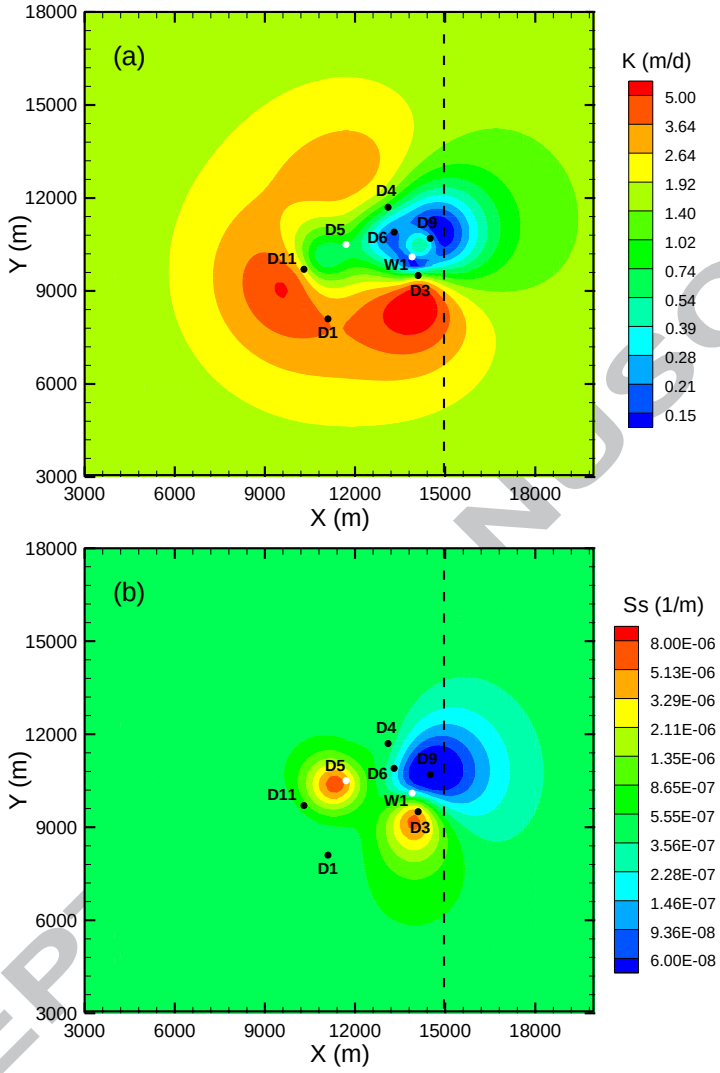

Figure 10: Estimated parameter distributions by removing the impermeable fault on the right boundary. (a) Hydraulic conductivity distribution and (b) Specific storage distribution. The dashed lines is the location of impermeable fault *F2*. Compared with the simulation domain in *x* direction of Fig. 7, the right boundary was extended from 15 km to 20 km.

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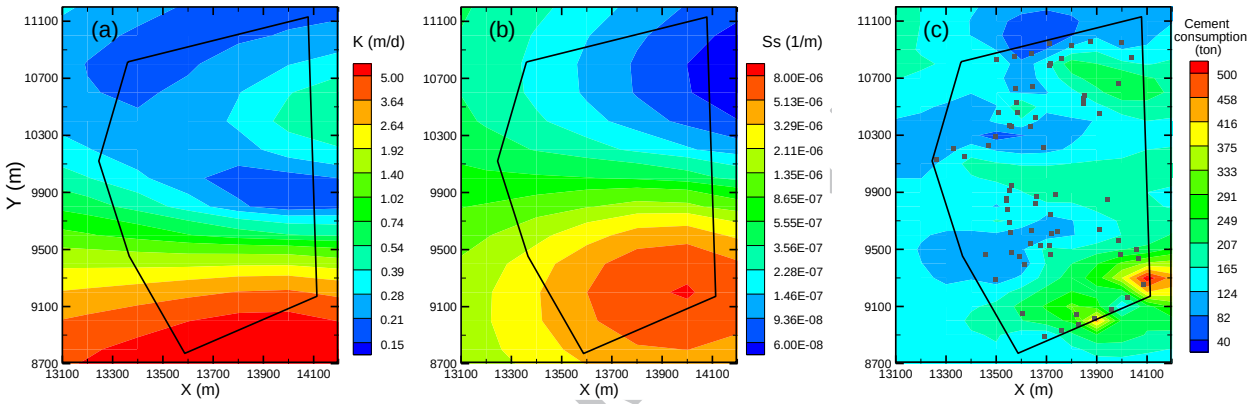

Figure 11: Comparison of estimated hydraulic parameters with cement consumption. (a) Hydraulic conductivity, (b) Specific storage, and (c) Cement consumption. The black frames in the three figures indicate the grouting range. The dots in the last figure show the cement recording locations.

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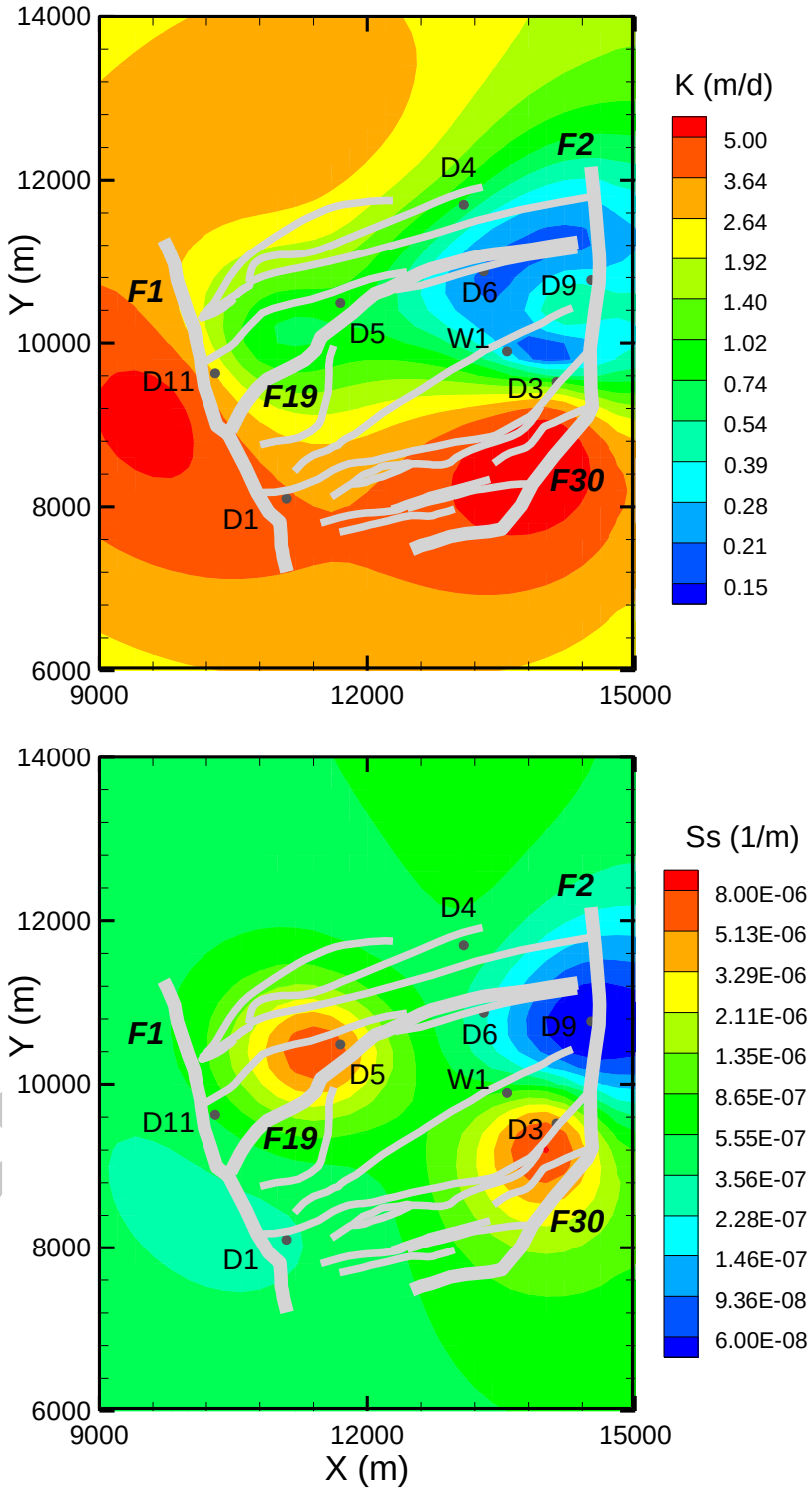

Figure 12: Overlay of estimated hydraulic conductivity and specific storage over the faults in the range of Xingdong coal mine.

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Highlights

1 A water inrush incident was conceptualized as an active pumping test

2 A traditional pumping test and a water inrush incident were combined for a hydraulic tomography survey

3 This hydraulic tomography allows mapping large-scale geologic features at distances about 10 km.

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04_其他合作论文科学背景包.pdf

文献 2/9

Development of Laboratory Procedure for Evaluating Microcracking Technology on Cement-Modified Soil Subgrade

Mahdi Al-Naddaf; George A. Tannoury; Jie Han; Robert L. Parsons; Chi Zhang

International Journal of Geomechanics (2019); DOI: 10.1061/(ASCE)MT.1943-5533.0002982

证据用途：其他合作论文背景；张弛老师角色：末位合作者；证据等级：C

Technical Note

Development of Laboratory Procedure for Evaluating Microcracking Technology on Cement-Modified Soil Subgrade

Mahdi Al-Naddaf, Ph.D., A.M.ASCE[1]; George A. Tannoury, P.E., A.M.ASCE[2]; Jie Han, Ph.D., P.E., F.ASCE[3]; Robert L. Parsons, Ph.D., P.E., F.ASCE[4]; and Chi Zhang, Ph.D.[5]

Abstract: Cement modification of soils has been widely practiced for the last few decades. Recently, cement has become a more economical binder to modify in situ subgrade soil because other binders, such as fly ash, have become less available and therefore their prices have gone up significantly. In addition, a much higher percentage of fly ash needs be used when compared with cement to achieve the same subgrade strength and stiffness. However, cement-modified subgrade may have an issue with shrinkage cracking, which can eventually reflect through the asphalt pavement layers to the surface after construction. For some subgrade soils, a high cement content is needed to meet the unconfined compressive strength requirement without jeopardizing durability. A higher cement content will result in higher shrinkage cracking potential. To overcome this problem, microcracking technology has been developed and adopted in the field. This technology involves recompaction of cement-modified soil (CMS) with a roller 24–48 h after initial compaction to induce microcracks in the CMS and minimize the potential for large shrinkage cracks. Microcracking of CMS is not expected to significantly reduce the strength and stiffness of CMS, but it is expected to increase its permeability and reduce the potential for large shrinkage cracks. Unfortunately, the procedure to simulate microcracking of CMS in the laboratory and evaluate its effect on properties of CMS has not been established yet. This note documents the development of such a procedure and discusses the effect of microcracking on the properties (strength and modulus) of CMS specimens. **DOI: 10.1061/(ASCE)MT.1943-5533.0002982.** *© 2019 American Society of Civil Engineers.*

Author keywords: Cement-modified soil; Microcracking; Modulus; Shrinkage cracking; Strength.

Introduction

Cement is commonly used in practice to modify soils. Cement-modified soil (CMS) often has a high shrinkage potential. Shrinkage associated with cement-modified soils can be divided into two categories: autogenous shrinkage (resulting from hydration of cement) and drying shrinkage. George (1968) and Bofinger et al. (1978) studied the shrinkage cracking phenomenon extensively and concluded that drying causes the majority of shrinkage. The degree of shrinkage cracking depends on several influence factors: tensile strength of the modified soil (George 1968; Bofinger et al. 1978), restraint by friction between the modified soil layer and its underlying layer (Bofinger 1971; George 1973), creep characteristics of the modified soil (George 1969; Bofinger et al. 1978), temperature (Bofinger 1971; George 1973), amount and type of clay in the modified soil (George 1968), and pretreatment and moisture content at compaction (Bofinger et al. 1978).

Prior studies have documented well the factors that cause shrinkage in CMS; however, recent efforts have focused on understanding design and construction practices that can minimize the shrinkage cracking problem. George (2001) conducted a soil-stabilization field trial in Mississippi and concluded that precracking of stabilized base by roller compaction at 24 h after placement minimized its shrinkage cracking. The PCA (2003) recommends that the 7-day unconfined compressive strengths of CMS should be within the range of 2.1–2.8 MPa in the design phase. During construction, it recommends compaction of CMS at or slightly below its optimum moisture content with moist curing until a moisture barrier is placed (PCA 2003).

Although most early methods for reducing shrinkage cracking in CMS have focused on controlling desiccation using moist-curing or asphalt-curing membranes, more recent techniques explored the use of stress-relief layers, such as chip seals, geosynthetics, or thin unbound granular base layers, to reduce reflective cracking. These techniques can reduce the likelihood of reflective cracking but require additional steps in the construction process. An innovative concept originated in Austria to use a vibratory smooth-drum roller to create a microcracked CMS layer during the early curing stage (Litzka and Haslehner 1995). The study reported that this microcracking process prevented the development of large stress cracks (Litzka and Haslehner 1995). According to Litzka, microcracking is typically performed by three passes of a roller 24–48 h

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Note. This manuscript was submitted on November 2, 2018; approved on June 21, 2019; published online on September 30, 2019. Discussion period open until February 29, 2020; separate discussions must be submitted for individual papers. This technical note is part of the ***Journal of Materials in Civil Engineering***, © ASCE, ISSN 0899-1561.

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after compaction (Sebesta 2005). Brandl (1999) reported that the microcracking technology was most suitable among the available options for minimizing cracking on the Austrian–Hungarian Highway. The objective of microcracking is to induce hundreds of tiny cracks to accommodate the need for shrinkage, rather than large individual cracks that have the potential to reflect up into flexible surface layers. After the initial successful microcracking work in Austria, this construction technique was tried in Texas as early as 2000 to evaluate this microcracking concept. Scullion (2002) described these efforts, including the test sites on residential streets, and presented a specification for the microcracking process in the field. This study showed favorable results using the microcracking procedure.

Although previous research has confirmed the applicability of this construction method in the field, no literature was found describing a procedure or method to simulate the microcracking process in the laboratory. This note proposes a laboratory method that can be used to simulate the microcracking process on CMS in the field using unconfined compression (UC) specimens. The UC specimens were used to evaluate the properties of CMS with and without microcracking, including their strength and modulus.

Laboratory Investigation

Before the development of the procedure to generate microcracking of CMS specimens in the laboratory, several test methods were adopted in the laboratory to characterize the properties of native soil and CMS.

Native Soil

The native soil was collected from an ongoing project located at 330th Street, between K-15 and the Marion-McPherson County Line, west of Tampa, Kansas. The soil was excavated to a depth of 0.3 m along a 1.6-km road (1-mi) segment of the 12.9-km (8-mi) roadway that was planned to be built on cement-modified subgrade soils.

Sample Preparation

The collected soil was first mixed thoroughly, and a portion of the mixed soil was dried at 60°C in a large oven. To determine the Atterberg limits of this soil, the dry soil was washed through a No. 40 (0.425-mm) sieve using water in accordance with ASTM D4318 (ASTM 2017c). The portion of the soil that passed the No. 40 sieve was dried again at 60°C and then broken down with a mortar and pestle. To determine the moisture–density relationship and the unconfined compressive strength, the dry soil was crushed, pulverized, and passed through the 3/4-in. (19-mm) sieve.

Soil Classification

The grain-size distribution, Atterberg limits, standard Proctor compaction curve, and laboratory California bearing ratio (CBR) were determined. Sieve analyses were performed using a wet sieve method in accordance with ASTM D422 (ASTM 2007) to obtain the grain-size distribution of the native soil. Two specimens (approximately 1,300 g of oven-dried soil each) were washed through a No. 200 sieve (0.075 mm). The portions of the soil retained on and passing the sieve were dried and weighed. Based on the percent of soil passing the No. 200 (0.075 mm) sieve, it was classified as a coarse-grained soil. Fig. 1 presents the grain-size distribution curves of these two specimens.

The liquid limit, plastic limit, and plasticity index of the native soil were determined to be 30, 15, and 15 according to ASTM D4318. The liquid limit was determined using the multipoint liquid limit method (Method A) described in ASTM D4318 (ASTM 2017c). Based on the Atterberg limits, the soil can be classified as a clayey sand, with gravel (SC) and lean clay (CL) of the fine portion according to the Unified Soil Classification System (USCS) described in ASTM D2487 (ASTM 2017a).

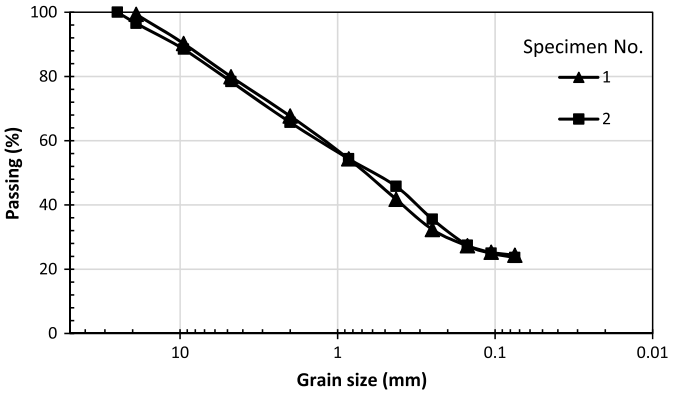

Fig. 1. Grain-size distribution curves of the native soil.

Soil Compaction Characteristics

The native soil was tested to determine its optimum moisture content and maximum dry density using the standard Proctor compaction method described in ASTM D698 (ASTM 2012). Method C was utilized to determine the moisture content dry density curves presented in Fig. 2(a). Immediately after the compaction tests, the same specimens were tested for their CBR following the procedure for unsoaked specimens described in ASTM D1883 (ASTM 2016). Fig. 2(b) presents the effect of the moisture content on the California bearing ratio, indicating the susceptibility of the subgrade strength with moisture. The two specimens tested for the maximum dry unit weight and CBR were compacted with two different methods, hand and mechanical compaction. Because mechanical compaction is more efficient (i.e., more compaction energy) than hand compaction, a lower optimum moisture content and a higher maximum dry density were obtained. However, the curve generated by the hand compaction was used in this research because subsequent work was mostly performed by hand. The optimum moisture content, maximum dry density, and CBR at the optimum moisture content for the native soil are $w_{opt} = 9.2\%$, $\gamma_d = 20.3$ kN/m^3, and CBR $= 4\%$ at w_{opt}.

Cement-Modified Soil

Portland cement (hydraulic cement) Type I/II was selected to modify the native soil and improve its properties. To design the soil-cement mix, a strength criterion based on UC strength of soil-cement specimens of 71 mm in diameter and 142 mm high cured for 7 days was specified. The selected UC strength was 2.1 MPa in this study. To find the appropriate cement content, three different cement contents, 3.5%, 5.0%, and 6.5%, were selected. For each cement content, five specimens were prepared in accordance with ASTM D1632 (ASTM 2017b) and tested for unconfined compressive strength following ASTM D1633 (ASTM 2017d). Soil and cement were mixed at $\pm 0.5\%$ of optimum moisture content as determined for the native soil. The soil-cement mix was placed and compacted in a mold, cured in the mold in a moist-curing room for 12–18 h, and then removed from the mold using a UC specimen extruder. The specimen was returned to the moist-curing room for continuous curing and protected from dripping water. After 7 days of the moist curing period, unconfined compression tests were performed on CMS specimens in the moist condition directly after their removal from the moist-curing room.

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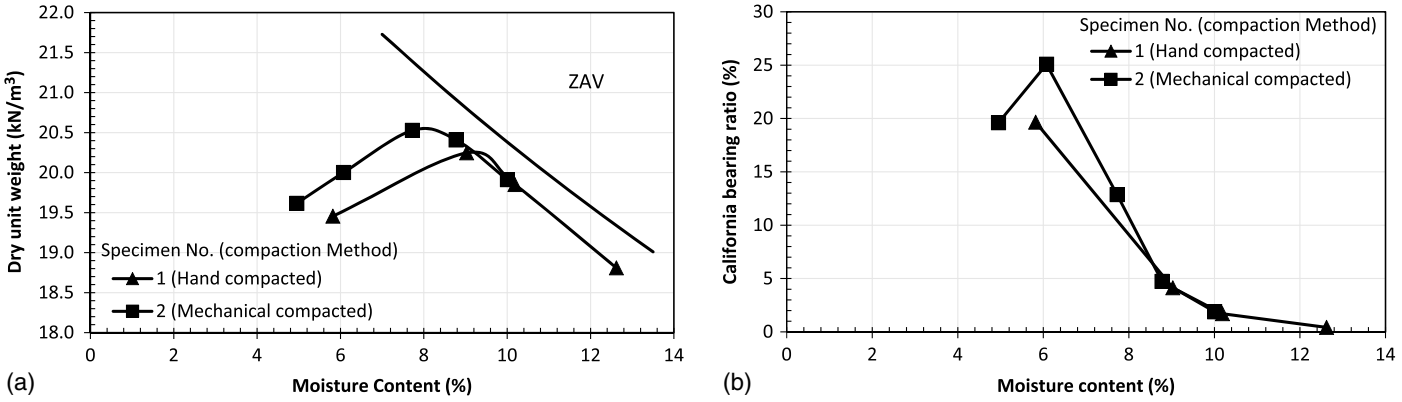

Fig. 2. Compaction characteristics of native soil: (a) compaction curves; and (b) effect of moisture content on CBR.

Table 1 provides the unconfined compression test results of three mixes of CMS after 7-day curing with cement contents of (1) 3.5%, (2) 5.0%, and (3) 6.5%. Fig. 3 shows the effect of the cement content on the UC strength of CMS. At the target UC strength, the cement content was determined as 5.0%. This cement content was used in the following investigations performed for this research.

Table 1. Unconfined compressive strength of CMS after curing for 7 days

Mix	Specimen	Cement content (%)	Compressive strength (MPa)	Average strength (MPa)
1	1	3.5	1.41	1.42
	2		1.46	
	3		1.80	
	4		1.43	
	5		0.99	
2	1	5.0	2.00	2.10
	2		2.11	
	3		2.41	
	4		2.32	
	5		1.67	
3	1	6.5	2.99	2.88
	2		2.72	
	3		2.74	
	4		3.05	
	5		2.89	

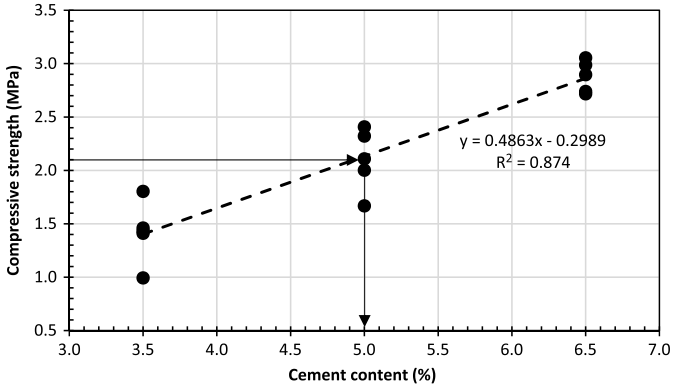

Fig. 3. Unconfined compressive strength of CMS versus cement content after 7-day curing.

Microcracking Simulation

The simulation of the microcracking process in the laboratory was intended to replicate the procedure used in the field. As mentioned previously, microcracks in a CMS layer are generated in the field by recompacting the CMS layer with a roller after the initial compaction and a short curing time of 24–48 h. In other words, the microcracks are generated by preloading the surface of the treated layer and inducing a series of tiny cracks. In practice, unconfined compression tests are conducted in the laboratory to determine the cement content in a CMS mix. In this research, the authors also utilized the unconfined compression tests and CMS specimens as the specimens began to yield to evaluate the effect of microcracking on UC strength of CMS.

Before developing and verifying the microcracking process, the UC strength of CMS after 48-h moist curing was determined. Six UC specimens of CMS at the design cement content (i.e., 5.0%) were prepared and tested after 48-h moist curing. Table 2 provides the UC test results of these specimens. The average UC strength of CMS after 48-h curing was 1.57 MPa, or 75% of the 7-day strength. Fig. 4 presents the stress-strain curves of the UC tests of the CMS specimens with 5.0% cement content after 48-h curing. It is shown that this material had a brittle behavior (i.e., stress sharply dropped after the peak).

To simulate the microcracking process in the laboratory by preloading the CMS layer after a short curing period, three trials were made. In these trials, three stress levels of preloading were applied to the UC specimens after the 48-h curing period: approximately 50%, 70%, and 90% of the ultimate UC strength of the specimens. Fig. 5(a) depicts the stress-strain curves of the UC specimens during preloading. For each preloading level, three CMS specimens were prepared and loaded. After the preloading stage, no sign of any cracks was observed on the UC specimens for all preloading levels. This phenomenon was also confirmed when the preloaded specimens were tested to failure after 2-h waiting time, showing no

Table 2. Unconfined compressive strength of CMS with 5.0% cement content after 48-h curing

Specimen	Compressive strength (MPa)	Average strength (MPa)
1	1.61	1.57
2	1.36	
3	1.56	
4	1.64	
5	1.60	
6	1.66	

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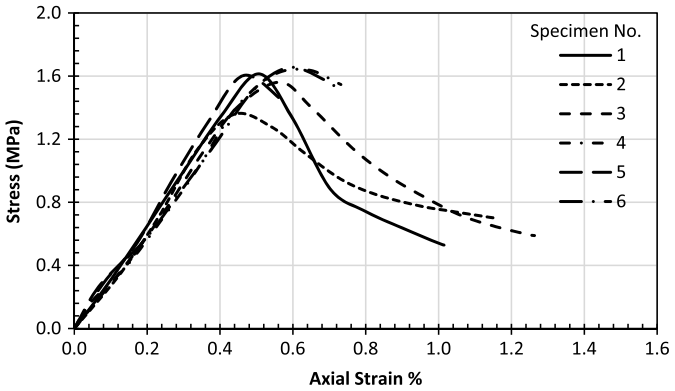

Fig. 4. Stress-strain curves of CMS with 5.0% cement after 2-day curing.

strength reduction. In fact, the UC strengths of the preloaded specimens were slightly higher than those cured for 48 h. Fig. 5(b) shows the stress-strain curves of the UC specimens after preloading. Table 3 summarizes the results of the UC tests for all preloading levels.

The specimens of Trial 3, which were preloaded to 90% of the ultimate UC strength, were microcracked and then tested for the UC strength after a 2-h waiting period. The UC tests of the second loading performed on failed or microcracked specimens from Trial 3 showed strength and modulus (E_{50}) reductions of approximately 50% and 42%, respectively, as indicated in Fig. 6 and Table 4. The secant modulus, E_{50}, was calculated as the slope of the secant line connecting from the origin to the point on the stress-strain curve corresponding to 50% the UC strength. These reductions were also found in the field after microcracking of CMS layers, which will be published in a future paper. Therefore, the procedure of loading the UC specimen to failure and stopping loading soon after reaching the peak strength would create a similar behavior of a microcracked CMS layer in the field.

In Trial 4, three additional CMS specimens were microcracked. Fig. 7(a) presents their stress-strain curves. After the microcracking process, the specimens of Trial 4 were returned to the moist room to continue the curing process and to allow healing of the specimens for strength gaining. After a total curing period of 7 days, the specimens of Trial 4 were tested for the UC strengths, and their stress-strain curves are presented in Fig 7(b). The average strength of these three CMS specimens was 1.59 MPa. In other words, the 7-day UC strength of the microcracked specimens were the same as the ultimate UC strength at 2 days of uncracked specimens. Moreover, the 7-day UC strength of the microcracked specimens were 77% of the design strength of the CMS mix (i.e., 2.1 MPa). This finding is similar to that for the strength of the CMS layer with microcracks in the field at the age of 7 days. Table 5 summarizes the UC strengths of the CMS specimens before and after microcracking for Trial 4.

Fig. 8 shows the CMS Specimen 3 of Trial 4 before microcracking, after microcracking at 48-h curing, and after UC testing at 7-day curing. Only microcracks were observed on the CMS specimens after the microcracking process. Based on the test procedure adopted and the test results obtained in this study, it is

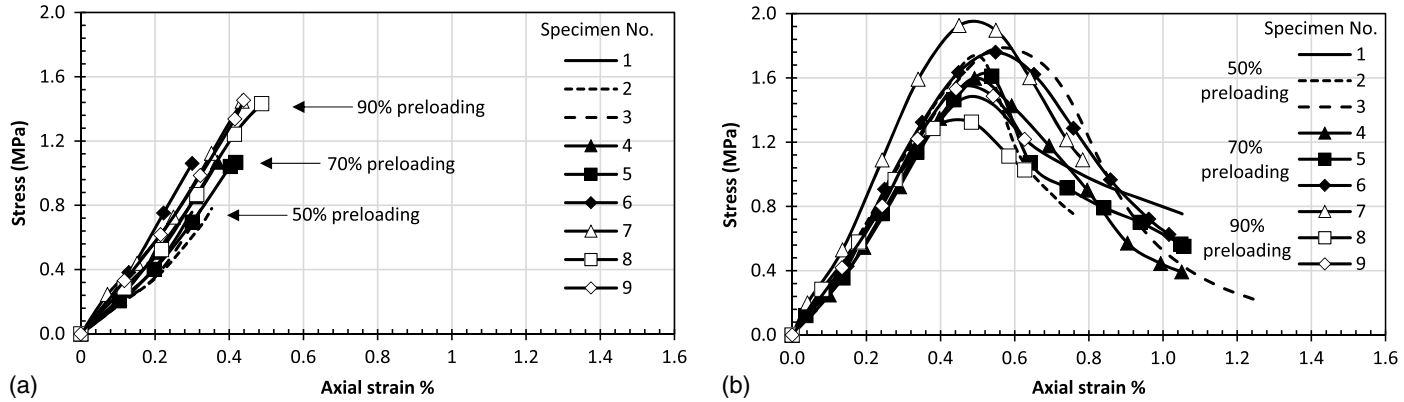

Fig. 5. Stress-strain curves of CMS with 5.0% cement after 2-day curing: (a) loaded to the preloading stress level; and (b) after preloading.

Table 3. Summary of UC test results of CMS specimens with 5.0% cement during and after preloading

Trial	Specimen	Preloading level (%)	Preloading stress (MPa)	Average preloading (MPa)	Compressive strength (MPa)	Average strength (MPa)
1	1	50	0.76	0.77	1.46	1.66
	2		0.78		1.73	
	3		0.77		1.79	
2	4	70	1.07	1.07	1.60	1.66
	5		1.07		1.61	
	6		1.06		1.76	
3	7	90	1.45	1.45	1.93	1.59
	8		1.44		1.32	
	9		1.45		1.53	
Specimen age			48 h		50 h	

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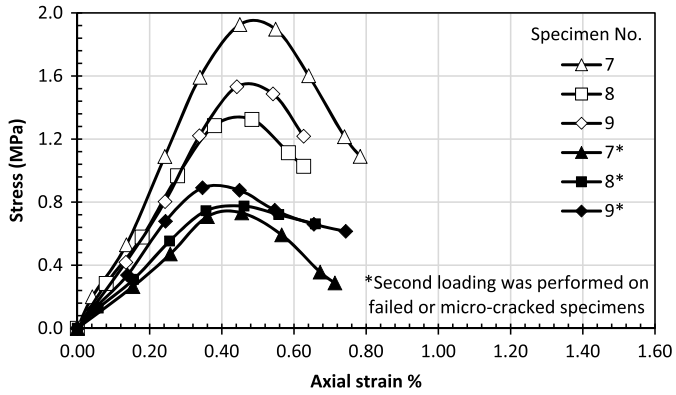

Fig. 6. Stress-strain curves before and after microcracking of CMS specimens in Trial 3.

recommended that the microcracking process in the laboratory should be controlled as loading of the CMS specimen passing the peak compressive strength by less than 0.1% axial strain and releasing the load soon after.

Table 6 presents the effects of the microcracks on the UC strength and modulus (E_{50}) after the 7-day moist-curing period. The uncracked specimens of Mix 2 had a higher UC strength and a lower modulus (stiffness) than those of the microcracked specimens of Trial 4. The average UC strength of the microcracked specimens was approximately 70% that of the uncracked specimens. However, the microcracked specimens had the moduli approximately 150% that of the uncracked specimens because the specimens were preloaded during the microcracking process.

Table 5. UC strengths of CMS specimens before and after microcracking

Trial	Specimen	Compressive strength (MPa)	Average strength (MPa)	Compressive strength (MPa)	Average strength (MPa)
4	1	1.64	1.63	1.58	1.59
	2	1.60		1.54	
	3	1.66		1.65	
Specimen age		48 h		7 days	
Specimen condition		After initial compaction		After microcracking	

Furthermore, the modulus to strength ratio of the uncracked specimens after 7-day curing was approximately 100, which is the same as suggested by Han (2015) for soil-cement. On the other hand, the ratio of modulus to strength for the microcracked specimens increased to approximately 200 because they were preloaded. These results indicate that the presence of the microcracks made the specimen weaker but stiffer.

Conclusions

This note has developed a laboratory procedure to simulate the microcracking process adopted in the field for CMS subgrade. The developed procedure utilized unconfined compression test specimens. The CMS specimens were preloaded to different stress levels compared with the unconfined compressive strength. It was found that when the preloading stress level was lower than the unconfined compressive strength, no microcracking was generated, and their compressive strengths were not degraded. To generate microcracks,

Table 4. Unconfined strengths and moduli of failed CMS specimens subjected to second loading

Trial	Specimen	Specimen age (h)	Compressive strength (MPa)	Average strength (MPa)	Modulus, E_{50} (MPa)	Average modulus, E_{50} (MPa)	Strength reduction (%)	Modulus E_{50} reduction (%)
3	7	50	1.93	1.59	454	372	50	42
	8		1.32		338			
	9		1.53		324			
3[a]	7[a]	52	0.73	0.80	179	216		
	8[a]		0.78		207			
	9[a]		0.89		261			

[a]Second loading (i.e., Trial 3) was performed on failed or microcracked specimens.

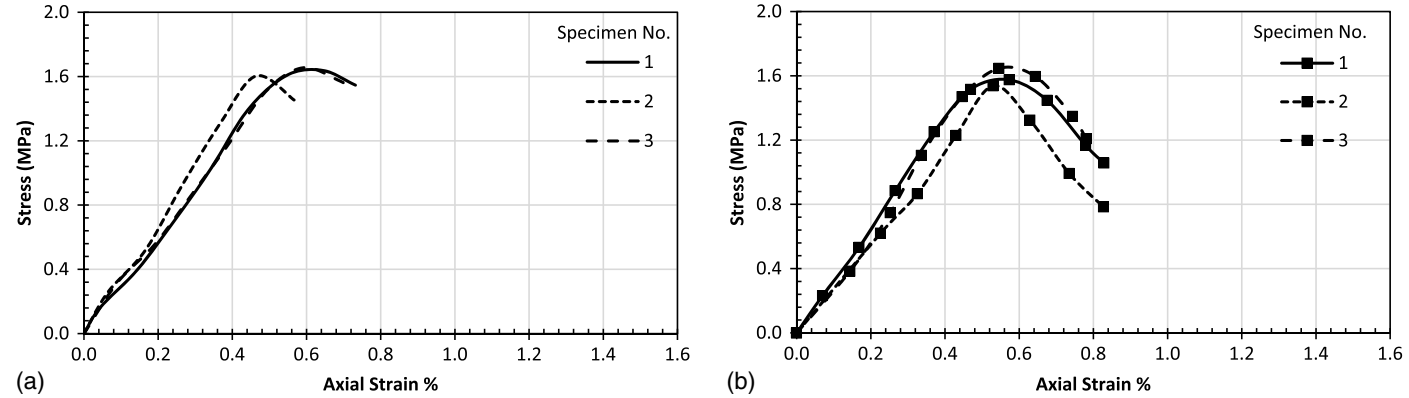

Fig. 7. Stress-strain curves of CMS specimens with 5% cement in Trial 4: (a) during microcracking process after 2-day curing; and (b) microcracking specimens at 7-day curing.

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(a) (b) (c)

Fig. 8. CMS Specimen 3 of Trial 4: (a) before microcracking; (b) after microcracking at 48-h curing; and (c) after UC testing at 7-day curing.

Table 6. Effects of microcracks in CMS specimens on the UC strength and modulus (E_{50}) at 7-day of curing

Specimen condition	Specimen	Compressive strength (MPa)	Average strength (MPa)	Modulus, E_{50} (MPa)	Average modulus, E_{50} (MPa)	Modulus to strength ratio
Uncracked specimens (Mix 2)	1	2.00	2.10	144	210	100
	2	2.11		220		
	3	2.41		274		
	4	2.32		186		
	5	1.67		224		
Microcracked specimens (Trial 4)	1	1.58	1.59	337	306	193
	2	1.54		273		
	3	1.65		308		

this study found that the preloading stress level must be equal to the unconfined compressive strength of the CMS specimen. The CMS specimens right after microcracking had UC strengths of approximately 50% of that before microcracking and a modulus of approximately 42% of that before microcracking. However, the specimens microcracked after a 48-h curing period had 70% of the strength of the uncracked specimens, and their moduli was approximately 150% of the moduli of the uncracked specimens. The microcracked specimens had a modulus to strength ratio two times the uncracked specimens.

Based on the finding of this research, it is recommended that to achieve a representative microcracking process in the laboratory, loading of the CMS specimen (cured for 48 h) should continue past the peak compressive strength by less than 0.1% axial strain and then the load should be released to zero. The microcracked specimens should be returned to the curing room for further curing to a desired curing period.

Acknowledgments

This research was jointly sponsored by Kansas Department of Transportation (KDOT) and Terracon Consultants, Inc. Mr. Isaac M. Ferguson, the Geotechnical Specialist at KDOT, is the project monitor. Their support and guidance are greatly appreciated.

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文献 3/9

Variable secondary porosity modeling of carbonate rocks based on μ-CT images

Xin Nie; Chi Zhang; Chenchen Wang; Shichang Nie; Jie Zhang; Chaomo Zhang

Open Geosciences (2019); DOI: 10.1515/geo-2019-0049

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Research Article

Xin Nie*, Chi Zhang, Chenchen Wang, Shichang Nie, Jie Zhang, and Chaomo Zhang

Variable secondary porosity modeling of carbonate rocks based on μ-CT images

https://doi.org/10.1515/geo-2019-0049
Received Dec 05, 2018; accepted Aug 27, 2019

Abstract: As an essential carbonate reservoir parameter, porosity is closely related to rock properties. Digital rock physics (DRP) technology can help us to build forward models and find out the relationship between porosity and physical properties. In order to prepare models for the rock physical simulations of carbonate rocks, digital rock models with different porosities and fractures are needed. Based on a three-dimensional carbonate digital rock image obtained by X-ray microtomography (μ-CT), we used erosion and dilation in mathematical morphology to modify the pores, and fractional Brownian motion model (FBM) to create fractures with different width and angles. The pores become larger after the erosion operation and become smaller after the dilation operation. Therefore, a series of models with different porosities are obtained. From the analysis of the rock models, we found out that the erosion operation is similar to the corrosion process in carbonate rocks. The dilation operation can be used to restore the matrix of the late stages. In both processes, the pore numbers decrease because of the pore surface area decreases. The porosity-permeability relation of the models is a power exponential function similar to the experimental results. The structuring element B's radius can affect the operation results. The FBM fracturing method has been proved reliable in sandstones, and because it is based on mathematics, the usage of it can also be workable in carbonate rocks. We can also use the processes and workflows introduced in this paper in carbonate digital rocks reconstructed in other ways. The models we built in this research lay the foundation of the next step physical simulations.

Keywords: carbonate, digital rock physics, secondary porosity, fracture, mathematical morphology, fractional Brownian motion

1 Introduction

Approximately 50% of oil and gas are stored in carbonate rocks around the world. Porosity is one of the critical parameters for reservoir evaluation. Unlike typical sandstone reservoirs with single-porosity systems, reservoirs in carbonate rocks commonly have multiple-porosity systems which cause petrophysical heterogeneity [1]. Porosity in carbonate rocks can be classified based on the timing of porosity evolution into primary pores and secondary pores [2]. Primary porosity includes pores supported by the particles when sediments were depositing. Secondary porosity includes pores form as a result of post-depositional dissolution. There are also natural fractures in carbonates. Due to the complexity of carbonate rock pore space, it's difficult to carry out the petrophysical interpretation for predicting the transport properties and producibility of carbonate reservoirs [3].

To evaluate the pore properties, first work we need to do is forward modeling which reveals the relationship between porosity and physical properties. The digital rock physics (DRP) technology has been helpful in the forward modeling in sandstones [4–8], carbonates [3], natural gas reservoirs [9], fractured rocks [10], thinly bedded rocks [11], and shales [12, 13] in recent years. With DRP, rock physical properties such as elastic moduli [14], acoustic properties [15], electrical properties [16], and fluid flow properties [17]. The digital rock models with different porosities are needed in the simulations. Arns *et al.* (2005) [3] characterized the carbonates in a pore-scale using X-ray mi-

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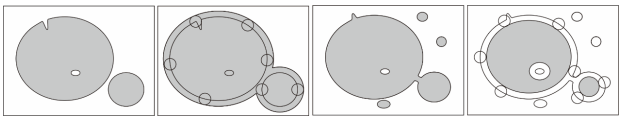
crotomography (μ-CT) which is an accurate way to build a digital rock model. However, for a single rock sample, the porosity is fixed. Therefore, lots of rock samples need to be scanned to build rock models with different porosities. First, it isn't economic to scan so many samples; second, for different carbonate rocks, the secondary porosities formed in the diagenesis are different, which causes the trend of simulation results not so visible. Other mathematical methods have been used to generate models with different porosities, including stochastic methods [18, 19] and process-based methods [20], etc. There are too many uncertainties using these stochastic methods. Recently, many studies have been done to simulate the CO2 injection, and numerical study of carbonate dissolution by CO2 and its effect on the transport properties are presented [21–25]. These process-based methods can provide us accurate models with different porosities after the present dissolution stage. However, the stages before present still cannot be acquired. Therefore, to get more models of the stages before present, we introduce dilation and erosion in mathematical morphology to process the μ-CT scanned carbonate rock image. We also introduce the application of fractional Brownian motion algorithm (FBM) to generate natural fractures in μ-CT scanned images in carbonate rock to get fractured models.

2 Methodology

2.1 Dilation and erosion in mathematical morphology

Mathematical morphology is a standard method used in image processing. The principle of mathematical morphology method is to use a structure element with a specific shape to measure and pick up the equal shape in the image to analyze the image. There are four basic operation modes: dilation, erosion, opening, and closing [26]. The opening and closing operations have been used on sandstones to generate fluid distributions in the pore space [27].

In mathematical morphology, dilation operation and erosion operation are opposite processes. The dilation operation enlarges the target image, while the erosion operation reduces the target image. The mathematical morphology is based on set theory. Sets in mathematical morphology represent the shapes which are manifested on binary or gray tone images. A and B are two sets; $A \oplus B$ means the dilation of A by B, while $A \ominus B$ means the erosion of A by B. Eq. 1 and 2 are expressions of dilation and erosion,



(a) Original image M (b) Dilation of M by B (c) Original image N (d) Erosion of N by B

Figure 1: Sketch of dilation and erosion

respectively.

$$A \oplus B = \{x \,|B_x \cap A \neq \emptyset\} \tag{1}$$

$$A \ominus B = \{x \,|B_x \subseteq A\} \tag{2}$$

In Eq. 1 and 2, B_x is the translations of set B by x; $\emptyset$ is null. Figure 1 is a 2D sketch of dilation and erosion. Figure 1(a) shows an original image M. In this case, B is a disc with a fixed radius, and the origin is the center of the circle. After the dilation of M by B, the image $M \oplus B$ is as the grey part shown in Figure 1(b). The original image is enlarged with a certain radius outward. Figure 1(c) is another original image N. The grey part shown in Figure 1(d) is $N \ominus B$, *i.e.* the erosion of image N by B. In this case, M and N are images, and B is structuring element. In our study, we use a sphere structuring element for 3D processing.

2.2 Fractional Brownian motion

Experimental evidence [28] shows that the roughness of natural fracture internal surfaces in rocks follows self-affine fractal statistics. In this paper, we adopt the idea of considering a rock surface without overhangs. The surface height is described by a function $h(x, y)$, with the coordinates x and y lying in the mean plane of the fracture [10]. Self-affinity of the rough surfaces implies scale invariance. When $x \rightarrow \lambda_x x$, $y \rightarrow \lambda_y y$, $h \rightarrow \lambda_h h$, we assume that $\lambda_x = \lambda_y = \lambda$ and $\lambda_h = \lambda_H$, and then we have the following equation:

$$h(x, y) = \lambda^{-H} h(\lambda x, \lambda y) \tag{3}$$

where λ is the scale factor, H is the roughness (Hurst exponent). For the surfaces of fractures presented in rocks, the roughness exponent is around H = 0.8, which has no relationship with the lithology and fracture mode [29].

Fractional Brownian motion (FBM) is a standard mathematical model for self-affine fractals, which fits the following equation:

$$E\left[\left(G_H(X + h) - G_H(X)\right)^2\right] = |h|^{2H}\sigma^2, (0 < H < 1) \tag{4}$$

where $G_H(X)$ is a random walking which satisfies the Fractional Brownian motion; E is mathematical expectation; H

is the roughness exponent; σ is standard deviation; σ_0 is a random initial standard deviation between 0~1; j is the times of iterative operations.

According to central-limit theorem, Eq. (4) can be written as:

$$\sigma^2(h) = \sigma^2(1)h^{2H} \tag{5}$$

standard deviation σ can be calculated as follow equation:

$$\sigma^2{}_j = \frac{\sigma^2{}_{j-1}}{2^H} = \frac{\sigma^2{}_0}{(2^H)^j}\left(1 - \frac{2^H}{4}\right) \tag{6}$$

There are many algorithms to generate FBM data. In this paper, we adopt a corrected successive random addition (CSRA) algorithm to generate the FBM surfaces [30, 31].

The workflow of this algorithm for 2D square surface is as follows (Figure 2):

(1) First select random numbers from and add them to the 4 corner points of the square, denoted by number 1, where stands for a Gaussian random number generator with a mean of 0 and a variance σ_0^2.
(2) Calculate the midpoint denoted by number 2 by linear interpolation based on the 4 corner point values.
(3) Select random numbers from (σ_1^2 can be derived from Eq. (6)) and add them to all the existing points;
(4) Calculate the midpoints on each side of the square denoted by number 3 by linear interpolation using two adjacent corner values obtained in step 3;
(5) Select random numbers from $N(0\sigma_2^2)$ (σ_2^2 can be derived from Eq. (6)) and add them to all existing points.
(6) Repeat the process, keep performing linear interpolations and adding random numbers from $N(0\sigma_j^2)$ up to the nth level, resulting in a total of $(2^n+1)^2$ points. Then the generation of the 2D rough surface is done (*e.g.*, Figure 3).

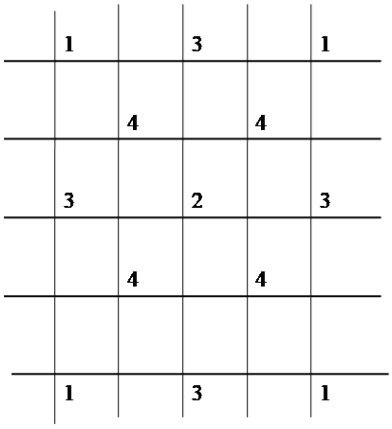


Figure 2: Point denoting and value assignment order of SRA

Figure 3: Rough surface created by using SRA algorithm

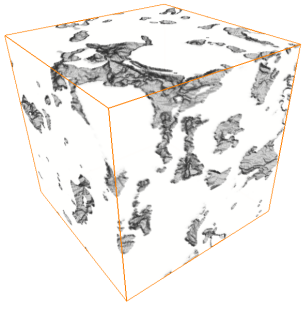


Figure 4: The μ-CT scanned digital rock sample C1 of carbonate rock used in this study

3 Carbonate rock secondary porosity modeling results

3.1 μ-CT based 3D digital rock

In this study, we use a 3D image of a carbonate rock (named C1) obtained using Microcomputed X-ray tomography (μ-CT) from Imperial College London [32]. The segmented 3D carbonate digital image is shown in Figure 4, which has 400×400×400 voxels. The resolution is 2.85μm/voxel, and the porosity is 17.6%.

3.2 Modeling of carbonate digital rock with different porosities

One most important component of carbonate reservoir pore space is secondary porosity formed by corrosion. Corrosion process in the carbonate rocks is similar to the erosion operation in mathematical morphology. Therefore, we use erosion to mimic the later stages in a corrosion process. Based on the digital rock model obtained by μ-CT scanning and using the erosion operation, we generated the digital carbonate rocks with different porosities.

Figure 5 shows an example result of digital carbonate rocks with different porosities after the erosion. In these digital rocks, pixels with value 0 (shown as black parts)

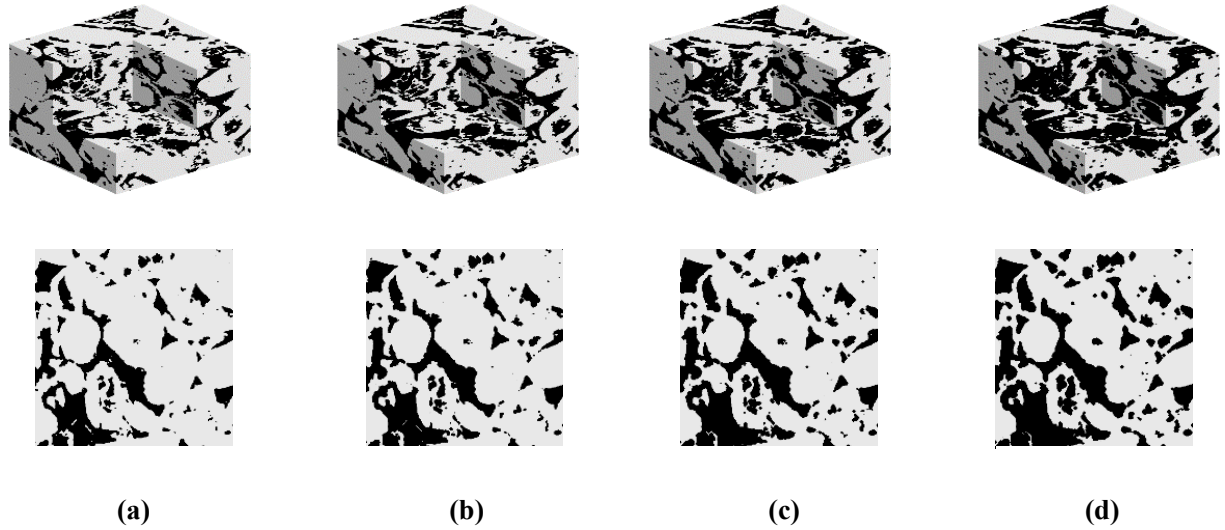


Figure 5: Digital rocks and their slices of carbonates with different porosities generated after the erosion process. (a) Original image (Φ= 23.26%); (b) Image after 1 erosion (Φ = 29.22%); (c) Image after 2 erosion (Φ= 34.93%); (d) Image after 3 erosion (Φ = 40.4%)

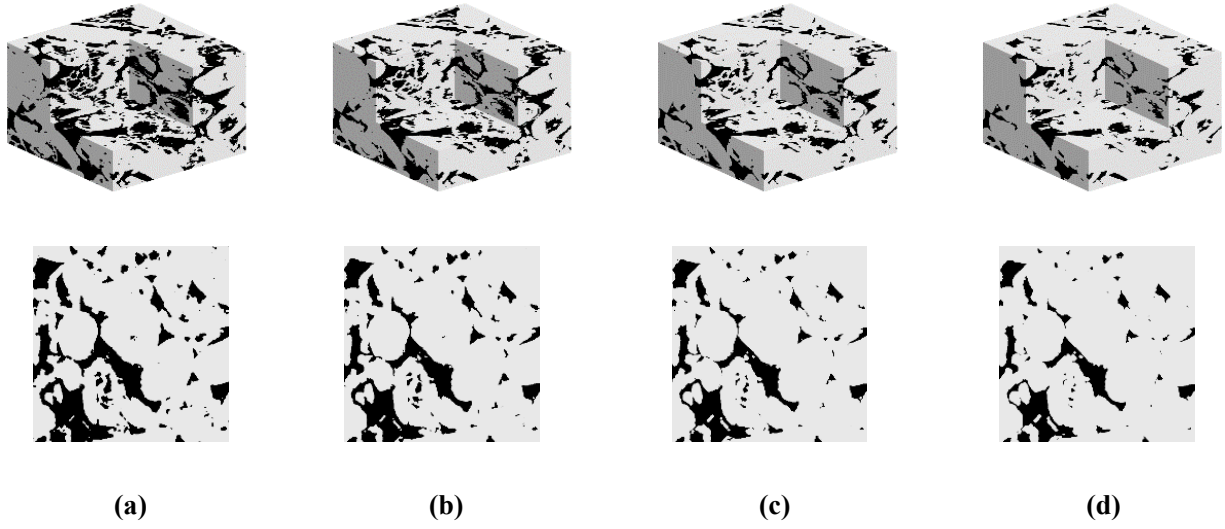


Figure 6: Digital rocks and their slices of carbonates with different porosities generated afterdilation process. (a) Original image (Φ= 23.26%); (b) Image after 1 dilation (Φ= 17.76%); (c) Image after 2 dilation (Φ= 13.5%); (d) Image after 3 dilation (Φ= 10.22%)

represent pore space, while those with value 1 (shown as lighter parts) represent the matrix. The structuring element is a sphere with a radius of 1 pixel. Figure 5a is the original core and its one slice, and Figure 5b~d is the cores and their slices after 1~3 times of erosion. As we can see in Figure 5, after the sphere goes around the pore space, the matrix is eroded, resulting in an increase of porosity. And during the process, pores used to be close to each other can merge as one larger pore. Through this way, we can increase the porosity of the carbonate digital rock models.

With the erosion, we generated digital rocks with higher porosities. Meanwhile, by using the dilation process to dilate the matrix, digital rocks with lower porosities can be acquired. The structuring element is the same as in the erosion process. Figure 6 shows the dilation results. Figure 6a is the original core and its one slice, and Figure 6b~d is the cores and their slices after 1~3 times of dilation. As we can see in Figure 6, after the sphere goes around the pore space, the matrix is dilated, resulting in a decrease of porosity. Some small pore space vanishes into the matrix. This process can be seen as reverse of a corrosion process. The cores with lower porosities can be seen as earlier stages of the corrosion process. Through this way,

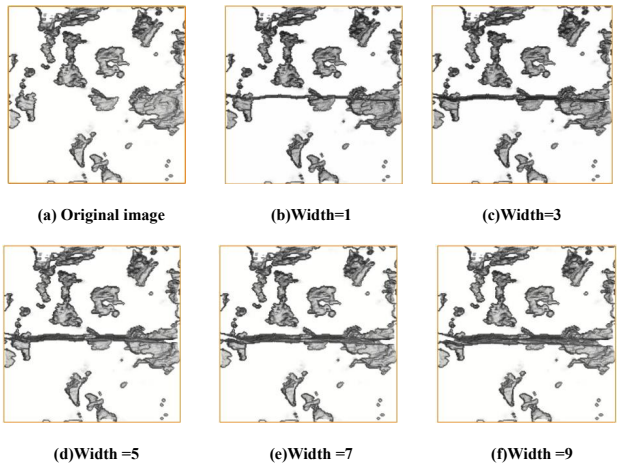


Figure 7: Sections of fractured carbonate digital rock models with different fracture widths

we can decrease the porosity of the carbonate digital rock models.

3.3 Modeling of fractured rocks

Fractures are another critical space for carbonate reservoirs. They can be both reservoir space and seepage channels. But due to the difficulty in coring and doing experiments for fractured formations, the research about fractured carbonate reservoirs hasn't been well done before. Digital rock technology can be introduced to provide fractured rock models which can be the basis of numerical simulations. In this study, we use fBm to generate fractures with different widths and angles.

For a digital rock with 400×400×400 voxels, we choose a 400×400 pixels area in the rough surface $z=f(x,y)$ generated by using FBM in Section 2. Then we set it as the lower surface of the fracture. Set the width of the fracture as d, then the upper surface $Z=z+d$. Because 0 represents pore space, we set the pixels between upper and lower surfaces as 0. By changing the value of d, we can get different fracture widths. By setting the pixels where the fractures are supposed to be as 0 in the original image, we can obtain the digital rock models with fractures different in widths (*e.g.*, Figure 7).

According to the attitudes of the fractures, they can be divided into "high angle" (dip angle > 75°), "bevel" (15° < dip angle < 75°) and "low angle" (dip angle < 15°) fractures. To mimic different attitudes, by rotating the fractures, we can obtain fractured digital rocks with varying angles of dip (*e.g.*, Figure 8).

4 Discussion

4.1 Pore geometry, numbers and porosity changes of models after erosion and dilation

Because the secondary corrosion pore space in carbonate rocks is formed through the corrosion process which is similar to dilation and erosion operations in mathematical morphology, the dilation and erosion processes are adopted in this research to simulate different stages of eroded pores. Our goal is to have variable porosity but retain a similar pore structure. Therefore, we use the maximal ball algorithm [32] to extract the pore network (Figure 9) and statistic the pore and throat numbers of all radiuses. The balls and sticks in Figure 9 represent pores and throats, respectively, and different colors show different pore sizes. From blue to red, the pore size increases.

From the network models, the pore/throat radius distribution curves of the digital rock models with different porosities can be obtained (Figure 10). In Figure 10, the frequency in y-axis stands for the count of the radius in x-axis divided by the whole count. C11~C15 are the names of digital rock models after erosion, and C1_1~C1_7 are the ones after dilation. As we can see in Figure 10, the peak of the curve slightly shifts right in the erosion process, while it moves left in the dilation process. This reveals the pores are somewhat enlarged or shrunken during the respective operations.

The numbers of pores are counted (Figure 11). In Figure 11, each dot represents a dilation or erosion processing point. In both processes of erosion and dilation, the pore numbers decrease. This also indicates that in the dilation process, the small pores vanish; and in the erosion process, the nearby pores merge into bigger ones. Figure 12 shows the porosity change rate during the operations. As we can see in Figure 12, the more the processes have been done, the slower the erosion and dilation process is. Because each process only adds on or peels off a fixed thickness of one voxel on the pore surface, the speed of change relates only to the surface area. As a result of small pores vanishing or nearby pores merging, the pore surface decreases in both processes.

4.2 Porosity-permeability relation

The relationship between porosity and permeability is always concerned by petroleum engineers and petrophysicists. Many studies reveal that in on a particular area or

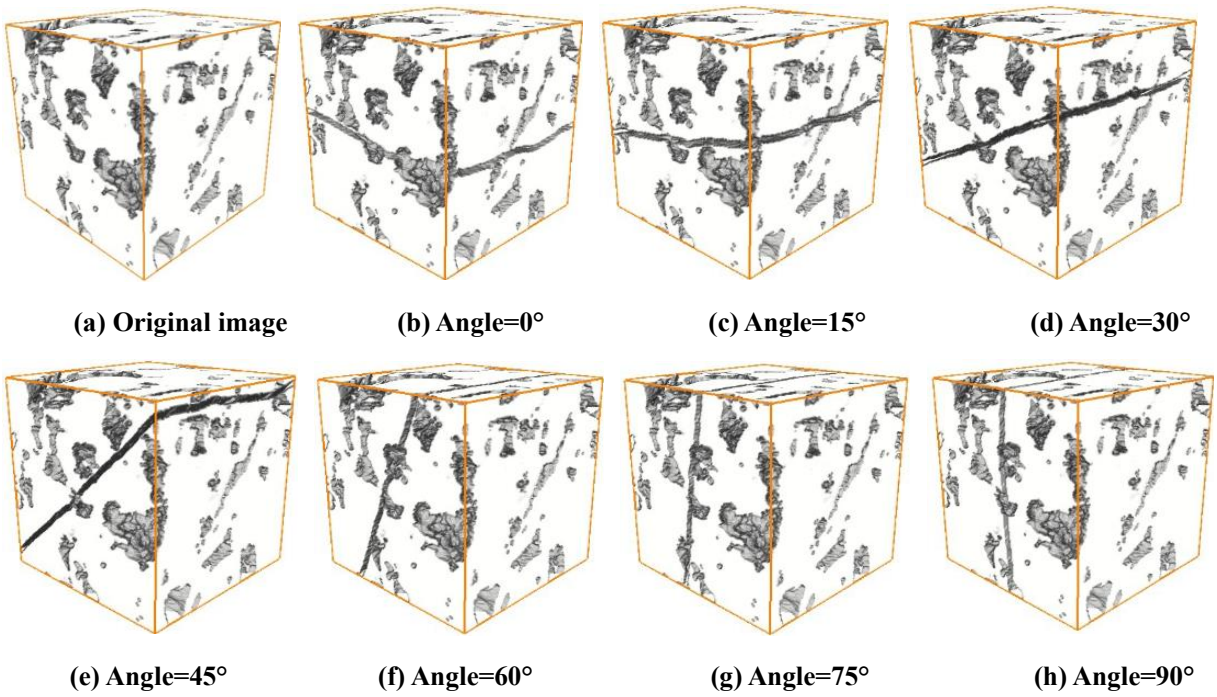


Figure 8: Models of fractured carbonate digital rock models with different fracture angles

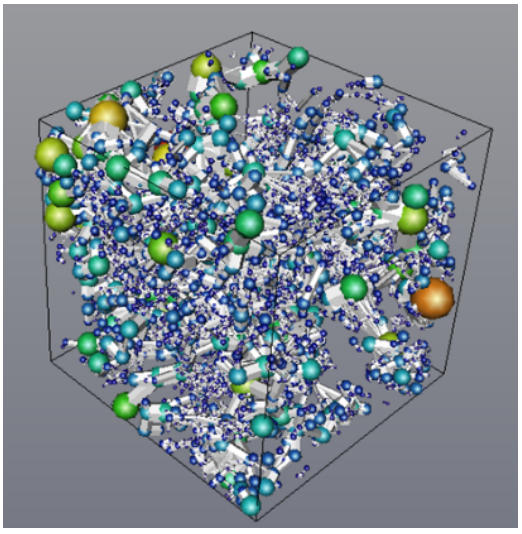
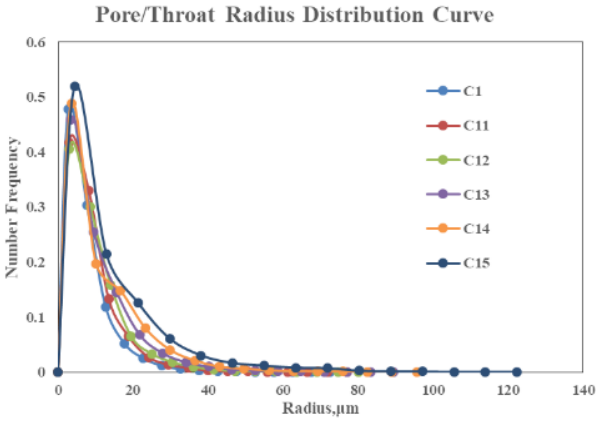


Figure 9: Pore network of C1 digital rock model

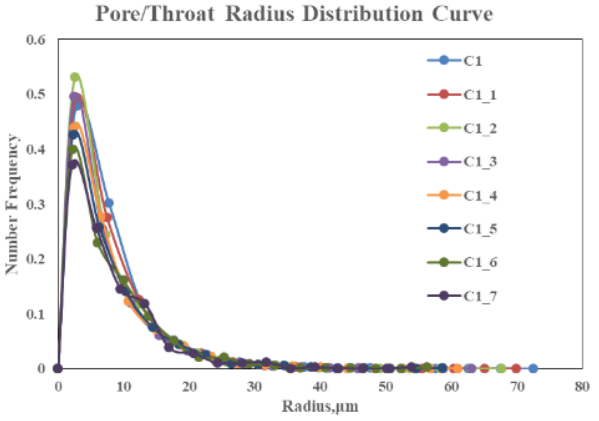
for one specific kind of carbonate, the permeability can be a power function of porosity (*e.g.*, Rashid *et al.*, 2015). We have some carbonate rock samples with laboratory experimental permeability results (Figure 13). The trend is also power exponential. As we can see, there are rock samples with minimal porosity and high permeability. This reveals there might be micro-fractures in those samples. Based on the network models, the permeabilities of the digital rock models are calculated by using the single-phase flow permeability calculation method based on Darcy's law [34]. The simulation results are shown in Figure 14. As we can see in Figure 14, the trend is similar to the experimental results of carbonate rock samples from a specific area acquired in the laboratory. Because the simulations are based on the same sample, the trend of the simulation results is more evident than the laboratory results. This indicates the method used to modify the porosity in this paper is reasonable. The models can be used in the next-step physical simulations.

4.3 The effect of *B*'s radius on porosity changing

In this study, we adopt a sphere with a radius of 1 voxel as the structuring element *B*. The porosity changes by one layer of voxel on the pore surface during each process. Because the resolution of this 3D digital rock is 2.85μm/voxel, the pores' radiuses change 2.85μm (by add or minus) each time. In a pixel image, processing with the sphere structuring elements with different radius can cause different results. We compare the dilation and erosion results of the digital rock by a sphere with a radius of one voxel for five times and by a sphere with a radius of five voxels for one time. The statistic results are shown in Table 1. C13 and C1_3 are the results of erosion and dilation by *B* with a



(a) Erosion results



(b) Dilation results

Figure 10: Pore/throat radius distribution curves of the digital rock models

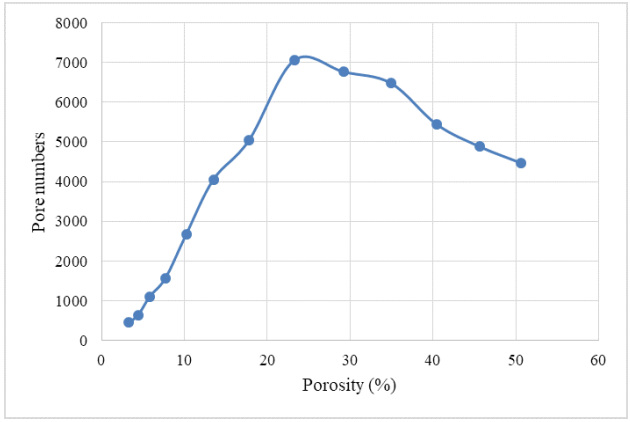


Figure 11: Pore numbers of digital rocks with different porosities

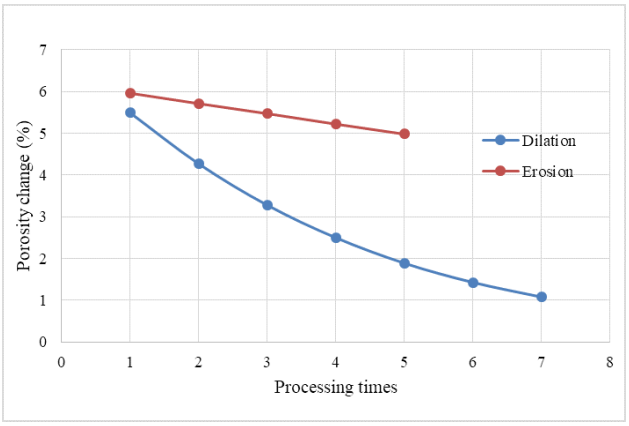


Figure 12: Porosity change variation with the processing times

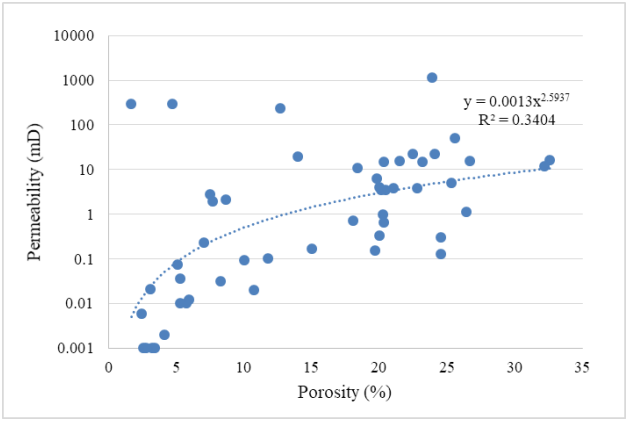


Figure 13: Laboratory experimental permeability results of carbonate rock samples

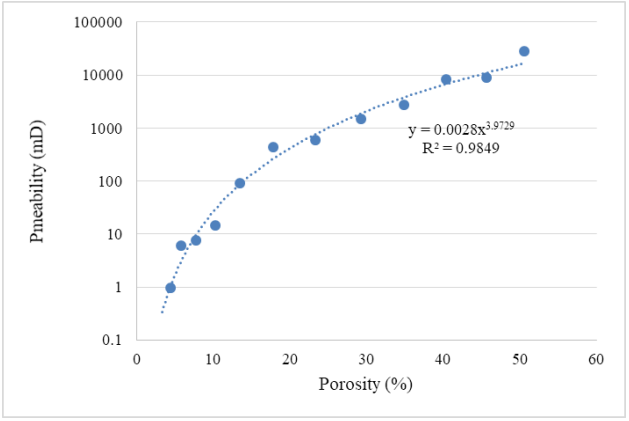


Figure 14: Permeability simulation results of the models generated after erosion and dilation

radius of one voxel for three times, and C13r and C1_3r are the ones by B with a radius of three voxels for one time. There are differences in porosity, permeability, and pore numbers. The porosity and permeability change more when B's radius is three voxels than one voxel. Figure 15 shows the pore networks of the four digital rocks. From the network images, we can see some differences in pore connection. In the dilation process, when B's radius is three voxels, the smaller throats and pores are less. This makes

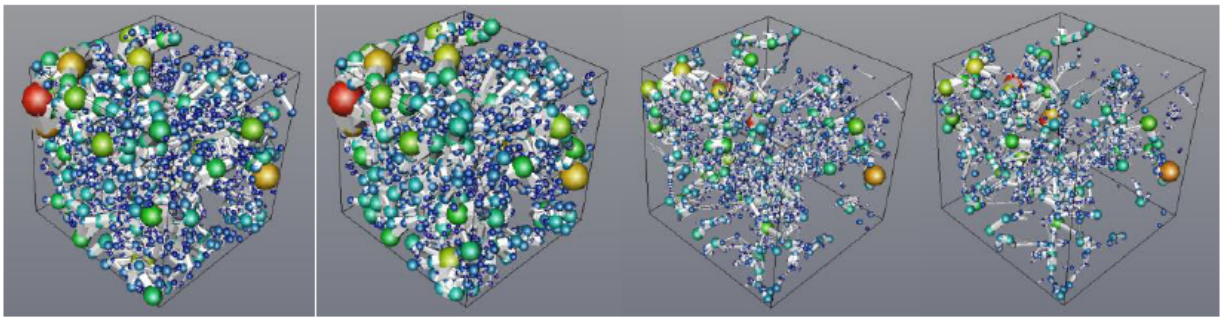


Figure 15: Pore networks of digital rock models with different porosities

Table 1: The results of processing with structuring elements *B* with different radius

Model	Porosity/%	Permeability /mD	Pore Numbers
C1_3	10.22	14.8	2674
C1_3r	8.21	6.87	1805
C1_13	40.4	8434.1	5446
C1_13r	44.75	13559.4	3656

the permeability lower. This indicates when a large radius is adopted for a sphere structuring element, more details will be erased earlier than using a one-voxel-radius structuring element.

4.4 Method advantage and limitation

Based on the μ-CT image, erosion and dilation operations are adopted to generate different porosities in this study. Other mathematical methods have been used to construct models with different porosities before, including stochastic methods [18, 19], process-based methods [20]. When using these methods, each time of the process, the porosities are different, and so are many other parameters. The pore shape of one model can't be guaranteed to maintain the same way as the others. In this study, operations are based on the same μ-CT image. Therefore, the basic pore structures and distributions of the models are the same. And based on the dissolution process, many researchers have done the numerical study of carbonate dissolution by CO2 and its effect on the transport properties [21–25]. These process-based methods can provide us accurate models with different porosities after the present dissolution stage. In this study, the erosion operation can approximately simulate the corrosion effect, while the dilation operations can restore the matrix before the current status. By using dilation and erosion, both higher and lower porosities can be generated.

However, dilation and erosion operations are used under assumptions, namely: (a) the rock corrosion happens evenly and simultaneously at each direction on each pore surface, and (b) the pore will be corroded no matter it is connected to the primary fluid channel or not. Therefore, these methods have nothing to do with physical properties such as the heterogeneous of minerals and gravity. More studies need to be done in the future.

4.5 Reliability of fractured models

We use the FBM method to model the fractures in carbonate rocks based on μ-CT images. The surfaces of natural fractures in rocks have very high self-correlation and self-affine property. The FBM is a standard self-affine model. By using this method, we can generate a fracture-like rough surface. Zhao *et al.* (2013) [10] applied the FBM method to create fractures in sandstones. And based on the fractured models, the electrical simulations were done. By comparing the simulation results with the experimental results, the FBM method was proved efficient. In this study, we adopt this same method into the carbonate rocks. Because the method is based on mathematics, and it has nothing to do with the rock properties, the usage of the FBM method in carbonate rocks is also workable. The fractured digital rock models can be used in the next step physical simulations.

μ-CT is one common but not the only way to build 3D digital rocks. Because the methods and workflows introduced in this paper are purely based on mathematics, they can also be used in carbonate digital rocks reconstructed in other ways, such as using process-based methods.

5 Conclusions

To generate multi-porosity and fractured digital rock models of carbonate rocks for the physical simulation, the erosion and dilation of mathematical morphology and FBM methods were performed based on a μ-CT image. The main results can be summarized as follows:

(1) Erosion operation is similar to the corrosion process in carbonate rocks. The dilation operation is a reverse to the erosion, which can be used to restore the matrix of the former stages. The pore radius distribution frequency curves show the pore space change in the models during the operations: pores are getting larger and smaller after the erosion and dilation, respectively. In both erosion and dilation operations, the pore numbers decrease because of the pore surface area decreases. The porosity-permeability relation is a power exponential function similar to the experimental results. The structuring element B's radius can affect the operation results.

(2) The fracturing method has been applied in sandstones and proved reliable. Because the method to generate fractures is only based on mathematics and has nothing to do with the rock properties, the usage of the FBM method in carbonate rocks is workable.

(3) The analysis results reveal the digital rock models with different porosities and natural fractures are reasonable. The methods and workflows introduced in this paper can also be used in carbonate digital rocks reconstructed in ways other than μ-CT based images such as using process-based methods. The models we build in this research can be used in the next physical simulations. However, the mathematical morphological methods are used based on assumptions of equal corrosion of each pore. Factors such as heterogeneity of minerals and gravity need to be considered in future studies.

Acknowledgement: This research is supported by National Natural Science Foundation of China (grant no. 41504094 and 51704033) and the National Science and Technology Major Project (No. 2017ZX05032003-005). We thank Imperial College for the carbonate digital rock sample. We thank four anonymous reviewers for their constructive suggestions for improvements to the manuscript.

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文献 4/9

Study on the characterization of pore structure and main controlling factors of pore development in gas shale

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Journal of Natural Gas Geoscience (2020); DOI: 10.1016/j.jnggs.2020.09.003

证据用途：其他合作论文背景；张弛老师角色：末位合作者；证据等级：C

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Original research paper

Study on the characterization of pore structure and main controlling factors of pore development in gas shale

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Received 29 November 2019; revised 18 August 2020; accepted 4 September 2020
Available online 17 September 2020

Abstract

Shale has a complex pore structure, which leads to complex petrophysical characteristics. The porosity, pore type, pore size, pore volume, TOC, and other petrophysical parameters of shale all affect the shale gas content. Therefore, using multiple methods to jointly characterize the pore structure of shale with different gas content and study on the main controlling factors of pore development have an important guiding role in the exploration and development of shale gas. In this study, we carry out gas shale samples in different areas. The samples collected from wells in the Sichuan Basin have high gas content, collected from wells near the Pengshui, and Tongren areas have medium gas content and collected from wells near the Guiyang area have low gas content. Shale in these areas with different gas production has been characterized, by the integration of geophysical and petrophysical data, in terms of pore type and pore connectivity, including X-ray diffraction, low-field Nuclear Magnetic Resonance (NMR), low-pressure N_2 and CO_2 adsorption isotherms, porosity, permeability, thin-section observations, scanning electronic microscopy (SEM) and mercury injection capillary pressure (MICP) measurement. The results show that the pore types are clay mineral intercrystalline pores and intergranular pores, rock skeleton mineral pores, organic pores, and micro-fractures. The high gas-bearing shale has a small pore size, a large specific surface area, and pore volume, and the T_2 distribution is bimodal, with a peak value around 1 ms. It shows that micropores and mesopores are widely developed in high gas-bearing shale, which provides the main place for adsorbing gas, especially shale with high TOC content. During the hydrocarbon generation process, a large number of organic pores favorable for gas adsorption will be produced. The low gas-bearing shale has a large pore size, a small specific surface area, and pore volume, and the T_2 distribution presents a bimodal shape with a wider peak and the lowest spectral value. This shows that low gas-bearing shale mainly develops macropores and micro-fractures, and the pore connectivity is good, especially the interconnection between small pores and fractures, which leads to gas escape. In the high gas-bearing samples, bone needles and radiolaria are distributed in large quantities, and the silicon content of biological sources is high, which provides a lot of organic matter. In the low gas-bearing samples, the horizontal layer is not developed and the mineral particles are evenly distributed.

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Keywords: Shale; NMR; N_2 and CO_2 adsorption isotherms; Thin-section observations; SEM; MICP

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Peer review under responsibility of Editorial office of *Journal of Natural Gas Geoscience*.

https://doi.org/10.1016/j.jnggs.2020.09.003

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1. Introduction

With the exploration and development of unconventional resources, research on shale gas is becoming more and more important [1]. According to the estimates issued by the Research Center of Oil and Gas Resources of the Ministry of Land and Resources in China [2], the technically recoverable shale gas resource in China is about 886.4 tcf (25.1×10^{12} m^3), which is even greater than that of the United States (24.4×10^{12} m^3). However, by comparing the geological conditions of the shale gas production areas in the United States and China. We found there is a big difference between them. The shale gas in the United States is mainly developed in the North American platform where the crust is stable and the structure is gentle. The tectonic movement is dominated by lifting and there is no obvious wrinkle change [3−5]. Shale gas reservoirs in China are buried deep and the structure is relatively complicated. The shale gas enrichment areas are mostly located in mountainous areas with complex surface topography [6,7]. Therefore, the research method of shale gas in the United States cannot be simply applied to the study in China. However, shale has complex diagenetic processes and heterogeneity characteristics, so the study of the petrophysical properties of shale is not comprehensive. Therefore, a comparative study of shale in different gas producing areas in China can provide a reference for the research of petrophysical properties of shale [8,9].

The pores and throats of shale are the main places for gas adsorption and dissociation, so the pore characteristics of shale have always been the main factors affecting gas content. The current research on shale pore characterization mainly uses gas adsorption and high-pressure mercury intrusion experiments to obtain data such as pore size, specific surface area, and pore volume. Carbon dioxide molecules are very small and can enter tiny pores. Therefore, CO_2 adsorption experiments are often used to study the micropore structure characteristics of shale, discuss the main controlling factors of micropore development, and analyze its influence on shale gas generation [10,11]. Both N_2 adsorption and high-pressure mercury intrusion experiments can reflect the distribution of shale from micropores to macropores, but in terms of data accuracy, N_2 adsorption data are often used to characterize mesopore structure [12,13], while high pressure The mercury experiment is often used to characterize the structural characteristics of larger connected pore throats [14,15]. Nuclear magnetic resonance technology is one of the important methods to characterize the microscopic pore structure of rocks and evaluate the effectiveness of reservoirs [16,17]. As NMR experiments can reflect the pore characteristics of shale non-destructively, quickly and accurately, more and more scholars have used them in the petrophysical research of shale [18,19]. The type and microstructure of shale pores and minerals can be directly described by thin-section observation and SEM observation [20]. Currently, the most widely used are FE-SEM and polarizing microscope. Through shale slices, we can observe the microscopic morphology of minerals, biological fossils, pore types, pore morphology, and pore development characteristics. The pore characteristics, microstructure, and mineral types of shale with different gas content are quite different, and they are also the key influencing parameters of shale gas content. Therefore, for the petrophysical comparative study of shale with different gas content, a research method combining multiple tests has gradually formed.

To characterize the petrophysical properties of gas shale with different gas content, we combine multiple methods to conduct research. In this study, (i) the pore type was characterized by FE-SEM observation, pore connectivity and pore size distribution characteristics of shale with different gas content were characterized based on N_2 and CO_2 adsorption isotherms, NMR, and MICP tests; (ii) we studied the influence of the mineral microstructure of shale samples on the gas content by XRD and thin-section observations; (iii) established the correspondence between reservoir space type and relaxation time range by NMR test and FE-SEM; and (iv) we studied the effects of mineral composition, TOC and R_O on the specific surface area, pore volume and pore diameter by XRD, thin-section observations, gas adsorption experiment and geochemical tests. Through the above joint research, the petrophysical characteristics of shale with different gas content have been explored, and the main controlling factors and influence mechanism of shale gas content have been clarified. It provides theoretical guidance for the exploration and development of high-yield shale gas in the future.

2. Materials and methods

2.1. Sampling

Thirteen core samples with 25 mm diameter and 50 mm long were collected from representative wells. The samples were collected from five different wells across Sichuan and Guizhou province (Fig. 1). Well A and well B is located in the Xiang'exi fold belt. According to the drilling data, the pressure coefficient (the ratio of the actual fluid pressure versus the corresponding normal hydrostatic pressure at the same depth) is between 0.8 and 1.0 at the bottom of S—∈ (Silurian-Cambrian). The gas production of the two wells is between $(9.0-10) \times 10^4$ m^3/d and belongs to medium production gas well [21,22]. Through analyzing the gas content of the 6 samples from the two wells, it was found that the gas content of the samples was between 2 m^3/t and 4 m^3/t. Therefore, the six samples A1, A2, A3, A4, B1, and B2 are medium gas-bearing samples. Well C and well D are located in the Sichuan Basin and drilling data reveals that the pressure coefficient is approximately between 1.2 and 1.4 at the bottom of O−S (Ordovician-Silurian). The gas production of the two wells is between $(18-21) \times 10^4$ m^3/d and belongs to high production gas well [23,24]. By analyzing the gas content of the five samples of the two wells, it was found that the gas content of the samples was greater than 4.3 m^3/t. Therefore, the five samples C1, C2, D1, D2, and D3 are high gas-bearing samples. Well E is located in the Qianzhong fold belt and a large number of faults developed. The terrain condition is

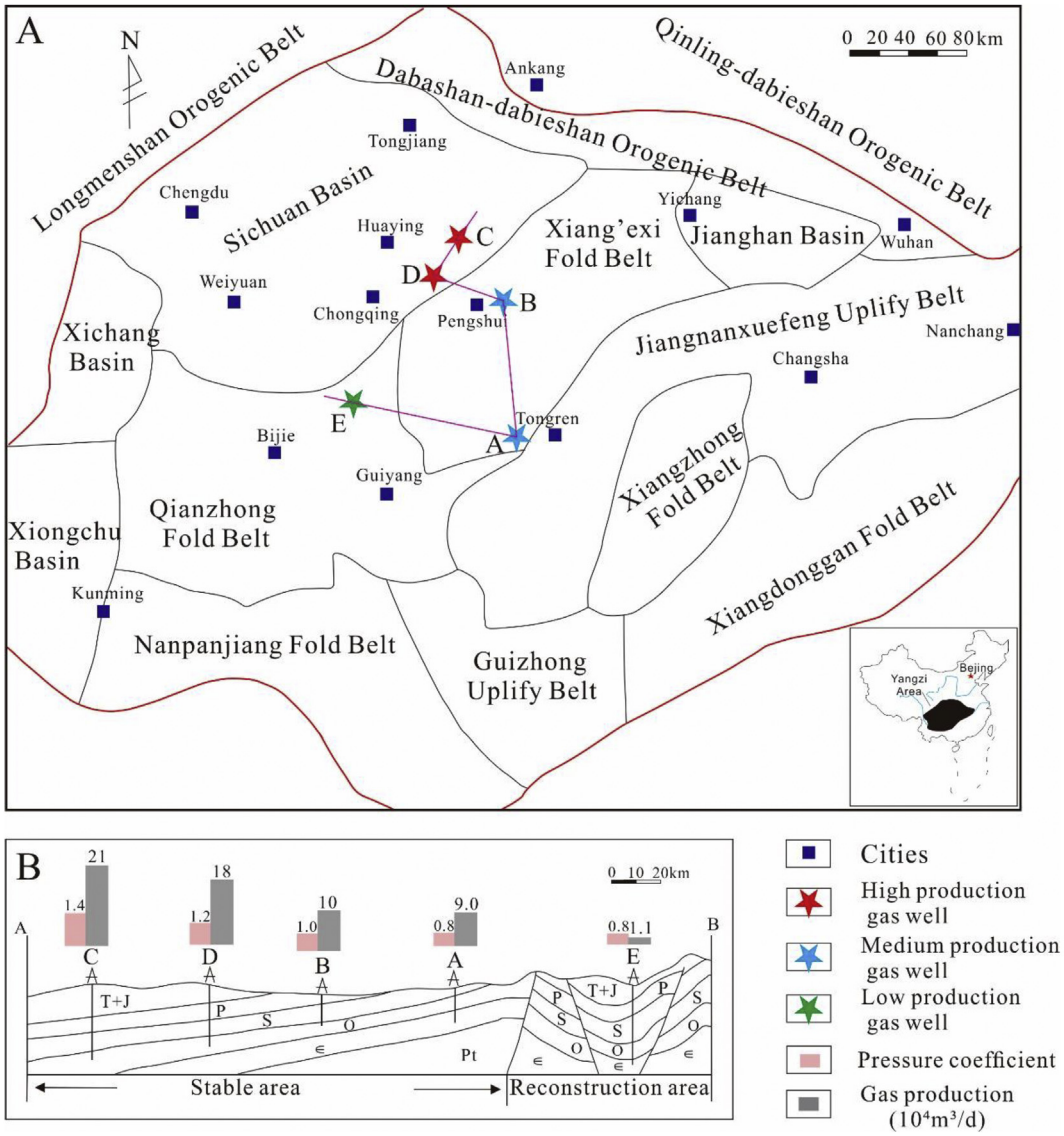

Fig. 1. (A) Geological map of the study area. The wells are marked on the map. (B) Cross-section of wells showing the structural framework and the main stratigraphic intervals. Formation abbreviations: O= Ordovician; S= Silurian; ∈ = Carboniferous; P= Permian; T = Triassic; J = Jurassic.

complex, the landform change is big, and the relief is more than 500 m. In this case, the gas production of well E is only 1.1×10^4 m^3/d and belongs to low production gas well [25]. By analyzing the gas content of the two samples of the well, it was found that the gas content was less than 0.5 m^3/t. Therefore, the two samples E1 and E2 are low gas-bearing samples (Table 1).

The shale gas layer drilled in Well A and Well E is the Cambrian Niutitang Formation. The shale gas production layer of well B, well C and well D is the Silurian Longmaxi Formation. The lithology of Longmaxi Formation is relatively simple, it is mainly composed of gray-black, black, and thin to thick layered carbonaceous shale, banded siltstone, and lenticular muddy limestone. A large amount of radiolarian, calthrop, and graptolite fossils are observed in the lower of Longmaxi Formation, indicating a deep-water shelf sedimentary environment [26,27]. The Niutitang Formation is mainly composed of gray-black mud shale and silty shale and is partially sandwiched with thin layered limestone. Thicker, continuously deposited black shale is developed in shale formations in the Tongren area. The shale near the Qianzhong fold belt is affected by tectonic movement and the burial thickness is shallow, and the folds and faults develop on a large scale [28,29].

2.2. Petrography

To study the sediment texture of shale samples with different gas contents, identify some pore types in minerals, 26 standard 30 μm thick thin sections were prepared [30]. JMicroVision image analysis was performed on natural light thin section images to quantify the proportion of particles (>20 μm) present in the particle matrix [31]. To highlight the porous network, thin sections were impregnated with a

Table 1
TOC and mineralogical composition.

Sample	Formation	Gas content/(m^3/t)	Description[a]	*TOC*/%	R_O/%	Mineralogical composition/%						Clay composition[b]/%		
						Quartz	Feldspar	Calcite	dolomite	Pyrite	clay	I/S	I	C
A1	Niutitang	2.56	M	1.47	2.6	30.9	7.7	4.5	19.7	2.8	34.4	33	56	11
A2	Niutitang	2.82	M	2.01	2.57	36.1	9.0	/	/	4.7	50.2	45	49	6
A3	Niutitang	2.21	M	1.83	2.54	42.2	7.9	4.1	4.4	2.7	38.5	53	43	5
A4	Niutitang	3.88	M	3.01	2.66	36.8	9.4	4.4	3.1	2.8	43.5	62	29	9
B1	Longmaxi	2.87	M	3.46	3.06	42.7	9.2	6.5	3.7	2.8	35.1	69	28	3
B2	Longmaxi	2.15	M	1.59	3.08	34.2	9.2	3.6	4.1	/	48.9	56	41	3
C1	Longmaxi	5.35	H	3.99	3.06	47.6	7.2	2.4	3.6	2.9	36.3	61	35	4
C2	Longmaxi	5.60	H	3.64	3.02	44.7	7.7	3.0	17.7	2.3	24.6	75	20	5
D1	Longmaxi	6.37	H	4.77	2.91	51.5	9.3	3.5	4.7	2.7	28.3	82	15	3
D2	Longmaxi	4.34	H	4.23	3.13	57.2	7.2	2.3	3.1	2.6	27.6	85	13	2
D3	Longmaxi	7.24	H	4.87	2.86	56.5	7.4	2.8	3.0	5.1	25.2	71	26	3
E1	Niutitang	0.21	L	1.11	3.45	25.2	8.7	/	/	3.5	62.6	44	52	4
E2	Niutitang	0.02	L	0.60	3.37	29.7	7.5	/	/	1.9	60.9	52	44	4

[a] H: high gas-bearing samples; M: medium gas-bearing samples; L: low gas-bearing samples.
[b] C: Chlorite; I: Illite; I/S: illite-smectite mixed layer.

fluorescent epoxy resin under vacuum. The identifiable pores of the optical microscope substantially correspond to the pores in the particles, and it is impossible to identify pores that are too small, such as micropores [32,33]. By performing image analysis on thin-section images, we can quantify the mesopores in the particles.

In the research of shale, scanning electron microscopy (SEM) is often used to identify the type of pores and to observe mineral characteristics. The instruments used in this experiment are: PECS II 685 precision etching coating system, Quanta 250 FEG field emission environment scanning electron microscope. When microscopic observation of the rock surface, the smooth and fresh rock surface was first selected using the PECS II 685 precision etching coating system for argon ion polishing. Two wide beam argon ion beams are used to polish the surface of the sample to remove the loss layer, thereby obtaining a high-quality sample. Besides, the two ion guns can be used for sputtering of the target material for the metal film deposition process to prevent the sample from being charged in the electron microscope. Then we analyzed high-quality samples using the Quanta250 FEG field emission environment scanning electron microscope with image magnifications from 1000 to 20,000 times.

2.3. Petrology

2.3.1. TOC and mineral composition

Each sample was cut several times into multiple pieces to analyze the mineralogical composition, organic petrography, and various porosimetry properties. Powdered shale samples (15−25 mg) were analyzed for TOC in the Key State Laboratory of Oil and Gas Reservoir Geology and Exploitation at Chengdu University with a CS-200 Analyzer (precision of 2%). The XRD experiment used an X'Pert PRO MPD diffractometer and the shale sample used a 300-mesh powder (5 g). The quantification and parameterization of mineral properties were achieved by interpreting diffraction patterns from random powder preparation using *Cu Kα* −radiation produced at 40 kV and 40 mA. The recording of the XRD traces used a scan rate of 2° (2θ)/min. The 2θ value of the recorded diffraction pattern ranges from 5° to 45°. All calculations are performed by a computer program.

2.3.2. Porosity and permeability

The total porosity of shale samples was measured at the Key State Laboratory of Oil and Gas Reservoir Geology and Exploitation at Chengdu University. Samples were oven-dried at 70 °C for 72 h until the weight is constant. Then based on Boyle's Law, the grain volume was measured under ambient condition and bulk volume was measured by Archimedes' principle with helium pycnometry. Porosity was calculated from the difference between bulk volume and grain volume. The gas permeability of shale samples was measured using a Vinci nitrogen permeameter. The plugs were mounted in a "Hassler" type core holder at a confining pressure of 400 psi and a steady-state gas flow was established through the samples. Klinkenberg's empirical correlation was used to correct for permeability errors due to gas slip.

2.3.3. Nuclear magnetic resonance (NMR)

The NMR experiments were carried out in the Department of Geology at the University of Kansas. The Magritek NMR Rock Core Analyzer System with a resonance frequency of 2.38 MHz and a magnetic field strength of 1200 Gauss was used to obtain the T_2 spectra of thirteen brine saturated samples. The number of echoes was set as 10,000, the number of scans was set as 256 during the test. In the sample preparation phase, the 13 samples were oven-dried at 70 °C for 72 h until the weight is constant. Then, the samples were placed in a vacuum dish and were saturated with 2% brine under vacuum for 72 h. The samples were assumed to be 100% brine saturated after the above-described process.

2.3.4. Low-pressure N_2 and CO_2 adsorption isotherms

Low-pressure N_2 adsorption and CO_2 adsorption were carried out using the Autosorb iQ apparatus. N_2 adsorption

was measured at −195 °C and the relative pressure (P/Po) was between 0.01 and 1. CO_2 adsorption was calculated within a pressure range of 4×10^{-4}−3.2×10^{-2} at 0 °C [34]. The volume of mesopores and macropores (2−150 nm) can be quantified by N_2 adsorption and the distribution of micropores (<2 nm) can be determined by CO_2 adsorption. Powdered samples (3−5 g) were outgassed at 110 °C for 12 h under high vacuum (<10 mm Hg) to remove adsorbed moisture and volatile matter. The hysteresis type of N_2 adsorption was determined by measuring the adsorption and desorption isotherms. Specific surface areas were determined using the BET method [35], and the pore volume was determined using the BJH method [36]. The density functional theory (DFT) was used to calculate the pore size distributions [37−39].

2.3.5. Mercury-injection capillary pressure (MICP)

Mercury injection capillary pressure was carried out using AutoPore IV 9510 to characterize pore structure, including the skeletal and bulk densities, pore-throat size distribution, pore surface area. For a cubic shale sample with 1 cm on each side, the MICP method is suitable for the analysis of the connectivity space of pores with pore diameters ranging from 2 nm to 50 μm and porosity from 0.1% to 5.0% (China standards press, 2012). As a non-wetting fluid relative to most porous materials, mercury (molecular size 0.31 nm) does not spontaneously enter the pore space of the shale. When mercury enters the pores, higher pressure is required to force mercury into the smaller pore throat. Therefore, the MICP method can generate a mercury intrusion curve by accurately detecting the volume change of mercury at each pressure.

3. Results

3.1. Petrography

According to different gas production, samples are divided into high gas-bearing samples, medium gas-bearing samples, and low gas-bearing samples. Based on the integration of both thin-section observations and SEM measurements of 13 shale samples, we described the sediment texture, quantify the proportion of particles (>20 μm) present in the particle-matrix, and identified the pore types in the shale.

3.1.1. Thin-section observations

In this study, 26 thin sections were observed according to the difference in shale gas content. Through observation, it was found that there are some differences in shale samples with different gas contents. In high gas-bearing shale samples, the distribution of radiolaria is more uniform, most of which is sucked in carbonaceous clay, and a small amount is still siliceous (Fig. 2A). The distribution of the large three-shot bone can be seen. The bone needle is mostly composed of calcite. Occasionally, the uniaxial double-shot siliceous bone needle is seen, and a few radiobugs are replaced by pyrite (Fig. 2B). Silica sand, mud powder crystal dolomite, and charcoal clay are uniformly mixed, the dolomite is mostly semi-automorphic crystal, and the silt and dolomite are aggregated in parallel strips. Some of the siliceous and flake minerals are in the shape of small needles. Powdery pyrite is unevenly distributed. In the medium gas-bearing shale samples, the horizontal texture is developed and a wavy non-parallel texture locally, and the thickness of texture is between 0.09 mm and 1.1 mm. The texture layer is an unequal thickness interbed between cloud-like or cloud-bearing sandstone and silty-soiled clay rock, with occasional interaction. The mud powder crystal dolomite is unevenly distributed. The quartz silt is evenly distributed in a scattered shape, and the maximum diameter of the silt remains as 0.06 mm. A small number of mica fines are arranged along with the texture layer, and the powdery pyrite is evenly distributed (Fig. 2C and D). Among the low gas-bearing shale samples, the distribution of quartz silt is relatively uniform, and the maximum diameter of the silt remains as 0.06 mm. Occasionally, the mica fine film and the mud powder crystal calcite are evenly distributed in a scattered manner. The powdery pyrite is evenly distributed. The wavy non-parallel texture is developed, and the thickness of the texture layer is between 0.2 mm and 2.75 mm. The texture layer is interbedded between carbonaceous and clay siltstone and silty carbonaceous clay, and a few interact. The content of the filler between particles is high, and its composition is mainly clay. There is a small amount of radiolarian debris (Fig. 2E and F).

3.1.2. Scanning electron microscopy (SEM)

A large number of nano-scale pores were developed in shale, and different scholars have different classifications of shale pores. Loucks et al. [17] studied nano-micron pores in the Barnett shale and divided the pores into micropores (diameter greater than 0.75 μm) and nanopores (diameter less than 0.75 μm). Zou et al. [40,41] observed the pore structure of the Paleozoic shale in the Sichuan Basin by SEM and summarized it as organic nanopores, nanopores in the particles, and micro-fractures in the organic matter. Through a large number of field emission environment scanning electron microscopy experiments, combined with the results of previous research, according to the pore development position and developmental genesis, the pores of shale reservoirs are divided into four types: clay mineral intergranular pores, rock skeleton mineral pores, organic nanopores, and micro-fractures. Besides, we describe the morphology, size, and shale gas storage significance of each pore.

Clay mineral is one of the most important mineral components of shale, its content can reach 24.6%−62.6% (Table 1), and it is mainly composed of illite and I/S. A large number of intergranular pores are developed in abundant clay minerals (Fig. 3A and B). In some areas, it is seen that a plurality of clay mineral dissolution pores is arranged in a line, connecting organic pores and mineral pores, and to some extent has the function of micro-fractures. The intergranular pores of clay minerals are concentrated, the cementation is complicated, the sorting is poor, and the clay minerals have small particle size and strong plasticity. After hydration expansion, migration is likely to occur, blocking the pores and reducing the permeability of the reservoir. If there is no strong

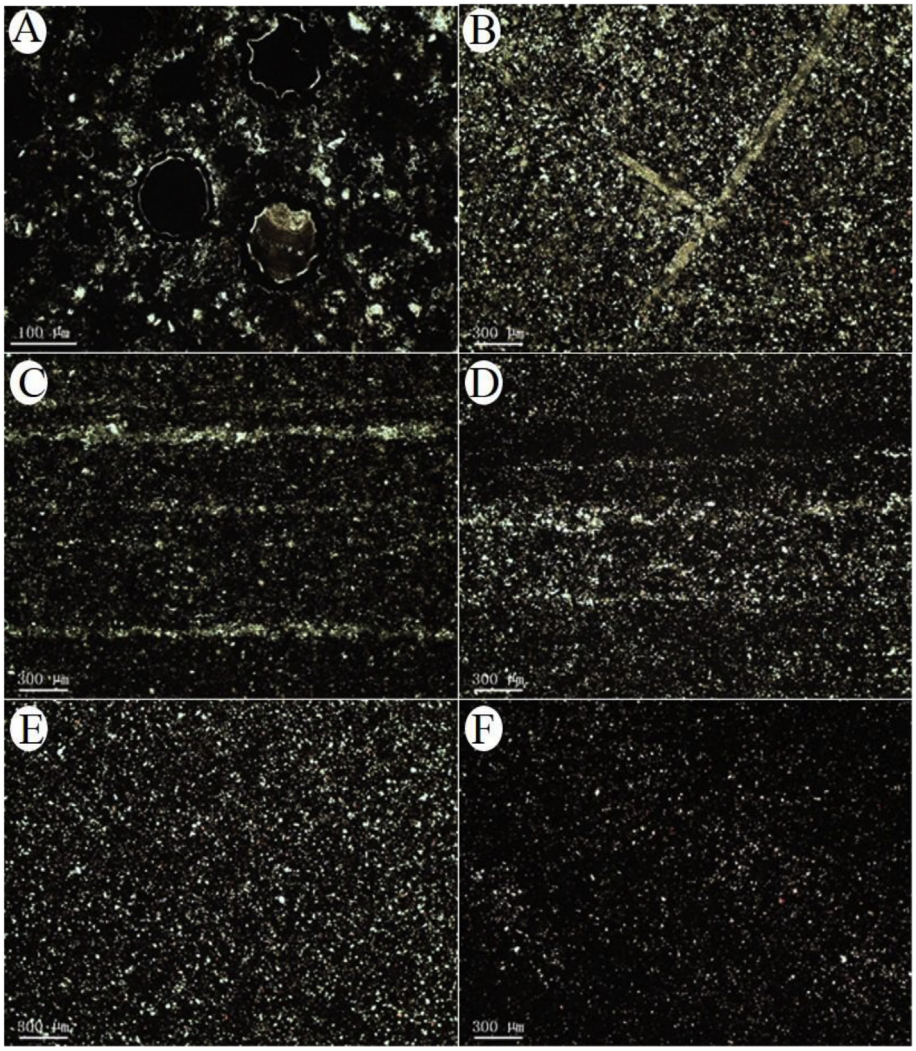

Fig. 2. Thin-section photomicrographs of the three different gas-bearing shale samples under plane-polarized light. (A, B) High gas-bearing samples C1 and D1 with a large number of bone needles and radiolarians. (C, D) Medium gas-bearing samples A2 and B1 with horizontal stratum development. (E, F) Low gas-bearing samples E1 and E2 with undeveloped horizontal stratum and uniform distribution of mineral particles.

transformation in the later stage of diagenesis, it is difficult for a single clay mineral interparticle pore to have good oil and gas migration capacity. The specific surface area of clay minerals is larger than that of quartz. The more the intergranular pores develop, the stronger the gas adsorption capacity [42], and the lower the organic carbon content of shale, the adsorption of clay minerals is very significant [43]. It can be seen that the intergranular pores of clay minerals mainly play a role in storage.

The pores formed by the rock skeleton minerals are mainly intergranular pores of pyrite, nano-scale pores formed by quartz dissolution, and mineral cleavage joints (Fig. 3C and D). These pores are usually scattered and the pores are not connected or the connectivity is extremely poor. Nano-scale dissolution pores, intercrystalline pores, and intergranular fractures are formed in the skeleton minerals. When the content of brittle minerals such as quartz and feldspar is high, natural fractures and induced fractures are easily formed [44], and the modification effect can improve the seepage ability of such pores.

There are abundant nano-scale pores inside the organic matter particles in shale. Organic nanopores are one of the most developed pore types in shale, and an organic particle with a diameter of a few microns can contain hundreds or thousands of nanopores. The pore size ranges from 5 nm to 600 nm and is mainly near-spherical or ellipsoidal. Besides, there are other irregular shapes such as cave-like and slit shape (Fig. 3E and F). A large number of pyrite particles are scattered near a part of the organic nanopores. However, not all organic matter develops nanopores, which are related to the maturity of organic matter, and low maturity organic matter particles have fewer pores. Organic nanopores are derived from the process of organic matter accumulation and thermal evolution. In this process, many tiny pores or fractures are developed due to changes in the geological environment. The organic matter is mainly distributed along the micro-layered or

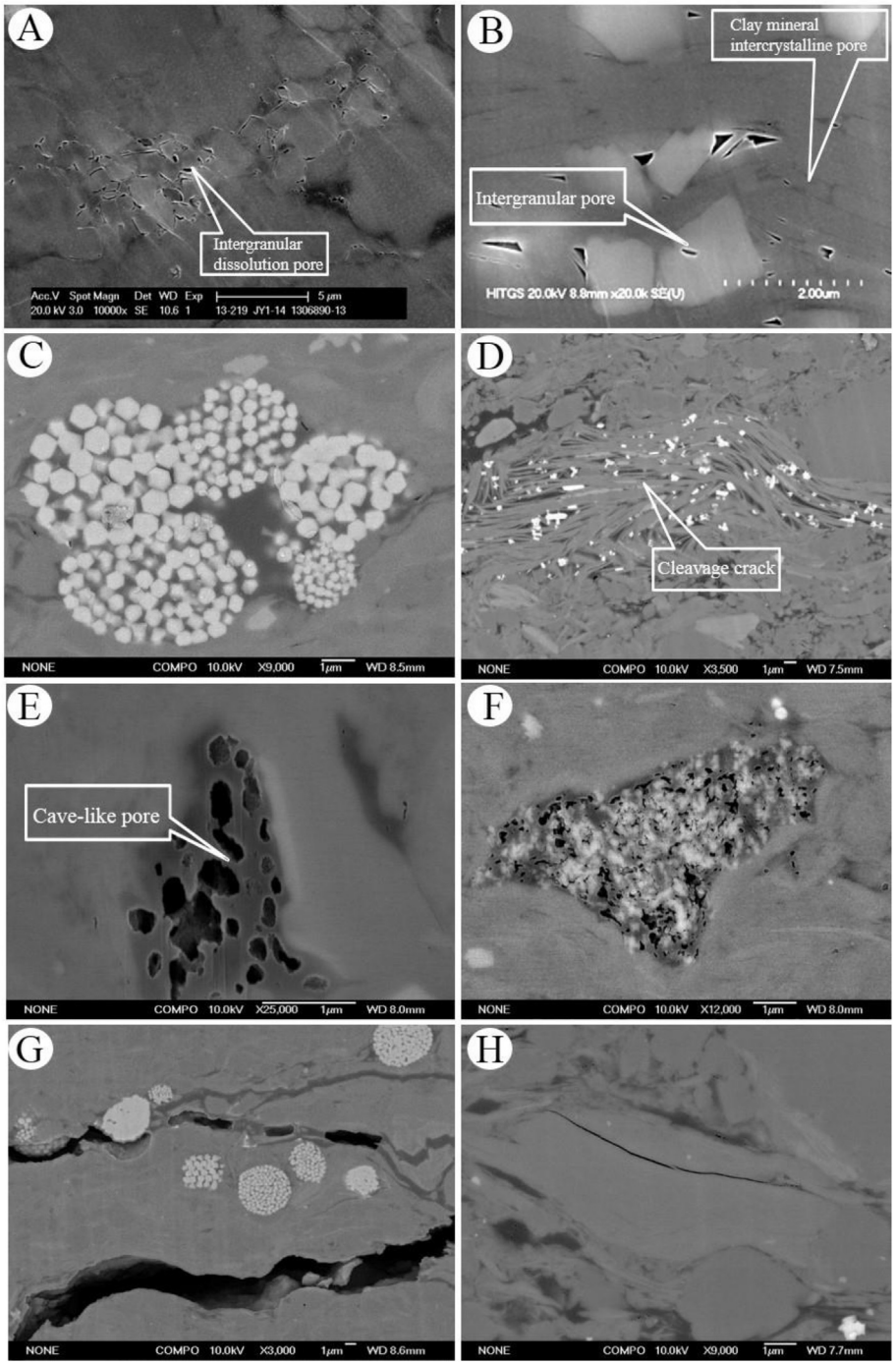

Fig. 3. Shale samples with different gas contents under the SEM: (A) Intergranular dissolution pore from C1; (B) Clay mineral intercrystalline pore and intergranular pore from C2; (C) Pyrite intergranular pore from B1; (D) Cleavage fracture of illite minerals from A1; (E) Cave-like pore from D2; (F) Complex pores in organic matter from C1; (G) Micro-fractures between minerals and structural fractures from E1; (H) Micro-fractures inside minerals from E2.

inter-deposited sections [45], which are easy to form an interconnected porous network and have good permeability. The oleophilic nature of the organic matter makes the organic nanopores an important storage space for adsorbing gas [44].

Micro-fractures play an important role in the seepage of shale gas, and are the bridge connecting microscopic pores with macroscopic fractures. There are two main types of micro-fractures in shale, one developed inside the particle, and the other developed at the edge of the crumb particle (Fig. 3G and H). The micro-fractures inside the particles are generally straight, with a low degree of tortuosity and few fillers. The micro-fractures between the particles are sawtooth curved. The micro-fractures have a length of 5−10 μm and a fracture pitch of 50 nm or more but rarely extend to the entire surface of the slice. In the region with micro-fractures, the rock brittleness index is high, and it is

easy to form a micro-fracture network, which becomes the main channel for oil and gas seepage on the micro-scale in shale. Fractures can effectively improve the seepage capacity in the shale reservoir, and the degree of fracture development is an important indicator for evaluating the quality of the reservoir. However, for shale reservoirs, having fractures is not necessarily a favorable factor. The large-scale development of natural fractures reduces the protective effect of shale as a caprock, resulting in the loss of gas. A good shale reservoir should have good crackability and development of micro-fractures. If short fractures are widely developed inside the shale, it will not only facilitate the mass storage of free gas but also significantly improve the permeability of the reservoir.

3.2. Pore structure

3.2.1. TOC and mineral composition

Analysis of organic geochemistry and petrology of shale are shown in Table 1. TOC and mineral content has a difference in samples with different gas contents. The TOC content of the high gas-bearing samples was greater than 3.6%, the medium gas-bearing samples had a TOC content between 1.4% and 3.5%, and the TOC content of the low gas-bearing samples was less than 1.1%. The mineral composition of the 13 shale samples is the same, the main mineral types are quartz, feldspar, calcite, dolomite, pyrite, and clay. Among them, clay is mainly composed of illite, I/S, and chlorite. The high gas-bearing samples have the highest quartz content, generally greater than 44%, and the lowest clay content, less than 36%. Among the clay minerals, I/S is the main composition, and the content is more than 60%. The quartz content is the least in the low gas-bearing samples, less than 30%. The clay content is the highest, greater than 60%. In medium gas-bearing samples, the quartz content is ranged from 30% to 43%, and the clay content is in the range of 34%−49%. In the medium gas-bearing and low gas-bearing samples, the content of illite and I/S is not much different, the illite content is between 28% and 56%, and the I/S content is ranged from 33% to 69%.

3.2.2. Porosity and permeability

As a very dense rock, the porosity and permeability of shale are generally low (Table 2). By analyzing samples with different gas contents, we found that the samples with high gas content have the highest porosity, generally greater than 4.5%, and the permeability is ranged from 0.007 mD to 0.132 mD. In medium gas-bearing samples, porosity is between 3.4% and 4.3%, and the permeability is ranged from 0.05 mD to 0.28 mD. The porosity in the samples with low gas content is the least, less than 3.1%, but the permeability is great than 0.135 mD.

3.2.3. Nuclear magnetic resonance (NMR)

The T_2 distribution reflects the distribution of pore volume and pore surface area, which includes the pore shape factor and the surface rugosity. The logarithmic mean of the T_2 distribution (T_{2ml}) for each sample was calculated. The shale has a relatively low T_{2ml} value of less than 3 ms due to the presence of dispersed clay flakes (Table 2) [46]. The T_{2ml} value of the high gas-bearing samples is relatively high, ranging from 1.8 ms to 2.2 ms (average 2.0128 ms). The T_{2ml} value for low gas-bearing samples is relatively low, generally less than 1.2 ms, with an average of 1.141 ms. The T_{2ml} value of medium gas-bearing samples is in the range of 1.3−1.7 ms, with an average of 1.527 ms.

Generally, micropores have a lower T_2 value than the macropores, and the waveform of the T_2 spectrum reflects the connectivity and number of the pores [47,48]. The bimodal T_2 spectrum of the high gas-bearing samples has two peaks, and the amplitude of the left peak is larger than that of the right peak. The T_2 value of the left peak is ranged from 1 ms to 5 ms, and the T_2 value of the right peak is between 70 ms and 100 ms (Fig. 4), which indicates that the micropores and the mesopores are more developed than macropores. However, it can be seen that the connectivity between them is poor.

Table 2
Summary of the petrophysical characteristics of the samples.

Sample	CO_2 isotherm analyses			N_2 isotherm analyses			MICP data				NMR data	Routine petrophysical data	
	Micropore surface area/(m^2/g)	Micropore volume/(ml/g)	Micropore diameter/nm	BET surface area/(m^2/g)	BJH pore volume/(ml/g)	Average pore diameter/nm	SHg_{max}/%	r_{max}/μm	r_{ave}/μm	r_{apex}/μm	T_{2ml}/ms	Porosity/%	Permeability/mD
A1	9.98400	0.00261	0.87000	6.68034	0.00795	9.69	67.4	2.11	0.517	0.89	1.328	3.49	0.1580
A2	9.91800	0.00264	0.86820	7.45985	0.00927	8.03	58.7	1.99	0.605	1.07	1.565	3.61	0.1500
A3	10.23500	0.00248	0.78720	6.67637	0.00781	9.82	74.2	2.19	0.613	0.66	1.606	3.83	0.0556
A4	/	/	/	/	/	/	56.7	2.18	0.447	1.29	1.422	3.73	0.2820
B1	/	/	/	/	/	/	40.5	0.76	0.158	0.26	1.552	4.34	0.0737
B2	11.98000	0.00315	0.77140	7.27181	0.00838	8.77	35.8	2.25	0.593	0.96	1.689	4.21	0.2063
C1	14.03500	0.00364	0.75190	9.06550	0.01054	7.04	33.6	2.77	0.714	1.21	2.083	4.56	0.0830
C2	/	/	/	/	/	/	46.1	3.25	0.823	1.27	2.034	4.81	0.1320
D1	16.57200	0.00420	0.72510	10.03365	0.01139	6.81	20.3	2.49	0.741	0.58	2.142	4.59	0.0069
D2	14.43700	0.00346	0.74670	9.92388	0.00983	6.92	22.6	2.49	0.614	0.57	1.873	4.36	0.0035
D3	14.47600	0.00373	0.74670	9.44292	0.01096	7.88	48.1	3.67	0.737	1.55	1.932	5.12	0.0560
E1	9.36800	0.00221	0.94160	6.23423	0.00696	9.87	68.9	0.41	0.093	0.07	1.158	3.09	0.2714
E2	9.55800	0.00237	0.85470	6.19221	0.00760	10.25	67.5	0.20	0.059	0.04	1.125	3.04	0.1359

Comparing the T_2 distribution curve with the MICP pore throat distribution curve, it was found that they were similar in morphology. The NMR data set for the medium gas-bearing samples exhibit a multi-peak behavior and showing three distinctive peaks in which the corresponding relaxation times are ranged from 0.8 ms to 1 ms, 20−50 ms, and 100−200 ms, respectively (Fig. 4). The distribution of the pore throat radius from MICP in the medium gas-bearing samples is three-peak except for the sample A2 which is bimodal. The T_2 spectrum of the low gas-bearing samples exhibits a bimodal shape with one broader wave (Fig. 4), indicating good connectivity between the smaller and larger pores. Compared the spectrum values of three different gas content samples, the high gas-bearing samples have the highest spectrum value, the medium gas-bearing samples are the second, and the spectrum value of low gas-bearing samples is the lowest (Fig. 4).

3.2.4. Low-pressure N_2 and CO_2 adsorption isotherms

Fig. 5A shows the N_2 adsorption isotherm of the samples of this study. According to the BET classification, the reversed S-shaped N_2 adsorption isotherms were type II [35]. The relative pressure is between 0 and 1. When the relative pressure is less than 0.3, the adsorption volume gradually increases with an increase in pressure, and the shape of the curve is slightly convex, indicting the transitional stage from the monomolecular to multimolecular layer. When the relative pressure is between 0.3 and 0.8, the adsorption volume increases slowly as the pressure increases, indicating a multilayer adsorption process. When the relative pressure is higher than 0.8, the adsorption volume suddenly increases rapidly, indicating that there must be large pores or fractures in the shale samples to allow a large amount of N_2 to enter. According to the IUPAC classification, the hysteresis loop formed by the adsorption and desorption branches is mainly of the H3 type [49]. H3 type hysteresis loops are usually observed in samples with slit-like pores. The BET surface area and BJH pore volume of shale samples with different gas contents were calculated from N_2 adsorption analysis at relative pressure between 0.05 and 0.30. The BET surface area and BJH pore volume of high gas-bearing samples were the largest, medium gas-bearing samples were the second, and low gas-bearing samples were the smallest (Table 2). Also, the specific surface area and TOC content show a strong positive correlation (Fig. 6), indicating that the organic matter provides mainly methane sorption sites.

According to the BET classification, the carbon dioxide adsorption isotherms are type I [35], indicating that the pore type is mainly microporous. Besides, the volume of CO_2 adsorbed of the shale samples increased with increasing TOC content (Fig. 5B), indicating that TOC is the main controlling factor for microporous development. The micropore volume from the carbon dioxide adsorption experiment was calculated using the DFT method. The micropore surface area and pore volume of high gas-bearing samples were the largest, medium gas-bearing samples were the second, and low gas-bearing samples were the smallest (Table 2).

3.2.5. Mercury-injection capillary pressure (MICP)

As a classical analytical method, MICP can better reflect the distribution of connected pores in rocks. The shape of the mercury intrusion curve reflects the development of pores in each throat section and the connectivity information between pores [50,51]. The maximum connected pore radius (r_{max}) of the samples is ranged from 0.20 μm to 3.67 μm and have an average of 2.14 μm (Table 2). The high gas-bearing samples

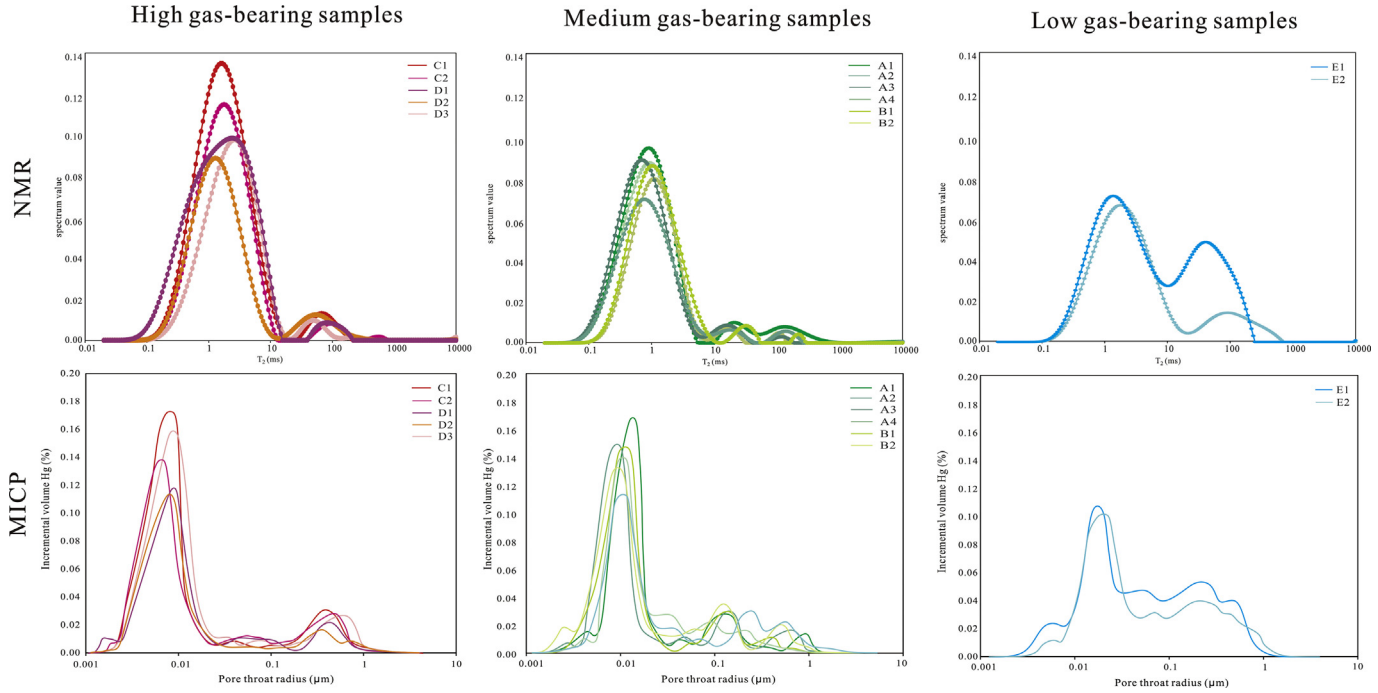

Fig. 4. Overview of the NMR and MICP results for three different gas-bearing shale samples.

have relatively high r_{max} values, greater than 2.4 μm. The r_{max} values of the medium gas-bearing samples are higher than the low gas-bearing samples (Table 2). The r_{apex} is the vertex of the curve on a plot of mercury saturation over capillary pressure vs mercury saturation, corresponding to the state when the effective storage space is completely occupied by the non-wetting liquid [52,53]. Through analysis, we found there is no correlation between porosity and permeability (Fig. 7A), but permeability has a strong correlation with the r_{apex} parameter, and with the correlation coefficient is 90.47% (Fig. 7B). Moreover, the r_{apex} parameter is also had a good relationship with the average connected pore radius (Fig. 7C), and a maximum connected pore radius (Fig. 7D). Therefore, the r_{apex} parameter can be used to evaluate the reservoir.

The pore throat radius distribution of high gas-bearing samples is bimodal, and the position of the two peaks is discontinuous (Fig. 4), indicating that the radii of the different pore throats vary widely. In medium gas-bearing samples, the pore radius distribution is a plurality of peaks, and the pore radius is small, indicating that the connectivity between the different pores is poor (Fig. 4). Although the pore throat radius distribution of low gas-bearing samples is also bimodal, the peaks toward high values, indicated that mercury intrusion is controlled by a series of pore throats. In this study, the major pore types are micropores and mesopores, while the macropores are relatively undeveloped.

4. Discussion

4.1. Identification of pore types from T_2 distribution

NMR technology is one of the important methods to characterize the microscopic pore structure of rocks and evaluate the effectiveness of reservoirs [54,55]. The shale sample of saturated brine has a T_2 relaxation time distribution ranging from 0.1 ms to 200 ms (Fig. 4). The range of T_2 relaxation time has a difference with previous studies [56,57], indicating that the difference in thermal maturity and diagenetic processes may result in differences in T_2 relaxation times. The shale samples of this study were from the Niutitang Formation and the Longmaxi Formation. The shale is in a stage of maturity, and the organic matter is highly developed, resulting in a large number of organic pores. As the proportion of pores and residual carbonaceous organic matter increases, the stability of the shale internal structure decreases, causing the organic pores to be compacted and filled. The inorganic pores are compacted under the influence of the overburden stress [58], resulting in a shortened T_2 relaxation time.

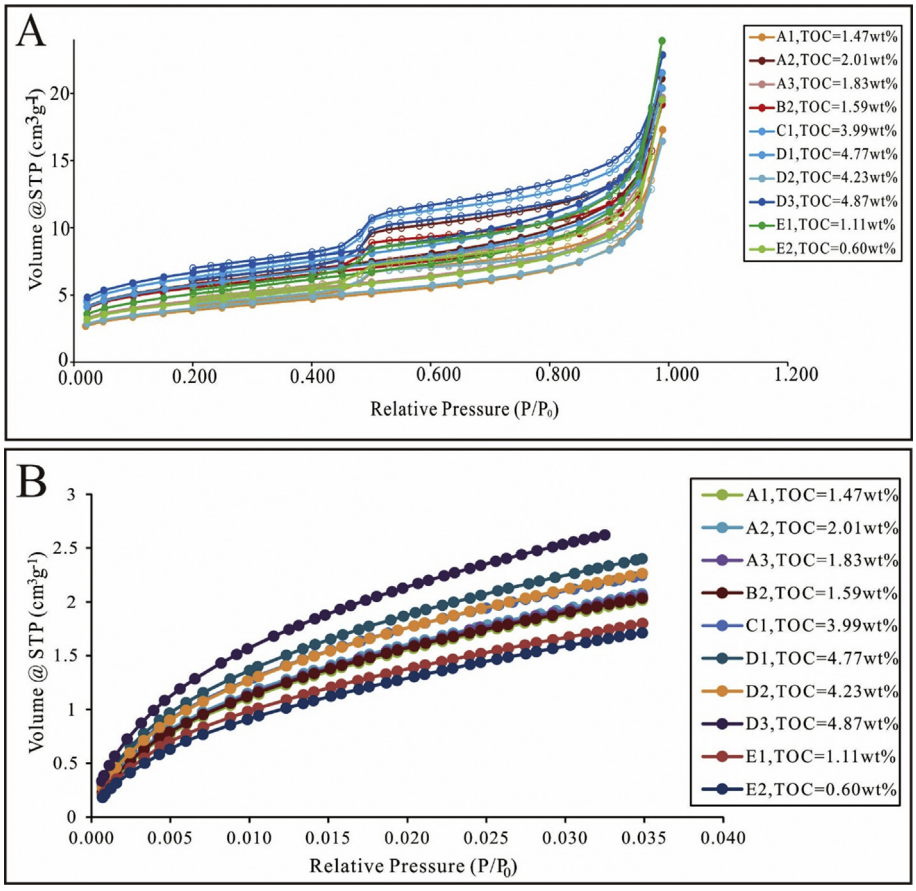

Fig. 5. (A) Low-pressure N_2 adsorption–desorption isotherms for three different gas-bearing shale samples. The curve connecting the solid points is the adsorption curve, and the curve connecting the hollow points is the adsorption curve. (B) Low-pressure carbon dioxide adsorption isotherms.

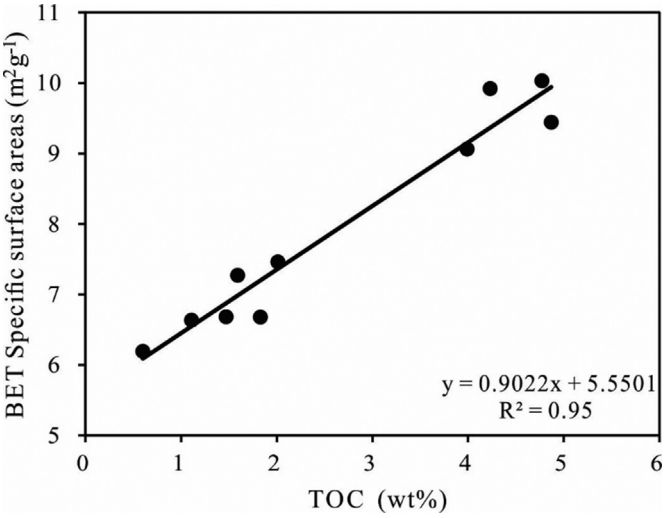

Fig. 6. Relationship between BET specific surface areas and TOC content.

According to the basic principles of NMR, the T_2 distribution of the hydrogen nucleus is positively corresponds with the pore size [47,59]. Therefore, through the T_2 distribution of samples with different gas contents, we can find that high gas-bearing and medium gas-bearing samples are rich in micropores, while low gas-bearing samples are rich in mesopores and macropores. Sondergeld et al. [33]. Calculated the pore throat radius of the shale using a surface relaxation rate of 0.05 μm/ms and compared it with the scanning electron microscopy (SEM) image of the sample. The short relaxation peak below 3 ms is considered to be the kerogen hydrocarbon produces pores with pore sizes between 3 nm and 100 nm. The high gas-bearing samples are rich in organic matter. In scanning electron microscopy, a large number of organic pores with pore sizes between 5 nm and 100 nm were found (Fig. 8A). A large number of intergranular pores are developed in medium gas-containing samples with a pore size between 5 nm and 50 nm (Fig. 8B). Odusina et al. [60] also used a surface relaxation rate of 0.05 μm/ms to calculate the T_2 spectrum of Barnett saturated shale according to the calculation model and compared with X-ray micro-computed tomography (X-CT). In comparison, it is concluded that the peak of 10−100 ms on the T_2 spectrum represents a micro-fracture of 1−10 μm, and the amplitude represents the volume of the micro-fracture. Through scanning electron micrograph observation, we found that in the low gas-bearing shale samples, a large number of micro-fractures were developed with a width of 3−6 μm (Fig. 8C). The sedimentary environment of the high gas-bearing samples is a deepwater shelf. Therefore, the source of organic matter is mainly algae and organisms in the ocean. During the late burial diagenesis, maturation and recrystallization mutually restrict and control the preservation of porosity. In contrast, the low gas-bearing samples have a high content of clay minerals, resulting in increased specific surface area, reduced pore volume, and reduced T_2 relaxation

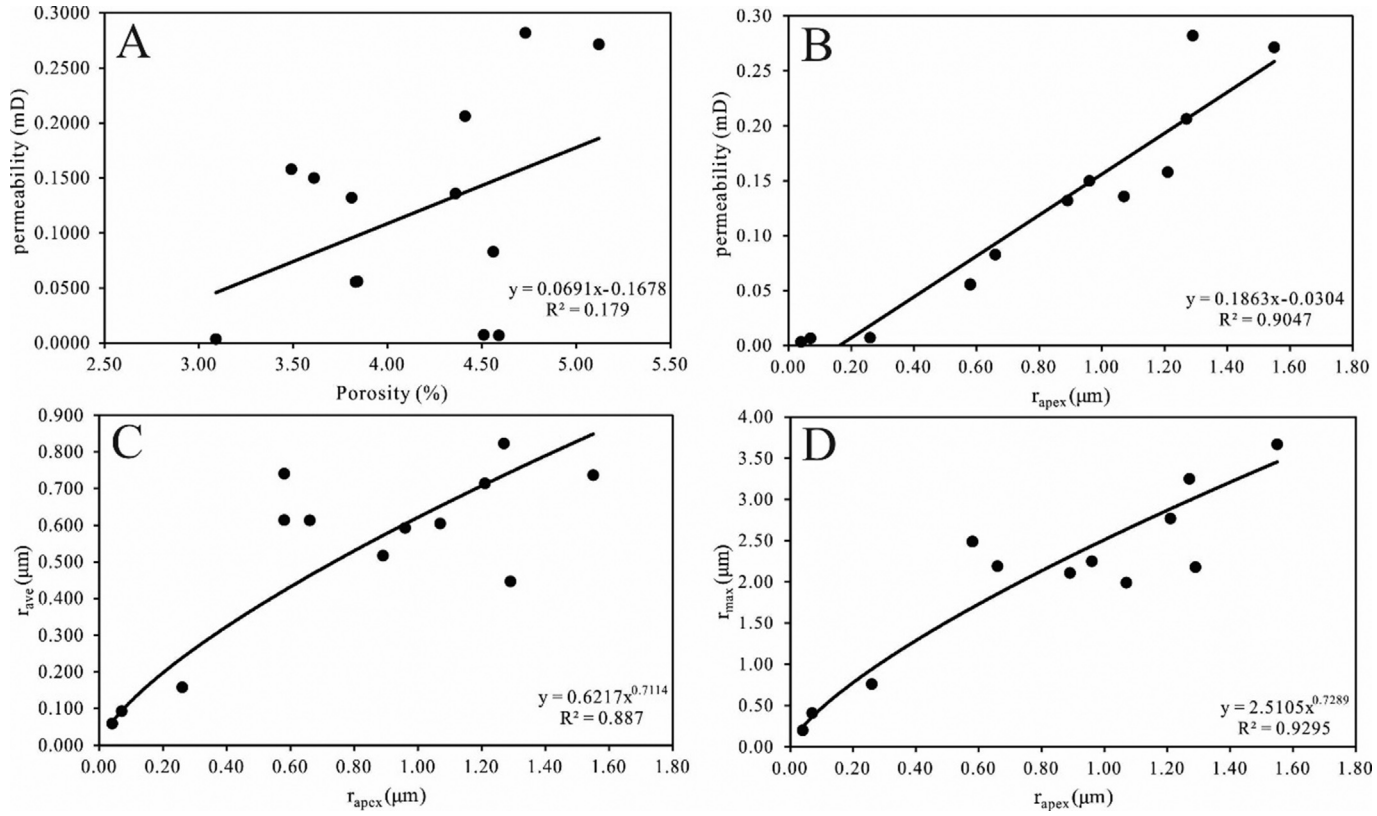

Fig. 7. (A) Relationship between porosity and permeability. (B) Relationship between r_{apex} parameter and permeability. (C) Relationship between r_{apex} parameter and average connected pore radius. (D) Relationship between r_{apex} parameter and maximum connected pore radius.

time [46,61]. The T_2 spectrum indicates that the micro- and mesoporosity of the low gas-bearing samples has fewer levels than that of the high gas-bearing and medium gas-bearing samples.

4.2. Pore genesis and development

The pore type of shale with different gas contents may be formed by different mechanisms during sedimentation and thermal maturation. This leads to great differences in the genesis and development of shale pores with different gas contents. The mudstone is mainly compacted in the early stage of diagenesis and then pyrolyzed to generate hydrocarbons in the later stage [62]. Affected by compaction, the pores gradually become dense, and a large number of organic pores begin to be produced under the influence of pyrolysis and hydrocarbon generation.

Micropores and mesopores associated with organic matter are abundantly developed in high gas-bearing samples. During the thermal decomposition of organic matter, as the volume of kerogen decreases, hydrocarbons are gradually produced, together with residual kerogen and increasing porosity. The organic pores formed by hydrocarbon generation and hydrocarbon removal are generally circular or elliptical [17]. However, we have identified in the SEM image that the shape of the pores is mostly irregular. This may be because large pores are easier to deformation than small pores during compaction [63].

In medium gas-bearing samples, a large number of intergranular pores are formed under compaction and chemical precipitation. Loucks et al., [17] and Milliken et al., [64] found that internal pores within the pyrite crumb may be native or diagenetic. The organic matter that fills the pores between the pyrite particles is generally softer than the mineral debris and facilitates the migration of hydrocarbons. During the migration process, some heavy components of the hydrocarbon remain in the pores, resulting in loss of pore space [65]. Some remnant material in the migration can produce micropores [62], which can be observed in the SEM image (Fig. 3C).

Low gas-bearing shale mainly develops macropores and micro-fractures, with relatively few micropores. The reason for this phenomenon may be that the low gas-bearing shale is strongly affected by tectonic processes, causing minerals to fracture and form mineral-related micro-fractures, while the fracture pores associated with the organic matter may be due to shrinkage of terrestrial organic matter. Fig. 9.

4.3. Control factors for pore development

The relationship between pore volume, specific surface area, and pore size of shale with different gas content is shown in Fig. 10. Overall, there is a positive relationship between the specific surface area and the total pore volume. The average pore size has a negative relationship with the specific surface area and total pore volume. The slope of the linear fit model indicates that the decrease in average pore size leads to an increase in total pore volume and specific surface area, and the effect of contrast surface area is greater. These relationships are the same as those of Marcellus, Barnett, Woodford, Haynesville, and Doig Phosphate shale in North America [20,66]. From the pore characteristics of shale with different gas content, high gas-bearing shale has a smaller average pore size and larger specific surface area and total pore volume; low gas-bearing shale has a larger average pore size and smaller specific surface area and total pore volume.

Combined with the previous research on pore genesis and development, the high gas shale is affected by compaction, and the large pores are compacted and become small pores. Besides, the pyrolysis of organic matter to generate hydrocarbons will produce a large number of organic micropores. These micropores are important adsorption sites for adsorbed gas. The low-gas shale is greatly affected by tectonic processes, destroying the pore structure and allowing gas to escape. Therefore, structure and diagenesis may be one of the influencing factors of pore development.

It was found by XRD analysis that the mineral composition of the shale is mainly quartz and clay (Table 1). The relationship between pore structure and shale minerals is shown in Fig. 10. Both specific surface area and pore volume are independent of quartz and clay content in different gas content shale. The average pore size is independent of clay content, but the average pore size shows a negative correlation with the quartz content (Fig. 10). Therefore, the mineral content has little effect on the shale pore development.

High gas-bearing shale has a higher TOC, while low gas-bearing shale has a lower TOC. The pore volume and specific surface area increase as the TOC content increases (Fig. 11A and B), the average pore size decreases as the TOC content increases (Fig. 11C), indicating that samples with higher TOC content may contain larger numbers of small pores and have a larger specific surface area and total pore volume. Previous studies of some North American basin shale have proposed these relationships [1,20,34,66]. It can be seen that high gas-bearing shale with higher TOC will form more organic micropores than low gas-bearing shale. At the same time, acidic substances generated during pyrolysis will dissolve feldspar, which increases the gas adsorption sites. Therefore, TOC may be one of the main controlling factors affecting pore development.

The shale in this research has a high maturity ($R_O > 2.5\%$, Table 1), so the organic matter in the shale has undergone varying degrees of deformation. High gas-bearing shale develops a large number of organic pores, to provide a large number of adsorption sites. The relationship of specific surface area, the pore volume, the pore size, and R_O in different gas content shale are shown in Fig. 12. The pore volume and specific surface area increase as the R_O value increases. Through studies of gas window mature samples from Fort Worth Basin and Barnett Shale, it was found that immature kerogen, which originally contained significant aliphatic components, can evolve into over-mature kerogen through a poorly condensed aromatic structure [67,68]. Kerogen pyrolysis and hydrocarbon generation during thermal maturation

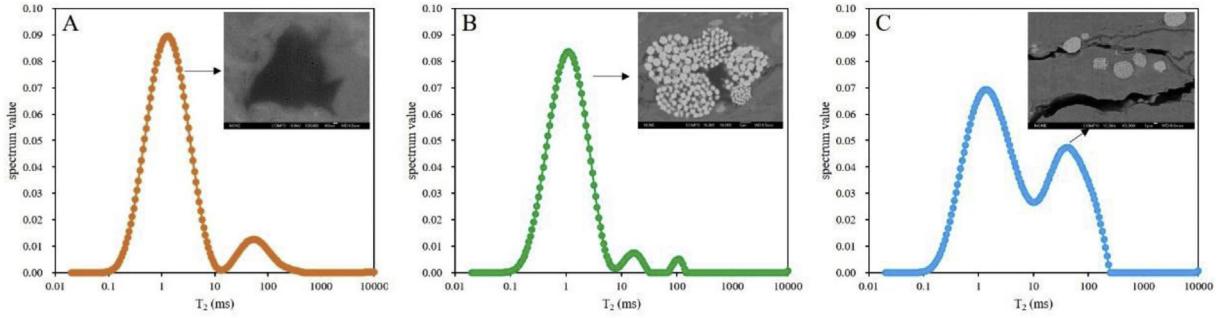

Fig. 8. Overview of the NMR and SEM for the (A) high-gas-bearing samples C1, (B) medium gas-bearing samples B1, (C) low gas-bearing samples E1.

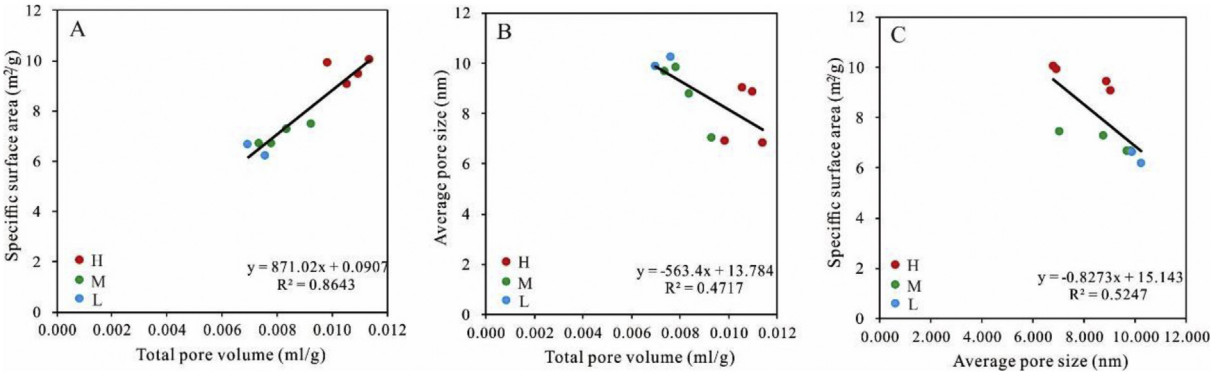

Fig. 9. (A) Relationship between total pore volume and specific surface area. (B) Relationship between total pore volume and average pore size. (C) Relationship between average pore size and specific surface area.

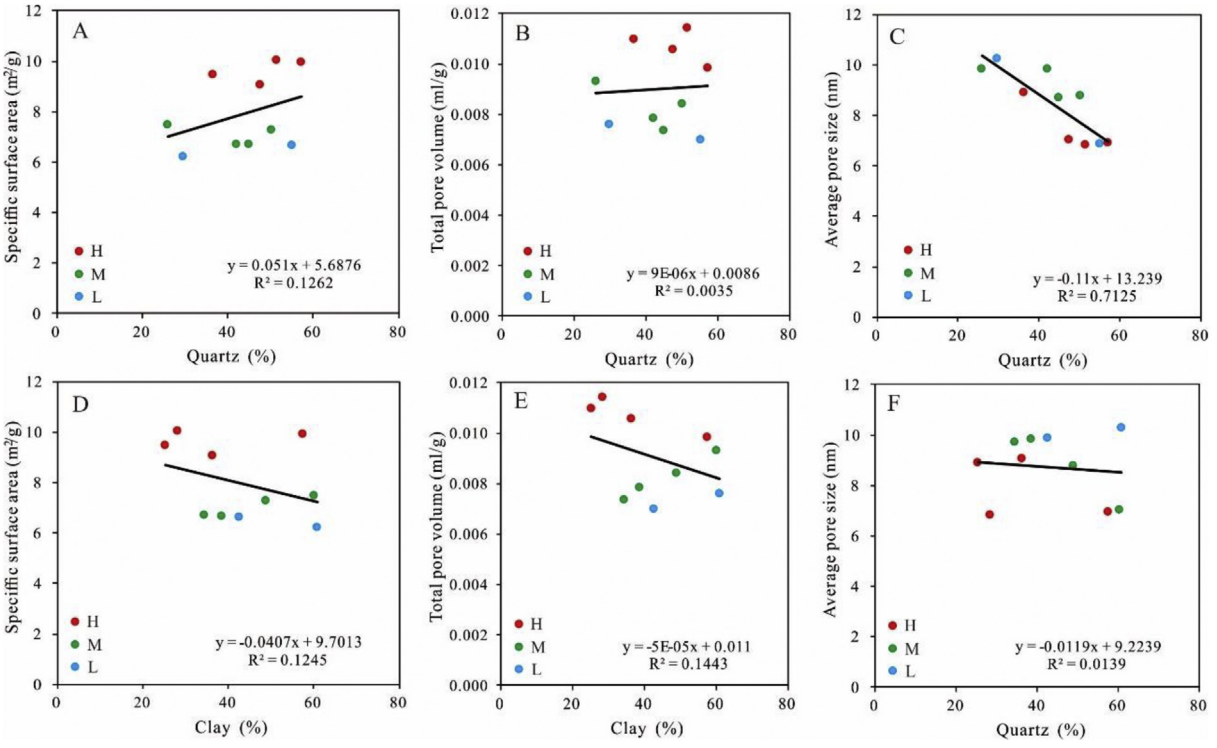

Fig. 10. (A) Relationship between quartz content and specific surface area. (B) Relationship between quartz content and total pore volume. (C) Relationship between quartz content and average pore size. (D) Relationship between clay content and specific surface area. (E) Relationship between clay content and total pore volume. (F) Relationship between clay content and average pore size.

have a major influence on the microstructure of shale. The difference in R_O shows the difference in maturity, which directly indicates the diagenetic evolution and pore evolution stage of shale [69]. It can be found from Fig. 12 that the R_O of the low gas-bearing shale is bigger than 3.3, indicating that it has reached the over-mature stage. At this time, the evolution is almost over and there are fewer pores for gas adsorption, so the gas content is small. The R_O of high gas-bearing shale is in the mature stage, and the organic matter generates a large number of organic micropores [70], resulting in a decrease in the average pore size (Fig. 12C). Besides, brittle minerals are dissolved by organic acids produced during the organic matter conversion process, resulting in increases in the gas adsorption area [71,72], thereby increasing the adsorption gas content. Therefore, R_O may also be one of the main controlling factors affecting pore development.

4.4. Gas shale storage and flow capacity

Shale gas is mainly stored in pores and fractures in free and adsorbed states. Free gas is mainly stored in open pores and fractures, while adsorbed gas is mainly stored on the surface of organic matter and clay. The flow of shale gas is mainly controlled by the pore network. The flow of gas in the larger connected pores and throat generally follows the Darcy equation. In our consideration, both sorption and diffusion dominate gas flow in nanopores. In mesopores such as interparticle pores, the gas flow is mainly diffusion and slippage (Fig. 13) [19,32].

Through the observation of scanning electron micrographs, the high gas-bearing shale samples develop a large number of organic pores with pore sizes in the nanometer and micrometer scales. There are also a small number of micro-fractures development. In the medium gas-bearing shale samples, a large number of intergranular pores and intragranular pores were found, and some micro-fractures developed. Micro-fractures and interparticle pores are commonly developed in low gas-bearing shale samples. The micropores and mesopores of shale provide the main specific surface area and pore volume, which are the main sites for gas adsorption and storage. Organic pores and intergranular pores are the two most widely developed pore types in shale, mainly storing adsorption gas, while micro-fractures store free gas [73,74]. Therefore, in high gas-bearing and medium gas-bearing shale samples, shale gas is mainly dominated by adsorbed gas, and the gas transport mode is mainly sorption and diffusion. The shale gas in the low gas-bearing shale samples is mainly free gas, and the gas migration mode is mainly Darcy (Fig. 13). According to the analysis of MICP, the low gas-bearing samples have the largest pore throat radius, the medium gas-bearing samples are the

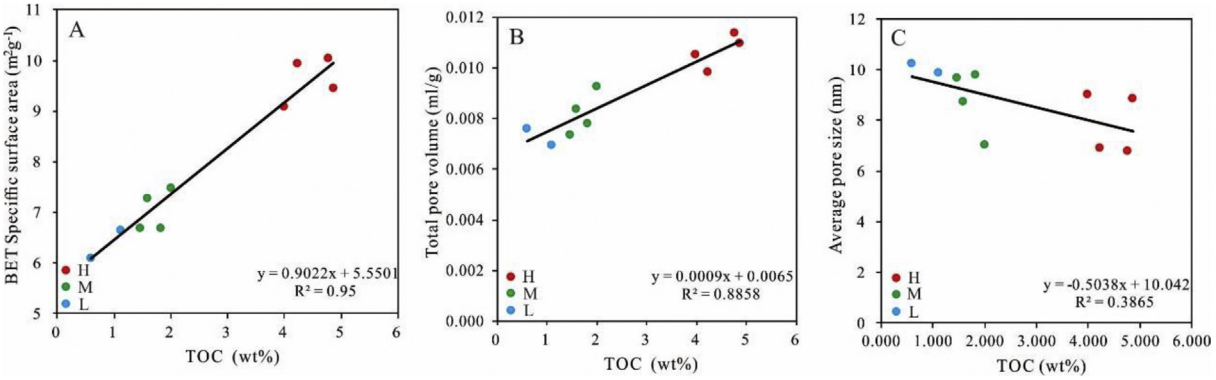

Fig. 11. (A) Relationship between TOC content and specific surface area. (B) Relationship between TOC content and total pore volume. (C) Relationship between TOC content and average pore size.

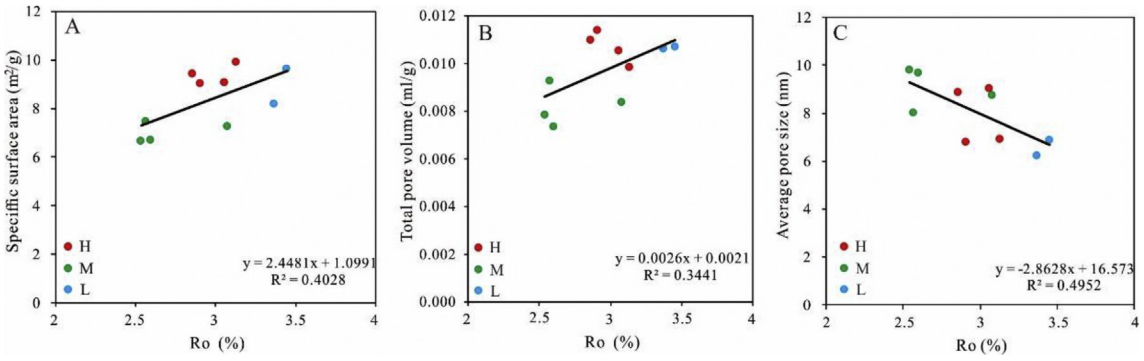

Fig. 12. (A) Relationship between R_o and specific surface area. (B) Relationship between R_o and total pore volume. (C) Relationship between R_o and average pore size.

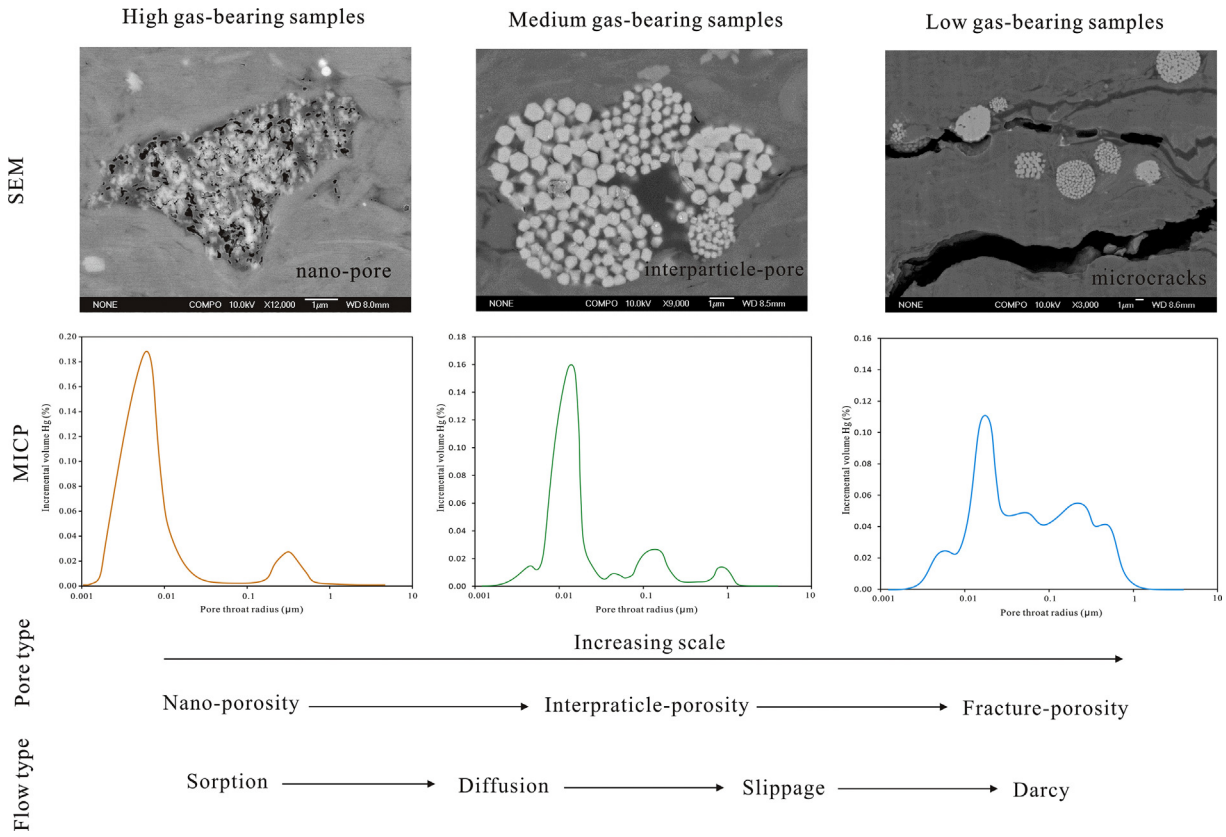

Fig. 13. Overview of the MICP and SEM of high gas-bearing sample C1, medium gas-bearing sample B1, and low gas-bearing sample E1. Different scale pore networks represent different pore types and flow types.

second, and the high gas-bearing samples are the smallest (Fig. 13). In high-gas samples, micropores, mesopores, and fractures have poor connectivity and fracture development is less. Therefore, the adsorbed gas can be well preserved, and less free gas is lost through the fracture. In medium gas-bearing samples, some mesopores are connected with the fracture and some of the fractures develop. Therefore, the adsorbed gas can be well preserved, and some of the free gas is lost through the fracture. In low gas-bearing samples, the mesopores have good connectivity to the fractures, and the fractures are generally developed. Therefore, shale gas is difficult to store and a large amount of gas escapes through the fracture.

5. Conclusion

Using a variety of methods to analyze different gas-bearing shale samples, the main conclusions of this work can be drawn as follow:

1. Observed by scanning electron microscopy, the main pore types of shale are clay mineral intercrystalline pore, intergranular pore, rock skeleton mineral pole, organic pore, and micro-fracture. In high gas-bearing shale samples, organic pores are generally developed, mainly in the nanometer-micron range. Among medium gas-bearing samples, intercrystalline pores and intragranular pores are mainly developed. Low gas-bearing samples mainly develop micro-fractures.
2. Through the observation of the thin-sections, we found that in the high gas-bearing samples, the bone needles, radiolaria, and bio-derived silicon were widely distributed. In medium gas-bearing samples, horizontal striations develop and the stratum is mostly carbonaceous. The content of quartz-based silt and particles is high. In the lower gas-bearing samples, the horizontal stratum is not developed, the quartz siltstone content is high, and the minerals such as calcite and pyrite are evenly distributed.
3. The T_2 distribution of the high gas-bearing samples is bimodal with a higher spectrum value. The pore throat radius is small and the pore connectivity is poor. The medium gas-bearing samples have a multimodal T_2 distribution. Both the mesopores and micro-fractures of the sample developed, but the connectivity was poor. The NMR characteristics of low gas-bearing samples exhibit a bimodal shape with one broader wave, and the spectrum value is the lowest. The pore throat radius is large and the pore connectivity is good. The T_2 distribution of NMR experiments can semi-quantitatively characterize the pore structure of shale.
4. The gas adsorption capacity, specific surface area, and pore volume of the shale are the largest in the high gas-bearing samples, the lower gas-bearing samples are the smallest. The high gas-bearing samples have the smallest

pore diameter, the low gas-bearing samples have the largest pore diameter. The main controlling factors affecting pore development are diagenesis, tectonics, TOC, and R_O.

Declaration of competing interest

The authors declare no conflict of interest.

Acknowledgments

We thank China Geological Survey (No. 12120114046901) for providing samples and partial data. This study was jointly supported by the National Natural Science Foundation of China (No. 41672134), the Shale Gas Survey of National Geological Survey (No. 12120114055801) and the Foundation of State Key Laboratory of Nuclear Resources and Environment. We thank Wanzhong Shi (China University of Geosciences) for many critical and constructive comments. We appreciate the enthusiastic support of G.Y. Zhai (China Geological Survey) for his help to provide the well logging data, and of Chi Zhang (The University of Kansas) for providing language help. We thank Nyujia Peng (East China University of Technology) for data collation.

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Journal of Natural Gas Geoscience
Volume 6, Issue 1, February 2021, Page 65

DOI: https://doi.org/10.1016/j.jnggs.2020.12.001

Available online at www.sciencedirect.com

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Journal of Natural Gas Geoscience 6 (2021) 65

http://www.keaipublishing.com/jnggs

Erratum to "Study on the characterization of pore structure and main controlling factors of pore development in gas shale" [J. Nat. Gas Geosci 5 (5) (October 2020) 255–271/JNGGS_165]

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Available online 30 December 2020

The publisher regrets "the affiliation of First Author Zhuang Xu of 'Study on the characterization of pore structure and main controlling factors of pore development in gas shale' should be School of Earth Sciences, East China University of Technology, Nanchang, 330013, China".

The publisher would like to apologise for any inconvenience caused.

DOI of original article: https://doi.org/10.1016/j.jnggs.2020.09.003.

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Peer review under responsibility of Editorial office of *Journal of Natural Gas Geoscience.*

https://doi.org/10.1016/j.jnggs.2020.12.001

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文献 5/9

Evaluating the potential of carbonate sub-facies classification using NMR longitudinal over transverse relaxation time ratio

Fan Zhang; Chi Zhang

Advances in Geo-Energy Research (2021); DOI: 10.46690/ager.2021.01.09

证据用途：其他合作论文背景；张弛老师角色：末位/通讯作者；证据等级：C

Advances in Geo-Energy Research

Vol. 5, No. 1, p. 87-103, 2021

Original article

Evaluating the potential of carbonate sub-facies classification using NMR longitudinal over transverse relaxation time ratio

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Keywords:
NMR logging
T_1/T_2 ratio
lithology indicator
carbonate lithofacies

Cited as:
Zhang, F., Zhang, C. Evaluating the potential of carbonate sub-facies classification using NMR longitudinal over transverse relaxation time ratio. Advances in Geo-Energy Research, 2021, 5(1): 87-103, doi: 10.46690/ager.2021.01.09

Abstract:
While the well log-based lithology classification has been extensively utilized in reservoir characterization, the classification of carbonate sub-facies remains challenging due to the subtle nuances in conventional well-logs. The nuclear magnetic resonance (NMR) log provides extra information of pore size and pore geometry features, improving differentiating carbonate sub-facies. Here we explore the feasibility of using the ratio between NMR longitudinal relaxation time and transverse relaxation time as a potential lithology indicator to determine carbonate sub-facies. We analyzed a series of logging data and corresponding core samples of Arbuckle Group carbonate containing mudstone, packstone, grainstone, incipient breccia, and breccia in northern Kansas for the characteristics of longitudinal relaxation times, transverse relaxation times, and longitudinal over transverse relaxation time ratios. The results show that mudstone, packstone, and grainstone exhibit high, intermediate, and low longitudinal over transverse relaxation time ratios, respectively, while incipient breccia and breccia have a wide range of longitudinal over transverse relaxation time ratios. Furthermore, we evaluated the potential of using longitudinal over transverse relaxation time ratios to classify carbonate sub-facies using multivariate analysis. By adding longitudinal over transverse relaxation time ratios to neutron porosity, total gamma-ray, and conductivity logs as inputs of automated facies classification, the prediction error decreased, especially for incipient breccia. On the contrary, when photoelectric log and computed gamma-ray are also available, adding longitudinal over transverse relaxation time ratios does not improve the accuracy of sub-facies classification. Our results suggest that longitudinal over transverse relaxation time ratio is an independent lithology indicator. However, it cannot replace other logs like gamma-ray and photoelectric logs in classifying carbonate sub-facies. Our study provided valuable evidence and credible elucidation of the importance and physicochemical mechanism of longitudinal over transverse relaxation time ratios, which is essential for deciphering NMR logging data in carbonate reservoirs.

1. Introduction

The primary task of petrophysical studies is to provide a full picture of the tightly coupled physical and chemical processes that govern the behavior of reservoir rocks and their constituent fluids. Petrophysical properties such as porosity, saturation, and permeability can be derived from well log responses, and represent the critical inputs of rock typing and subsurface geologic models (Busch et al., 1987; Delfiner et al., 1987). The classification of lithology from well logs is crucial to reservoir evaluation as the logs reflect the mineralogical and physical properties of the subsurface. Early attempts in the early 1980s to predict lithologies from log responses to augment core description used empirical geological facies-log response associations. Serra and Abbott (1982) developed the concept of electrofacies, defined as collective wireline logging responses that uniquely distinguish facies from one another, as a measure to extend the petrophysical properties identified in the core to the borehole scale. Numerous logs, including environment logs (e.g., temperature, pressure, and caliper logs), lithology logs (e.g., gamma-ray and density logs), neutron log, and resistivity log provide distinct information on the reservoir, can be grouped or clustered to define electrofacies, which are essential for reservoir characterization (Bucheb and Evans,

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Yandy Scientific Press

2207-9963 © The Author(s) 2021.
Received February 22, 2021; revised March 5, 2021; accepted March 5, 2021; available online March 8, 2021.

1994; Mathisen et al., 2001; Kim et al., 2005; Perez et al., 2005; Lim et al., 2006; Ali et al., 2008; Sharma et al., 2011).

Among these borehole well logs, nuclear magnetic resonance (NMR) wireline tools such as magnetic resonance imaging logging (MRIL) has created excitement in the well logging community. NMR is a geophysical method providing fluid content and porous structure of porous media due to its direct sensitivity to hydrogen (Coates, et al., 1999). In the past few decades, the NMR technique has been successfully applied in well logging and laboratory to estimate water content, porosity, pore size distribution, irreducible and free water content, and permeability (for detailed reviews, see Coates et al. (1999) and Dunn et al. (2002)). Combining with other logs, NMR log has been applied to zonation and lithology classification (or rock typing) for the last two decades. Previous studies use NMR log to identify the lithology by analyzing the partitioning of the pore size distribution (Doveton and Watney, 2015) and the variation of log mean relaxation time (T_{2ml}) (Skalinski et al., 2006; Rastegarnia et al., 2017). However, there are other information from NMR logs are not fully exploited in rock typing, such as the longitudinal (T_1) to transverse (T_2) relaxation time ratio–T_1/T_2. The T_1/T_2 ratio has been interpreted as an indicator of the molecular environment of hydrogen proton (Callaghan, 1991; McDonald et al., 2005). In bulk and unrestricted space, without the diffusion of the fluid, the T_1/T_2 value is equal to one (Kleinberg et al., 1993b). Within a confined environment, such as a pore, decreasing T_2 corresponds to the restricted motion of water molecules, whereas increasing T_1 indicates more ordered lattice structure and higher crystallinity (Callaghan, 1991). These results suggest that T_1/T_2 value would be higher in more ordered materials, and in materials with smaller pores.

Kleinberg et al. (1994) suggested an average T_1/T_2 =1.6 for 6 sandstone samples and the higher clay content and metal content will increase the T_1/T_2 ratio due to the higher content of the paramagnetic particle. Korb et al. (1997) and Godefroy et al. (2001) gave an explicit expression of the frequency dependent T_1/T_2 ratio. Given the fact that the NMR frequency of electron is way larger than the NMR frequency of proton, the expression of T_1/T_2 ratio can be simplified as

$$\frac{T_1}{T_2}=\frac{16\ln\dfrac{\tau_m}{\tau_s}+3\ln\dfrac{1+(\omega_p\tau_m)^2}{\dfrac{\tau_m}{\tau_s}+(\omega_p\tau_m)^2}}{6\ln\dfrac{1+(\omega_p\tau_m)^2}{\dfrac{\tau_m}{\tau_s}+(\omega_p\tau_m)^2}} \tag{1}$$

where τ_m is the surface diffusion correlation time describes the molecular jumps of adsorbed molecules, for instance, water molecules which diffuse across the pore surface, and τ_s is the surface residence time describes the time period of water molecules staying on the pore surface, and ω_p is the NMR frequency of proton (^{1}H in water molecules). The τ_m/τ_s has a range from 10^{-2} to 10^{-4} and $\omega_p\tau_m \gg \tau_m/\tau_s$ (McDonald et al., 2005). Both modeling and experiments support that T_1/T_2 ratio has the potential to be an indicator for lithology. For instance, rocks with various minerals and compositions should have distinct surface environments yielding various T_1/T_2 ratios.

The purpose of our study is to evaluate the potential of using T_1/T_2 ratio derived from NMR logs to determine carbonate sub-facies. We analyzed the well logs and core samples from Arbuckle Group in Wellington (#1-32) in Sumner County, Kansas. A dolomite interval contains five sub-facies (mudstone, packstone, grainstone, incipient breccia, breccia) is selected and tested. The core samples were also measured by lab NMR to validate the sensitivity of T_1/T_2 ratio to lithology. The T_1/T_2 ratio is integrated with other logs to perform an electrofacies classification using principal component analysis and fuzzy-C mean clustering to predict carbonate sub-facies. The prediction accuracies are compared among four different log sets to assess the contribution of T_1/T_2 ratio as lithology indicators. Our results confirmed that T_1/T_2 ratio is an independent lithology indicator, but it cannot replace other logs like gamma-ray and photoelectric log in determining carbonate sub-facies. Our study is the first attempt to exploit the full potential of NMR logging data in characterizing petrophysical properties of carbonate reservoir rocks by investigating T_1/T_2 ratio through linking NMR theoretical background with core experimental data and NMR log data. Our study provided valuable evidence and credible elucidation of the importance and physicochemical mechanism of T_1/T_2, which is essential for deciphering NMR logging data in carbonate reservoirs.

2. New theory

2.1 NMR fundamentals

The fundamental physical principle of NMR is the behavior of the nuclei. In the presence of a background magnetic field (B_0), the nuclear spins associated with the hydrogen nuclei are oriented in the direction parallel to the background field, which can be tipped out of alignment by an external radiofrequency (RF) magnetic field (B_1). After the removal of the external field, the nucleus relaxes back to equilibrium. Regular NMR measurement generates two main relaxation results: T_1 and T_2 relaxation time. The magnitude of the relaxation decay is proportional to water content, while the distribution of relaxation time (T_1 or T_2) can be used to estimate pore size distribution, irreducible and movable liquid saturation, surface area-to-volume ratios, and hydraulic conductivity (Coates et al., 1999). The relaxation time distribution is linked to three different mechanisms (Kleinberg et al., 1994; Westphal et al., 2005).

$$\frac{1}{T_1}=\frac{1}{T_{1B}}+\frac{1}{T_{1S}} \tag{2}$$

$$\frac{1}{T_2}=\frac{1}{T_{2B}}+\frac{1}{T_{2S}}+\frac{1}{T_{2D}} \tag{3}$$

where the subscripts B, S, D represent bulk fluid (free fluid) relaxation, surface relaxation, and molecular diffusion, respectively. Bulk fluid relaxation times (T_{1B} and T_{2B}) are the relaxation time of free fluid (3.82 s at 30 °C (Simpson and Carr, 1958)). In geologic media, we assume a fast diffusion regime, the bulk fluid contribution ($1/T_{1B}$ and $1/T_{2B}$) and the

diffusion relaxation ($1/T_{2D}$) can be neglected. Thus,

$$\frac{1}{T_{1,2}} \approx \frac{1}{T_{1,2S}} + \rho_{1,2}\frac{S}{V} \tag{4}$$

where $\rho_{1,2}$ is the surface relaxivity corresponding to the capability of relaxation enhancement on the surface, S and V are the surface area and volume of pore space, respectively. Hence, $T_{1,2}$ is proportional to the surface area to volume ratio S/V (will be presented as S_{por} below) and pore size. Accurate determination of surface relaxivity $\rho_{1,2}$ is crucial in estimating pore size distribution and is usually resolved by calibrating NMR relaxation time using mercury intrusion capillary pressure or/and gas adsorption experiments.

2.2 Calculation of T_1/T_2 ratio

From the basic principles of NMR, T_1 and T_2 spectra are very similar in shape. Therefore, the T_1/T_2 ratio can be simply considered as a scaler of longitudinal relaxation time distribution versus transverse relaxation time distribution. The intensity function $a_{1,2}$ of both longitudinal and transverse relaxation time (Khetrapal et al., 1975; Kleinberg et al., 1994) can be derived from the following equations:

$$M_1(t) = \sum_{i=0} a_1(T_{1i})\left(1 - 2e^{-\frac{t}{T_{1i}}}\right) \tag{5}$$

$$M_2(t) = \sum_{i=0} a_2(T_{2i})e^{-\frac{t}{T_{2i}}} \tag{6}$$

where t is the time for one pulse train, M_1 and M_2 are the magnitude of magnetizations. To calculate T_1/T_2 ratio, both a_1 and a_2 are required (Kleinberg et al., 1993a). For instance, a_1 and a_2 are intensity functions for discrete relaxation times T_i ($i = 4, 8, \ldots, 2048$ ms). The zero-normalized cross-correlation coefficient, P, can be calculated as

$$P_{a_1a_2}(\omega) = \sum_{i=0} \frac{\left[a_1(T_{1i}) - a_1\left(\overline{T_1}\right)\right]\left[a_2(\omega T_{2i}) - a_2\left(\omega\overline{T_2}\right)\right]}{\sigma_{a_1}\sigma_{a_2}} \tag{7}$$

where ω is the T_1/T_2 ratio, σ_{a_1} and σ_{a_2} are the standard deviation of amplitudes of T_1 and T_2 spectrums, $\overline{T_1}$ and $\overline{T_2}$ are the mean value of T_1 and T_2. This simplified correlation function is constrained by the shape and range of T_1 and T_2 distributions. Although ω can be obtained by maximizing $P_{a_1a_2}(\omega)$, the effective range of i should be carefully selected with fully considering the saturation, saturation fluids, and the development of microfractures (Kleinberg et al., 1993b; Song et al., 2002).

3. Litholofacies prediction

3.1 Petrophysical recognition of mineralogy

Single variable methods generally are used for early-stage and current electrofacies recognition. For example, Flaum and Pirie (1981) applied spectral gamma-ray to quantitatively subdivide sandstone, clay, limestone in strata of the Cotton Valley Group (Jurassic) in eastern Texas, and Lucia (2007) successfully subdivided mud-supported carbonate and grain-supported carbonate in San Andres and Grayburg formations (Permian) in western Texas using computed gamma-ray log. However, the single variable analysis is not applicable to clean reservoir rocks with no clay, e.g., sandstone, or highly heterogeneous lithologies, e.g., carbonate.

In most strata, integrating more logs is necessary for electrofacies recognition. Litho-porosity cross-plots like the M-N plot (Burke et al., 1969), which combines neutron log, density log, and sonic log, is another frequently used electrofacies discriminator. However, in this approach, 'M' and 'N' are created without correlations with physical properties. To improve the litho-porosity method, Clavier and Rust (1976) developed the matrix-identification (MID) plot, which is also called U_{maa} vs. ρ_{maa} plot. It uses two quasi-physical quantities: apparent matrix density (ρ_{maa}) in g/cc and apparent volumetric cross section of photoelectric absorption (U_{maa}) in barns/cc calculated by:

$$\rho_{maa} = \frac{\rho_b - \rho_f\varphi_t}{1 - \varphi_t} \tag{8}$$

$$U_{maa} = \frac{P_eF\rho_b - U_f\varphi_t}{1 - \varphi_t} \tag{9}$$

where ρ_b is bulk density, ρ_f is the fluid density, φ_t is true volumetric porosity, P_e is photoelectric absorption measurement, P_eF is the photoelectric absorption factor, and U_f is the fluid volumetric photoelectric absorption. More recently, third-generation density logs (litho-density log) provide an additional photoelectric measurement. P_e, which reflects the average atomic number of the formation, is a reliable indicator of the formation lithology (Dewan, 1983). In carbonate reservoir logging analysis, three end-member minerals (quartz, calcite, dolomite) can be plotted in a triangle on the U_{maa} and ρ_{maa} cross plot based on standard U_{maa} and ρ_{maa} values from various minerals. End-member ρ_{maa} values for pure quartz, calcite, and dolomite are 2.65, 2.71 and 2.87 g/cc, respectively, whereas the end-member values U_{maa} for quartz, calcite, and dolomite are 4.78, 13.80 and 8.98 barns/cc, respectively (Poupon et al., 1971). The artificial or formalized parameters are defined to be independent of porosity (Doveton, 2014). If the pore characteristics dominate the log signals, the MID and $U_{maa} - \rho_{maa}$ triangle approaches are less accurate.

3.2 Automated electrofacies classification

To obtain more representative electrofacies from well logs, an alternative method is proposed to use multivariate data analysis to correlate well logs with core descriptions. The multivariate data analysis is expected to improve petrophysical properties estimation by clustering the logs with similar characteristics into groups (electrofacies). With the blooming of computational power, the use of mathematical procedures and computer programs to automatically classify electrofacies from even more complex set of variables has become more common (Doveton and Cable, 1979; Busch et al., 1987; Delfiner et al., 1987; Grimm, 1987; Bucheb and Evans, 1994). One complete multivariate statistical analysis includes analysis of variance, nonparametric regression, clustering, and classification. In our study, the analysis follows three steps: 1) Principal component

analysis (PCA); 2) Fuzzy c-mean clustering; and 3) Classification. The logs were first transformed into PCA scores, which are then subjected to model-based clustering, and lastly, to be categorized into groups that can be interpreted as electrofacies. The details of each step are discussed below.

3.2.1 Step 1: PCA

PCA is a multivariate technique in which several related variables are transformed into a set of uncorrelated variables (Jolliffe, 2011). The most significant advantage of PCA is the simplification of data by reducing its dimensionality, and it facilitates extraction of the structure of variables. This procedure is vital in the modern well logging analysis as logging suites provide closely correlated logs, challenging the multivariate analysis.

In PCA, all logs first should be normalized with zero-score standardization, which subtracts the mean and standardizes data to the unit deviation of each log to ensure the equal weight of each response. The first two principal components (PC1, PC2) are selected to address the score plot. Since most of the dataset variability is contained in the first two PCs, noise collected in the observations can be extracted and eliminated from the original data suite. However, even if well logs can be projected into uncorrelated variables, it is unrealistic to establish connections between principal components and actual petrophysical properties, different from the capabilities of core descriptions and logs.

3.2.2 Step 2: Clustering

Cluster analysis is used as a primary stage for unsupervised learning to group the dataset into subsets so that similar instances are grouped, whereas different instances belong to different groups (Rokach and Maimon, 2005). The purpose of the cluster analysis is to separate the data from a suite of logs into meaningful groups that can be interpreted in terms of electrofacies. Standard clustering algorithms use a hard boundary, only allowing a component to be in one cluster. The dataset can be divided into several subgroups using centroid-based algorithms (K-mean clustering, mean-shift clustering), density-based algorithms, hierarchical development, or neural network algorithms (see Jain (2010) for a review of details of current pattern recognition and clustering methods). However, since the underlying algorithms are intended to be efficient for pattern recognition instead of revealing physical mechanisms, these methods do not perform well if the clusters are not drastically separated. Sub-facies with very similar petrophysical properties like packstones and mudstones usually overlap in the score maps and are challenging to be isolated. Therefore, hard boundaries are not preferred for electrofacies classification. Fuzzy c-means (FCM) is a method of centroid-based clustering (Dunn, 1973; Bezdek et al., 1984) which employs fuzzy membership and allows one data point to belong to many (or even all) clusters with degrees of membership from 0 to 1 (e.g., the total membership = 1) (one can refer to Appendix A for the detailed theory of FCM algorithm). Unlike hard boundary clustering, FCM allows a changeable clustering result by adjusting the membership threshold to each cluster, giving the best result for the overlapped dataset. Usually, the FCM is better than the k-means algorithm in recognizing members with characteristics of multiple groups (Bezdek, 1981). For instance, mudstone, packstone, and grainstone can have very similar PC1 and PC2 values, closing cluster centers. Therefore, FCM would be less successful in recognizing sub-facies that include a mudstone surrounded by the packstone and grainstone in the score plot.

3.2.3 Step 3: Classification

The last step is to assign lithologies to clusters to provide them meaningful electrofacies. This procedure is usually performed by either applying the lithologies of centroid to the whole cluster universally (Doveton, 2014) or assigning lithologies based on their corresponding logs and log interpretations (Roslin and Esterle, 2016). However, those two methods rely on 1) a relatively large number of clusters; and 2) distinct log values among clusters and thorough understanding of these log values. These two conditions are not satisfied in this study as our carbonate samples are with a fixed number of clusters and have relatively small changes in log responses. To avoid human bias while assigning the lithologies to clusters and make this interpretive procedure into an unsupervised way, we applied a statistical method to minimize the overall objective function S:

$$S = \phi_m + \beta\phi_n \tag{10}$$

where ϕ_m is data misfit and $\phi_m = \sum_{i=1}^{n}\left[(d_i^e - d_i^l)/\sigma\right]^2$, d_i^e and d_i^l are the electrofacies and lithologies for ith data point respectively, σ is the standard deviation of the dataset, and n is the total number of data points. ϕ_n is representing the model norm or regularizer and

$$\phi_n = \alpha\sum_{i=1}^{n-1}\left(d_{i+1}^{elec} - d_i^{elec}\right)^2 + (1-\alpha)\|misprediction\ rate\|_2^2 \tag{11}$$

Electrofacies that do not match lithologies would be considered as mispredicted. The *misprediction rate* is simply the number of mispredicted data points over the total number of data points. Therefore, ϕ_n contains the information of the smoothness of rock type transactions and the success rate of prediction. Weighting parameter α and β are used to adjust the weight among terms and can be optimized, α is the weight between the smoothness of rock type transactions and the success rate of prediction and β is the weight between data misfit and model norm.

4. Material and method

4.1 Geological setting, well logs, and cores

Well logs and core plug samples used in this study were from the deep subsurface (4900 to 5000 ft) from the well Wellington #1-32 in Sumner County, Kansas (Fig. 1). Wellington #1-32 was logged with photoelectric (PE), neutron porosity (NPHI), 90 inch conductivity (CT90), gamma-ray total (GRTO), potassium (POTA. %), uranium (URAN. ppm), thorium (THOR. ppm), and MRIL. Our target interval of the well Wellington #1-32 is the lower Cambro-Ordovician Arbuckle

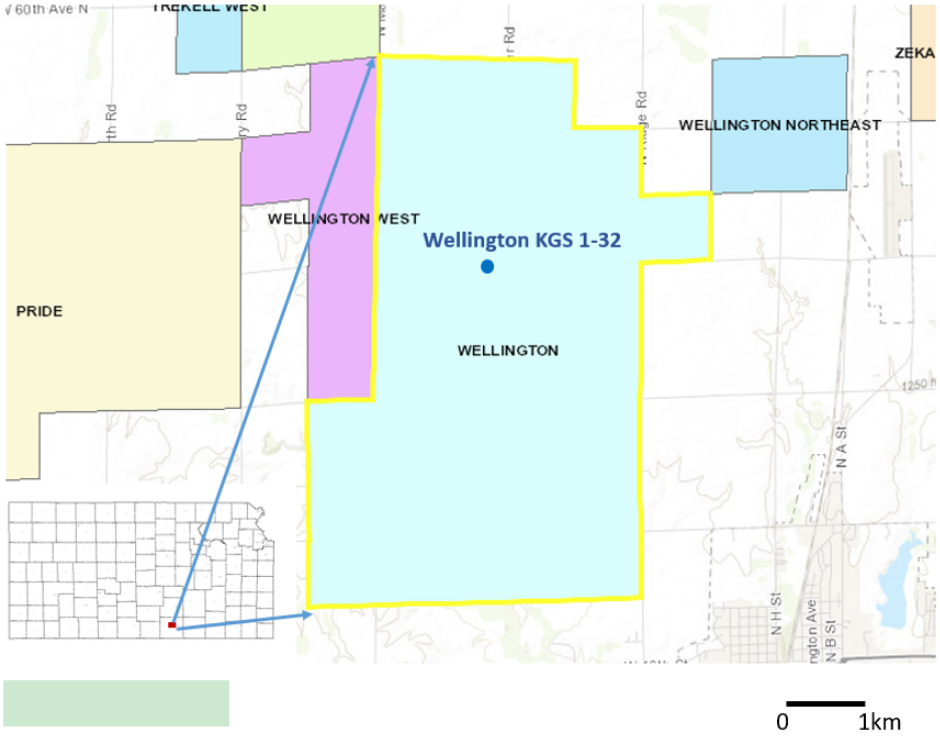

Fig. 1. The location of well Wellington KGS #1-32 is presented on a Kansas map. The light blue area with yellow outlines is the Wellington oil and gas field. The blue dot marks the location of Wellington KGS #1-32.

Table 1. Porosity, volumes, permeabilities, T_1/T_2 ratios from core and logs for each sub-facies.

Sub-facies (Based on core)	Porosity	Volume (ml)	Permeability (mD)	T_1/T_2 core	T_1/T_2 log*	T_1/T_2 log average †
MS	2.3	38.5	2.95	3.9	3.220	2.556
PS	8.5	22.3	0.06	2.6	2.300	2.502
GS	9.72	38.3	202.61	1.6	1.346	2.339
IB	3.36	21.1	88.9	1.3, 2.3, 3.4	1.274	1.834
BC	9.77	41.9	3.7	2.5	2.793	2.820

Note: Sub-facies are discriminated based on core observations where MS=mudstone, PS=packstone, GS=grainstone, IB=incipient breccia, and BC=breccia.
*log data are from the same depths as core plug samples.
† averaged over the entire depth.

Group. The Arbuckle Group is recognized as strata dominated by cherty dolostone, dolostone, and bedded limestone (Barwick, 1928; Zeller, 1968; Cole, 1975) and was developed in the southern Great American Carbonate Bank. The lithologies of the Arbuckle Group reflect a cyclical depositional environment from shallow subtidal to intertidal conditions (Franseen, 1994). The sub-facies of Arbuckle Group in well Wellington #1-32 include mudstone, packstone, grainstone, incipient breccia, and breccia. Grainstone consists of intraclasts, ooids, peloids, and skeletal fragments. It represents high-energy deposition in subtidal to peritidal environments. The well-cemented pores lead to the low permeability of the samples. Packstone and mudstone are mainly massive to horizontally laminated with porosities from 3% to around 20%. Breccias are commonly associated with subaerial exposure. Seawater evaporation leads to evaporite deposition, and the dissolution of evaporite, in turn, induced collapse horizons in Arbuckle strata. In short, due to its heterogeneous texture, breccia exhibits a wide range of porosity and permeability. Franseen (1994, 2000) and Steinhauff et al. (1998) identified lithological characteristics of the Arbuckle Group in Kansas. They concluded that although Arbuckle carbonates have been extensively dolomitized, their original depositional facies and textures are much preserved as the dolomitization is non-fabric destructive. We carefully selected five core plug samples, one from each carbonate sub-facies. Table 1 shows the lithologies, porosities volumes, and air permeabilities with Klinkenberg correction (K_{air}) of core plug samples.

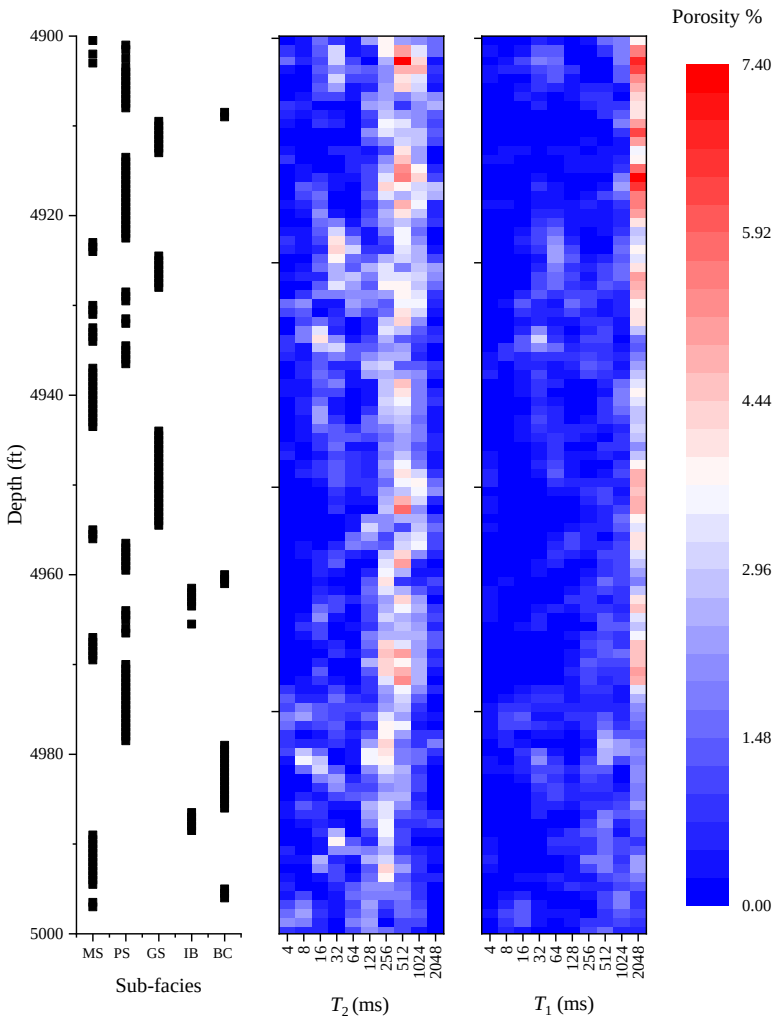

Fig. 2. Sub-facies of samples (where MS = mudstone, PS = packstone, GS = grainstone, IB = incipient breccia, BC = breccia) from the interval of interest (4900-5000 ft) are listed in the order of depth along with the corresponding T_2 and T_1 distributions. Note that NMR relaxation data are subdivided into bins with 2^n (n $\in$ [2,10]) ms (as provided by the service company). Also, note some samples are missing from the lithofacies (4901, 4920-4923, and 4997-5000 ft), which will be removed from the subsequent analyses.

4.2 Laboratory NMR measurements

Five samples are selected for lab experiments to present each lithology. Mudstone, packstone, grainstone, and incipient breccia samples are from the target interval. The breccia sample tested in our experiment is picked out of our target interval from the depth of 3172 ft, as no breccia sample is recovered in the target interval. All core samples were first oven-dried at 70 °C for 48 hours and placed under a vacuum with deionized water for 48 hours to adequately saturate the effective pore space. All samples were wrapped with Teflon tape after saturation to prevent water run-off. Laboratory NMR experiments were taken at 27 °C using a 2 MHz Rock Core Analyzer (Magritek Ltd). The core samples were tested in the lab using a 2 MHz Rock Core Analyzer (Magritek Ltd) for 1) T_1 measurements using Inversion Recovery (IR) pulse sequence (Kenyon et al., 1988) with 20 recovery time τ_1 from 0.1 to 5000 ms; 2) T_2 measurements using Carr-Purcell-Meiboom-Grill (CPMG) pulse sequence (Carr and Purcell, 1954; Meiboom and Gill, 1958) with echo times of 200 μs, and inter-experiment time of 15000 ms; and 3) $T_1 - T_2$ 2D correlation measurements using an IR-CPMG pulse sequence (Song et al., 2002). A minimum of 32 scans were stacked together to improve the signal-to-noise ratio (SNR) for all three measurements. The recorded T_1 and T_2 were inverted using the Laplace inversion algorithm, and $T_1 - T_2$ correlations were inverted using a fast iterative shrinkage-thresholding algorithm (FISTA).

5. Reslts and discussion

5.1 T_1 and T_2 distributions

Fig. 2 shows lithologies and corresponding T_2 and T_1 distributions within our target depth interval of 4900 to 5000 ft. Overall, the transverse relaxation rate $1/T_2$ is larger than the longitudinal relaxation rate $1/T_1$. It is noteworthy that most of the porosities in T_1 distributions are accumulated in the bin T_1 = 2048 ms which is due to the detecting cap of the equipment is set to 2048 ms.

T_1, T_2, and $T_1 - T_2$ measurements were performed for

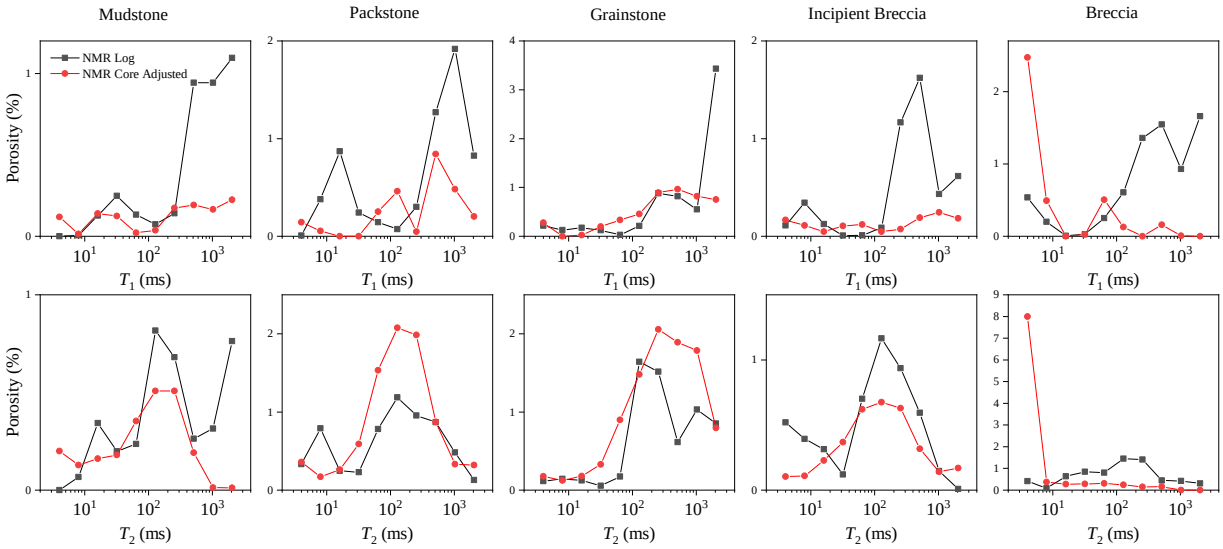

Fig. 3. The comparison of T_1 (upper row) and T_2 (lower row) spectra generated from NMR log and NMR core data for five sub-facies.

all five core-plug samples to obtain the characteristics of relaxation time distributions and T_1/T_2 ratios of each lithology. The T_1 and T_2 distributions from core measurements are compared with NMR logs in Fig. 3. In order to compare these two datasets on the same scale, incremental porosities from 200 relaxation steps (0.1 to 5000 ms) are congregated to 8 data points (2 to 2048 ms) for core measurement (NMR Core Adjusted in Fig. 3). Overall, T_2 distributions between NMR logs and NMR core measurements show better correlation than T_1 distributions. The shapes and positions of major peaks of T_2 distributions are very similar in NMR logs and core measurements. By contrast, T_1 distributions exhibit a general gap of porosity at higher relaxation time where NMR logs captured more porosities than NMR core measurements. While the NMR core measurement captures a higher porosity than the log results below 2 ms in the breccia sample.

The major discrepancy at high relaxation time could be explained by two possible reasons. Firstly, the effective collecting area of NMR logging tools is 61 cm in diameter and 1.07 m in length, two to three times larger than the core plug sample size (20-40 cm^3). Such a broader detecting range of NMR logging tools potentially increases the signals capture area that exceeds the core plug size. Additionally, the long measurement time of T_1 logging can capture signals from high relaxation time in multiple aperture volumes if the sonde moves fast (Kleinberg et al., 1993b; Coman et al., 2015). Due to these reasons, discrepancies of T_1 distributions between core experiments and NMR logs are more significant than that of T_2 distributions (Fig. 3). The discrepancy at low relaxation time, on the other hand, is caused by long echo time used in NMR log, which is 1.2 ms (Coates et al., 1999; Doveton and Watney, 2015). With a long echo time, signals from micropores which have relaxation times below 1.2 ms are not collected. Therefore, signals of samples with abundant micropores can be underestimated by NMR logging. The discrepancy of relaxation time distributions between core experiments and NMR logs can induce error while calculating the T_1/T_2. As shown in Eq. (7), we calculate T_1/T_2 by maximizing the zero-normalized cross-correlation $P_{a_1a_2}(\omega)$ where the intensity functions a_1 and a_2 are weighted by the standard deviation of σ_{a_1} and σ_{a_2}. Therefore, as long as the major peak positions are the same in core experiments and logs (e.g., our mudstone, packstone, grainstone and breccia samples), the difference between intensity of the distribution spectra will not affect the T_1/T_2. However, samples that show unclear peaks (e.g., incipient breccia sample) might have very different core T_1/T_2 and log T_1/T_2. Therefore, the weight of the intensity functions a_1 and a_2 have to be manually calibrated.

5.2 T_1/T_2 ratio

Due to the limitation mentioned above, the log derived T_1/T_2 ratios might deviate from the actual values. A comparison between T_1/T_2 derived from logging data and core experiments can elucidate the relationship between lithology and T_1/T_2. The lab T_1-T_2 2D mapping results are presented in Fig. 4. The essential criteria for picking the T_1/T_2 value is to find the best fitting line should cross the peak amplitude for most pore families, which are defined as compartments of signals that are distinguishable from each other (Fleury and Romero-Sarmiento, 2016). According to these criteria, the best fitting T_1/T_2 for mudstone, packstone, grainstone, and breccia are 4.1, 2.6, 1.6, and 2.5, respectively. Incipient breccia, however, reveals at least three distinct pore families that associate with three different T_1/T_2 values as 1.3, 2.3, and 3.4. The change of T_1/T_2 is usually due to abundant free water that reduces the T_1/T_2 ratio at high relaxation time (d'Eurydice et al., 2016), and/or the existence of other fluid components (Fleury and Romero-Sarmiento, 2016). However, none of these attributions applies here since incipient breccia was only saturated by only water, and its T_1/T_2 ratio increases with the relaxation time. Therefore, we concluded the observed three pore families in incipient breccia T_1-T_2 2D maps are associated with different mineral compositions of the autoclasts within incipient breccia.

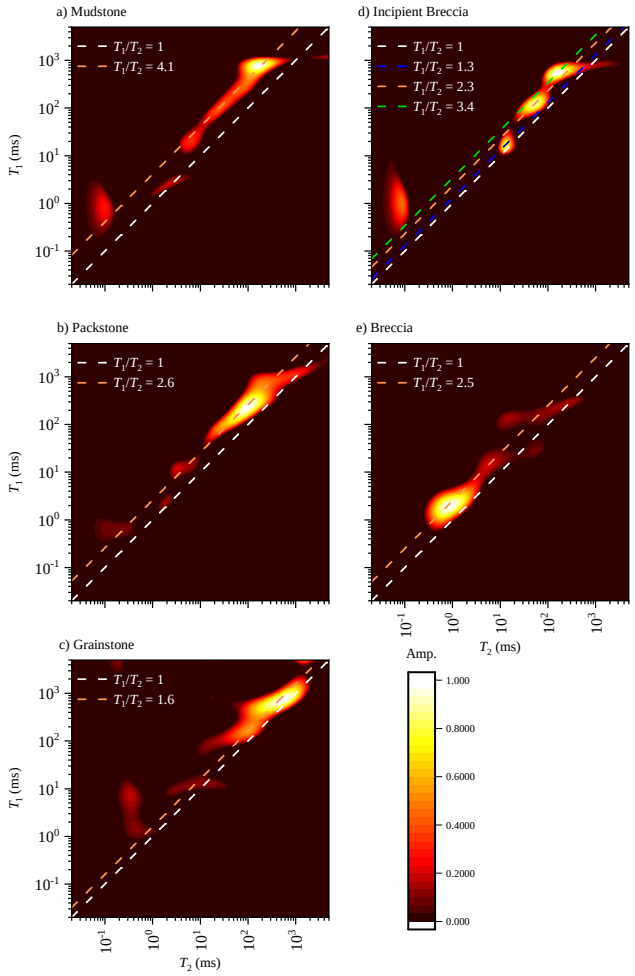

Fig. 4. The $T_1 - T_2$ correlation maps for five core-plug samples. In maps associating with mudstone (a), packstone (b), grainstone (c), and breccia (e), the white dashed lines are for $T_1/T_2 = 1$ and the orange dashed lines are for the best fit of T_1/T_2. Incipient breccia (d) has three distinct T_1/T_2 ratio where the white dashed line is for $T_1/T_2 = 1$, the blue dashed line is for $T_1/T_2 = 1.3$, the orange dashed line is for $T_1/T_2 = 2.3$, and green dashed line is for $T_1/T_2 = 3.4$.

The logging T_1/T_2 values ranges from 0.66 to 7.44, which are consistent with the measured T_1/T_2 values for carbonate samples (Kenyon et al., 1988; Cheng et al., 2017). Table 1 presents the T_1/T_2 for 5 core samples from both lab NMR and logging NMR, as well as the average T_1/T_2 values for five lithologies in the target interval. The T_1/T_2 decreases as the lithologies transit from mudstone to packstone, and then to grainstone. Breccia and incipient breccia have mixed T_1/T_2 values, but overall breccia has higher T_1/T_2. From these consistent evidence shown in both core and log data, T_1/T_2 has shown the potential to be an independent lithology indicator.

We use a ternary plot of quartz, calcite, and dolomite abundance to represent the mineral constituents of a carbonate reservoir (Fig. 5). The plot indicates that mineral components within the target interval are mostly quartz and dolomite, combined with a limited amount of calcite. High ρ_{maa} and U_{maa} indicates more dolomite content, while lower ρ_{maa} and U_{maa} reflects increasing silica (quartz) content. In Fig. 5(b), relatively high T_1/T_2 values (>5) and relatively low T_1/T_2 values (<2) correspond with highly dolomitic samples, whereas intermediate T_1/T_2 (2-5) samples are broadly distributed. The correlation between dolomite percentage and T_1/T_2 value suggests that T_1/T_2 can be affected by the mineral composition. The overlapping of low T_1/T_2 and high T_1/T_2 clusters highlights the fundamental shortcoming of MID plot, which limits the information only to 2D space.

5.3 Electrofacies classification

5.3.1 PCA results

Table 2 lists five lithologies and associated log responses, including T_1/T_2, apparent matrix density (ρ_{maa}), matrix volumetric photoelectric absorption (U_{maa}), total porosity (PHIT), lateral log CT90, total gamma-ray reading (GRTO), THOR, and POTA averaging over the target interval. Mudstone, packstone, and grainstone are high in NPHI, GTRO, ρ_{maa}, U_{maa}, T_1/T_2 and low in CT90 comparing with incipient breccia and

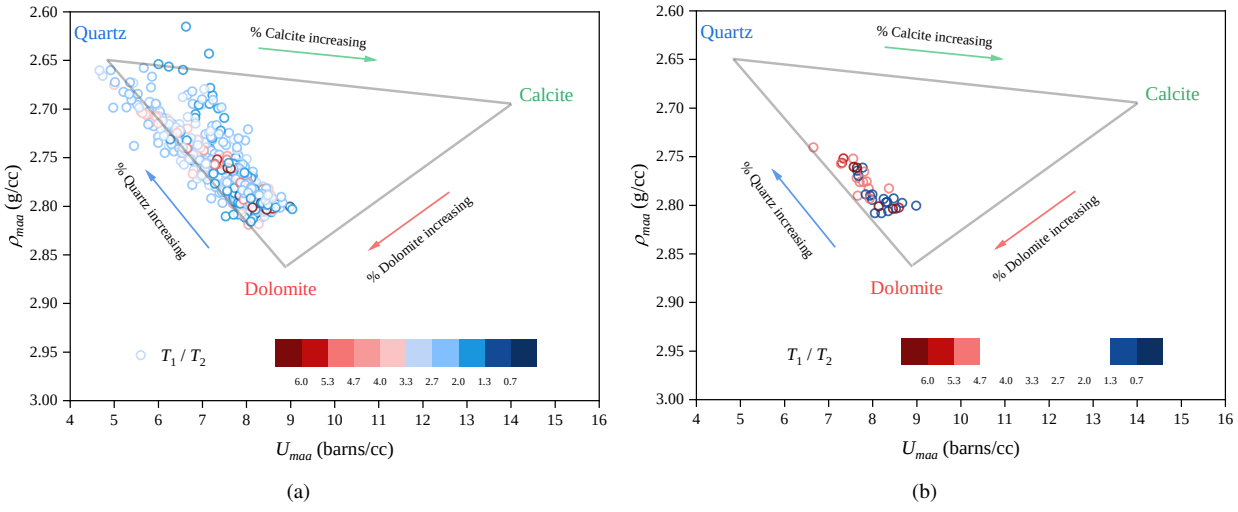

Fig. 5. a) $U_{maa} - \rho_{maa}$ crossplot (MID plot) with superimposed T_1/T_2, b) MID plot with only T_1/T_2 higher than 4.7 and lower than 1.3. T_1/T_2 are presented by circles in color with a range from 0.66 to 7.44. Gray lines link mineral end members of a ternary mixture of quartz, calcite, and dolomite. Arrows indicate an increasing percentage of the end-member minerals. Note the extreme values of high and low T_1/T_2 appear in high dolomite end.

Table 2. Average values of calculated log responses.

Sub-facies (Based on core)	NPHI (%)	CT90 (.mmo/m)	GRTO (.api)	POTA (.ppm)	THOR (.ppm)	ρ_{maa} (gm/cc)	U_{maa} (barns/cc)	T_1/T_2
MS	7.553	62.935	9.274	0.128	1.229	2.770	7.596	2.556
PS	7.847	63.429	9.659	0.144	1.096	2.774	7.786	2.502
GS	7.267	50.465	11.743	0.192	1.308	2.784	7.855	2.339
IB	6.272	76.566	9.238	0.150	1.381	2.764	7.492	1.834
BC	6.990	83.632	8.063	0.118	1.000	2.741	6.839	2.820

Table 3. Correlations (or loadings), eigenvalues and variances of Principal Component 1 to 8 referenced to logs.

	Principal Components							
	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8
NPHI	0.070	0.846	0.352	-0.087	0.337	0.169	0.033	-0.057
GRTO	-0.008	-0.083	-0.049	0.087	0.372	0.040	0.350	0.849
CT90	-0.295	0.111	0.271	0.869	-0.195	-0.174	0.064	0.002
ρ_{maa}	0.570	-0.157	0.122	0.277	-0.160	0.721	0.106	-0.039
U_{maa}	0.754	0.008	0.082	0.135	0.098	-0.621	-0.107	0.029
POTA	-0.011	-0.188	-0.080	0.085	0.464	-0.077	0.683	-0.513
THOR	-0.042	-0.088	-0.341	0.321	0.624	0.168	-0.586	-0.102
T_1/T_2	-0.113	-0.443	0.810	-0.160	0.264	0.004	-0.199	-0.015
Eigenvalue	2.893	0.732	0.509	0.294	0.219	0.124	0.084	0.006
Variance	0.595	0.151	0.105	0.060	0.045	0.026	0.017	0.001
Cumulative variance	0.595	0.746	0.850	0.911	0.956	0.982	0.999	1.000

breccia. Meanwhile, similar to the observed trend of T_1/T_2, the GRTO, ρ_{maa}, and U_{maa} data are showing either descending or ascending trends following the lithologic succession of mudstone, packstone, and grainstone.

The result from PCA provides eigenvalues, eigenvector loadings, and variance of each principal component referenced to eight logs (Table 3). The Pareto distribution of eight eigenvalues for each eigenvector (Fig. 6) illustrates that the first two principal components account for 74.6% of the total variability. Fig. 7 shows the biplot of PC1 and PC2, where loading vectors show the contribution of original variables (logs) to PC1 and PC2 and the correlations among original variables. If two vectors are close and form a small angle, the two input logs are well correlated (Martinez and Kak, 2001;

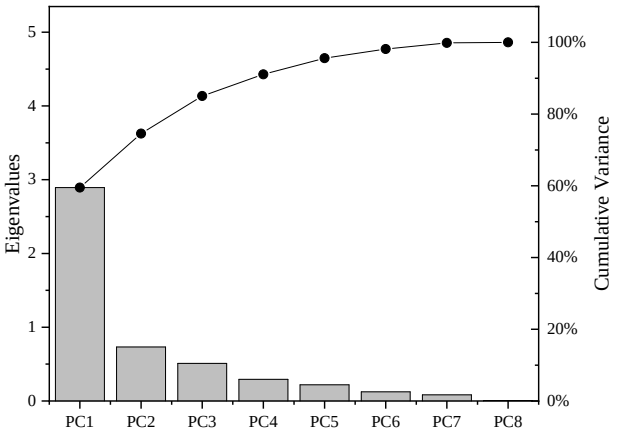

Fig. 6. Pareto plot for PCA. Bars represent eigenvalues and relative variances of the principal components.

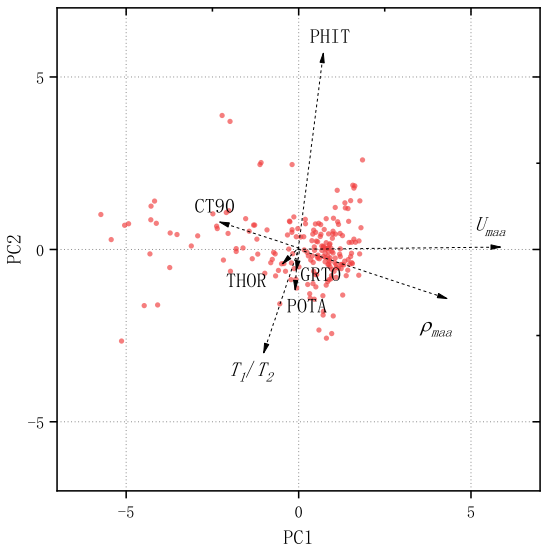

Fig. 7. Biplot of PC1 versus PC2 from PCA. Arrows represent eigenvectors of corresponding logs. The loadings determine the length and position of eigenvectors on PC1 and PC2. Note that a smaller angle between two loading vectors indicating the closer correlation between the input variables (logs) corresponding to loading vectors.

Doveton, 2014). The length of vectors represents the variance of each principal component gained from original logs. Indicated in Table 3 and Fig. 7, THOR, POTA, and GRTO all have very low weights on the first two principal components. The ρ_{maa} and U_{maa} vectors point to high positive loadings toward PC1 and low positive loading toward PC2, which are negatively correlated with CT90. The PHIT and T_1/T_2 show high loadings towards PC2.

The PCA results provide some preliminary links between logs and lithologies. For instance, the low loadings of THOR, POTA, and GRTO reveal the minor impact of clay content on differentiating carbonate sub-facies in this case. Also, the fact that CT90, ρ_{maa}, and U_{maa} contribute to most variance on PC1 while PHIT and T_1/T_2 count more for PC2, gives the non-uniqueness correlation between porosity and permeability (Babadagli and Al-Salmi, 2004; Ehrenberg and Nadeau, 2005; Doveton, 2014).

5.3.2 Cluster analysis results

We need to determine the number of expected clusters as a priori information when defining clusters. In this study, since the clustering algorithm–FCM is an unsupervised method, an accurate number of clusters is required during the self-training process (Bezdek et al., 1984; Zadeh, 1996). The number of clusters is fixed as five, corresponding to five primary sub-facies (mudstone, packstone, grainstone, incipient breccia, and breccia). Initially, five cluster centers are inserted randomly on the Euclidean space of the PC1-PC2 cross plot, and self-training starts with an initial guess of cluster membership values μ_{ij}. We then calculated and updated the objective function of all five clusters until the stopping criteria is met. At iteration 11, the objective function meets the minimum threshold, and the updated centers (crosses in color) are superimposed with clusters of principal component scores in Fig. 8. The PC1 reasonably maps EFs as EF1-3 plot in positive PC1 regime, whereas EF4-5 have negative PC1 values. EF4 tends to be more distinctive with lower PC1 values and can be easily differentiated from other EFs. Meanwhile, EF1, EF2, and EF3 are heavily overlapping, elucidating the importance of implementing a fuzzy membership clustering algorithm instead of hard-boundary algorithms.

5.3.3 Classification results

To achieve the unsupervised classification of electrofacies generated in the last step, we followed the following steps. First, five electrofacies are assigned with five sub-facies with no repetition, then S in Eq. (10) is calculated. This step is repeated with different permutations of lithologies k times, where k is 120 for 5 clusters. Finally, the analysis picks the combination that minimizes S as the lithologies matching electrofacies. As a result, EF1-5 clustered in Fig. 8 are assigned to mudstone, packstone, grainstone, incipient breccia, and breccia, respectively. The classification shows that EF1, EF2, and EF3 with positive PC1 scores suggest non-breccia carbonate (mudstone, packstone, grainstone), and EF4, EF5 with negative PC1 scores suggest breccia carbonate (incipient breccia and breccia).

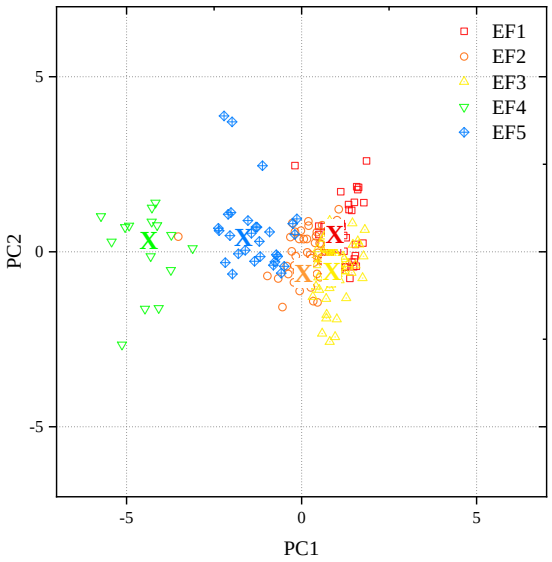

Fig. 8. Score plot of PC1 and PC2 with classified clusters and cluster centers as the results of cluster analysis. Five clusters are marked in different colors and symbols. The cluster centers are presented as 'X' marks with the same color of belonging clusters. Interpreted electrofacies by the classification analysis are assigned to corresponding clusters. Note the clusters of EF1, EF2, and EF3 overlap considerably.

The clear separation of breccia and non-breccia validates the distinct average log values in Table 2. Three non-breccia carbonate clusters overlap and have very close cluster centroids, due to their similar textures and petrophysical responses. In contrast, incipient breccia and breccia can be distinguished very well, as PC1 and PC2 scores are more distinctive as a result of the dissimilarity of petrophysical responses (Table 2). The breccia samples in the Arbuckle Group were formed due to the collapse of paleokarst features, and they were locally cemented with micrite (Franseen, 2000; Franseen et al., 2003). Brecciation and fracturing fabricate vuggy porosity, whereas the autoclasts (typically from mudstones and wackestones) were better cemented. As a result, despite the low PHIT, breccia samples are more conductive due to the well-connected vuggy pores and intercrystalline pores.

5.3.4 Predicted carbonate sub-facies

To quantitatively assess if T_1/T_2 can improve the accuracy of prediction of carbonate sub-facies, a series of electrofacies discrimination was conducted with different variable sets. Most borehole-based carbonate reservoir characterizations utilize density, neutron, conductivity, and gamma-ray logs (Dorfman et al., 1990; Lee et al., 2002; Perez et al., 2005; Asgari and Sobhi, 2006). To set the control group, we use PHIT, total gamma-ray, and lateral log 90 inches conductivity as the primary input variable set, namely Set 1, for electrofacies classification. T_1/T_2 was then added to the primary input dataset to form the second input variable set Set 2. Set 3 contains the full suite of log except T_1/T_2, and it was tested as a third input variable set. Finally, the full log suite, including T_1/T_2 serves as the fourth input variable set (Set 4). Each variable set was used as an input for PCA, clustering analysis, and classification, generating four facies profiles.

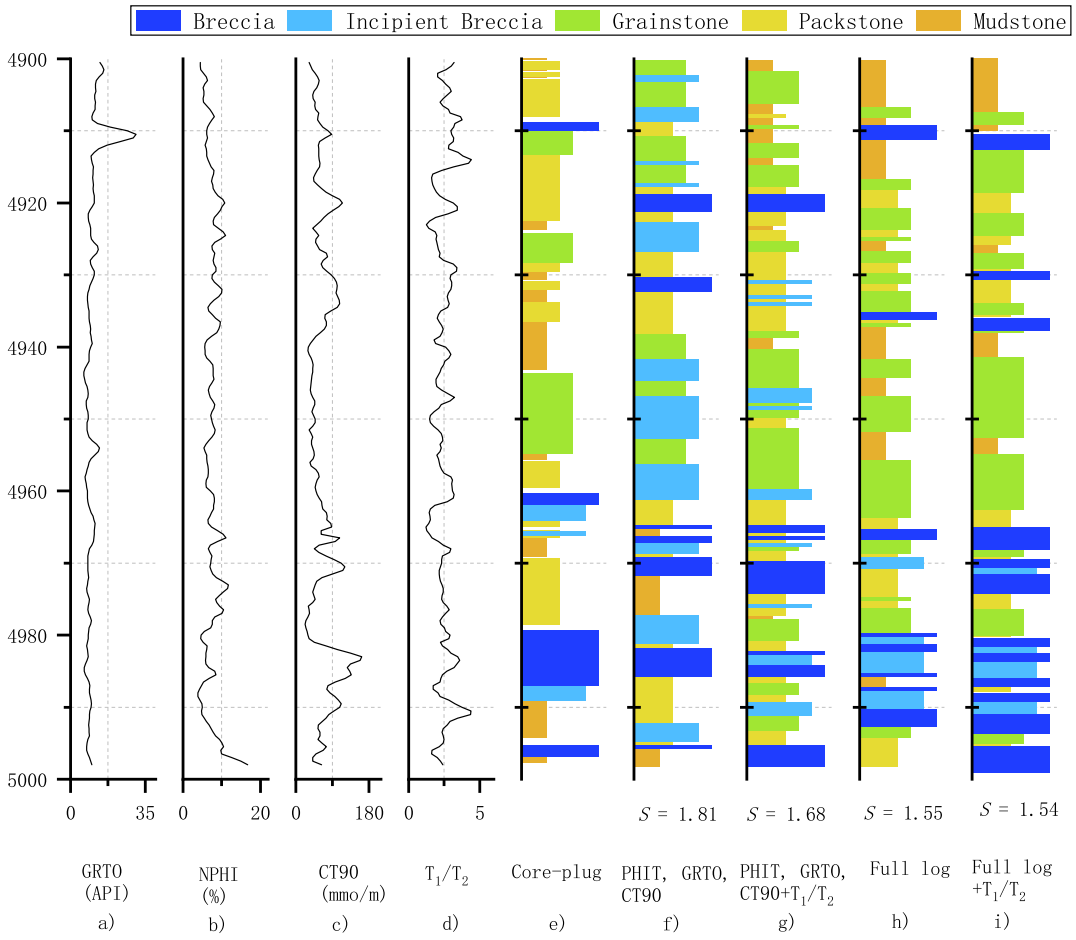

Fig. 9. Comparison of logs, lithologies, and eletrofacies predicted by four different datasets, plotted against depth. a)-c) GRTO, NPHI, and CT90 logs collected from well Wellington KGS #1-32 (4900 – 5000 ft), d) T_1/T_2 curve, e) Carbonate sub-facies determined from core observation, f)-i) Electrofacies predicted with Dataset 1 including PHIT, GRTO, and CT90 with a recognition error S = 1.81, dataset 2 (PHIT, GRTO, CT90, T_1/T_2) with S = 1.68, dataset 3 (PHIT, GRTO, CT90, THOR, POTA, ρ_{maa}, U_{maa}) with S = 1.55, and dataset 4 (PHIT, GRTO, CT90, THOR, POTA, ρ_{maa}, U_{maa}, T_1/T_2) with S = 1.54.

Fig. 9 is the predicted facies from each variable set. For additional reference, GRTO, NPHI, CT90, T_1/T_2 curves (Fig. 9(a)-9(d)), and an actual sub-facies profile (Fig. 9(e)) from core-plug observations are also provided in Fig. 9. The core plug lithologies are identified from both full core description and special core analysis.

The value of objective function S provides a means to compare the prediction accuracy of each variable set and a lower S value indicates better accuracy on prediction lithologies. Set 1, 2, 3, and 4 have S = 1.81, 1.68, 1.55, and 1.54, respectively. The outcome of Set 1 (Fig. 9(f)) serves as a baseline of the prediction and it has the lowest accuracy, especially in differentiating breccias from other lithologies. Adding T_1/T_2 to Set 1 builds Set 2 (Fig. 9(g)) and its S value decreases from 1.81 to 1.68. Moreover, Set 2 lowers the misprediction rate of lithologies from 0.79 to 0.66 and also decreases the number of incorrect predictions between non-breccia carbonates and breccia carbonates. Therefore, the introduction of T_1/T_2 increased the prediction accuracy relative to the control group Set 1. The results from Set 3 (Fig. 9(h)) and Set 4 (Fig. 9(i)), which include more mineralogy related parameters (THOR, POTA, ρ_{maa}, U_{maa}), outperform Sets 1 and 2 with minimized objective function values. Both Set 3 and Set 4 successfully discern a breccia layer (4910 ft) that is missed in Set 1 and 2 which is characterized by a relatively high GRTO value (Fig. 9(a)). However, comparing the S values of Set 4 and Set 3, the inclusion of T_1/T_2 in Set 4 does not improve the assessment of Set 3 remarkedly. The outcomes indicate that 1) T_1/T_2 does not significantly correlate with clay content; 2) T_1/T_2 provides information partly redundant with other mineralogy related well logs; 3) T_1/T_2 can improve carbonate sub-facies determination when the gamma-ray and photoelectric logs are missing. However, T_1/T_2 is not essentially a substitute for the gamma-ray and photoelectric logs.

5.4 Limitation of NMR logging tools

The sensitivity of apparent T_1/T_2 values to lithology is based on the correlation between T_1/T_2 and other lithology indicators such as porosity, conductivity, gamma-ray, and the ρ_{maa} versus U_{maa} plot. However, our sub-facies identification including in carbonate strata does not provide satisfactory results. Here we discuss the possible factors that affecting logging T_1/T_2.

The MRIL tool collects relaxation time distribution as a scattered 10 points spectra ranging from 2 ms to 2048 ms. First, there is no recorded signals under 2 ms, resulting the missing of major peaks for formations are dominated by small pores. Additionally, 10 points spectra is too sparse to recover the actural relaxation time distribution and peaks might have been merged. Due to both reasons, T_1/T_2 valued could be skewed from Eq. (7). Furthermore, considering MRIL's large peneatration area, it suffers from so-called 'bed boundary effect,' which describes the the merging of geophysical responses from distinct thin layers/beds in a single NMR measurement (Kleinberg et al., 1993b). This leads to the discrepancy of the resolution between T_1 and T_2 measurement, as well as discrepancies between MRIL and other logs. Since T_1 measurement requires an longer acquisition time than T_2 measurement, allowing more signal merging, therefor it affects the T_1/T_2 values. Lastly, as we are linking lithologies determined from small scale core plugs with moving average log data from a much larger inverstigation area, such cross-scale projection might introduce errors to the prediction of lithologies due to the scaling issue. Future works can help eliminate such systimatic errors in T_1/T_2 by improving MRIL instrumentation to provide shorter deadtime and finer recording intervals.

5.5 Impact of fluid type on T_1/T_2

T_1 and T_2 are dependent not only on lithology but also on the reservoir fluid types. Since the NMR signal is directly sensitive to the hydrogen content in either water or hydrocarbon and the fluid's environment, T_1 and T_2 could capture the fluid properties and fluid-rock interactions. Viscosity, one of the most important fluid properties, turns to play a vital role in controlling the hydrogen motion and variations of T_1/T_2. Initially, the T_1/T_2 ratio was considered as a constant in brine-saturated rock samples under low-frequency magnetic field (2 MHz) (Kleinberg et al., 1993b; Freedman and Morriss, 1996). Later studies by (Straley, 1997; Coates et al., 1999) demonstrated T_1/T_2 is sensitive to fluid types in the presence of mixed fluids in oil shale and gas shale samples. Comparing to T_1/T_2 value of water which is 1-2, highly viscous hydrocarbon fluids have higher T_1/T_2 value (up to 100), whereas light oils and gas with low viscosity have lower T_1/T_2 (4-20) (Coates et al., 1999; Freedman et al., 2001). The results conducted by Washburn and Birdwell (2013) and Fleury and Romero-Sarmiento (2016) show that bitumen, heavy oil, and methane have high T_1/T_2 and their presence in the sample could increase the overall T_1/T_2.

Wellington #1-32 is not a producing well, and the Arbuckle Group in this area is extensively water-flooded, left with bare residual oil. However, gas may still be a confounding factor driving T_1/T_2 values in Arbuckle. The dependency of fluid in deriving T_1/T_2 is also notable in the PCA results. According to the loadings of each component referenced to input variables (Table 3), the T_1/T_2 vector has relatively low loadings on both PC1 and PC2 compared with other logs, and the maximum correlation between T_1/T_2 and principal component appears in PC3 with a very high loading of 0.810. This indicates that the change in T_1/T_2 values heavily depends on the parameters that are not strongly associated with other input variables. However, T_1/T_2 is the only input variable sensitive to both fluid types and lithology, resulting in similar sub-facies classification for Set 3 and Set 4.

Even with such non-unique sensitivity, T_1/T_2 still has the potential to be an rock type indicator as long as the effects of fluids are well calibrated. Researchers have developed new insights into multi-phase flow in carbonate rocks under low magnetic field, for example, the application of 2-D T_1-T_2 map is an effective way to recognize multi fluid phases simultaneously and it has become a popular means for fluid typing (Fleury and Romero-Sarmiento, 2016; Mailhiot et al., 2017). Engaging such methods, we believe future studies can derive a modified T_1/T_2 value that is exclusively representative of the lithology.

5.6 Other factors that may fail electrofacies recognition

Lithologies sometimes cannot be tied to electrofacies (petrophysical facies) due to 1) the applied statistical analysis is not sensitive to the subtle changes of logs between facies (Brandsegg et al., 2010; Doveton, 2014); and/or 2) scaling and resolution issues arose from well log and laboratory measurements (Hurlimann et al., 2005).

In this case study, the targeting zone is restricted within 100 feet interval. The change of lithology is limited within this interval, thus leading to an insignificant change of log data for electrofacies recognition. The gamma-ray log shows a very typical response for clean sedimentary rocks. Except for 4955-4961 ft., where gamma-ray values reach 45 API units, the section has the uniformly low readings (Fig. 9(a)), indicating clean carbonate with low clay content. Similarly, the trace of deep resistivity (RDEP) changes only slightly, within one order of magnitude, suggesting relatively homogenous pore attributes and mineral composition at the resolution scale of the log. This interpretation is consistent with the readings in $U_{maa}-\rho_{maa}$ MID plot (Fig. 5). Therefore, although eight different logs are applied to the electrofacies recognition, they may not provide enough variation that the statistical methods need to differentiate sub-facies of carbonate.

The application of conventional multivariate statistical methods (e.g., PCA, cluster analysis, and discriminant analysis using well logs) have succeeded in predicting electrofacies and permeability in many carbonate reservoirs (Burke et al., 1969; Chang et al., 1997; Qi and Carr, 2006; Doveton, 2014). Those studies documented that, without core calibration, the accuracy of recognizing sub-facies of carbonate is low, compared to the accuracy of distinguishing between sandstone and shale. Our study is consistent with the results from Dorfman et al. (1990), Lee et al. (2002), and Criollo et al. (2016), and El Sharawy and Nabawy, (2016). El Sharawy and Nabawy (2016) documented their statistical methods failed to discern the difference between dolomitic limestone and limestone, although they did successfully recognize sandstone, shale, and limestone. Brandsegg et al. (2010) and Doveton (2014) suggest that since PCA is a linear dimentionality reduction

method, the distance between and within lithologies are considered with the same scale. As a result, variability of sub-facies in a fairly uniform lithologies will be masked by the varibility between major lithologies. To solve that problem, one should consider either a structured PCA which analyzes sub-facies independently (Brandsegg et al., 2010) or a non-linear dimensionality reduction method (e.g., T-distributed Stochastic Neighbor Embedding (t-SNE) method (Van Der Maaten, 2009)) to identify the variablity within one lithologies at different scales.

To improve the accuracy of electrofacies recognition and need to be constrained by more a priori information like depth. Furthermore, the infucluence from fluid should be eliminated by using $T_1 - T_2$ map and also permeability should be calculated to compare with core-permeability which provides a more practical evaluation of the reservoir quality.

6. Conclusion

To fully exploit the potential application of the NMR log in reservoir characterization, this study investigates the feasibility of T_1/T_2 in lithologies recognition within a dolostone reservoir. Logging data and core plugs from of a total 100 ft interval were analyzed and measured. We established the sensitivity of T_1/T_2 to lithology by comparing T_1, T_2, and T_1/T_2 characteristics from both log and lab NMR measurements. To validate the inferences of T_1/T_2, electrofacies were analyzed and defined through PCA, clustering, and classification. The results show that among five electrofacies, incipient breccia and breccia are more distinct while mudstone, packstone and grainstone are heavily overlapping. The comparison of classification results between Set 1 (PHIT, GRTO, and CT90), and Set 2 (Set 1 with T_1/T_2) indicates that addition of T_1/T_2 reduced the recognition error from 1.81 to 1.68 (unitless) whereas the errors of Set 3 (Set 1 with THOR, POTA, ρ_{maa}, U_{maa}) and Set 4 (Set 3 with T_1/T_2) are close (1.55 and 1.54). These outcomes suggest that adding T_1/T_2 can improve the classification accuracy, especially for incipient breccia, when THOR, POTA, ρ_{maa}, U_{maa} data are not available. However, T_1/T_2 failed to improve the classification accuracy when the entire suite of logs are available. Collectively, our results show: 1) T_1/T_2 is a lithology indicator but cannot substitute other logs like gamma-ray and photoelectric log in carbonate sub-facies determination; 2) instrumental limitation of MRIL and lacking of 2-D $T_1 - T_2$ map make T_1/T_2 ratio from log less representative; 3) variance of T_1/T_2 inside one sample should be considered as another lithology indicator.

This study exploits the potential of borehole NMR data, which focuses on investigating the inferences from NMR longitudinal and transverse relaxation time ratio on electrofacies classification and lithologies identification. The analysis suggests beyond traditionally considered parameters, NMR log can characterize the petrophysical properties of reservoirs. Nonetheless, challenges remain in the interpretation of data, such as how to connect the physical meanings of relaxivity change to mineralogy change, and how to eliminate the impact of hydrocarbon to get a more representative T_1/T_2 for certain lithology. Our study provided valuable evidence and credible elucidation of the importance and physicochemical mechanism of T_1/T_2, which is essential for deciphering NMR log in carbonate reservoirs.

Acknowledgement

This is an extended project of individual study with Dr. John Doveton (Kansas Geological Survey). We thank Dr. John Doveton for providing log data and essential instruction on data processing. We thank Dr. Lynn Watney (Kansas Geological Survey) for providing core samples from well Wellington #1-32. The authors are grateful to the two anonymous reviewers for their detailed reviews and constructive comments which helped to improve this manuscript.

Conflict of interest

The authors declare no competing interest.

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Appendix

Appendix A: Fuzzy-C mean algorithm

A fuzzy c-mean process tries to minimize the primary objective function:

$$J_m = \sum_{i=1}^{N}\sum_{j=1}^{C} \mu_{ij}^{m} \|x_i - c_j\|^2 \tag{A.1}$$

where m is a real number $\in [1,\infty]$ called fuzziness index or factor, μ_{ij} is the degree of membership of x_i in the cluster j, x_i is the ith of measured data in Euclidean space, c_j is the center of the cluster in Euclidean space, and $\|*\|$ is the Euclidean distance between data x_i and the cluster center c_j. Fuzzy partitioning is calculated by updating of fuzzy membership μ_{ij}.

$$\mu_{ij} = \frac{1}{\sum_{j=1}^{C}\left(\frac{|x_i - c_i|}{|x_i - c_j|}\right)^{\frac{2}{m-1}}} \tag{A.2}$$

moreover, the cluster centers c_i by:

$$c_{ij} = \frac{\sum_{i=1}^{C} x_j \mu_{ij}^{m}}{\sum_{i=1}^{C} \mu_{ij}^{m}} \tag{A.3}$$

Repeating above steps through an iterative operation until the minimum objective function J_m has been achieved or the termination criterion has been reached.

This study considers two-dimensional FCM analysis of PC1 and PC2, as derived from the log data. If only two possible electrofacies (A and B) have been assigned, a membership partitioning matrix for both k-means and FCM algorithm as M_{KM} and M_{FCM} includes:

$$M_{KM} = \begin{bmatrix} 0 & 1 \\ 0 & 1 \\ 1 & 0 \\ \vdots & \vdots \\ 0 & 1 \end{bmatrix}, \quad M_{FCM} = \begin{bmatrix} 0.3 & 0.7 \\ 0.4 & 0.6 \\ 0.8 & 0.2 \\ \vdots & \vdots \\ 0.1 & 0.9 \end{bmatrix} \tag{A.4}$$

Numbers indicate the degree of membership that belongs to each electrofacies. For sample 1 in the first row, a data point belongs exclusively to electorfacies B with a degree of membership equal to 1 in the k-means algorithm, whereas in the FCM algorithm it belongs to electrofacies A with a membership coefficient of 0.3 and belongs to electrofacies B with a membership coefficient of 0.7. As an outcome, every data point will have a membership partitioning for each cluster and that membership represents the possibility of belonging to a cluster.

Appendix B: Introduction to logs

The most informative lithology log is spectral gamma-ray (SGR), which is based on the gamma-ray response to identify clay minerals from radioactivity emitted by isotopes of POTA, thorium, and uranium (K, Th, and U, respectively) series. Since uranium sometimes is high in minerals other than clays (for example, in dolomite), calibrated gamma-ray reading (CGR) which only includes POTA and thorium is used more commonly in determining clay content. Units with high CGR suggest shaly lithologies and low permeability zones, whereas low CGR commonly indicates clean sandstone or carbonate. The GR log usually is analyzed together with additional logs like density log, neutron log, or sonic slowness.

Like the GR log, both density and neutron logs are nuclear tools responding to gamma-ray scattering and neutron scattering effects caused by the formation matrix and pore-filling fluids. Porosity, bulk density and hydrogen content are the common outcomes of density and neutron logs. As a supplementary measurement of density log, photoelectric factor (PeF) provides a means to discriminate lithologies by analyzing the low-energy PE absorption which is mostly independent of porosity. The sonic log is based on propagation of acoustic waves through the section. It not only is used for porosity estimation and cement volumetric evaluation, but is also combined with density to produce impedance logs and synthetic seismic profiles which can be used to correct depth and seismic travel times.

Resistivity is another petrophysical property that indirectly reflects the permeability and fluid saturation of reservoirs which is challenging to measure. To solve the challenge of accurate measurement, numerous resistivity and induction logs have been introduced in modern wireline logging probes, including microlog (ML), medium and deep resistivity (MRT/DRT), conductivity

(90 inches conductivity (CT90) is used in this thesis) and spontaneous potential (SP). These logs measure the resistivity of mudcake, flushed zone, uninvaded zone and formation fluid, respectively.

Additionally, recently developed electromagnetic logging tools provide additional information about pore structure and fluid type in the surrounding reservoir environment. Among those tools, magnetic resonance imaging logging (MRIL) has created excitement in the well logging community. Based on the same physical principle as its laboratory-scale counterpart, MRIL is used to reveal total fluid-filled porosity, pore-size distribution, fluid types, and fluid contents that can be used subsequently to determine permeability. Since the 1970s, significant advances have been made in NMR wireline tools. The new acquisition schemes and processing methods significantly improved data acquisition, data quality, and interpretation.

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文献 6/9

Nuclear Magnetic Resonance Investigation of Forced Imbibitions in Longmaxi Shales: Consideration of Different Boundary Conditions

Yong Liu; Yanbin Yao; Dameng Liu; Chi Zhang

Energy & Fuels (2023); DOI: 10.1021/acs.energyfuels.3c00404

证据用途：其他合作论文背景；张弛老师角色：末位合作者；证据等级：C

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Article

Nuclear Magnetic Resonance Investigation of Forced Imbibitions in Longmaxi Shales: Consideration of Different Boundary Conditions

Yong Liu, Yanbin Yao,* Dameng Liu, and Chi Zhang

Cite This: *Energy Fuels* 2023, 37, 5853−5866

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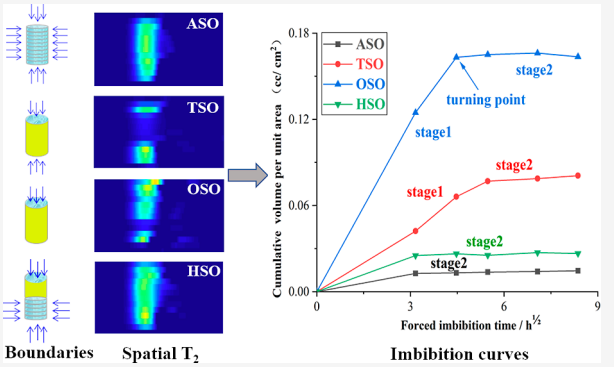
ABSTRACT: The forced imbibition in shale reservoirs plays a significant role in achieving high-performance recovery of shale gas. Forced imbibition is the main reason for bringing low flowback efficiency (<30%) of fracturing fluid and some reservoir damages in shale reservoirs. When a large amount of fracturing fluid is injected into the reservoir to enhance the production of gas, the effect of the imbibed liquid on the fracturing fluid leak-off remains poorly understood. In this study, we conducted systematic forced imbibition experiments combined with 1D and 2D nuclear magnetic resonance (NMR) to investigate imbibition behaviors in Longmaxi shales with four boundary conditions: all-side-open (ASO), two-side-open (TSO), one-side-open (OSO), and half-side-open (HSO). The NMR imbibition experimental results (T_2 spectra, fluid saturation profiles, and spatial T_2 images) showed that imbibition dynamics of OSO and TSO are more complicated than those of HSO and ASO, which illustrates that the shale bedding structure is the key to determine different imbibition behaviors and imbibition patterns of four boundaries. For four boundaries, ASO imbibition is composed of the dominant imbibition parallel to beddings and the affiliated imbibition against beddings, and HSO is a combination of the imbibition parallel to beddings (the open part) and the imbibition perpendicular to beddings (the sealed part). In contrast, OSO imbibition and TSO imbibition are unidirectional cross-layer imbibition and bidirectional cross-layer imbibition, respectively. Six imbibition parameters including water saturation, gas recovery, imbibition capacity index, diffusion capacity index, imbibition rate, and imbibition potential are adopted to quantificationally analyze the different forced imbibition dynamics. The understanding of how these boundary conditions affect imbibition dynamics in shale gas reservoirs can be applied to interpret fracturing liquid retention and to optimize the hydraulic fracturing design.



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1. INTRODUCTION

Shale gas accumulated in the southern Sichuan Basin has become the primary target for unconventional hydrocarbon resources in China, and the marine Longmaxi shale is the major shale play for shale gas exploration and exploitation.[1−4] Horizontal well drilling technology combined with large-scale multistage hydraulic fracturing operations is the most effective stimulation technique to achieve commercial shale gas productivity. During this operation, reasonable well shut-in is adopted to improve the cleanup of fracturing fluid and to enhance fracturing performance. In the shut-in period, matrix imbibitions including forced imbibition and spontaneous imbibition are primarily responsible for low flowback efficiency (<30%) of fracturing fluid and severe formation damages such as water sensitivity and water blocking in shale gas reservoirs.[5−12] Subsurface multiphase-flow phenomena during hydraulic fracturing such as fracturing fluid distribution, hydrocarbon percolation, and fluid−rock interactions can also be influenced by imbibition behaviors in shut-in.[8,13−17] Therefore, it is essential to investigate the matrix imbibition dynamics in the shut-in stage, which could be instructive to fracturing design optimization and shale gas exploitation.

Matrix imbibitions during shut-in can be classified as forced imbibition and spontaneous imbibition, depending on whether there exists an external pressure acting on the fracturing fluid which drives the fluid phase into the rock matrix.[18−22] From the perspective of the hydraulic fracturing operation in shale gas reservoirs, the fracturing fluid imbibition from fracture networks into the shale matrix under the effects of fluid pressure and reservoir pressure can be defined as forced imbibition,[18] whereas when the fluid pressure in fractures reaches equilibrium with the reservoir pressure, the fracturing fluid imbibition driven by capillary force (also called leak-off of

Received: February 9, 2023
Revised: March 17, 2023
Published: March 30, 2023

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fracturing fluid) can be considered as the spontaneous imbibition process. Many research studies have focused on matrix imbibition behaviors have been carried out by using numerical analysis, numerical modeling, and experimental investigations such as dynamic spontaneous imbibition, nuclear magnetic resonance (NMR) technology, computed tomography (CT) scanning, and so on.[12,13,18,19,22,23] Takahashi and Kovscek[22] adopted a combination of X-ray CT scanning and numerical simulation to investigate the flow characteristics of spontaneous and forced imbibition with various brine formulations in siliceous shale. Dehghanpour et al.[12] found excess brine intake than oil intake in Horn River shale due to water adsorption and induced microfractures by comparing the spontaneous imbibitions of aqueous and oleic phases. Lan et al.[23] conducted comparative and systematic spontaneous imbibition experiments on core plugs and powder packs to examine the correlations between the imbibition dynamics and shale wettability. Deng and King[19] presented a semianalytic methodology to describe the transient imbibition flow from spontaneous to forced imbibition. Zhong et al.[18] developed an analytical model of forced imbibition to describe the forced imbibition process that hydrofracturing fluid imbibed into the shale matrix. Roychaudhuri et al.[13] carried out forced imbibition experiments with Marcellus shale cores to analyze shale−fluid interactions during the hydraulic fracturing stimulation. Especially, these research studies mainly focus on spontaneous imbibition behaviors of shales, and so far, there is limited experimental research to investigate the forced imbibitions during hydraulic fracturing due to extremely complex experimental procedures and complicated imbibition mechanisms. Thus, conducting systematic experiments on forced imbibition to explore its imbibition behaviors is meaningful for determining shut-in time and enhancing fluid cleanup.

Various boundary conditions of imbibition processes correspond to different fluid−rock contact relationships in the fracture-matrix system of fractured shale formations. For example, typical imbibition conditions of all-side-open (ASO) and two-side-open (TSO) have significant influences on shale imbibition performances, which can be used for fluid simulations of complex subsurface fluid−rock contact relationships and production performance prediction in unconventional reservoirs.[24−26] Yildiz et al.[26] identified that spontaneous imbibition rates on kerosene/brine/rock systems can be promoted due to the increase in total contact surface areas by comparing ASO, TSO, two-side-closed (TSC), and one-side-open (OSO) conditions. It was noted by Lyu et al.[24] that all-face-open (AFO) spontaneous imbibition has a complex 3D counter-current flow mode consisting of a radial counter-current imbibition in the cylindrical face and a linear imbibition in the two end faces. For the TSO condition in an oil−water system, there exists a combination of cocurrent and counter-current imbibitions, both contributing to flow dynamics and oil recovery in different imbibition stages.[27,28] As for OSO, purely linear counter-current imbibition is the dominant flow pattern and the piston-like displacement driven by capillary pressure occurs inside the core.[29] Kim and Kovscek[30] adopted CT scanning technology to quantificationally investigate the fluid distribution and saturation curves of spontaneous imbibition processes under ASO, TSO, and OSO conditions. Umeobi et al.[31] compared imbibition dynamics and flowback of fracturing fluids in sandstones by NMR imbibition experiments with AFO, TEC, and two-end-open (TEO) boundary conditions. Through the NMR imbibition experiments on fractured shales under three boundaries, Yang et al.[32] found that boundary conditions mainly affect the imbibition by changing the movement relationship and contact area between the core and water. Wang et al.[33] also conducted NMR experiments to study the imbibition laws of glutenite with ASO, OSO, TSO, one-side-closed, and TSC conditions. Despite considerable publications on the boundary effects for oil−water imbibitions, there are relatively few research studies focused on the boundary effects for gas−water forced imbibition of gas shale. Hence, the forced imbibitions of shales under different boundaries need more experimental investigations to obtain an in-depth understanding of boundary effects on imbibition dynamics.

The core-scale NMR experiment has been widely adopted for reservoir characterization, subsurface fluid dynamic detection, and reservoir stimulation evaluation in unconventional reservoirs due to its unique advantage in monitoring fluid dynamics in different pore networks.[1,3,6,34−36] Fernø et al.[37] used magnetic resonance imaging technology to monitor the fluid displacement processes of spontaneous imbibitions under one-end-open (OEO) and TEO conditions. Karimi et al.[38] studied wettability change and fluid transport of spontaneous imbibition and forced imbibition in Middle Bakken cores with the help of NMR T_2 relaxation and NMR saturation profiles. Based on NMR T_2 results of imbibition and drainage experiments, Shen et al.[39] compared the fluid seepage characteristics of different water saturations in high-rank coal. Lyu et al.[24] compared the flow processes of spontaneous imbibition on five boundary conditions in tight sandstones by using NMR T_2 and 2D-NMR images. With the help of NMR measurement, Zhang et al.[40] proposed a new method to calculate capillary pressure and relative permeability from spontaneous imbibition experiments. It is worth noting that most of the current NMR imbibition experiments are conducted to investigate the spontaneous imbibitions of gas shale instead of forced imbibitions. Compared with spontaneous imbibition, there exists a significant external fluid pressure in forced imbibition for the in situ condition. What is more, the influence of boundary conditions on forced imbibition has not been systematically studied. Thus, in the present work, forced imbibition experiments combined with NMR techniques are applied to monitor fluid dynamics and flow patterns for accurate characterizations of forced imbibition dynamics corresponding to different boundaries.

In this paper, imbibition patterns and imbibition dynamics of Longmaxi shales under four boundary conditions are compared and analyzed by the systematic forced imbibition experiments combined with advanced NMR techniques. The objectives of this study are twofold: (1) to analyze the imbibition patterns corresponding to four different boundary conditions and (2) to compare the imbibition dynamics during forced imbibitions. Besides, the potential application of this study for shale hydraulic fracturing was discussed at the end of the paper.

2. EXPERIMENTAL SECTION

2.1. Materials. Marine shale samples used in this study were obtained from the Lower Silurian Longmaxi formation in the Pengshui Block, southeast of the Sichuan Basin, China. To avoid damage during transportation, big blocks were sealed with multilayer polyvinyl chloride plastic wrap and packed with foam and bubble bags after being cut from the field profile. Five cylindrical core plug samples

https://doi.org/10.1021/acs.energyfuels.3c00404
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(nos. 1, 2, 3, 4, and 5) were drilled perpendicularly to the bedding planes from a large shale block. This procedure ensures no significant difference in the individual core samples. During the coring process, cooling airflow was adopted to cool the bit without liquids making contact with the shales.

Routine characteristic parameters including the helium porosity, pulse decay permeability, bituminous reflectance (R_o), total organic carbon (TOC) content, and mineral composition were analyzed for these samples. Fragments from no. 1 shale were processed into thin slices (20 × 15 × 5 mm^3) for scanning electron microscopy (SEM) observation. The surface of the sample was polished by an argon-ion cross-section polisher (Gatan 697) after mechanical polishing but was not coated with gold to keep the pore morphology noticeable. A cold field-emission scanning electron microscope (Quanta FEG 450) in secondary electron mode was used to capture the characteristics of the pore structure and mineral composition from high-resolution SEM pictures (theoretically, 3 nm pixel).

2.2. Imbibition Experiments. *2.2.1. Experimental Setup.* Forced imbibition experiments considering four boundary conditions were carried out in a specially designed high-pressure imbibition experimental system (Figure 1). The apparatus consists of three

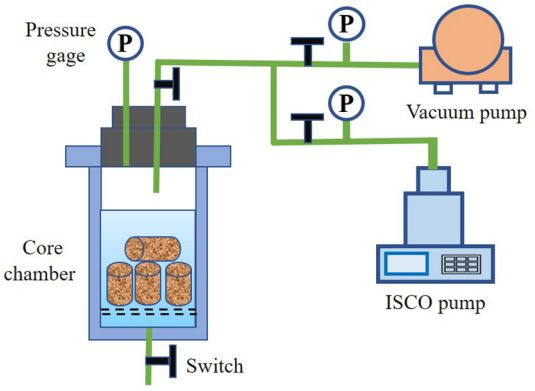

Figure 1. Diagram of the forced imbibition experimental system.

subsystems: a high-pressure imbibition vessel (a sealed core chamber) with a size of Φ 80 × 160 mm and a maximum working pressure of 35 MPa, a fluid injection pump (Teledyne ISCO 260D), and a vacuum pump with an ultimate vacuum pressure of 6 × 10^{-2} Pa. This apparatus can ensure simultaneous imbibition experiments with four boundary conditions under the selected fluid injection pressure to simulate the forced imbibition process during hydraulic fracturing.

2.2.2. Boundary Conditions for Imbibition Experiments. Boundary conditions have significant influences on imbibition behaviors and fluid dynamics of unconventional reservoir rocks. As illustrated in Figure 2, four parallel samples (nos. 2, 3, 4, and 5) were processed and sealed with the nonmagnetic heat-shrink tubes (0.16 mm thickness) to acquire four corresponding boundaries including ASO, TSO, OSO, and HSO. Heat-shrink tubes are made of polytetrafluoroethylene (PTFE) plastics and cannot generate any base NMR signal noise during imbibition experiments. Moreover, the tubes can confine the selected boundaries of plugs and protect cores due to good waterproofness and pressure resistance.

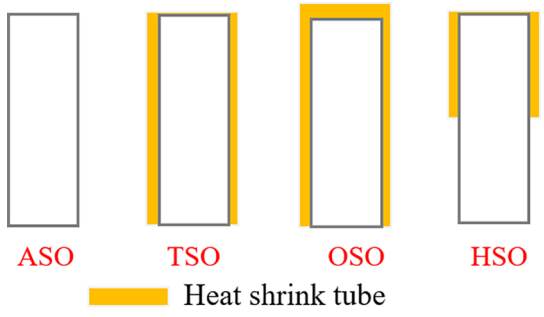

Figure 2. Schematic of four boundary conditions for imbibition experiments.

2.2.3. Experimental Procedures of Forced Imbibitions. The force imbibition experiment needs a higher injection pressure to ensure that the fluid enters the shale quickly and thoroughly. However, too much higher pressure should be avoided because the investigated shale has a high proportion of water-sensitive clay minerals (illite 39 wt % and chlorite 7 wt %), and the shale is easy to be damaged by the effects of clay hydration, mineral dissolution, or microfracture generation under extra-high pressure.[21,41] As a result, the injection pressure was set as 10 MPa for all imbibition experiments in this study. Moreover, considering the influence of salinity on imbibitions, distilled water rather than brine was used as imbibition fluid during imbibition experiments. The matrix imbibition rate of shale rock is relatively low, and imbibition data can be easily influenced by experimental conditions and setup accuracy. The traditional mass method for shale imbibition is not sensitive to a small amount of fluid variation. Thus, an NMR spectrometer with a fluid detection accuracy of 0.01 cc was adopted to monitor fluid dynamics during imbibitions.

Experimental procedures of forced imbibitions are described as follows

(1) Shales were classified into two groups: group 1 (no. 1) for full-scale pore size distribution (PSD) measurement and group 2 (nos. 2, 3, 4, and 5) for forced imbibition measurement. Samples were dried in a dry oven at 100 °C for more than 72 h until the weight remained unchanged.

(2) No. 1 sample was fully saturated with distilled water at 30 MPa for 72 h in a ZYB-II vacuum-pressurizing saturation device. NMR measurement was performed on a 100% water-saturated sample to acquire the full-scale PSD.

(3) Plugs of group 2 were processed and sealed with PTFE plastics to obtain four boundaries for imbibition tests (Figure 2). NMR tests on plugs before and after boundary processing were carried out to check the influence of the sealing material on NMR signals. T_2 spectra after boundary treatment were basically consistent with that before boundary treatment, which meant that the PTFE material has a negligible influence on the results of NMR imbibition experiments. Finally, the weights of the plugs were recorded.

(4) Before imbibition experiments, the pressure tightness of the experimental apparatus (Figure 1) was tested and checked. Samples were placed into the core chamber and the system was vacuumed to 10^{-1} MPa for 6 h. Next, distilled water was pumped to the chamber via an ISCO pump in a mode of constant pressure to submerge samples. Injected fluid pressure applied on samples was kept at 10 MPa and a pressure transducer at the top of the core chamber was used to monitor the pressure of the imbibition liquid being dispensed. The pressure was removed at selected measurement time points (10, 20, 30, 50, and 70 h), and the sample was taken out for NMR tests (T_2 spectra, saturation profiles, and spatial T_2 distribution) at each time interval. Finally, plugs were dried at 100 °C for 120 h and the dried samples were weighed again. All NMR data were collected and saved by the NMR data-acquisition unit in GIT systems for imbibition dynamic analysis.

2.3. NMR Tests of Imbibition Experiments. NMR tests were performed on an NMR core analysis device (Geospec 12-53, Oxford Instruments, UK) equipped with the data acquisition and analysis software (GIT Systems Advanced version, Green Imaging Technologies, Canada) to obtain T_2 spectra, NMR saturation profiles, and spatial T_2 images. The NMR core analyzer adopted a low magnetic field of 0.3 T with a resonance frequency of 12.76 MHz. NMR test parameters are as follows: an echo spacing of 0.12 ms, a waiting time of 1500 ms, a number of scans of 64, a number of echoes of 12,000, a field of view of 6 cm, and a maximum gradient strength of 70 gauss/

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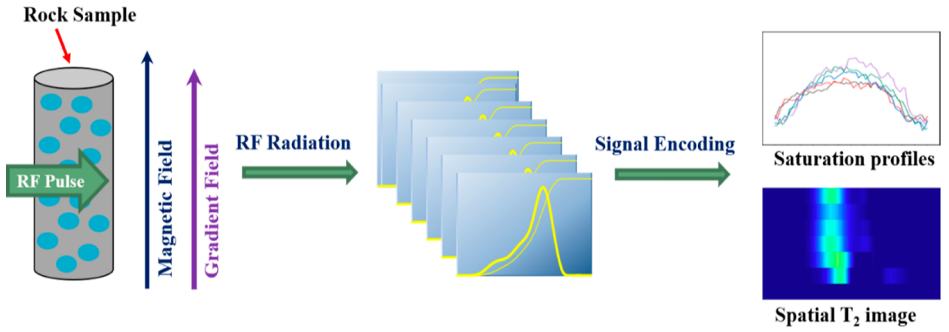

Figure 3. Schematic diagram of the NMR saturation profile and NMR spatial T_2 image.

cm. The laboratory temperature and permanent magnet temperature of NMR systems were 27 and 35 °C, respectively.

Different from T_2 relaxation measured by the Carr−Purcell−Meiboom−Gill (CPMG) pulse sequence in a uniform magnetic field, NMR saturation profiles and spatial T_2 images are obtained by an inbuilt Oxford Instruments spin-echo single-point imaging pulse sequence in varying gradient magnetic field (SE-SPI). Overall T_2 decay acquired in an NMR experiment is the sum of all T_2 decays from all the pores; in other words, the observed T_2 spectrum is an average of the signal observed from all the hydrogen nuclear spins in the core. Spatial T_2 technology provides a method to add spatial resolution to T_2 distribution and to yield spatially resolved PSD as a function of position along the shale core. Generated from spatial T_2 curves, the NMR saturation profile is where the total volume derived from the area under each spatial T_2 distribution is plotted as a function of the position along the core sample.

To achieve a spatially resolved T_2 image, the applied magnetic field is varied by applying a second magnetic field known as the gradient magnetic field. The gradient field is tuned in such a manner so that only a small section of the core still has spins aligned with the applied magnetic field. For example, the applied magnetic field might be zero for the bottom of the core and then be more and more positive toward the top of the core. This means that only the nuclear spins at the bottom of the core are still aligned with the applied magnetic field. The other spins will be tipped to different angles. As a result, the RF pulse will only excite those spins in the bottom of the core and the observed T_2 spectrum will only be from the bottom of the core. If the procedure is repeated multiple times and the gradient field is tuned, then the subsequent sections are the ones with their nuclear spins still aligned with the applied magnetic field. A series of PSDs will be obtained as a function of position along the core (Figure 3). Detailed NMR theories of T_2 distribution, NMR saturation profiles, and spatial T_2 can be found in many pieces of literature.[3,42−44] These two technologies enhance ways of imbibition analysis by the NMR method.

3. RESULTS

3.1. Characteristics of Shale Samples. Detailed characteristic parameters, petrophysical properties, and mineral composition of parallel samples are summarized in Table1. Pengshui shales have similar porosity (~4.39%) and permeability (~697 nd), which demonstrates that the effects of shale heterogeneity can be neglected and five cylindrical plugs can be used as parallel specimens.

Table 1. Characteristic Parameters of Parallel Samples

sample	nos. 1	nos. 2	nos. 3	nos. 4	nos. 5
diameter (cm)	2.53	2.52	2.52	2.53	2.52
length (cm)	5.46	5.54	5.52	5.51	5.55
volume (cm^3)	27.43	27.62	27.52	27.69	27.67
boundary condition		ASO	TSO	OSO	HSO
porosity (%)	4.38	4.32	4.42	4.36	4.46
permeability (nd)	736	678	692	669	712

All samples are at the high-mature stage (exhibited by R_o = 1.38%) and have a high TOC content (2.80 wt %). Quantitative mineral composition analysis shows the presence of quartz (28 wt %), feldspar (7 wt %), plagioclase (11 wt %), calcite (7 wt %), a small quantity of pyrite (1 wt %), and a high proportion of clay minerals (illite 39 wt % and chlorite 7 wt %) in marine shale samples.

Longmaxi shale reservoirs have complicated pore networks from the nanoscale to the microscale. Pore morphologies and mineral components of shale were analyzed based on FE-SEM images (Figure 4).[3] Pores in marine shale reservoirs can be divided into four categories: organic pores, microfractures, interparticle pores, and intraparticle pores. A large number of organic pores were found with diameters of tens of nanometers and distributed on the surface of the black porous organic matter particle (Figure 4b). SEM images revealed that there existed a certain amount of interparticle pores occurred between quartz and other crystals in the matrix and intraparticle pores occurred within minerals (Figure 4a,c,d). Additionally, microscale microfractures can be observed within minerals and the matrix (Figure 4c,d). Several banded and cotton-shaped clay minerals with a light-gray color were detected associated with organic matter and matrix grains (Figure 4b−d).

3.2. PSD Measurement by NMR. PSD of shale formation is an essential petrophysical parameter in unconventional reservoir evaluation, reserve calculation, and fracturing design. The T_2 relaxation time is directly proportional to the pore size of shale rock, and the T_2 spectrum of a 100% saturated sample can be converted to a full-scale shale PSD by eqs 1 and 2.[3,42,45]

$$\frac{1}{T_2} = \rho_2 \frac{S}{V} = \rho_2 \frac{F_S}{r} \tag{1}$$

where ρ_2 is the transverse surface relaxivity factor, which is 0.0265 μm/ms for Pengshui shale,[3] and S/V is the ratio of pore surface area to pore volume, μm^{-1}. The pore shape factor F_S is determined as 2 because the pore morphology of Longmaxi shale is assumed to be cylindrical.[46] Thus, the pore radius (r, μm) can be calculated by a corresponding T_2 by eq 2

$$r = 0.053 T_2 \tag{2}$$

The T_2 spectrum (blue solid line) and cumulative T_2 distribution (blue dotted line) for shale are presented in Figure 5. The NMR pore radius ranges from 0.0058 to 5.3 μm. As shown in the top of Figure 5, PSD can be divided into three regions: unrecoverable fluid pores P_U ($T_2 < T_{2C2}$), capillary bound fluid pores P_C ($T_{2C2} < T_2 < T_{2C1}$), and movable fluid

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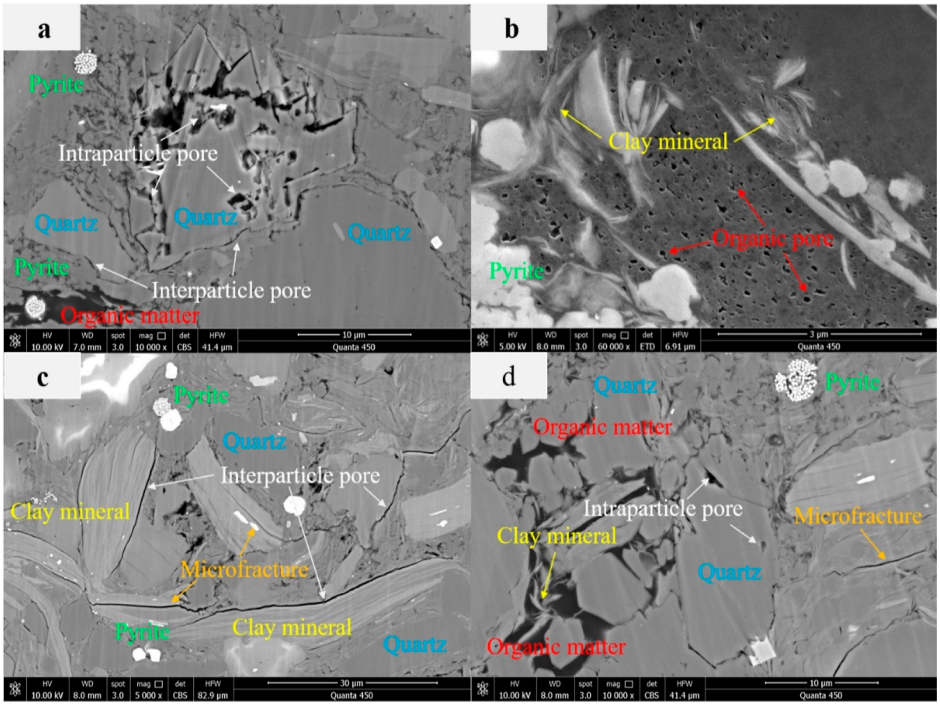

Figure 4. FE-SEM images show complicated pore structures and mineral compositions in Longmaxi shales.

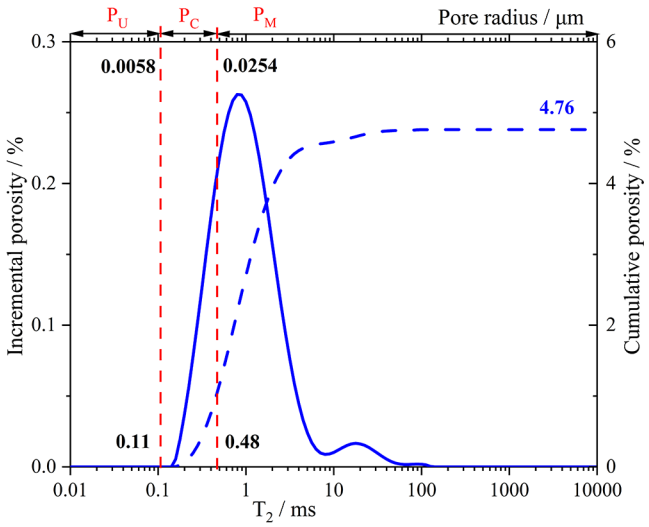

Figure 5. NMR T_2 spectrum and PSD results of no. 1 shale. Note that the red dashed lines from left to right represent the movable fluid T_2 cutoff (T_{2C1}) and unrecoverable fluid T_2 cutoff (T_{2C2}), respectively.

pores P_M ($T_2 > T_{2C1}$).[3] The T_2 curve exhibits a continuous bimodal distribution with two obvious peaks: the left large peak (0.1−8 ms) accounts for 96% of T_2 spectral area and the right small peak (8−100 ms) accounts for only 4%, which reflects abundant capillary bound fluid pores and movable fluid pores of the shale formation. Shale with a large percentage of movable fluid porosity has a relatively high matrix permeability, implying that reservoir fluids including gas and water can have more flowing spaces within the shale matrix to contribute to gas production and to increase the flowback efficiency of fracturing fluid.

3.3. NMR T_2 Spectra of Forced Imbibitions. NMR T_2 distributions can reveal fluid saturation change, water transportation, residual gas distribution, and imbibition dynamics in imbibition processes. Figure 6 shows T_2 relaxation changes of forced imbibitions under four boundaries. For ASO imbibition (Figure 6a), T_2 curves are all characterized by continuous bimodal distribution patterns: one large left peak at 0.1−10 ms and another small right peak at 10−100 ms, which illustrates that most of the water is imbibed into nanopores of the shale sample. For TSO imbibitions (Figure 6b), T_2 distributions have a discontinuous trimodal distribution pattern with one large peak (0.1−10 ms) and two small peaks (10−100 and 100−10000 ms). Based on NMR PSD results in Section 3.2, the second and third small peaks at 100−10000 ms of T_2 curves represent the generated microfractures during forced imbibitions. The change of NMR T_2 curves of OSO imbibition (Figure 6c) is similar to that of the TSO case (Figure 6b). However, for OSO imbibition, the signal area of the second right peak (100−10,000 ms) is bigger than that of TSO, illustrating that more microfractures are generated in OSO imbibitions. For HSO imbibition, there also exist three peaks in T_2 distributions: one big left peak (0.1−10 ms), one small right peak (10−100 ms), and the other small right peak (100−10,000 ms). T_2 signals of HSO (Figure 6d) have similar variation characteristics with those of ASO imbibitions (Figure 6a), but T_2 peak values of HSO are bigger than that of ASO, and these T_2 curves are close to each other. The cumulative imbibition water volume of HSO is the highest among these four cases, followed by OSO, TSO, and ASO.

As depicted in Figure 6, the right small peaks (>100 ms) represent the induced microfractures that can be observed in TSO, OSO, and HSO imbibitions. While in ASO imbibition, no obvious microfractures were found in the T_2 distributions. The differences in the cumulative imbibition volumes (HSO > OSO > TSO > ASO) are more obvious. There are certain differences in the generated microfractures during forced imbibition with different boundaries. Induced fractures provide more imbibition spaces for water imbibition and increase the

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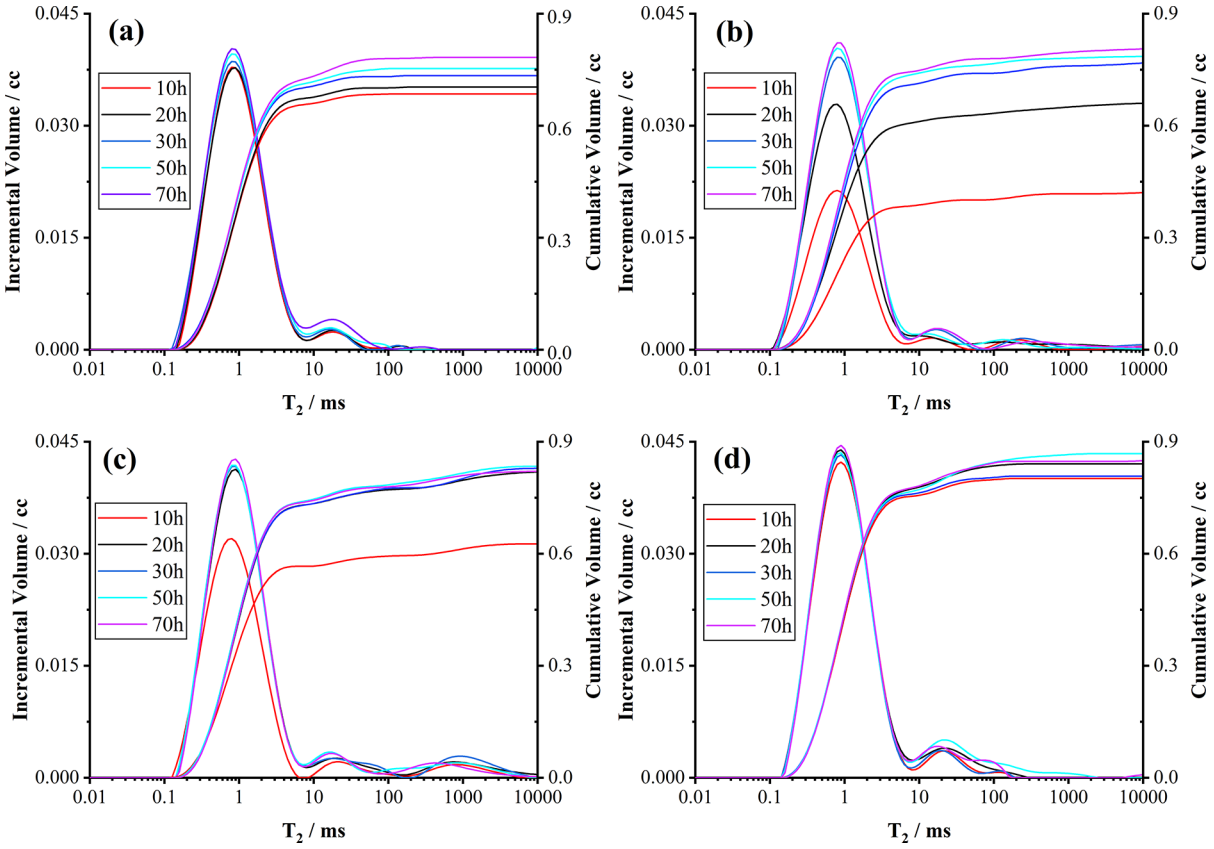

Figure 6. T_2 spectra of forced imbibitions under (a) ASO, (b) TSO, (c) OSO, and (d) HSO boundaries.

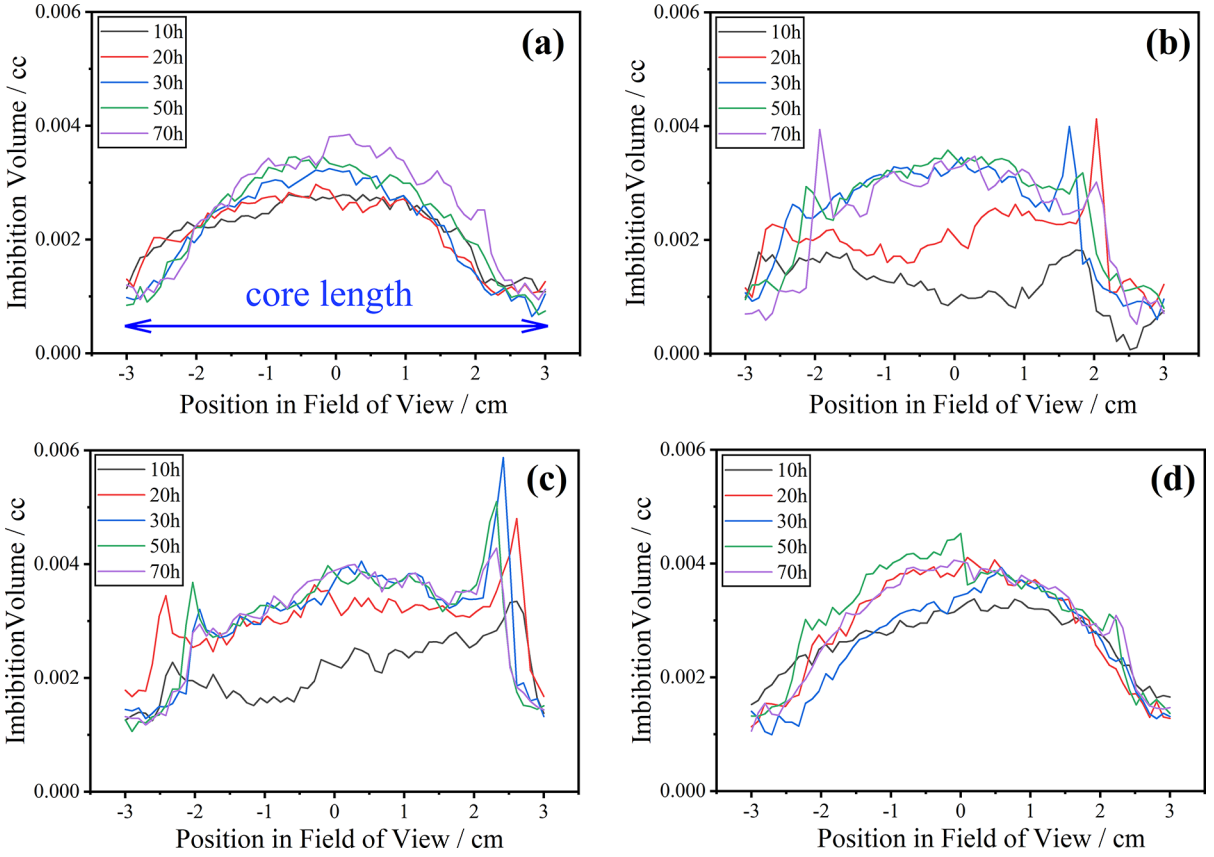

Figure 7. NMR fluid saturation profiles of forced imbibitions under (a) ASO, (b) TSO, (c) OSO, and (d) HSO boundaries.

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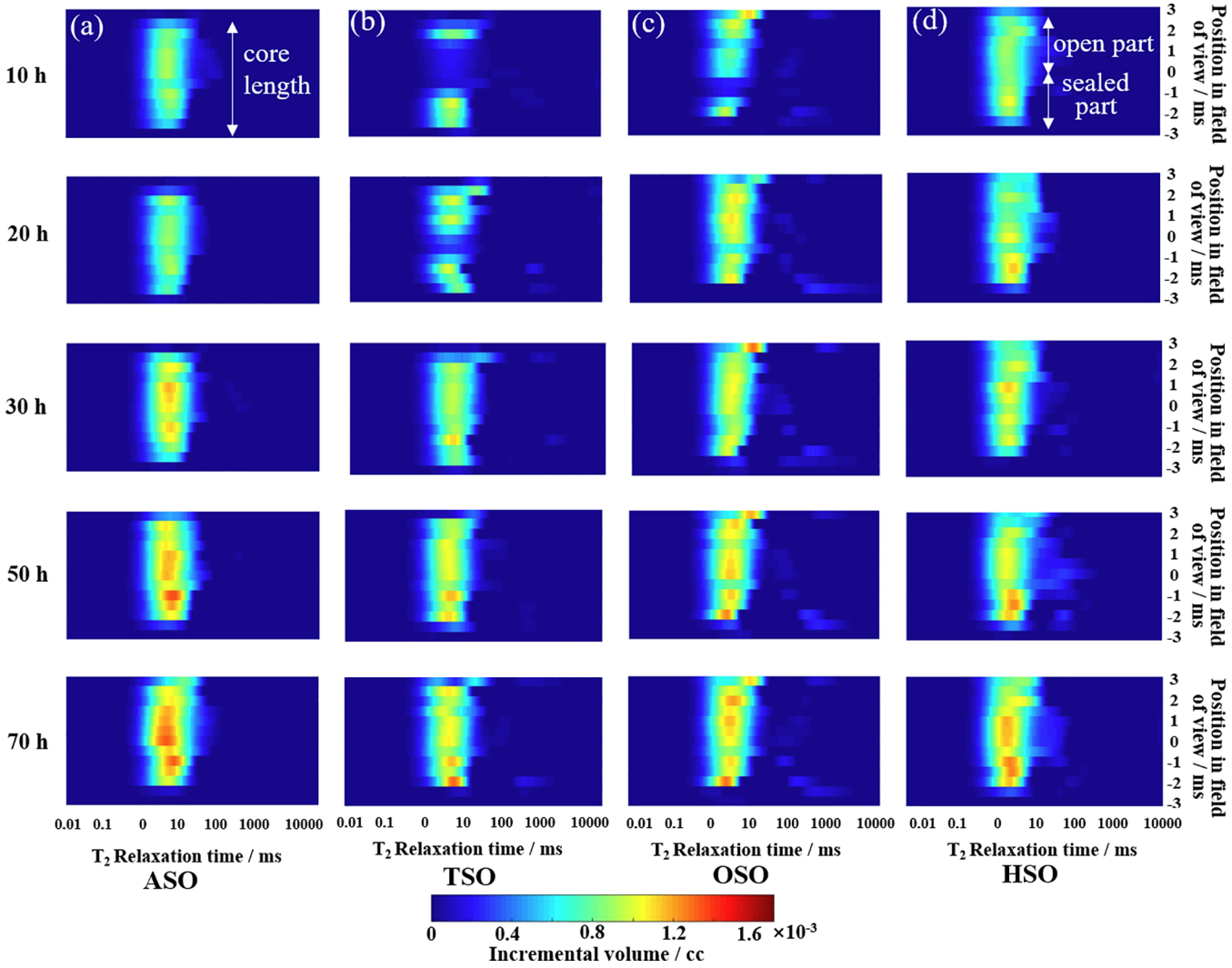

Figure 8. NMR spatial T_2 images of ASO, TSO, OSO, and HSO boundaries.

matrix permeability, which promotes forced imbibition to a certain degree.

3.4. NMR Fluid Saturation Profiles of Forced Imbibitions. NMR fluid saturation profiles were measured for imbibitions with four boundaries, which show imbibition fluid volumes along core samples at different times. For ASO imbibition (Figure 7a), saturation profiles are convex curves, implying that the cylindrical surface of the core imbibed more fluids than the two end faces. There is a complex 3D imbibition pattern in ASO imbibition that two kinds of imbibition processes from the cylindrical surface and from two end faces occurred simultaneously. NMR fluid saturation profiles of TSO (Figure 7b) have two peaks located in the two end faces of the sample making contact with water. Peak values and signal amplitudes of the saturation curves increase gradually with increasing imbibition time, illustrating that more fluids are imbibed into the sample from two end faces. The imbibition pattern of TSO imbibition is a bidirectional linear imbibition along the sample. Saturation curves of OSO imbibition (Figure 7c) have one big peak in the open inflow end face of the core, and the signal amplitudes of saturation curves decrease from the open end to the close end along the core. These features of saturation profiles illustrate that OSO imbibition is a unidirectional linear imbibition along the core. Based on curve shapes, NMR saturation profiles of HSO (Figure 7d) can be divided into two parts: the left part similar to ASO curves and the right part similar to the center part of OSO curves. The left part of saturation curves corresponds to the open imbibition parts of the core and has the same imbibition behavior as ASO imbibition (Figure 7a), and the imbibition pattern of the right closed part is the linear imbibition identical to OSO imbibition (Figure 7c). These two-stage profiles illustrate that HSO imbibition is a complicated combination of ASO imbibition and OSO imbibition.

3.5. NMR Spatial T_2 Images of Forced Imbibitions. NMR spatial T_2 images can qualitatively reflect fluid distributions and fluid transportations during forced imbibitions (Figure 8). The horizontal axis is the T_2-axis, and the right vertical axis is the position in the field of view which reflects the core position in the magnetic field. The bottom part is a graduated color bar representing incremental fluid volume. Figure 8 shows fluid distributions of ASO, TSO, OSO, and HSO imbibitions under 10 MPa conditions. Due to the largest contact area with water, ASO achieves a rapid increase in water content and has a uniform fluid distribution along the core sample (Figure 8a). Also, NMR T_2 signals are smaller than 10 ms, illustrating that imbibed water exists in nanoscale

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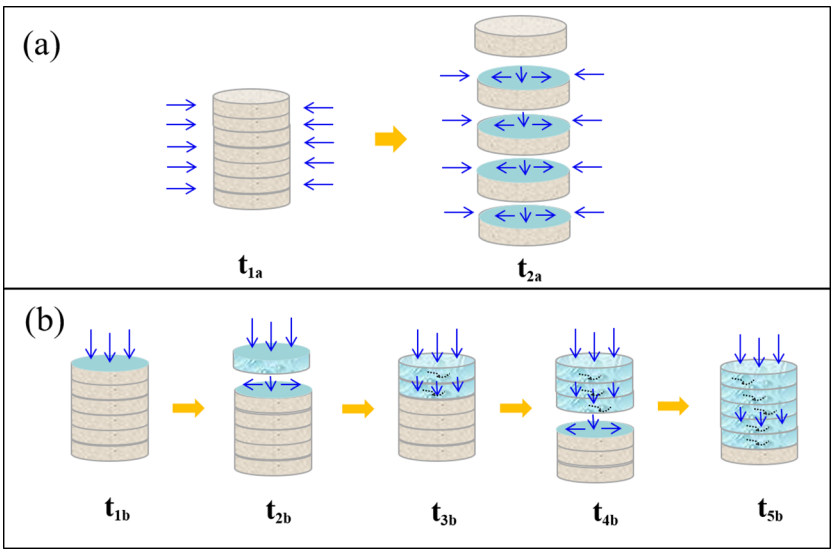

Figure 9. Conceptional imbibition dynamic models of shale: (a) imbibition parallel to beddings; (b) imbibition vertical to beddings.

matrix pores. For TSO imbibition, color changes of spatial T_2 images show that water is imbibed into the sample from two end faces and then transport along the plug (Figure 8b). Besides, the existence of a small number of T_2 signals (>100 ms) illustrates that some microfractures are generated during the TSO imbibition case. As for OSO (Figure 8c), water imbibes into the sample from the open-end face to the sealed end face under applied water pressure. Compared with the TSO scenario, more microfractures are generated in this unidirectional imbibition process. With regard to HSO imbibition (Figure 8d), 2D NMR images can be divided into two parts corresponding to the open part (the lower part) and the sealed part (the top half part). Water is imbibed into shale from the open section and gradually migrates into the sealed part, and there exist considerable differences in water contents for the two different parts of HSO imbibition.

The 2D NMR T_2 images also illustrate the existence of induced microfractures during forced imbibitions with different boundaries. There are some obvious light-blue spots (T_2 > 100 ms) in the lower-right regions of spatial T_2 images for OSO imbibition, which show a certain amount of generated microfractures during forced imbibition processes. For TSO imbibition, only a very small number of vague light-blue spots can be found in the spatial T_2 images. For HSO, no obvious light-blue spots can be recognized due to the limited resolutions of NMR spatial T_2 image techniques. These 2D NMR findings on the induced microfractures of forced imbibitions are consistent with the experimental results of T_2 spectra, which are attributed to the different imbibition patterns with respective boundaries.

4. DISCUSSION

4.1. Imbibition Pattern Analysis—Imbibitions Parallel and Vertical to Beddings. Shale rock is a thin-sheet sedimentary rock formed by the compaction of clay minerals as well as organic matter and other inorganic minerals. During the deposition of mineral particles in shale reservoirs, fluids (water and hydrocarbons) flow between rock layers, which indirectly leads to the generation of bedding layers.[47] One essential petrophysical characteristic of gas shale is its natural bedding planes.[48] It is common knowledge that bedding planes and their orientations of shale have important consequences for permeability anisotropy, fluid migration, cracking characteristics, mechanical behavior, and reservoir pressure distribution. Due to the complex bedding structure of shale, there are two imbibition processes during forced imbibition, namely, imbibition parallel to bedding and imbibition vertical to bedding, which brings challenges to make a thorough inquiry into imbibition mechanisms. In view of the complex imbibition dynamics in laminated shale, it is imperative to fully understand the significance of bedding planes on imbibition parallel to bedding and imbibition vertical to beddings, which can lay the foundation for further investigations on forced imbibition under different boundary conditions. Gas shale is a fine-grained and layered unconventional reservoir with strong porosity anisotropy and permeability anisotropy. SEM images revealed that porosity measured perpendicular to the bedding direction is smaller than that in parallel to the bedding direction, and the radius of the throat parallel to bedding is likely to be larger than perpendicular to bedding.[49−51] Moreover, Revil et al.[52] confirmed that the tortuosity parallel to bedding is much smaller than in the direction perpendicular to bedding in shales due to compaction. Considerable publications have proved that permeability in the direction parallel to the bedding plane is much larger than that vertical to the bedding plane.[53−55] Strong porosity anisotropy and permeability anisotropy of layered shale can control fluid seepage in the shale matrix and are the key factors responsible for different imbibition dynamics of two imbibitions.

As shown in Figure 9, the conceptional imbibition dynamic models of imbibition parallel to bedding and imbibition vertical to bedding are proposed to compare the differences between the two imbibitions. Imbibition parallel to beddings means that the imbibing direction of water is parallel to the bedding plane of shale, and this corresponds to the situation that water imbibes into pore spaces between beddings from the cylindrical surface of the shale sample (Figure 9a). With the advantage that porosity and permeability parallel to beddings are much larger than those vertical to beddings, water can be imbibed into pore spaces along shale beddings to reach an equilibrium state within a relatively short period of time (t_{2a} in Figure 9a). Imbibition vertical to bedding means that the

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imbibition direction of water is vertical to bedding planes, and it refers to that water imbibes into pore networks from the end face of the plug (Figure 9b). Imbibition behaviors of imbibition vertical to bedding are more complicated than that of vertical imbibition due to interbedded flow resistances resulting from the bedding structures of shale. For this situation, water is imbibed through bedding planes from the end face of the sample and then imbibed into pore networks between bedding planes ($t_{1b} \sim t_{2b}$ in Figure 9b). When water imbibition between beddings reaches the equilibrium state, water continues to be imbibed through the next bedding plane and imbibed into pore networks between lower beddings ($t_{2b} \sim t_{3b}$ in Figure 9b). The layered water imbibition will continue until it ultimately reaches an equilibrium state for a quite long time ($t_{5b} \gg t_{2a}$) or water arrives at the other end face of the shale. In summary, the imbibition dynamics of the former is characterized by imbibition along bedding planes without less interbedded flow resistance, but the latter is featured by cross-layer imbibition vertical to beddings with interbedded imbibition resistance. Publications have proved that water imbibition parallel to the shale lamination is higher than that against the lamination.[11,56] Therefore, shale beddings are particularly important for analyzing forced imbibitions under different boundaries.

4.2. Comparison of the Imbibition Pattern of Four Boundaries. Imbibitions with various boundary conditions have different imbibition dynamics, fluid distributions, and gas−water−rock interactions due to diverse imbibition patterns. Based on analyses of imbibition dynamics mentioned above, NMR imbibition experiment results including T_2 spectra, saturation profiles, and spatial T_2 images are combined to investigate the imbibition patterns of four selected boundaries (Figure 10).

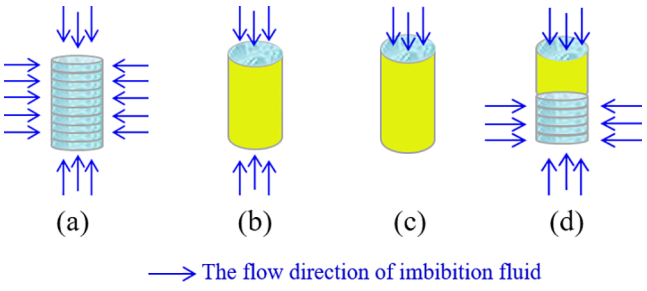

Figure 10. Comparison imbibition patterns under four boundaries: (a) ASO, (b) TSO, (c) OSO, and (d) HSO.

For OSO imbibition, only one end face of the sample is open and the other faces (the other end face and the cylindrical surface) are sealed (Figure 10c). Perpendicular to the shale bedding plane, imbibition fluid is imbibed through beddings into the core from the open side to the sealed side, which is proved by NMR fluid saturation profiles with one big peak in the open end of the core and one small peak in the other sealed end (Figure 7c). Besides, the NMR spatial T_2 image also verifies this imbibition fluid distribution accumulated on the open side of the sample (Figure 8c). NMR responses of OSO imbibition (Figure 6c) illustrate that it is a unidirectional cross-layer imbibition for OSO imbibition (Figure 10c). Compared with OSO, TSO imbibition with two open-end faces has double imbibition areas making contact with fluid as shown in the NMR spatial T_2 image (Figure 8b). Based on the morphology of NMR fluid saturation profiles that two peaks located at the two ends of the core (Figure 7b), we found that water is imbibed through beddings into the core from two end faces to the central section. The change of T_2 distribution of the TSO imbibition process (Figure 6b) is different from that of the OSO scenario (Figure 6c), which shows that there exist certain flow resistances in the bidirectional imbibition. Fluid−fluid momentum transfer can cause higher flow resistance when fluids flow in opposite directions as compared to the same direction. All NMR experimental results illustrate that TSO imbibition is a bidirectional cross-layer imbibition (Figure 10b).

For ASO imbibition, fluid is imbibed into the core from two end faces and a cylindrical surface (Figure 10a). Note that there coexist two kinds of imbibition processes in the ASO condition: the imbibition parallel to the bedding occurs on the cylindrical surface and the imbibition perpendicular to beddings occurs on two end faces. Compared with the TSO case which only has one imbibition perpendicular to beddings (Figure 10b), the imbibition pattern of ASO can be summarized. As shown in Figure 6a, both the signal amplitude and signal peak value of T_2 distributions of ASO is much higher than that of the TSO case (Figure 6b). Moreover, comparisons of spatial T_2 images between ASO and TSO (Figure 8a,b) show that considerable water is imbibed into the core from the cylindrical surface rather than the end faces. These differences in NMR responses illustrate that imbibition perpendicular to bedding occurring on end faces is a dominant position during the whole ASO imbibition process. Based on the discussions above, it can be found that the ASO imbibition pattern is composed of the dominant imbibition parallel to beddings and the affiliated imbibition perpendicular to beddings (Figure 10a).

For the HSO boundary, two end faces and half of the cylindrical surface are open to make contact with imbibition fluid (Figure 10d). NMR T_2 curve variation of HSO (Figure 6d) is similar to those of ASO (Figure 6a), but the cumulative NMR signal amplitudes of HSO imbibition are bigger than those of ASO imbibition. Moreover, NMR fluid saturation profiles (Figure 7d) and spatial T_2 images (Figure 8d) show the two-stage imbibitions corresponding to the sealed part and the open part of the HSO sample. HSO imbibition (Figure 10d) can be divided into two different parts: the open part has the same imbibition pattern as ASO imbibition (Figure 10a) and the sealed part has the same imbibition pattern as OSO imbibition (Figure 10c). The HSO imbibition pattern is a combination of the imbibition parallel to beddings (the open part) and the imbibition perpendicular to beddings (the sealed part). Besides, T_2 distributions (Figure 6d) show that some microfractures (T_2 > 100 ms) are generated during this process. Among these four cases, it can be concluded that the imbibition perpendicular to the beddings (TSO, OSO, and HSO) is more likely to generate microfractures than the imbibition parallel to the bedding (ASO).

The results discussed above are consistent with the recognition of other scholars.[24,32,57] For ASO imbibition, a radial imbibition occurs on the cylindrical side, and a linear imbibition occurs on two ends. The radial imbibition on the cylinder side meets the linear flow on the two end faces and interferes with each other in the forced imbibition process. Thus, complex three-dimensional imbibition with mutual interference of linear imbibition and radial imbibition in

https://doi.org/10.1021/acs.energyfuels.3c00404
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different directions occurs in ASO imbibition.[24] Forced imbibition with the OSO boundary is a one-dimensional linear imbibition flow. Compared with the OSO, TSO has an additional end face, but TSO imbibition cannot be simply regarded as the superposition of two open-end faces. The imbibition rate of the two end faces of the joint water imbibition under TSO conditions is not twice as big as OSO imbibition due to the existence of shale bedding structures.[32,57] Therefore, it is a reasonable explanation that the imbibition pattern of TSO is a complex one-dimensional linear flow compared with OSO imbibition.

4.3. Forced Imbibition Dynamics Analysis. Forced imbibitions under different boundaries are the fluid-pressure-dominated processes that occur in fractured shale reservoirs, which is one of the major patterns of aqueous phase migration during shale fracturing.[13,18,19] Based on the Handy equation,[58] Makhanov et al.[11] characterized the imbibition capacity of tight shales by the curve of cumulative imbibition volume per unit area changing with the square root of imbibition time, and the imbibition rate can be estimated from the slope of cumulative imbibed volume per unit surface area versus square root of time. In this study, forced imbibition curves can be obtained by eq 3, which is used to compare imbibition dynamics of boundary conditions with various imbibition areas. Equation 3 can be expressed in the following form

$$q_{imb} = Q_{imb}/A_{imb} = \sqrt{\frac{2P_c\varphi KS_{wf}}{\mu_w}}\sqrt{t} \quad (3)$$

where q_{imb} is the cumulative imbibition volume per unit area, cc/cm^2; Q_{imb} is the cumulative imbibed volume, cc; A_{imb} is the contact imbibition area, cm^2; P_c is the capillary pressure, MPa; φ is the porosity of the sample, %; K is the liquid effective permeability, Darcy; S_w is the water imbibition saturation, %; μ_w is the water viscosity, mP s; t is the imbibition time, s.

Figure 11 depicts the variation trends of imbibition curves versus imbibition time under four boundary conditions at 10 MPa. The imbibition curves of both shales appear convex, which illustrates that the forced imbibition rate is quick in the beginning and becomes slow gradually in the late period. A general observation of imbibition rate changes shows that the imbibition curves of four cases can be divided into two parts: the initial imbibition stage (stage 1) and the late imbibition stage (stage 2). The slope of the imbibition curve at stage 1 is defined as the initial imbibition rate, and the slope at stage 2 is defined as the late imbibition rate. The intersection points of stage 1 and stage 2 under each boundary can be defined as the turning point of forced imbibition. Theoretically, the most accurate value of the turning point for each boundary condition should be exactly determined at the intersection of the initial imbibition stage and the late imbibition stage based on the changes of slopes. However, because of the limited five measurement time points selected in the complex forced imbibition experiments, we chose the actual experimental time that is approximate to the calculated imbibition time as the turning point for each imbibition case. In this study, the turning points of OSO and TSO are assigned as 20 and 30 h, whereas the turning points of HSO and ASO are the same value (10 h) because the imbibition curves have reached a state of equilibrium with the facts that late imbibition rates are close to zero.

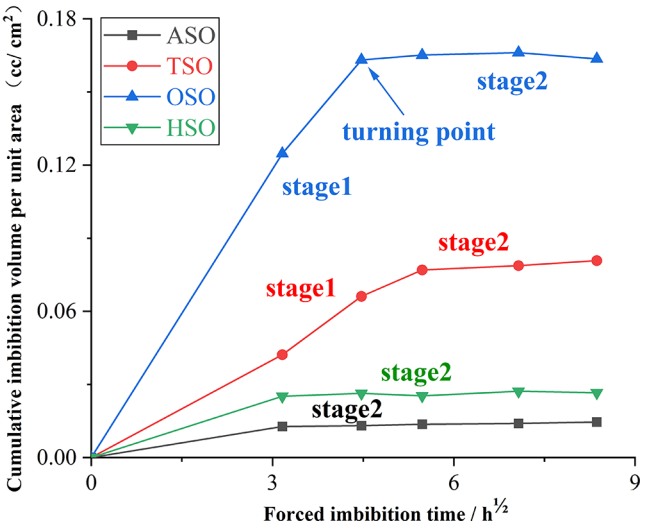

Figure 11. Comparison of forced imbibition curves of four boundaries.

There exists an obvious difference between the four types of forced imbibitions. As depicted in Figure 11, the imbibition curves of OSO and TSO are obviously different from those of HSO and ASO. Both OSO and TSO show the obvious stage 1 and stage 2, whereas the HSO and ASO only show stage 2 because they have reached the imbibition equilibrium before 10 h. These characteristics of forced dynamics can be explained by the differences in imbibition patterns that under the same injection water pressure, the cross-layer imbibition patterns of OSO and TSO need more time in stage 1 to overcome the interbedded flow resistances to reach the imbibition equilibrium. Due to flow interference of bidirectional imbibition from two end sides, the initial imbibition rate of TSO is much lower than that of OSO, and TSO imbibition needs more time to reach an imbibition equilibrium state. From the analyses mentioned above, it is concluded that the imbibition dynamics of cross-layer imbibition vertical to beddings (OSO and TSO) is more complicated than that of imbibition parallel to beddings (HSO and ASO).

Imbibition rates of stage 1 and stage 2 for forced imbibition curves play an important role in analyses of fluid migration and fracturing performance during shale fracturing. The initial imbibition rate of stage 1 represents how quickly can water imbibe into the different matrix-fracture systems of shale under injection fluid pressure, which illustrates the rate of fracturing fluid invading the fractured shale reservoir during the shut-in period of hydraulic fracturing. At shut-in, the higher the initial imbibition rate of the stimulated reservoir is, the larger the volume of fracturing fluid invading into shale formation will be, which accounts for a large amount of fracturing fluid retention in shale reservoirs. As for stage 2, the late imbibition rate is close to zero and the imbibition curves reached an equilibrium state, illustrating that the micro−nanopore networks of shales are filled completely with water. If shale formation has a higher initial imbibition rate but a lower late imbibition rate, it reveals that the shale stratum can imbibe a mass of fracturing fluid into matrix-fracture systems, but imbibed fluid has less flowback capacity back to the surface, which may cause serious aqueous phase trapping after fracturing. However, if the shale reservoir has a higher initial imbibition rate and a higher late imbibition rate, it implies that massive slick water can be invaded into formation during fracturing, causing certain fracturing fluid retention. Meanwhile, water retention can be removed to some

https://doi.org/10.1021/acs.energyfuels.3c00404
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extent due to subsequent water migration from fractures to matrixes. This scenario can explain the confusing well performances that a number of gas wells have serious fracturing fluid trapping after fracturing, but this fluid trapping can be autoremoved with considerable gas production after reasonable shut-in operations. In addition, if a shale reservoir has a lower imbibition rate of imbibition curve during fracturing, it illustrates that shale has fewer fracture networks due to unsatisfying fracturing performance, and commercial gas production of shale play may not be achieved.

The forced imbibition dynamics of shale with different boundaries in this study are quite different from the spontaneous imbibition dynamics under ASO, TSO, and OSO boundaries obtained by Umeobi et al.[31] and Yang et al.[32] Umeobi et al.[31] found that the shale spontaneous imbibition flow in all boundary conditions are contributions of both the capillary pressure and diffusion. Imbibition in AFO is predominantly by the capillary pressure with negligible diffusion, but imbibitions of TSO and OSO are predominantly by diffusion imbibition. Yang et al.[32] demonstrated that there is not much difference in the imbibition rate between ASO and TSO boundaries, and they are both higher than that of OSO. The main reason behind these imbibition dynamic differences is that there are great differences in imbibition patterns of various boundary conditions between spontaneous imbibition and forced imbibition. Moreover, the essential distinctions of shale imbibition mechanisms in spontaneous imbibition and forced imbibition also contribute to the imparities in imbibition behaviors. The research findings of this paper can improve the understanding of shale forced imbibition dynamics and are also applicable to forced imbibition in other tight rocks with various boundaries.

4.4. Comparison of the Imbibition Parameters of Four Boundaries. To quantificationally evaluate the forced imbibition behaviors under four boundary conditions, six imbibition parameters including water saturation, gas recovery, imbibition capacity index, diffusion capacity index, imbibition rate, and imbibition potential are calculated and compared. Imbibition saturation (S_w) is obtained from the volume of water imbibed into the shale sample as given by eq 4.

$$S_w = \frac{V_w}{V_p} \times 100\% \tag{4}$$

where V_w is the water volume imbibed into samples, cm^3; V_p is the pore volume of shale samples, cm^3.

Gas recovery (R_g) is the percentage of gas phase expelled from the shale sample. When the gas volume is matched by the water volume imbibed into the shales, the gas recovery is equivalent to the imbibition saturation

$$R_g = \frac{V_g}{V_p} \times 100\% \tag{5}$$

where V_g is the gas volume replaced by water imbibition, cm^3.

The imbibition capacity index (I_{imb}) is the ratio of the maximum imbibed volume ($V_{imb,max}$) to the pore volume (V_p), which can be expressed as follows

$$I_{imb} = \frac{V_{imb,\,max}}{V_p} \tag{6}$$

The diffusion ability index (I_{dif}) was calculated as the ratio of the cumulative fluid imbibed within the late imbibition stage (stage 2) to the total fluid imbibed.

$$I_{dif} = \frac{V_{dif}}{V_{imb,max}} \tag{7}$$

where V_{dif} is the cumulative fluid imbibed within the late imbibition stage (stage 2), cm^3.

The imbibition rate (R_{imb}) across the forced imbibition period is the average volume flow rate, as shown in eq 8.

$$R_{imb} = \frac{dN_{wt}}{dt} \tag{8}$$

where N_{wt} is the cumulative fluid imbibed during the whole imbibition period, cm^3; t is the forced imbibition time, h.

Imbibition potential (P_{imb}) is the ratio of the imbibition rate in the late imbibition stage (stage 2) to the imbibition rate in the initial imbibition stage (stage 1), which can be acquired using eq 9.

$$P_{imb} = \frac{dN_2/dt_2}{dN_1/dt_1} \tag{9}$$

where N_1 is the cumulative fluid imbibed in the initial imbibition stage, cm^3; where N_2 is the cumulative fluid imbibed in the late imbibition stage, cm^3; t_1 is the imbibition time of the initial imbibition stage, h; t_2 is the imbibition time of the late imbibition stage, h.

Table 2 listed the calculated results of six imbibition parameters under ASO, TSO, OSO, and HSO boundaries.

Table 2. Parameters of the Forced Imbibitions with Different Boundaries

parameters	ASO	TSO	OSO	HSO
S_w (%)	0.6562	0.6626	0.6799	0.6881
R_g (%)	0.6562	0.6626	0.6799	0.6881
I_{imb} (pu)	0.6562	0.6626	0.6799	0.6881
I_{dif} (pu)	0.1248	0.0468	0.0018	0.0560
R_{imb} (cm^3/h)	0.0118	0.0115	0.0117	0.0121
P_{imb} (pu)	0.0238	0.0367	0.0007	0.0099

During the water forced imbibition in the shale reservoir, the gas volume recovered by water imbibition is equivalent to the water volume imbibed into the shale formation. Thus, comparing eqs 4−6, the gas recovery and imbibition capacity index are equivalent to imbibition saturation. The values of imbibition saturation, gas recovery, and imbibition capacity are such that HSO > OSO > TSO > ASO, which implies that the shale with the HSO boundary had the largest imbibition ability, while ASO had the lowest imbibition ability. Generally, imbibition saturation and imbibition capacity are different in shales because of the different forced imbibition patterns corresponding to four boundaries. Higher imbibition saturation illustrates a greater gas displacement effect and improves the production of gas after hydrofracturing.

The diffusion ability index represents the contribution of the water diffusion process during the late imbibition stage when the forced imbibition reached equilibrium under each boundary condition. Higher diffusion ability also depicts a greater gas displacement effect and enhances gas production.[31] As is depicted in Table 2, the diffusion ability is 0.1248, 0.0468, 0.0018, and 0.0560 for ASO, TSO, OSO, and OEO shales,

https://doi.org/10.1021/acs.energyfuels.3c00404
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respectively. Therefore, the gas displacement effect in the late forced imbibition stage will be the highest in ASO imbibition, followed by HSO and TSO, and the lowest under the OSO case. Regarding the imbibition rates across forced imbibitions, they are not much different in these boundaries, which may be due to the approximate porosity, permeability, and total imbibition volume of these four parallel samples.[32,33] Moreover, the values of imbibition potential are such that TSO > ASO > HSO > OSO, illustrating the different contributions made by late imbibition stages to the whole forced imbibition process. However, these findings are contrary to the previous studies,[31,32,59] which can be attributed to the intrinsic differences of imbibition mechanisms between spontaneous imbibition and forced imbibitions coupled with the boundaries.

4.5. Clay Hydration during Forced Imbibitions. Due to the hydrophilicity of Longmaxi shale, under the action of injection fluid pressure up to tens of MPa, water-based fracturing fluid will be pumped into a shale formation, namely, forced imbibition, and water−shale interactions will occur. Water−shale interactions have a great influence on forced imbibition by triggering various behaviors including clay swelling, clay hydration, mineral dissolution, fine migration, crack generation, wettability alteration, and water sensitivity damage. Clay minerals are predominantly flat 2D structures with shared corners of silica tetrahedra or alumina octahedra. Illite is a clay mineral that contains two silica tetrahedrons and one alumina octahedron as the structure of smectite. Chlorite is a kind of hydrous aluminosilicate with one alumina octahedron sandwiched between two tetrahedral sheets of silica. Illite is the most common mineral component in shale gas reservoirs, clay minerals of marine Longmaxi shale in this study are illite (39 wt %) and chlorite (7 wt %) without montmorillonite, and thus, illite is the key factor to determine the clay hydration during fracturing. The nature of water−shale interactions of clayey shale is the clay hydration, and shale with a significant proportion of illite and chlorite will cause severe damage to pore structures. Compared with smectite, illite has a low cation exchange, poor hydration ability, and low dispersion. Illite-rich Longmaxi shale is a low-hydration swelling rock, but it has a high hydration stress with low strain and low critical water content. Therefore, the hydration of Longmaxi shale occurs within a relatively short time with very little water and produces a large hydration stress with a small expansion.

Clay hydration from water-sensitive clay minerals has been proven to be a dominant factor that influences the reduction in the strength of shale after water imbibition by altering the microstructures of clayey shale. With water soaking, clay minerals with interlayers like smectite and illite have a tendency to hydrate exchangeable interlayer cations to force the clay layers apart. The unsynchronized expansions among clay minerals result in nonuniform stress, cause local stress concentration, and further drive the initiation of microfractures in shale. Moreover, water can weaken the cementation between mineral particles and reduce cohesive strength, causing serious fine migrations of mineral particles to block the narrow pore throats with huge permeability loss. During forced imbibitions of TSO and OSO, shale hydration occurs during the cross-layer imbibitions under the increasing injection water pressure. Hydration of clay minerals causes pore pressure to be greater than the strength of the bond between clay minerals and results in the exfoliation of clays into particles.[60] These dispersed clay particles can transfer during imbibition and block narrow throats of fracture networks, causing great permeability loss of main seepage channels in shale. Meantime, a large proportion of fracturing fluids cannot be removed from stimulated shale formation, having a negative influence on gas production.

5. IMPLICATIONS FOR HYDROFRACTURING OF THE SHALE GAS RESERVOIR

Hydraulic fracturing plays a significant role in achieving commercial exploitations of tight shale reservoirs by generating complex fracture networks to improve gas seepage capacity. With the operation that large-scale slick water is pumped into a shale formation, forced imbibition occurs under applied fluid pressures, which causes both positive and negative effects on shale reservoirs. On the positive side, induced secondary fractures are conducive to the generation of more complex fracturing networks. On the negative side, fracturing fluid retention and reservoir damage are triggered to reduce gas production capacity. Based on our research, three targeted suggestions for engineering application are put forward to improve fracturing performances. For the first one, before stimulation operation, laboratory experiments to simulate the subsurface forced imbibition processes should be conducted to obtain the imbibition dynamics of fracturing fluid, which is beneficial for optimizing fracturing design and predicting the fracturing results. According to comparisons of imbibition curves corresponding to various fracture-matrix systems, fracture generation, fluid transport, flow back of fracturing fluid, and reservoir damages during shale fracturing can be evaluated for future fracturing design. For the second one, bedding effects should be fully considered, especially choosing a reasonable injection direction of slick water. When water-based fracturing fluid is injected parallel to shale layers, more fractures can be generated along layers. As it is injected against layers, severe reservoir damages including fine migration and water sensitivity should be prevented with appropriate measures. For the last one, it is essential to analyze detailed mineral compositions of clays for target shale formations before fracturing. For example, due to clay hydration, an illite-rich shale reservoir can produce clay particles to block narrow throats of pore networks with huge permeability loss. While for shales with abundant smectites, it is likely to cause serious water sensitivity damage due to clay swelling.

6. CONCLUSIONS

In this study, systematic forced imbibition experiments combined with 1D and 2D NMR techniques (T_2 curves, saturation profiles, and spatial T_2 images) were conducted to investigate the imbibition dynamics with ASO, TSO, OSO, and HSO boundaries in Longmaxi shales. The main conclusions are summarized as follows.

(1) Imbibition parallel to shale beddings is characterized by imbibition along bedding planes, whereas imbibition vertical to beddings is cross-layer imbibition with interbedded imbibition resistances. Forced imbibition patterns under four boundaries are different combinations of both imbibitions.

(2) imbibition pattern of OSO imbibition is a unidirectional cross-layer imbibition and that of TSO is a bidirectional cross-layer imbibition. ASO imbibition is a complex 3D imbibition composed of the dominant imbibition

https://doi.org/10.1021/acs.energyfuels.3c00404
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parallel to beddings and the affiliated imbibition against beddings. HSO imbibition is a combination of the imbibition parallel to beddings (the open part) and the imbibition perpendicular to beddings (the sealed part).

(3) Imbibition curves of four boundaries can be divided into two parts: the initial imbibition stage (stage 1) and the late imbibition stage (stage 2). OSO and TSO show obvious stage 1 and stage 2, whereas HSO and ASO only show stage 2.

(4) From the calculated imbibition parameters, HSO imbibition has the largest imbibition ability, followed by OSO imbibition and TSO imbibition, and ASO has the lowest imbibition ability. However, imbibition potentials from the highest to the lowest are such that TSO > ASO > HSO > OSO.

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Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

We acknowledge financial support from the National Natural Science Foundation of China (42125205; 41872123).

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https://doi.org/10.1021/acs.energyfuels.3c00404
Energy Fuels 2023, 37, 5853−5866

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Water partitioning and migration in unsaturated bentonites by low-field NMR characterization

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Vadose Zone Journal (2023); DOI: 10.1002/vzj2.20284

证据用途：其他合作论文背景；张弛老师角色：末位/通讯作者；证据等级：C

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Received: 22 June 2023 | Accepted: 13 September 2023

DOI: 10.1002/vzj2.20284

Vadose Zone Journal

ORIGINAL ARTICLE

Water partitioning and migration in unsaturated bentonites by low-field NMR characterization

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Assigned to Associate Editor Indra Sen.

Funding information
State Key Laboratory of Geomechanics and Geotechnical Engineering, Institute of Rock and Soil Mechanics, Chinese Academy of Sciences, Grant/Award Number: Z019003; University of Vienna's Open Access

Abstract

Water behavior in bentonite clay pores is influenced by soil–water interaction mechanisms such as capillary and adsorptive forces. Quantitative measurement of these water statuses remains challenging, leading to the adoption of advanced techniques. This study uses low-field nuclear magnetic resonance (NMR) technique to investigate water partitioning dynamics and changes in the water state in sodium-rich Wyoming bentonite and calcium-rich Denver bentonite under various humidity conditions. NMR T_2 relaxation and T_1–T_2 mapping techniques, along with a multi-Gaussian decomposition method, enable a quantitative analysis of capillary and adsorptive water in both bentonites. A conceptual water partitioning model is derived to explain water molecule trajectories of water molecules under unsaturated conditions. Our findings indicate distinct transitions in hydrated layers for Na^+-smectite and Ca^{2+}-smectite at different relative humidity (RH) ranges. Characteristic T_2 ranges are identified for capillary and adsorptive water in both clays and provide valuable insights into their water behavior. This study advances our understanding of soil properties at different RH environments and highlights the potential of low-field NMR techniques in characterizing capillary and adsorptive water in bentonite clays.

1 | INTRODUCTION

Unsaturated soils are commonly encountered in many natural and engineered systems, such as slopes, embankments, as well as the vadose zone (Augusto Filho & Fernandes, 2019; Fredlund & Rahardjo, 1993). Investigating unsaturated conditions is particularly crucial for clay minerals such as bentonite, due to their unique properties, including high surface area, adsorption capacity, and swelling potential. The behavior of bentonite clays under unsaturated conditions has a direct impact on their performance across a wide range of applications. For instance, in the field of waste containment, unsaturated bentonite clays are commonly utilized as barriers to prevent the migration of hazardous substances into surrounding soil and groundwater (Han et al., 2019). Moreover, unsaturated bentonite clays play a critical role in nuclear waste disposal, where they are employed as engineered barriers to restrict the migration of radionuclides from repositories into the environment, thereby ensuring the long-term safety of the groundwater and the subsurface environment (Pusch, 1983; Pusch et al., 2012; Villar & Lloret, 2004). These behaviors primarily rely on the mineral–water interactions within bentonite clays (Tessier, 1984). A variety of interaction mechanisms result in distinct and diverse physical properties of the water state within clay, leading to a better understanding of the pore pressure regime, viscosity, and density of bentonite barriers. Consequently, a comprehensive understanding

Abbreviations: NMR, nuclear magnetic resonance; RH, relative humidity; SWRC, soil water retention curve; XRD, X-ray diffraction.

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of the mineral–water interaction mechanism in unsaturated bentonite clay and the corresponding water state is vital for devising effective engineering solutions and promoting sustainable water resource management.

Generally speaking, two primary water states can be distinguished in clay minerals: interlayer water and interparticle water. The interlayer water exists in interlayer voids, while the interparticle water occupies the interparticle voids. Common approaches for studying interlayer water include X-ray diffraction (XRD) techniques (e.g., Holmboe et al., 2012; Mooney et al., 1952) and small angle X-ray scattering (e.g., Muurinen et al., 2013), which measure the basal spacings (i.e., d_{001}) or the corresponding interlayer distances of montmorillonite at a certain hydration level. In contrast, most investigations of interparticle water focus on its relationship with the dry density of compacted bentonite (e.g., Muurinen, 2009; van Loon et al., 2007; Wersin et al., 2004); for example, Suzuki et al. (2004) estimated the volume of fraction of interparticle water to be 61% for dry density of 0.9 g/cm^3 and 22% for 1.35 g/cm^3. However, the clear physical meaning of interlayer water and interparticle water, as well as their corresponding mineral–water mechanisms, remains ambiguous.

As an alternative, some researchers consider thermodynamic energy and recommended capillarity and adsorption as two distinct mechanisms to describe the clay mineral–water interactions (e.g., Lu, 2016). Capillarity refers to the reduction of free energy due to the presence of a curved water–air interface, which is commonly quantified using the Young–Laplace equation (e.g., Lu, 2016; van Honschoten et al., 2010). Capillarity occurs in soil pores when curved air–water interfaces exist under unsaturated conditions. In contrast, adsorption is frequently quantified via the molecular forces (or van der Waals force), which are part of the disjoining pressure (e.g., Derjaguin et al., 1987; Starov et al., 2007; Tuller et al., 1999). Adsorption always occurs on or within soil particles under both saturated and unsaturated conditions. From a molecular-scale physicochemical mechanism's perspective, capillarity is exclusively dominated by the surface tension, whereas adsorption may be related to four known different physicochemical mechanisms including hydrated layer adsorption, ion hydration or solvation, surface hydration, and interlamellar cation hydration (Zhang & Lu, 2018). As such, the water within clays can be physically divided into capillary water and adsorptive water, which are subject to capillary force and adsorptive force, respectively.

Capillary water and adsorptive water play very different roles in physical processes occurring in clay, such as fluid flow, heat transfer, electron conduction, stress, and deformation (e.g., Frydman & Baker, 2009; Grünberg et al., 2004; Iwata et al., 1995; Lu & Likos, 2006; Tuller & Or, 2005; van Genuchten, 1980). Thus, it is crucial to identify and differentiate capillary water and adsorptive water within clay soils. Experimentally, the centrifugation experiment and displacement experiment are considered the common experiments to determine the capillary water and adsorptive water. Capillary water is separated by a centrifugation experiment by spinning samples at high speed. However, achieving complete separation of the capillary water in clay samples is challenging due to the extremely high matric potential of clay (e.g., 10^8 Pa). Furthermore, there is no universally applicable spinning speed for various samples, since sample size, density, and viscosity can vary significantly. The displacement experiment is achieved by injecting air or the other non-wetting phase into the pore at a given pressure to displace the water (wetting phase) of the samples. The discharged water is considered capillary water. However, determining the appropriate displacement pressure is difficult for unconsolidated samples, such as soil samples (e.g., Lu & Likos, 2004). Recently, several models (e.g., Lu, 2016) and experimental techniques have emerged to quantify capillary water and adsorptive water. For instance, Lu (2016) proposed a soil water retention curve (SWRC) model that quantitatively obtains the capillary and adsorptive water under the assumption of local thermodynamic energy equilibrium. Zhang and Lu (2018) employed a helium gas pycnometer to directly measure the soil water density at varying water contents and quantified adsorptive water content values using the SWRC model. However, the governing forces underlying soil–water interactions within each water retention regime in the SWRC models remain unclear.

The low-field nuclear magnetic resonance (NMR) method has been proposed as a promising technique for characterizing the retention and transport of water in clay soils due to its unique sensitivity to water. This method provides important information on proton mobility (Colomban, 1992) based on the magnetic spin properties of hydrogen atoms forming pore water. Recently, there has been a growing interest

Core Ideas

- We utilized low-field NMR techniques to analyze water behavior within bentonite clays, shedding light on the dynamics of moisture within these complex materials.
- Our research developed a conceptual model that details water partitioning in bentonite clays under unsaturated conditions.
- We identified distinct transitions in hydrated layers of different types of bentonite clays based on saturation degree, highlighting the unique properties and responses of these clays to varied environmental moisture conditions.

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in employing low-field NMR techniques to study the water properties within clays (e.g., Hougardy et al., 1976; Nicot et al., 2016; Singer et al., 2016; Slade et al., 1987; Whittingham, 2004). For instance, Hougardy et al. (1976) conducted the ^{1}H NMR study of adsorbed water in the layered silicate Na-vermiculite, reporting that the lifetime of adsorbed water molecules at room temperature would be between 10^{-5} and 10^{-4} s, which is 10–100 times shorter than in the free liquid. Slade et al. (1987) investigated protonic conduction in Na^+, Li^+, and Ni^{2+} montmorillonite clays with 13 wt% water content using conductivity measurements and ^{1}H NMR spin-lattice (T_1) and spin–spin (T_2) relaxation time measurements. Their funding suggested that hydrogens outside the hydration shell undergo translation through chemical exchange. Chitale and Sigal (2000) and Fleury et al. (2013) showed that NMR T_2 distribution is capable of providing complementary information about the size and connectivity of the water-filled pores. Despite these advances, studies on the water state and its migration in clays using the NMR method are still scarce.

In this study, we conducted NMR T_2 and T_1–T_2 experiments on sodium-rich Wyoming bentonite and calcium-rich Denver bentonite specimens at varying humidity levels, ranging from 11.29% to 100%. We employed a peak decomposition method to quantitatively analyze the measured NMR T_2 spectra, characterizing capillary water and adsorptive water, as well as the water transport trajectory of each other. In addition, we identified the characteristic T_2 ranges of capillary and adsorptive water for the two bentonite clays. Our study opens a new window to facilitate the development of theoretical models of fundamental soil properties at unsaturated conditions. Furthermore, our study provides a fast and non-invasive mean to assess the integrity and performance of clay soils as protective/sealing buffer material for radioactive wastes containment.

2 | THEORETICAL BACKGROUND

2.1 | Low-field ^{1}H NMR

Low-field ^{1}H NMR measures the nuclear spin signals of hydrogen protons resided in porous media. Under the influence of the background field (B_0), the nuclear spins associated with hydrogen nuclei orient themselves parallel to the field. When perturbed by an energizing radio frequency magnetic field (B_1), the alignment of the nuclear spin disrupted. Upon removal of the external field, these nucleus spins re-equilibrate to their initial states, a process characterized by the so-called relaxation time. The magnitude of relaxation decay is proportional to the total water content.

Brownstein and Tarr (1979) observed the multiexponential decay of water in cells and proposed three relaxation regimes to elucidate this phenomenon: (1) the fast diffusion regime in which the exchange rate between different states is much faster than the difference in resonance frequencies of the exchanging spins; (2) the slow diffusion regime, characterized by an exchange rate between different molecular environments that is considerably slower than the difference in resonance frequencies of the exchanging spins; and (3) the intermediate diffusion regime, where the exchange rate is comparable to the difference in resonance frequencies of the exchanging spins. The transversal relaxation time T_2 of porous material is attributed to three different mechanisms: surface relaxation T_{2S}, bulk relaxation T_{2B}, and diffusive relaxation T_{2D}. Surface relaxation occurs when fluid molecules in the pores interact with the solid surface, leading to a loss of transverse magnetization. Bulk relaxation arises from the intrinsic relaxation of fluid molecules in the absence of any interaction with the solid matrix, while diffusive relaxation results from molecular diffusion within the porous material contributing to the relaxation process (Kleinberg et al., 1994). In the case of natural rock or clay materials, the dominant relaxation is surface relaxation (Korb et al., 2003), and bulk relaxation and diffusive relaxation can be omitted. Thus, the transversal relaxation time T_2 can be simplified as:

$$\frac{1}{T_2} = \frac{1}{T_{2S}} + \frac{1}{T_{2B}} + \frac{1}{T_{2D}} \approx \frac{1}{T_{2S}} = \rho\frac{S}{V} \propto \rho\frac{1}{r}, \qquad (1)$$

where S represents the surface area of the water-filled pores, V represents the volume of the water-filled pores, ρ is the surface relaxivity, and r is the radii of the water-filled pores. Equation (1) implies that the relaxation time is positively correlated with the pore size for the water-filled pores, and amplitudes of T_2 spectra are related to the abundance of water content with the corresponding pore size (Banavar & Johnson, 1987; Kleinberg & Horsfield, 1990). The adsorptive water or the water in the smaller pores relaxes faster than the capillary water or the water in larger pores. Each classified peak would represent one water-filled pore environment. However, extracting the information of characteristic water-filled pores from T_2 spectra can be difficult, as the T_2 spectra are an assemblage of the possibility density function. In addition, the information of certain water-filled pore size might be omitted, particularly when neighboring peaks merge to a broad distribution (e.g., Grunewald & Knight, 2009). Therefore, in order to characterize the characteristic T_2 ranges of the capillary water and adsorptive water, the multiple peaks or modes need to be quantitatively analyzed in order to isolate capillary water and four possible hydrogen-bonding scenarios of the adsorptive water.

In addition to T_2 relaxation distribution, the two-dimensional NMR techniques, such as T_1–T_2 mapping, are employed to obtain more comprehensive information on various proton contributions. These contributions are affected by the proton's mobility and their interaction with other species, such as paramagnetic impurities. In nano-meter

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scales pores, very short relaxation times (<0.1 ms) are obtained due to processes related to proton mobility. Water proton I repeatedly encounters paramagnetic species S due to paramagnetic impurities, and as a result, the local dipolar field is adjusted and relaxation arises. Hence, the dominant contribution of proton nuclear relaxation comes from the heteronuclear dipole–dipole interaction with the electronic spins. It depends on both surface and liquid characteristics (Abragam, 1961; Bloembergen et al., 1948). This framework is the so-called Bloembergen–Purcell–Pound model, which is related to the correlation time τ of the dipolar interactions and gives an expression for the ratio of T_1/T_2 as follows:

$$\frac{T_1}{T_2} = 2\frac{\left\{2\ln\left(\frac{\tau_s}{\tau_m}\right) + \frac{3}{4}\ln\left[\frac{1+\omega_I^2\tau_m^2}{\left(\tau_m/\tau_s\right)^2+\omega_I^2\tau_m^2}\right] + \frac{13}{4}\ln\left[\frac{1+\omega_S^2\tau_m^2}{\left(\tau_m/\tau_s\right)^2+\omega_S^2\tau_m^2}\right]\right\}}{\left\{3\ln\left[\frac{1+\omega_I^2\tau_m^2}{\left(\tau_m/\tau_s\right)^2+\omega_I^2\tau_m^2}\right] + 7\ln\left[\frac{1+\omega_S^2\tau_m^2}{\left(\tau_m/\tau_s\right)^2+\omega_S^2\tau_m^2}\right]\right\}}, \quad (2)$$

where ω_I and ω_S are the Larmor frequencies for proton species I and S, respectively, and τ_m and τ_s are the translational correlation time and surface residence time, respectively.

The T_1/T_2 ratio is related to the translational correlation time, which is associated with the individual molecular jumps at the surface, and to the surface residence time, which is limited by the molecular desorption from the surface layer, controlling the duration that proton species I and S remain correlated at pore surfaces. As such, the T_1/T_2 ratio serves as a characteristic parameter for molecular mobility (Godefroy et al., 2001; Korb, 2018). For example, the pseudo-solid water (e.g., hydrogen-bonded water) exhibits a very short T_2 relaxation time (e.g., <0.1 ms) but a very long T_1 relaxation time, resulting in a high T_1/T_2 ratio (Fleury et al., 2013). In contrast, free water molecules undergoing isotropic molecular tumbling without interacting with a solid surface have similar T_1 relaxation time and T_2 relaxation time (T_1/T_2 ratio ~1.0) (Bloembergen et al., 1948). The reduced mobility of the water molecule when interacting with the solid surface results in a T_1/T_2 ratio greater than 1.0. For instance, Tian et al. (2020) reported T_1/T_2 ratios ranging from 1 to 3 for two clay samples during the drying process, with the ratio values decreasing as the water content diminished. Korb et al. (2018) suggested that the T_1/T_2 ratio is around 1.4 for free water due to the heteronuclear dipole–dipole interaction with the electronic spins. Additionally, methane can be distinguished with T_1/T_2 ~10 (Fleury & Romero-Sarmiento, 2016). Consequently, adsorptive water exhibits a lower T_2 relaxation time value and higher T_1/T_2 ratio value than that for capillary water.

2.2 | Peak decomposition methods

Alba et al. (1996) illustrated Gaussian-like peaks in NMR relaxation time distribution by analyzing the NMR spectra of molecular sieves with well-defined pore sizes. Based on these findings, it is reasonable to assume that NMR T_2 spectra exhibit a multi-Gaussian distribution, where each Gaussian component represents a specific water-filled pore size environment.

The possibility density function of a typical Gaussian distribution is as follows:

$$f(t,\mu,\sigma) = \frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{1}{2}\left(\frac{t-\mu}{\sigma}\right)^2}, \quad (3)$$

where the t is the T_2 values in the log scale, and μ and σ stand for peak position and peak width, respectively.

The multi-Gaussian fitting problem has the following objective function:

$$\min \sum_{i=1}^{N}\left[m(t_i) - \sum_{j=1}^{K}\omega_j f(t_i,\mu_i,\sigma_i)\right]^2, \quad (4)$$

which corresponds to constraints

$$\begin{aligned} &\mu_i,\sigma_i > 0 (j = 1,2,\ldots,K),\\ &\text{and } \sum_{j=1}^{K}\omega_j = 1 (j = 1,2,\ldots,K). \end{aligned} \quad (5)$$

where N is the number of data points, K is the number of peaks, $m(t_i)$ is the measured amplitude at the ith relaxation time t, and ω_j is the weight of the jth peak.

In this study, we sought to identify the best parameters for fitting the multi-Gaussian model. To achieve this, we employed the computationally efficient Levenberg–Marquardt algorithm to find minimums. The detailed fitting procedure is presented in Figure 1. To ensure the convergence of the optimization process, we set the maximum iteration number at 1500 (F. Zhang & Zhang, 2021). Our approach began with an initial value of $K = 2$, representing the number of Gaussian peaks, and proceeded to iteratively update K to find the optimal solution that provides a reasonable number of peaks. Upon updating the value of K, we calculated the corresponding cost function:

$$C(K, m_{opt}) = \frac{1}{N}\sum_{i=1}^{N}\left|\frac{m_{opt}(t_i,K) - m_{opt}(t_i,K-1)}{m_{opt}(t_i,K-1)}\right|, \quad (6)$$

where m_{opt} (t_i, K) *is* the optimized amplitude at relaxation time t_i with K peaks. The iterative process of updating K continues until the cost function reaches a value below 0.1. Ultimately, the final outputs include the optimum number of peaks, optimized peak positions, peak widths, peak weights for individual peaks, and the optimized T_2 distribution.

3 | EXPERIMENTAL PROCEDURE

In this study, two types of natural soils were selected for investigation, namely, Denver bentonite (DBt) and Wyoming bentonite (WBt). The mineralogical composition of both dried soil samples was determined using XRD analysis,

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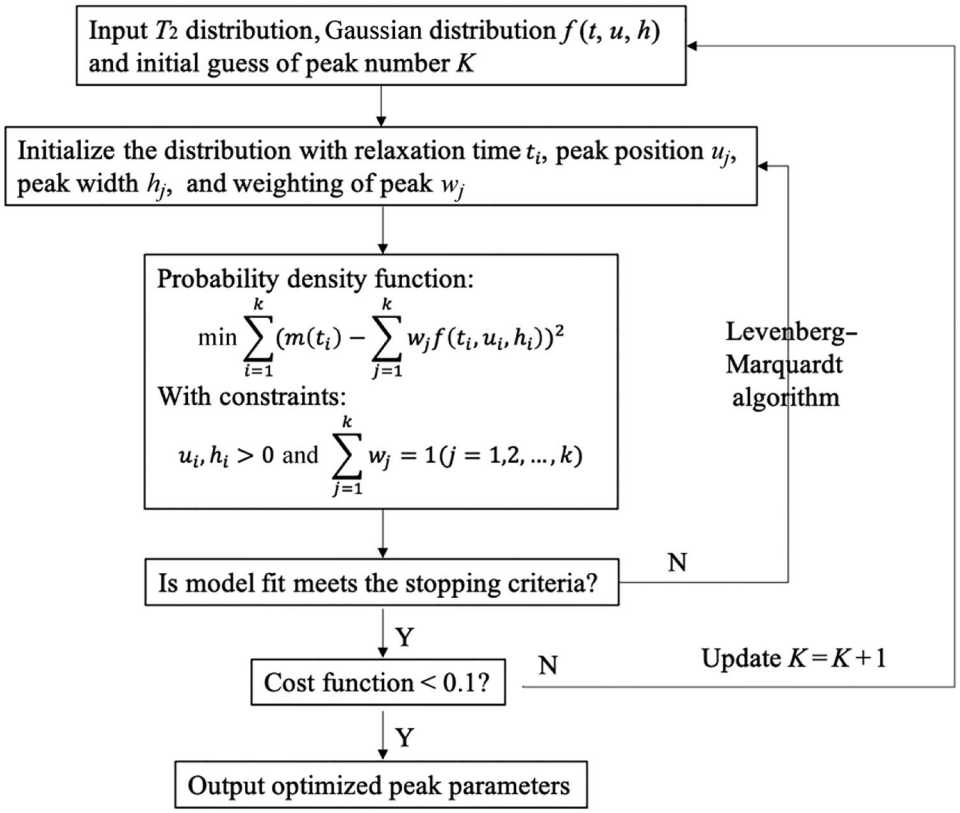

FIGURE 1 Flowchart of the iterative multiple Gaussian fitting.

TABLE 1 Summary of salt solution used for humidity control in our study.

	1	2	3	4	5	6	7	8	9	10
Salt solution	LiCl	$MgCl_2$ ($6H_2O$)	K_2CO_3	$MnCl_2$ ($4H_2O$)	$NaNO_2$	NaCl	KCl	KNO_3	K_2SO_4	Distilled water
Relative humidity (%)	11.29	32.64	44.30	56	64.40	75.21	84.20	92.70	98.18	100

employing the D2 PHASER benchtop diffractometer. The samples were scanned over a range of 5°–80°2θ with a step size of 0.02°2θ.

The clay soils were first pulverized to pass through a 0.15-mm (#100) sieve and then partitioned into 10 parts for equilibrate purposes. Ten oversaturated salt solutions were utilized to control the saturated water vapor pressure within sealed desiccators at ambient temperature. The samples were equilibrated with different oversaturated salt solutions for a minimum duration of 6 months. The relative humidity (RH) levels within each desiccator ranged from 11.29% to 100%, as indicated in Table 1.

Once equilibrium was reached, the specimens were packed into clear cylindrical plastic tubes (average inner diameter: 19.20 ± 0.2 mm, height: 50 ± 0.2 mm) using a mechanical jack with a dry density of 1.5 g/cm^3. As a result, a total of 20 unsaturated compacted specimens were prepared for NMR measurements in our study. In parallel, we prepared another set of 20 specimens for gravimetric water content measurement to compare the results with those obtained from the NMR measurements. These samples were obtained from the equilibrium state and dried in the laboratory oven at 105°C for 1 week to achieve a constant mass. Gravimetric water content ($w = m_w/m_s \times 100\%$, where m_w is the mass of water and m_s is the mass of dry soil) of each sample was computed at steady state by measuring the mass of the specimens with a precision of 0.001 g.

NMR T_2 and T_1–T_2 measurements were conducted on a 2 MHz Rock Core Analyzer (Magritek Ltd.) equipment at room temperature (~25°C). The Carr–Purcell–Meiboom–Gill pulse sequence (Carr & Purcell, 1954; Meiboom & Gill, 1958) was employed to measure T_2 relaxation. To eliminate the influence of the echo time, we conducted measurements with four different echoe spacings: 100, 120, 150, and 200 μs, respectively. Each measurement was stacked 128 times to enhance the signal-to-noise ratio. An inter-experiment time

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of 10 s was utilized to optimize the measurements, and each specimen was measured three times to minimize potential system errors. The inversion algorithm for T_2 data, provided by Magritek Ltd., is based on the non-negative least squared inversion algorithm. The 2D T_1–T_2 correlation spectra were obtained through the application of the Inversion recover—Carr–Purcell–Meiboom–Gill sequence, which can be displayed as follows:

$$P_{180} - \tau_1 - P_{90} - (\tau - P_{180} - \text{echo}-)_n, \quad (7)$$

where τ_1 is the recovery time for T_1, τ is half of the echo spacing, n is the echo number, which is adjusted in order to reach zero signal. T_1–T_2 2D distribution was obtained using a 2D inverse Laplace transform. In our test, $\tau = 50$ μs and number of echoes is 300. All the measurements were noise-corrected to minimize the impact of environmental noise on the NMR response. To ensure the highest data quality, we compared the NMR-derived water content from four distinct echo spacings to the gravimetric water content. Our findings revealed that for both Denver and Wyoming bentonite samples, the NMR-derived water contents were in closest agreement with the gravimetric value (minimum root-mean-square deviation) when the echo-spacing was set at 100 μs. Consequently, we utilized the NMR measurements obtained at this specific echo spacing for further analysis.

4 | RESULTS

4.1 | Soil moisture sorption isotherm curve

The XRD analysis revealed that Denver bentonite primarily comprises montmorillonite (92.7%), along with minor constituents of kaolinite (3.25%), calcite (1.93%), gypsum (1.78%), and quartz (0.97%). In contrast, Wyoming bentonite is predominantly composed of montmorillonite (84.52%), accompanied by quartz (9.31%), sylvite (2.22%), cristobalite (2.19%), and goldfieldite (1.76%). Based on the findings of Likos (2000), Denver bentonite and Wyoming bentonite can be simplified and referred to herein as Ca^{2+}-smectite and Na^{+}-smectite, respectively.

Figure 2 illustrates capillary water and four possible hydrogen-bonding scenarios for adsorptive water. Figure 2(I–III) depicts water molecules bonded to the silica surface, while Figure 2(IV) presents water molecules bonded to the interlayer cation. It should be noted that adsorptive water includes hydrogen-bonded water and hydrated layer water, which may adsorb on external surfaces within the larger scale pore space and/or interstratification of hydration states among various interlayer and interlayer groups (e.g., Bérend et al., 1995; Likos & Lu, 2006). The thickness of the adsorptive water layer varies, ranging from a single molecular layer up to several hundreds of molecular layers, depending on the water content and the type of exchange cations of clay (Tessier, 1990). Figure 2a,b displays the schematic representations of the smectite particle system for Ca^{2+}-smectite and Na^{+}-smectite, respectively.

Soil moisture sorption isotherm curve and further analysis for two bentonites are shown in Figure 3. Figure 3a indicates that the Denver bentonite (Ca^{2+}-smectite) powder absorbs more water than the Wyoming bentonite (Na^{+}-smectite) throughout the entire measured RH range. Additionally, the water content of compacted samples is lower than that of powder samples for both types of clays. These results are consistent with the sorption behavior typically observed in expansive clays (Likos & Lu, 2006). Moreover, the water absorption capacity of Denver bentonite (Ca^{2+}-smectite) surpasses that of Wyoming bentonite (Na^{+}-smectite) under the same RH atmosphere, reflecting the relatively high forces of initial hydration associated with the Ca^{2+} system (e.g., Keren & Shainberg, 1975).

The slope of the water-vapor sorption isotherms serves as an indicator of the amount of water absorbed in response to a given change in humidity (Rouquerol, 1999). Peaks observed in Figure 3b,c indicate that the clay mineral adsorbed a relatively larger amount of water for a given change in humidity. This behavior has been interpreted as a manifestation of crystalline interlayer swelling. The relatively steep portions indicate transitions from one hydration state to an adjacent hydration state, while the relatively flat sections represent a stable hydration state. Undefined transition regions may reflect concurrent adsorption on external surfaces within the larger scale pore space and/or interstratification of hydration states among various interlayers and interlayer groups (e.g., Bérend et al., 1995; Likos & Lu, 2006).

For Wyoming bentonite (Figure 3b), the powder sample exhibits three distinct peaks at an RH value of ∼30%, 65%, and 95%. These peaks are delineated by RH ranges of 20%–40%, 60%–75%, and RH > 85%, respectively, with the peak magnitude of about 0.39, 0.22, and 0.78, respectively. In accordance with the finding of Ohkubo et al. (2008), samples with high dry density (>1.4 g/cm^3) primarily exhibit two hydrated layers. Consequently, we interpret the first peak as corresponding to the transition of adsorptive water in interlayer spacing from a zero-layer state to a one-layer hydrate state, while the second peak signifies the transition from a one-layer state to a two-layer hydrate state. These transitions occur within RH ranges of 10%–40% and 50%–90%, respectively. The interlayer pore volume is minimal for dry clay, resulting in a high value for the main drive of water adsorption in the interlayer. As the amount of adsorptive water increases, the suction decreases, leading to a peak magnitude for the two-layer state that is lower than the first

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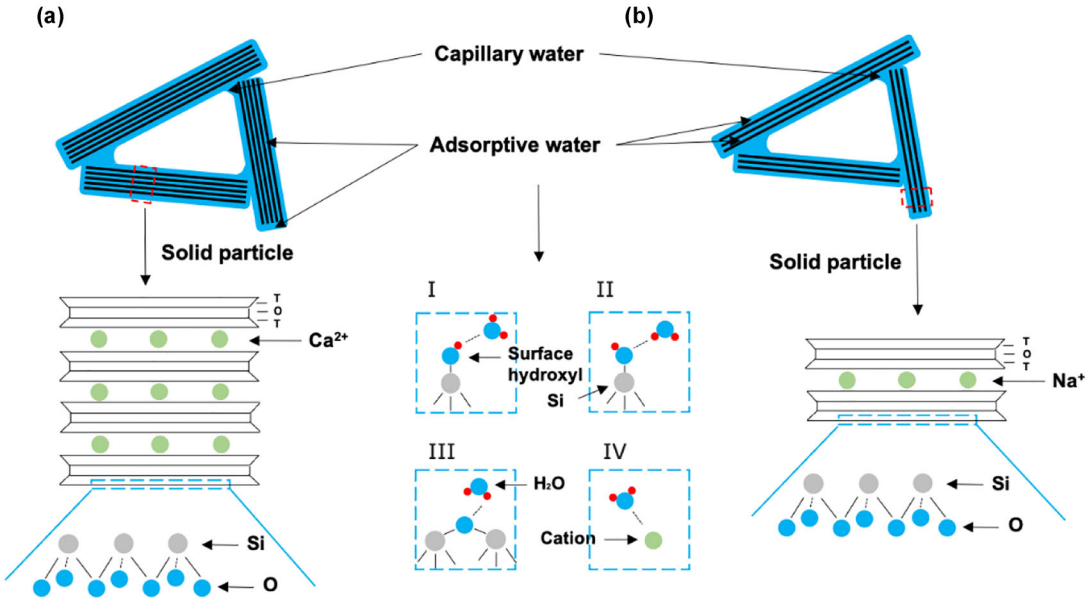

FIGURE 2 Sketch of unsaturated smectite particle system for (a) Ca^{2+}-smectite and (b) Na^{+}-smectite. Possible scenarios of the –OH group in soil samples: (I–III) the –OH group is bonded to the silica surface, (IV) the -OH group is bonded to the interlayer cation.

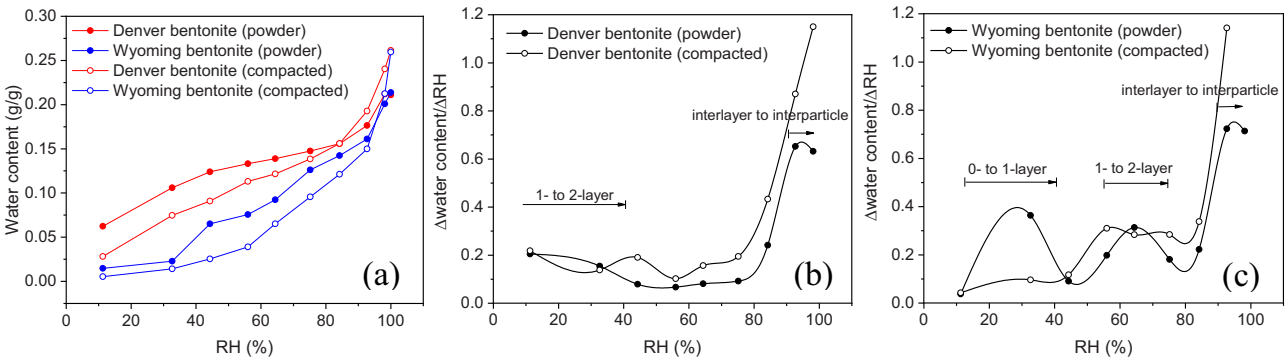

FIGURE 3 Comparative analysis of Denver and Wyoming bentonite: (a) Soil moisture sorption isotherms under varying relative humidity conditions, (b) association between relative water content/relative humidity and relative humidity (RH) for Denver bentonite, and (c) association between relative water content/relative humidity and RH for Wyoming bentonite.

peak, which is consistent with our observations. As the environmental RH rises, the primary driving force for water shifts from adsorption to capillarity, causing water to be hosted in the larger scale interparticle pore space. This phenomenon results in the substantially larger magnitude of the third peak, corroborating the conclusions drawn by Likos and Lu (2006).

For Denver bentonite (Figure 3c), the powder sample exhibits a clear peak trend at 11% RH and a peak at RH 95%. These peaks are delineated by RH <45% and RH >85%, respectively, with the first peak magnitude of ~0.21 and the other one around 0.7. Based on previous XRD studies (e.g., Chipera et al., 1997), Ca^{2+}-smectite has been shown to retain one water layer at RH values even approaching 0%, and transition to a two-layer state by 25% RH. This two-layer state remains stable up to RH values approaching 90%. Therefore, we posit that the first peak represents the water associated with the transition from a one- to a two-layer hydrated state, while the second peak indicates a transfer from the interlay space to larger scale pore space, such as interparticle pore space.

The compacted samples of Wyoming bentonite share a peak trend similar to that of the powder samples. However, the magnitude of the first peak is relatively lower in the compacted samples, which could be due to the confined state and longer preparation time at this humidity. Compacted samples for both clays show the distinct high value of Δwater content/ΔRH when RH is >85%. This observation may imply that the water will be relatively well interconnected because of compaction.

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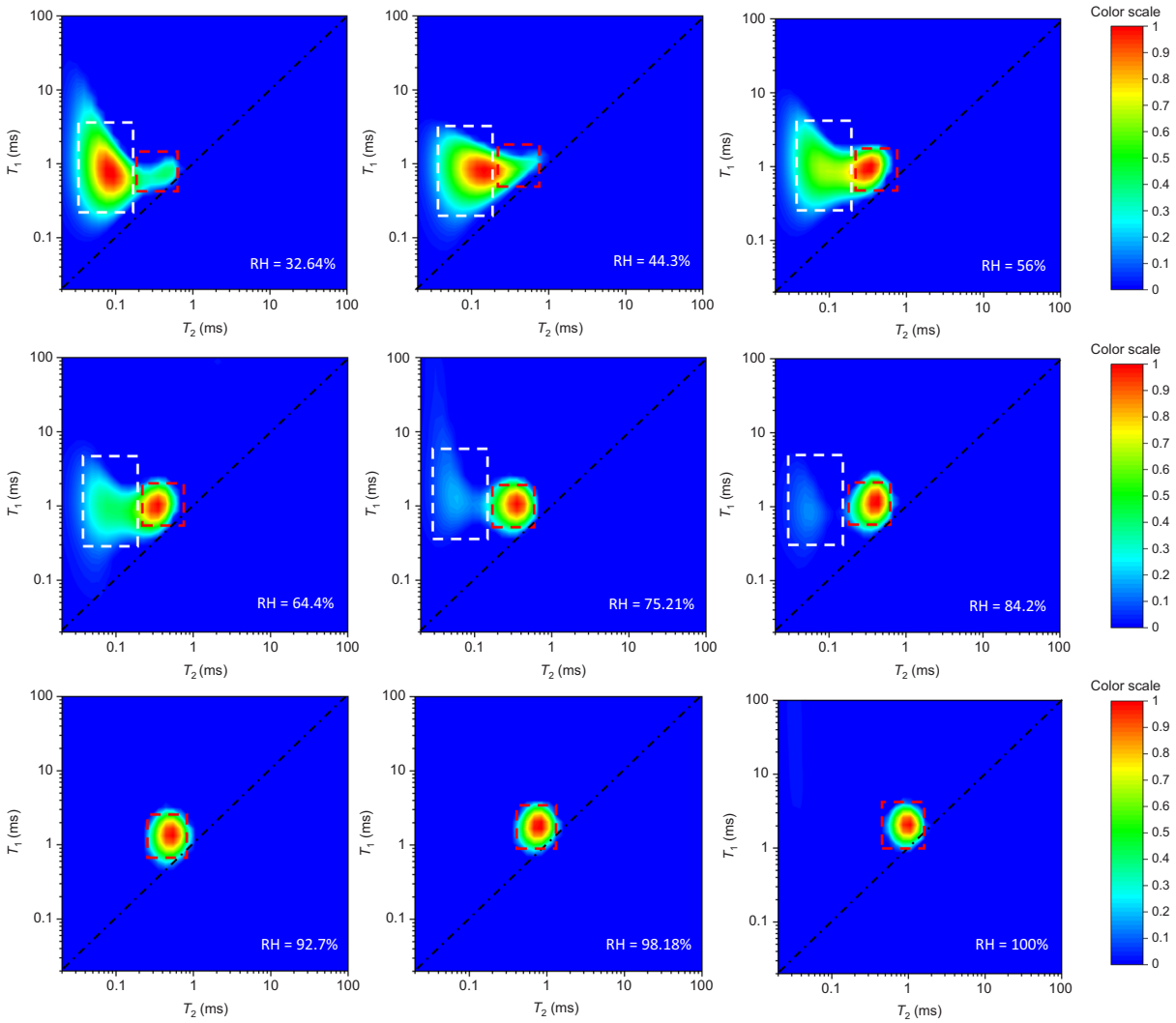

FIGURE 4 T_2 distribution of compacted unsaturated clay samples under 10 different relative humidity (RH) environments for Denver bentonite. The black dashed line corresponds to T_1–$T_2 = 1$, the white dashed area corresponds to adsorptive water, and the red dashed area corresponds to capillary water.

4.2 | T_1–T_2 distributions

T_1–T_2 distributions are useful in distinguishing protons mobility and rely on specific T_2 and T_1/T_2 ratio values: in general, the higher T_1/T_2 ratios correspond to lower protons mobility. In Figure 4 and Figure 5, we have mapped these distributions as two-dimensional T_1–T_2 maps. These maps visualize the positions of various proton populations, such as hydroxyls from the clay and water, across different RH environments. For a more focused analysis, Figures 5 and 6 show the T_1/T_2 ratios of the dominant peak against the RH, offering insights into the proton mobility and water partitioning behavior in the clay materials.

From Figures 4 and 5, two peaks are observed for Denver bentonite in the RH range of 32.64%–84.2% and for Wyoming bentonite in the range of 32.64%–75.21%. Additionally, a single dominant peak is present for all samples. To facilitate a more comprehensive analysis of the T_1–T_2 map, we have plotted a black dashed line for $T_1/T_2 = 1$, a white dashed area for $T_2 \leq 0.1$ ms, and a red dashed area for $T_2 \geq 0.1$ ms in Figures 4 and 5. We assume that the white dashed lines correspond to hydrogen-bonded water or hydrated layer water, while the red dashed area corresponds to capillary water. Moreover, we observe a dwindling peak area as RH increases, suggesting that in a high RH environment, the exchange time between adsorptive and capillary water is shorter than the relaxation time. In Figure 6, a considerable decrease in the T_1/T_2 ratio for both clays are observed, which can be attributed to the diminishing contribution of signal from surface adsorbed water. Notably, the T_1/T_2 ratio of Denver bentonite is much higher than that of Wyoming bentonite, indicating that the interaction between water molecules and the surface is stronger in Denver bentonite for each RH environment.

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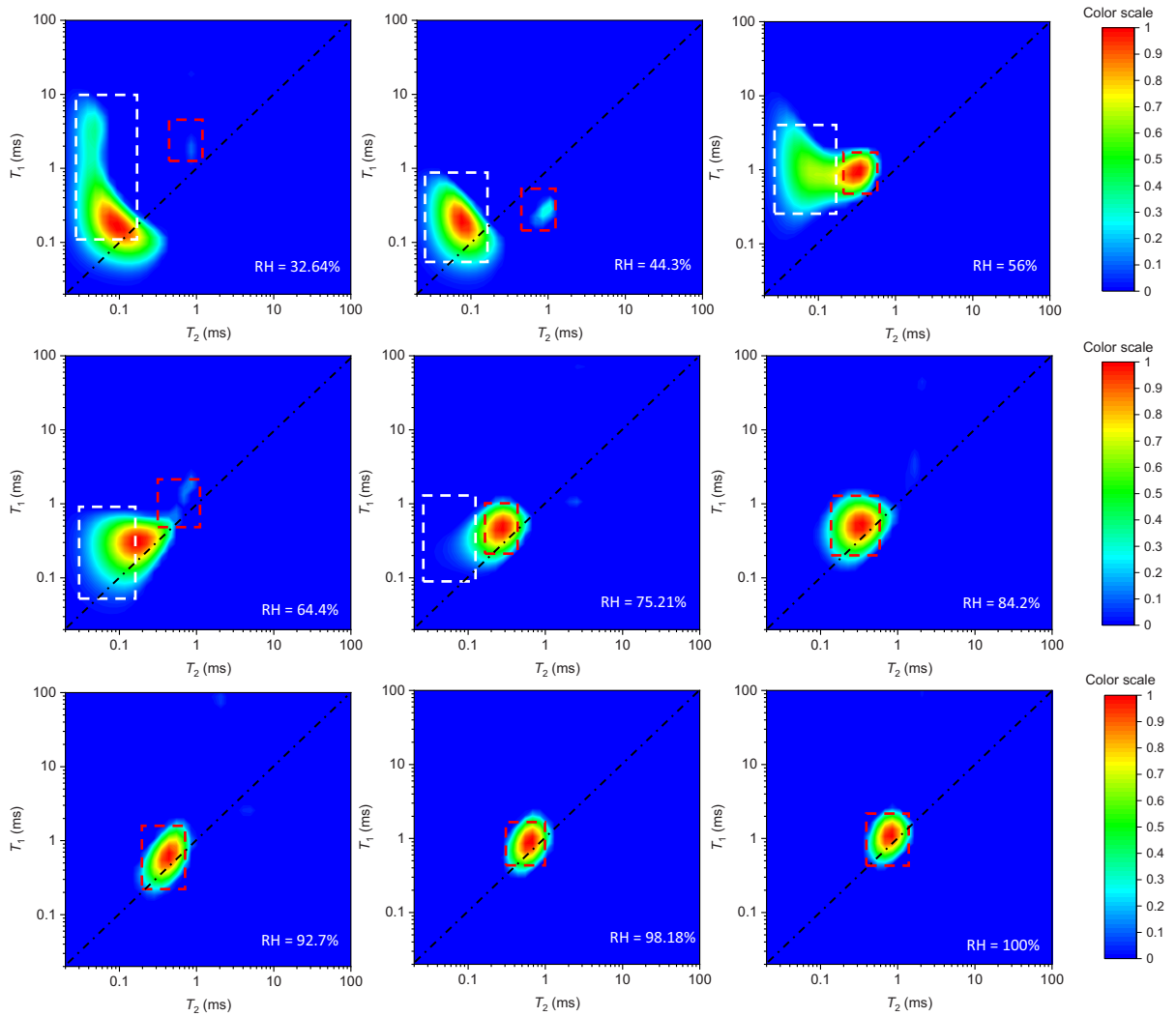

FIGURE 5 T_2 distribution of compacted unsaturated clay samples under nine different relative humidity (RH) environments for Wyoming bentonite. The black dashed line corresponds to T_1–T_2 = 1, the white dashed area corresponds to adsorptive water, and the red dashed area corresponds to capillary water.

In the case of Denver bentonite (Figures 4 and 6), a dominant peak with $T_2 < 0.1$ ms is observed at an RH of 11.29% (not shown). Based on the T_1–T_2 results for dry samples and the study of Fleury and Romero-Sarmiento (2016), the signal at $T_2 = 0.06$ ms and within the range of $0.01 < T_1 < 100$ ms is associated with hydroxyl. Considering the position of this peak, the signal that behaved as a pseudo-solid proton ($T_1/T_2 = 6$) is recognized as arising from hydrogen-bonded water or hydrated layer water, rather than the hydroxyl group, which is considered adsorptive water. Water molecules in this environment are predominantly influenced by the interaction with the solid surface and intra-molecular interactions. At an RH of 32.64%, the T_2 distribution becomes distinctly bimodal: the primary peak at shorter T_2 values ($T_2 < 0.1$ ms, $T_1/T_2 = 7.30$) corresponded to hydrogen-bonded water or hydrated layer water, while the higher T_2 peak likely corresponds to adsorptive water-filled in interparticle pore space ($T_2 > 0.1$ ms, $T_1/T_2 = 1.56$). This observation confirms that adsorptive water molecules existed in two different pore environments, namely, interlayer pore space and interparticle space, at a low RH environment. As the RH increases, the peak shifts from adsorptive water to capillary water, with the latter becoming dominant. This transition highlights the evolving pore environments and water interactions as humidity levels change.

For Wyoming bentonite (Figures 5 and 6), no signal can be obtained at an RH of 11.29% due to artifacts. At an RH of 32.64%, although the T_1/T_2 ratio equals 2, the peak occurs at $T_2 = 0.1$ ms, and T_1 is within the range of $0.1 < T_1 < 10$ ms. Consequently, this peak is recognized as hydrogen-bonded water or hydrated layer water rather than capillary water. The trend for the peak to shift with increasing RH is similar to that seen for Denver bentonite. The peak fully shifts to the

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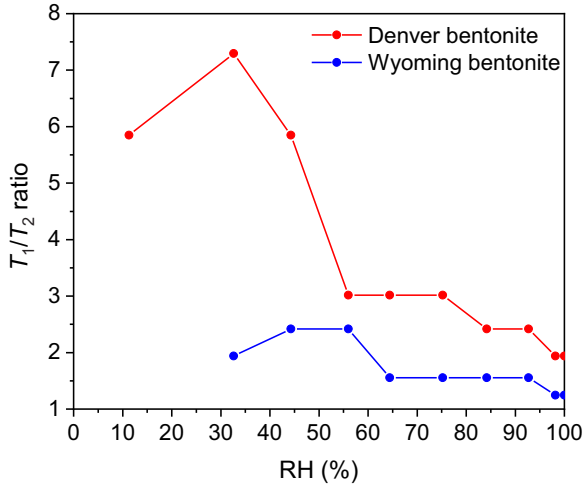

FIGURE 6 Correlation of T_1/T_2 ratio and relative humidity (RH) value for Denver bentonite and Wyoming bentonite.

red line area at RH 64.4%, indicating that the dominant interaction mechanism changed to capillarity at this RH level for Wyoming bentonite. When the RH exceeds 84.2%, the T_1/T_2 ratio is ~1.5, with T_2 = 1 ms, suggesting that the capillary water becomes dominant under these environments.

4.3 | Peak decomposition

NMR spin–spin relaxation spectra for unsaturated compacted samples of two clays under each different RH values are shown in Figure 7. The T_2 spectrum position for Denver bentonite resembles that of Wyoming bentonite, and the overall amplitude of Denver bentonite samples is obviously higher than that of Wyoming bentonite samples. This observation suggests that the water environment of Denver bentonite is similar to that of Wyoming bentonite, and the Denver bentonite can absorb more water than Wyoming bentonite under the same environment. This result implies that the water content change for Ca^{2+}-smectite specimens is greater than that for Na^{+}-smectite specimens across the entire range of RH considered, which is consistent with our previous observations. As the RH increases from 11.29% to 100%, the observed major T_2 peaks for both clays consistently shift to higher T_2 values.

For Denver bentonite, two peaks are observed when the RH value exceeds 84.2%. The left smaller peaks maintain the same position, while the right peaks shift to higher T_2 values as RH increases. At 32.64% RH, two peaks are observed, which then merge into one peak, and the peak shifts as RH increases. For Wyoming bentonite, two peaks are observed when the RH value is greater than 92.7% and when the RH value lies between 32.6% and 44.3%. In high RH environments, the left smaller peaks share the same position as

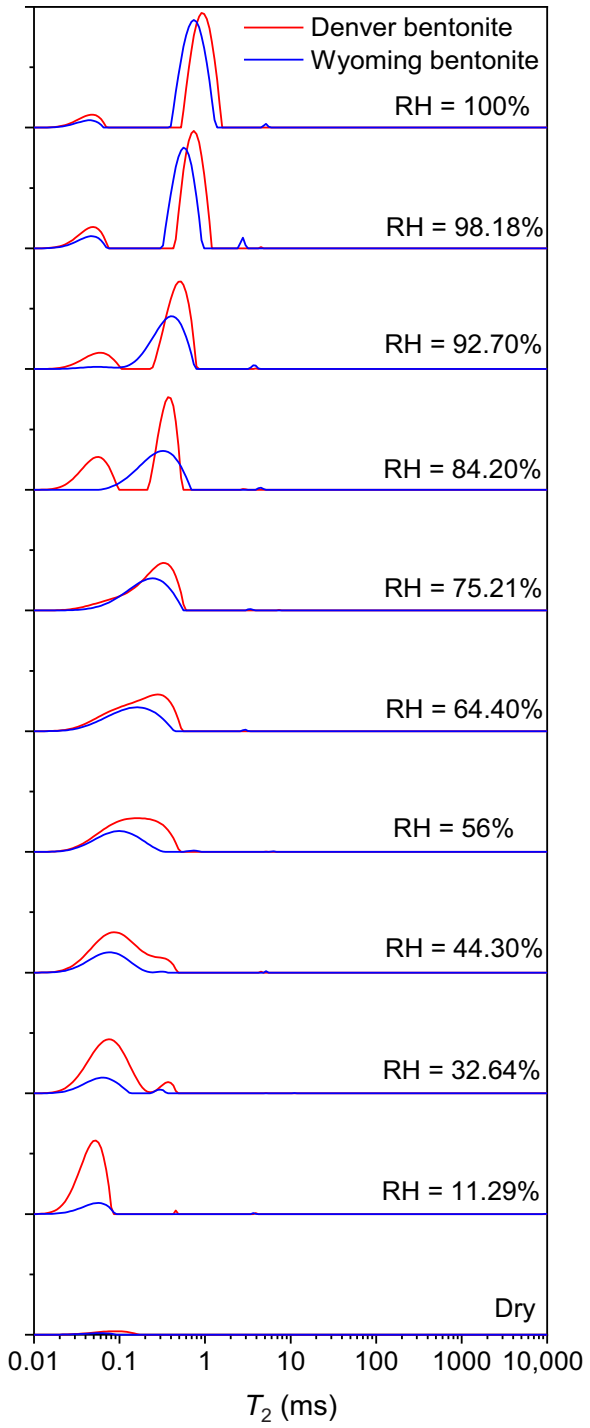

FIGURE 7 Normalized T_2 distribution of Denver bentonite (red) and Wyoming bentonite (blue). RH, relative humidity.

those of Denver bentonite but with a reduced amplitude. When RH values range from 56% to 84.2%, only one peak is observed. The NMR spectra shown above clearly demonstrate the presence of various water environments within the clays. Therefore, a single characteristic T_2 value (e.g., the logarithmic mean of the T_2 distribution, T_{2ml}; T_2 peak value) cannot

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serve as the parameter is involved in the analysis of the pore information.

Considering the clay properties, a peak decomposition method was applied to extract the water-filled pore space information from the T_2 distribution. This approach is a more quantitative, repeatable, and important method for analyzing the expansive clay under the unsaturated condition, allowing for the discernment of subtle differences at ambient temperature. By employing this method, we identified two peaks for both types of bentonite samples.

The peak decomposition results under 10 different relative humidity environments for Denver and Wyoming bentonite are displayed in Figures 8 and 9, respectively. For Denver bentonite (Figure 8), T_2 peak of peak 1 curve slightly shifted toward a higher relaxation time from 0.06 to 0.13 ms and then shifted back toward a shorter relaxation time to the same value as the lowest RH environment (0.05 ms). The trend of the T_2 distribution range for peak 1 curve is similar to its T_2 peak. T_2 peak of peak 2 curve initially shifts slightly toward a lower relaxation time from 0.32 to 0.24 ms and then shifts toward a higher relaxation time of 0.91 ms as RH increases. For Wyoming bentonite (Figure 9), T_2 peak of peak 1 curve shifts from 0.05 to 0.09 ms and then back to 0.05 ms, while the peak 2 curve shifts from 0.3 to 0.09 ms and then to 0.74 ms. Interestingly, both the Denver bentonite peak 1 curve and Wyoming bentonite peak 1 curve show a constant peak at 0.05 ms. To elucidate this peak, we measured the T_2 relaxation time of dry samples, which were dried at 105°C for over 3 months before being packed for testing, for the two clays with the same dry density (1.5 g/cm^3). The all T_2 peaks of both dry samples are at 0.05 (+0.01) ms, and the T_2 distribution lies within the range of 0.01 ms < T_2< 0.1 ms (see Figure 7). In this case, there is no water in the samples; thus, we are supposed to detect nothing but hydroxyls. Fleury et al. (2013) used a D_2O exchange experiment on kaolinite samples to determine the T_2 peak of hydroxyl at 0.06 ms and the signal within the range of 0.01–0.1 ms. This finding is consistent with our observations. To minimize the effect of the hydroxyl signal on T_2, we obtained the equivalent water from the T_2 distribution area of dry samples and will exclude it in the subsequent analysis.

The water masses of the decomposition curves for Denver bentonite and Wyoming bentonite are presented in Figure 10. For Denver bentonite, the water mass of peak 1 increases from 0.59 to 1.9 g as RH increases to 56% and then decreases to 0.62 g at RH 100%. Meanwhile, the water mass of peak 2 increases from 0 to 5.05 g with the increase in RH. For Wyoming bentonite, the water mass of peak 1 increases from 0.11 to 0.89 g as RH increases to 64.4% and then decreases to 0.35 g at 100% RH. The water mass of peak 2 increases from 0 to 5.29 g as RH increases. In light of previous studies, peak 1 is recognized as arising from hydroxyl and adsorptive water, while peak 2 can be interpreted as capillary water.

5 | DISCUSSION

In this section, we discuss and compare the water partition results from soil water retention, T_1–T_2 distributions, and peak decomposition. Subsequently, we propose a conceptual water partitioning model to elucidate the movement trajectories of water molecules under varying unsaturated conditions. The soil water retention results suggest that for Wyoming bentonite (see Figure 3b), the RH range of 10%–40% is associated with the transition of adsorptive water in interlayer spacing from a zero-layer state to a one-layer hydrate state, while the RH range of 50%–90% reflects the transition from a one-layer state to a two-layer hydrate state. For Denver bentonite (see Figure 3c), the transition to a two-layer state occurs by an RH of 45% and remains stable at the two-layer state up to an RH of 90%. The compacted samples for both clays show the distinct high value of Δwater content/ΔRH when RH is greater than 85%. This may imply that the water becomes relatively well interconnected due to compaction, and capillary water may occur at an RH greater than 85%. However, due to the limited RH data, only semi-qualitative water partition results can be obtained from the soil water retention curve.

In the T_1–T_2 maps, the positions of the signal for both clays under different RH environments (see Figures 4 and 5) suggest the various protons mobility (e.g., hydroxyls from the clay and water). As RH increases, the peak shifts from a low T_2 area to a high T_2 area, along with decreasing T_1/T_2 ratio, and indicates increasing proton mobility, which may correspond to the transition from adsorptive water to capillary water for both bentonite samples. However, although the two segments assumed to correspond to the adsorptive water and capillary water (i.e., white dashed area for $T_2 \leq 0.1$ ms, and red dashed area for $T_2 \geq 0.1$ ms) can explain water migration, quantitative results for water partitioning cannot be obtained.

In contrast, peak decomposition is a more quantitative and repeatable method for partitioning water in clays from T_2 distribution. Two peaks obtained by this method are recognized as adsorptive water and capillary water, respectively, for both bentonite samples. For Denver bentonite, the T_2 range of adsorptive water varies from 0.06 to 0.13 ms, and the water mass of adsorptive water increases from 0.59 to 1.9 g as RH rises to 56% and then decreases to 0.62 g at RH 100%. For Wyoming bentonite, the T_2 range of adsorptive water spans from 0.05 to 0.09 ms, and the water mass of adsorptive water increases from 0.11 to 0.89 g as RH reaches 64.4% and then drops to 0.35 g at RH 100%. The phenomena of first increase and then decrease in adsorptive water may imply the water adsorption sequence at a low RH environment. The water contained in the interlayer space or interparticle space does not exist independently. It is simultaneously present within the clay and transforms between forms under different environmental conditions. Moreover, the characteristic T_2 values of

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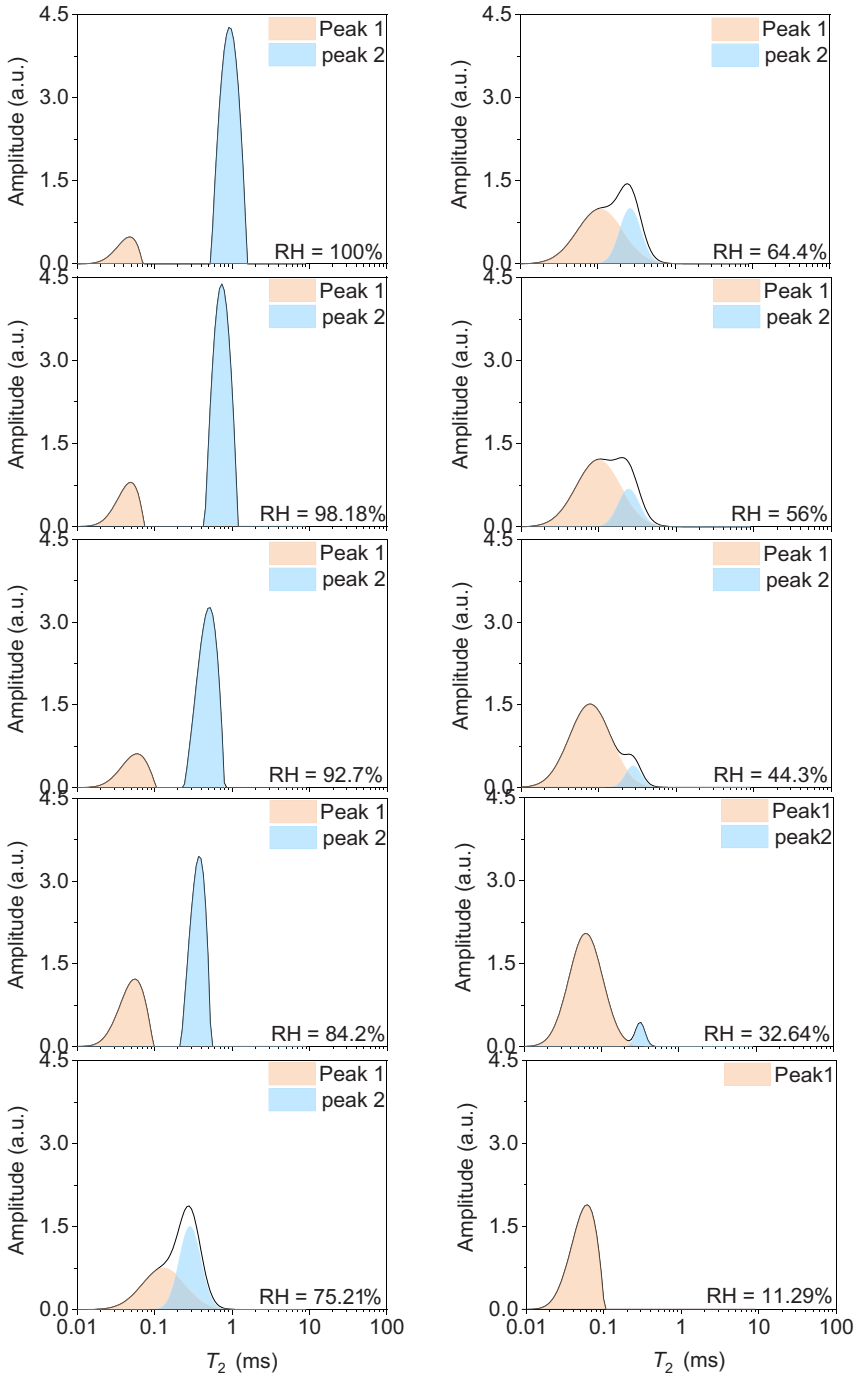

FIGURE 8 Illustration of peak decomposition results for Denver bentonite. RH, relative humidity.

capillary water are higher than 0.3 ms for both bentonite samples. Additionally, capillary water coexists with adsorptive water at low RH environments, such as 32.64% in our study.

A conceptual water partitioning model was developed to elucidate the various trajectories of water molecules under distinct unsaturated conditions, as illustrated in Figure 11. The diverse scenarios for water molecules encompass (a) hydrogen-bonded water on the surface, (b) hydrated layer water, and (c) capillary water.

In the T_2 spectrum for Denver bentonite, a T_2 peak value of 0.06 ms was observed for the specimen at an RH of 11.29%, which is hypothesized to correspond to hydroxyl groups. The significantly higher T_2 signal compared to hydroxyl implies the presence of hydrogen-bonded water. Taking into account previous XRD results (Chipera et al., 1997; Likos & Lu, 2006), we propose that one-hydrated layer water coexists with hydrogen-bonded water, as illustrated in Figure 11b. As the RH increases, most of the mineral layers become "saturated"

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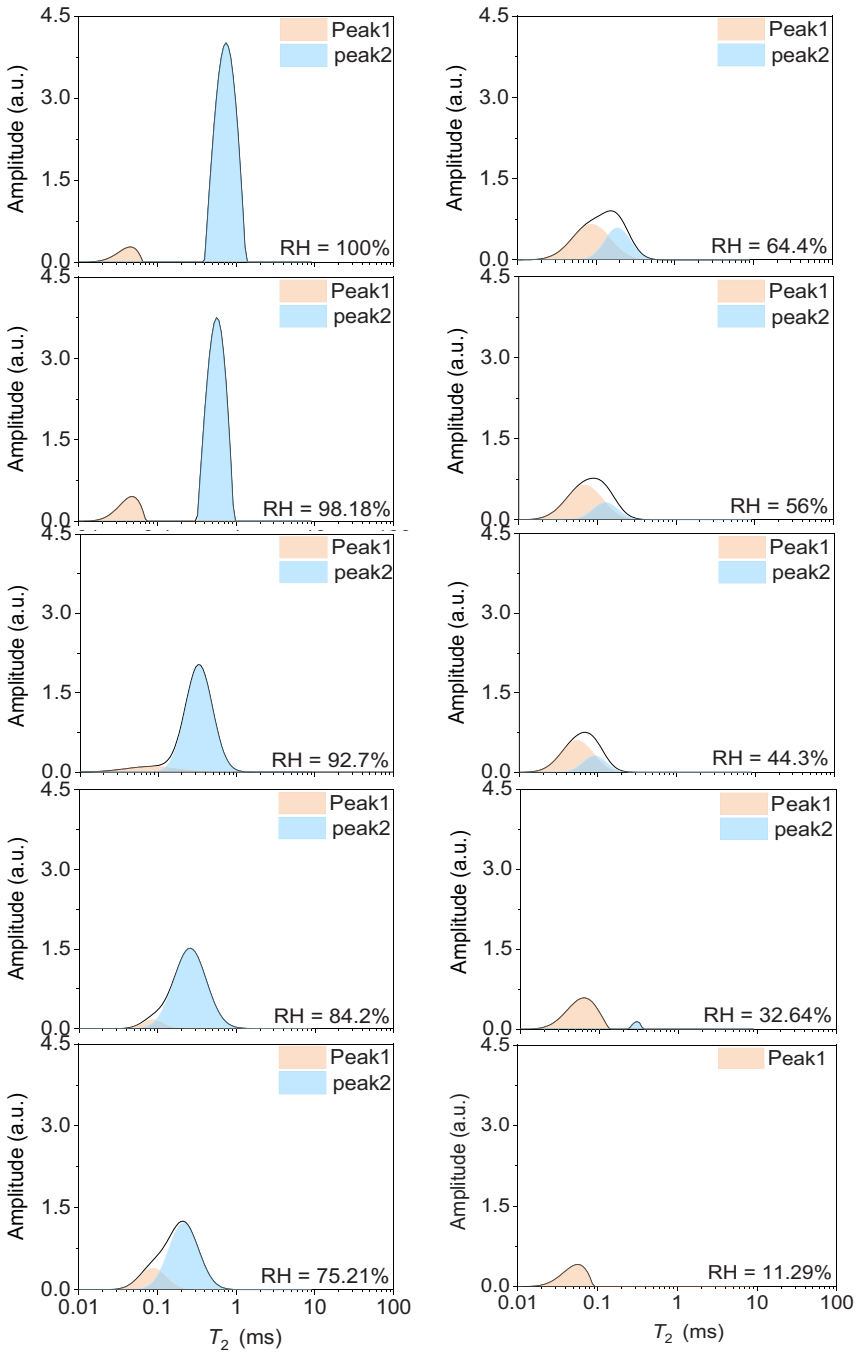

FIGURE 9 Illustration of peak decomposition results for Wyoming bentonite. RH, relative humidity.

with one-hydrated layer water, resulting in swelling behavior, an increase in the water mass of peak 1, and a slight T_2 peak shift. Subsequently, water molecules are gradually adsorbed as two-hydrated layer water. Simultaneously, some water molecules remain on the surface of the interparticle pore space, leading to an increase in the water mass of peak 2. At an RH of 56%, the maximum interlayer water content indicates that two-hydrated layer water predominantly occupies the mineral layers, as shown in Figure 11c. When RH exceeds 56%, the water mass of peak 1 decreases, suggesting that capillary water gradually becomes sufficient to exchange with the multiple hydrated layer water in the interlayer space (Figure 11d). When the water mass of peak 2 equals the amount of two-hydrated layer water, the two pore spaces appear to merge into a larger one, resulting in a higher T_2 peak. As RH continues to increase, a greater proportion of mineral layer two-hydrated layer water equates to the surrounding capillary water; at an RH of 75.21%, the water masses of peaks 1 and 2 are relatively equal. Due to the higher T_2 peak values, two-hydrated layer water is characterized as peak 2 from

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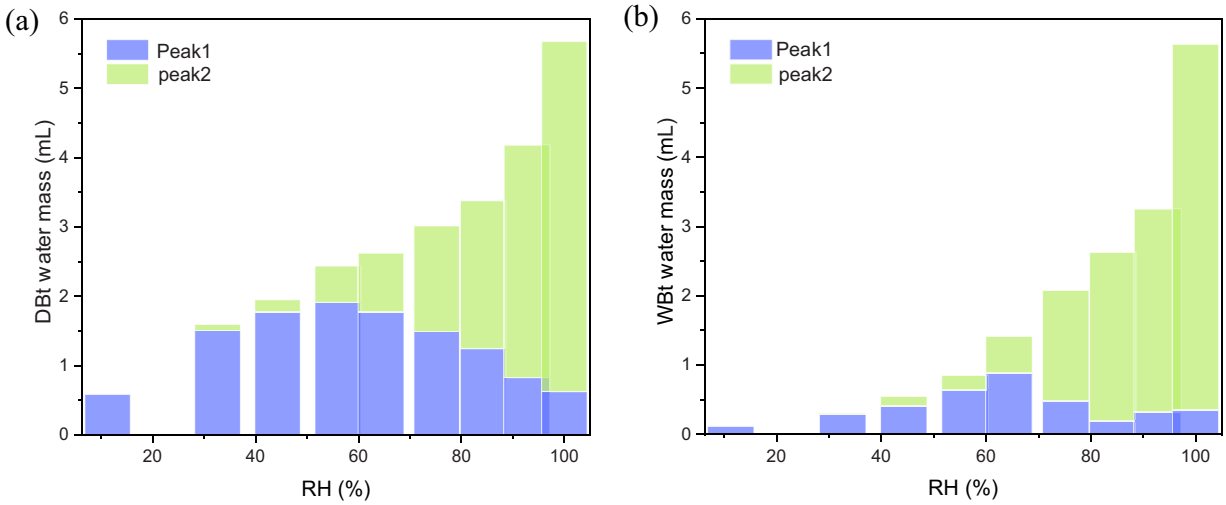

FIGURE 10 Two types of water masses determined by the peak decomposition method. DBt, Denver bentonite; RH, relative humidity; WBt, Wyoming bentonite.

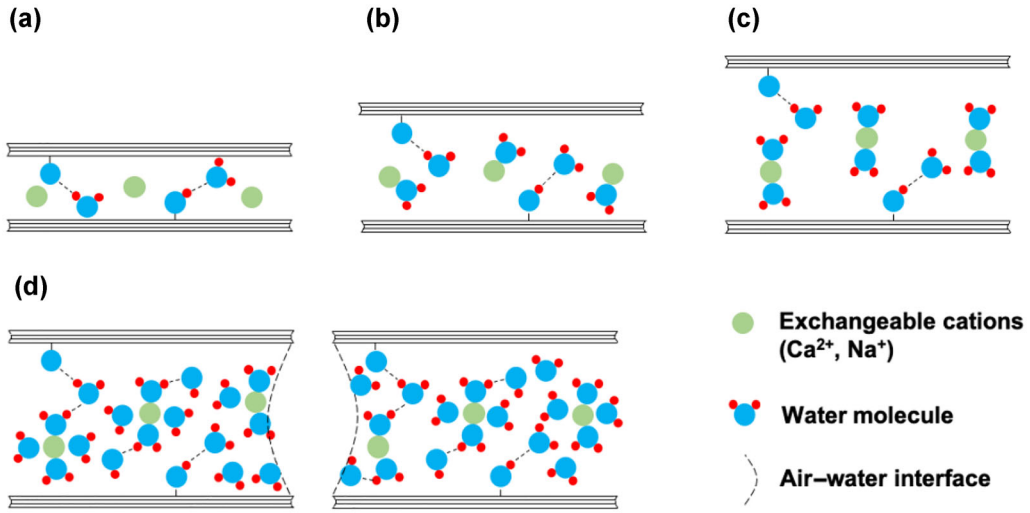

FIGURE 11 A water partitioning conceptual model for unsaturated bentonite clays. (a) Dry state, water molecules mainly adsorbed by the hydroxyl. (b) One-hydrated layer dominant, water molecules mainly adsorbed by exchange cations. (c) Two-hydrated layer dominant, water molecules mainly adsorbed by exchange cations. (d) Two-hydrated water exchange with capillary water, water molecules gradually mainly exist in the interparticle.

an RH of 56%. It should be noted that the decrease in peak 1 water mass does not imply the migration of adsorptive water to the interparticle space. When the water mass of peak 2 is greater than that of peak 1, two T_2 peaks are observed in the original T_2 spectrum when RH is at or above 84.2%, with T_2 peaks shifting toward longer relaxation times. Interestingly, the water mass of peak 1 at 100% RH is relatively similar to that at the lowest RH of the experiment, and both peaks share the same T_2 peak values (~0.05 ms), indicating that the water molecules of peak 1 at 100% RH are in the same state as those at 11.29% RH. This observation also confirms the notion that Ca^{2+}-smectite retains a water layer at low RH environments (Chipera et al., 1997; Likos & Lu, 2006). From the above analysis, we conclude that the T_2 relaxation time for adsorptive water in Denver bentonite ranges from 0.05 to 0.13 ms, while for capillary water, it is greater than or equal to 0.32 ms. It is worth noting that another factor contributing to the T_2 peak shifts of peaks 1 and 2, apart from the merging of two-hydrated layer water with capillary water, maybe the coupling behavior of adsorptive and capillary water (e.g., Qi et al., 1996).

In the case of Wyoming bentonite, a 0.05-ms T_2 peak value for the specimen was observed at an RH of 11.29%. By combining the previously reported XRD results (Chipera et al., 1997; Likos & Lu, 2006) with the relatively low water mass observed, we infer that water molecules primarily

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exist as hydrogen-bonded water, as depicted in Figure 11a. Between RH levels of 20% and 40%, an increasing amount of water is adsorbed as one-hydrated layer water, as illustrated in Figure 9b. Simultaneously, some water molecules are retained on the surface of the interparticle space, resulting in an increase in the water mass of peak 2 starting at an RH of 32.64%. From RH levels of 60%–75%, water is adsorbed as two-hydrated layers (Figure 11c), leading to a sustained increase in interlayer water mass. At an RH of 64.4%, the maximum adsorptive water content indicates that two-hydrated layer water dominates the mineral layers, as shown in Figure 11c. The capillary water content continuously increases.

A comparison of the adsorptive water and capillary water contents of the two bentonites when the interlayer displays the maximum water mass reveals interesting insights. The water mass of peak 2 for Denver bentonite is 0.52 g, while that of Wyoming bentonite is 0.53 g. The water mass of peak 1 is 1.91 g for Denver bentonite and 0.89 g for Wyoming bentonite. This suggests that the primary difference in water content between the two bentonites is due to the adsorption forces in the interlayer exchangeable cations, that is, the adsorption force of Ca^{2+} is greater than that of Na^{+}. These conclusions align with findings from clay science studies (e.g., Kittrick, 1969; MacEwan & Wilson, 1980). As the RH increases, the water mass of peak 1 decreases for Wyoming bentonite, similar to the situation observed for Denver bentonite. However, at an RH of 84.2%, the water mass of peak 1 decreases significantly and then slightly increases as the RH rises. The differing behavior of peak 1 in Wyoming bentonite when compared to Denver bentonite at RH levels above 84.2% is primarily due to variations in suction pressure within the interlayer space caused by the exchangeable cations (e.g., Kittrick, 1969; MacEwan & Wilson, 1980). Moreover, the water mass of peak 1 for Denver bentonite is consistently higher than that of Wyoming bentonite across all RH levels. This difference can be attributed to both variations in the suction pressure within the interlayer space and the number of mineral layers. Generally, Ca^{2+}-smectite has a larger particle size than Na^{+}-smectite, and the number of layers increases with the increasing suction pressure, reaching up to 400 face-to-face oriented layers for Ca^{2+}-smectite and ~20 layers for Na^{+}-smectite at 1000 bar (Likos & Lu, 2006; Tessier, 1990). From this analysis, we conclude that the T_2 relaxation time for adsorptive water in Wyoming bentonite ranges from 0.05 to 0.09 ms, while that for capillary water is equal to or greater than 0.3 ms.

Upon comparing the peak decomposition results of Denver bentonite and Wyoming bentonite, we can draw the following conclusions regarding the T_2 peaks of adsorptive water: for Ca^{2+}-smectite, the range is between 0.05 and 0.13 ms, whereas for Na^{+}-smectite, it lies between 0.05 and 0.09 ms. The T_2 peak for capillary water is higher than approximately 0.3 ms for both types of bentonites. Consequently, the state of water under ambient environmental conditions, even in unsaturated scenarios, can be effectively and intuitively discerned.

6 | CONCLUSIONS

This study investigates the primary water state inherent in two types of expansive clays, Denver bentonite (Ca^{2+}-smectites) and Wyoming bentonite (Na^{+}-smectites), by employing soil–water retention and NMR relaxation measurements at ambient temperature under various relative humidity environments. Peak decomposition methods were utilized to quantify the changes in adsorptive water and capillary water. The significant difference in capillary water values and the similarity in adsorptive water values under the same condition suggest that the primary distinction between these two bentonites stems from the differing adsorption force of interlayer exchangeable cations, with the adsorption force of Ca^{2+} being greater than that of Na^{+}. The developed conceptual water partitioning model provides a comprehensive understanding of the behavior and distribution of water molecules in unsaturated environments, thereby facilitating a more in-depth analysis of the underlying mechanisms and interactions that govern their movement and partitioning.

This study demonstrates the considerable potential of using low-field NMR measurement in unsaturated compacted expansive clays to directly evaluate the relative proportion and range of the T_2 relaxation time of adsorptive water and capillary water, as well as to understand water transition processes. In future research, the impact of pore coupling in clays will be explored to enhance the interpretation of NMR data and to advance the application of NMR methods in unsaturated expansive clays.

AUTHOR CONTRIBUTIONS

Ling Peng: Data curation; formal analysis; investigation; methodology; project administration; visualization; writing—original draft; writing—review and editing. **Fan Zhang**: Software; visualization. **Yi Dong**: Conceptualization; resources. **Chi Zhang**: Conceptualization; funding acquisition; investigation; resources; supervision; writing—review and editing.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

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How to cite this article: Peng, L., Zhang, F., Dong, Yi., & Zhang, C. (2023). Water partitioning and migration in unsaturated bentonites by low-field NMR characterization. *Vadose Zone Journal*, *22,* e20284. https://doi.org/10.1002/vzj2.20284

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Unveiling coastal dynamics using telecommunication fiber optic cables

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Science Bulletin (2025); DOI: 10.1016/j.scib.2025.02.023

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Short Communication

Unveiling coastal dynamics using telecommunication fiber optic cables

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ARTICLE INFO

Article history:
Received 22 September 2024
Received in revised form 17 January 2025
Accepted 22 January 2025
Available online 17 February 2025

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Rising sea levels and climate change are reshaping coastal landscapes, exacerbating erosion, and increasing flood frequency, posing serious threats to the ecological balance, economic growth, and the livelihoods of coastal city residents [1]. This necessitates advanced monitoring to mitigate climate impacts and strengthen coastal resilience. While geophysical methods for first-order coastal properties monitoring have gained traction in academia [2,3], traditional monitoring techniques are limited by sparse and unevenly distributed observation stations, failing to meet the needs for detailed regional monitoring. Moreover, the complex social and cultural environments of coastal cities challenge sustained and intensive monitoring efforts. Therefore, developing new generations of coastal dynamics monitoring methods suitable for complex urban environments is urgently needed.

Distributed Acoustic Sensing (DAS) is a sophisticated technology that leverages Rayleigh backscatter in fiber optics to measure local properties such as vibrations. Recent advancements in fiber optic communications and network expansion have significantly propelled the growth and application of DAS [4–7]. This technology enables continuous, distributed detection of acoustic vibrations, transforming urban telecommunication cables into extensive seismic sensor arrays. Researchers have recently applied urban telecommunication cables in various studies, such as imaging shallow subsurface structures, conducting geothermal investigations, monitoring seismic activities, and tracking traffic [8,9]. These applications demonstrate the effectiveness of urban telecommunication cables for both short-term geophysical observations and long-term monitoring. Despite its growing application, there has yet to be any reported use of urban telecommunication cables for coastal dynamic monitoring.

The Qiantang River, connecting the inland to the East China Sea, offers a unique perspective for studying the impact of tides on coastal geological structures due to its distinct tidal bore phenomenon, making the area a natural laboratory for coastal dynamics research. To explore the feasibility of using urban telecommunication cables for long-term monitoring of nearshore geological structures and coastal dynamics, we partnered with Hangzhou Junyun Technology Co., Ltd. to conduct a fiber optic sensing experiment along the Qiantang River. This study reports the preliminary findings and discusses the prospects of urban telecommunication cables for coastal dynamics monitoring and disaster risk assessment.

We utilized a 62 km stretch of urban telecommunication cable located along Zhijiang Road by the Qiantang River in Hangzhou for our experiment. This included a 27 km southern line and a 35 km northern line (Fig. 1a). The cable, housed within an underground pipeline (Fig. 1b), contained 128 standard single-mode fiber cores; only one spare fiber core (Fig. 1c) was connected to interrogator unit (Fig. 1d) for monitoring. The experiment was conducted independently on both lines with monitoring beginning on the southern line on May 8, 2023, and continuing to present, and on the northern line from September 17 to October 22, 2023. To maintain consistency throughout the monitoring process, the observation parameters for both lines were kept constant, with a time sampling interval of 2 ms and a spatial sampling interval of 4 m. Using DAS technology, we demodulated the 62 km cable into 15,500 seismic sensing channels, generating approximately 2.4 TB of observational data daily. This configuration achieves meter-scale spatial resolution across large-scale coverage, significantly enhancing our capability to effectively monitor extensive coastal regions.

To assess the data reliability from telecommunication cables, a nodal seismometer array was installed alongside the cable near Fuxing Bridge on Zhijiang Road. In comparing urban traffic noise

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https://doi.org/10.1016/j.scib.2025.02.023
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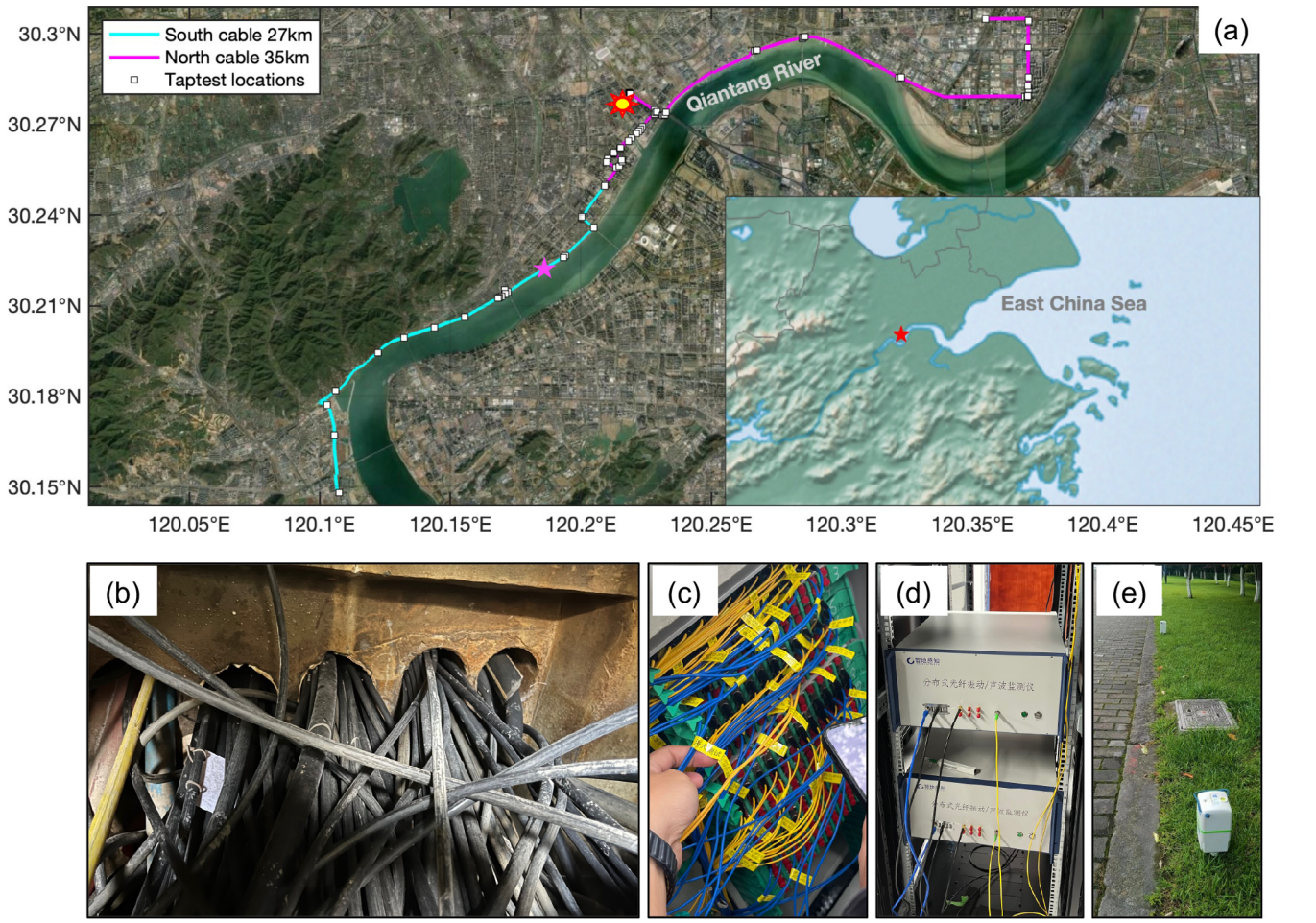

Fig. 1. Geometry map of the telecommunication cable along the Qiantang River. (a) Two telecommunication cables along Zhijiang Road by the Qiantang River, with the southern line (black) approximately 27 km long and the northern line (blue) about 35 km long. The red and yellow stars indicate the locations of the DAS demodulators, the green squares represent points along the line where tapping tests were conducted for geometry calibration, and the magenta pentagrams indicate co-located observation points of DAS and three-component node seismometers. (b) Coupling of the telecommunication cable within the urban pipeline channel. (c) Fiber optic patch cords in the telecommunication cable junction box. (d) DAS interrogator unit from Hefei Zhidi Sensing Technology Co., Ltd. (e) Smartsolo IGU-16HR 3C nodal seismometers positioned along the pipeline.

data collected simultaneously by both systems (Fig. S1 online), we observed that the signal quality from the telecommunication cable was slightly lower than that of the vertical component of nodal seismometers, particularly for high-frequency vehicle signals; however, it exceeded the quality of the horizontal component of the nodal seismometer, which usually suffered from issues related to installation and calibration. Furthermore, we utilize ambient noise interferometry technology to transform urban background noise into coherent seismic signals, which can provide detailed insights into the Earth's subsurface structures [10]. The quality of seismic signals extracted from the telecommunication cable mirrors the pattern observed in the collected urban traffic noise data.

The extract seismic signals can be utilized for rapid seismic velocity imaging, which is crucial for high-resolution coastal dynamics monitoring. We employed Multichannel Analysis of Surface Waves (MASW) method to ensure the reliability of our velocity imaging [11]. Our results revealed that one hour of telecommunication cable recordings could extract high-quality seismic surface wave signals, displaying clear and continuous dispersion energy from 1.5 to 15 Hz (Fig. S2 online).

By analyzing continuous and stable observations from telecommunication cables, we are able to determine the hourly seismic velocity changes beneath the cables. Due to the dispersive nature of surface waves, velocities at various frequencies can reveal subsurface structures at different depths [12]. Using 5 Hz seismic surface waves as an example, these seismic waves predominantly reflect the stability of geological structures approximately 15 m deep according to the half-wavelength rule [13]. During our initial experiment period, we observed distinct velocity changes in the surface wave dispersion energy spectrum (Fig. 2a) and captured the relative velocity changes of the 5 Hz surface waves (Fig. 2b). Incorporating public tidal data from the Qiantang River Zakou station, our results revealed a distinct anti-correlation between tidal variations and the subsurface (~15 m) seismic velocity changes along the Qiantang River. Similar periodic tidal shifts were also noted in subsequent telecommunication cable recordings (Fig. S3 online), corroborated by dominant periodic peaks associated with semidiurnal (M2), fortnightly (Mf), and diurnal (K1) tides (Fig. 2c). Comparisons of seismic velocity changes with environmental parameters such as air temperature, air pressure, and net solar radiation revealed no significant correlations (Fig. S4 online). This suggests that the near-shore fiber-optic cable effectively captures subsurface seismic responses related to tidal dynamics.

From May 8th, decreasing tidal levels led to reduced pore pressure and soil saturation along the riverbank, causing seismic velocities to rise by up to +5%. After May 17th, as tidal levels rose, seismic velocities sharply fell within 30 h, with the largest drop reaching –4.5%. This pattern is attributed to the combined effects

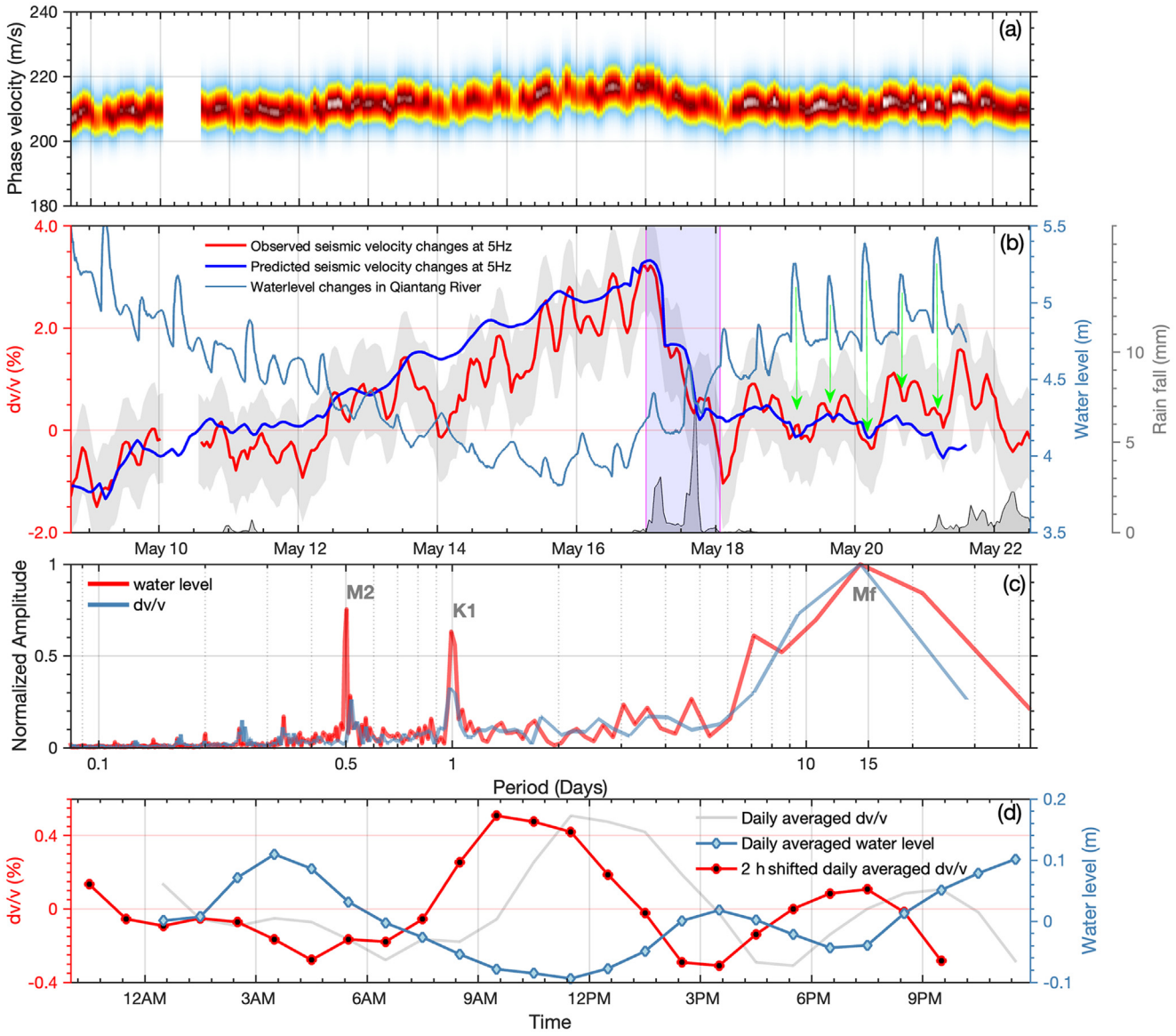

Fig. 2. Dynamic changes in subsurface seismic velocity monitored by telecommunication cable. (a) Hourly changes in the dispersion spectrum of seismic surface waves around 5 Hz recorded by the telecommunication cable from 8 to 22, May 2023. (b) Correlation of seismic velocity changes with tidal levels and rainfall events. The red curve represents the relative changes in seismic velocity obtained by extracting peak energy values from the dispersion spectrum and subtracting the median velocity within the observation period; the background gray patch indicates the picking error; the steel-blue curve shows water level recorded at the Qiantang River Zakou station; the grey histogram indicates daily rainfall changes near the Qiantang River; the blue curve depicts predicted seismic velocity changes based on rock physics modeling, which involves modeling pore pressure perturbations from observed hydraulic data (water level, precipitation, net radiation flux) using the SEEP/W module of GeoStudio software, converting these perturbations into effective stress states, and then translating these into seismic velocity variations through stress-velocity relationships. The rock physics model incorporates the geometrical configuration of the Qiantang riverbank, with flow dynamics characterized by the Richards equation and evaporation effects described using the Penman-Wilson equation. (c) Fast Fourier Transform (FFT) spectral analysis of seismic velocity changes (the steel-blue line) and tidal levels (the red line) with period peaks at approximately 0.5, 1, and 14 d, likely corresponding to periodic tidal components: M2, the principal lunar semi-diurnal component, K1, influenced by the sun's gravity, and Mf, the lunar fortnightly tide associated with lunar declination, respectively. (d) Daily averaged changes of seismic velocity (the gray line) and water level (the blue line) in (a) after the corresponding linear trends removed; the red dotted line represents the 2-hour-shifted daily averaged seismic velocity changes.

of rising tides, which increase pore pressure and soil saturation, and a significant rainfall event in Hangzhou on the 17th (depicted in the grey histogram in Fig. 2b). Preliminary rock physics modelling using GeoStudio software [14] quantifies these dynamics by linking changes in tidal levels and rainfall with variations in pore pressure and soil saturation (the blue line in Fig. 2b). This model demonstrates how increased pore pressure during high tides and heavy rainfall leads to decreased effective stress and subsequently lower seismic velocities, suggesting that short-term seismic velocity fluctuations are driven by both tidal changes and weather events. This interplay markedly accelerates velocity decreases and commonly triggers coastal disasters in Eastern China, particularly those caused by typhoons and heavy rains [15].

Following May 19th, as tidal fluctuations stabilized, seismic velocity changes showed a distinct anti-correlation, with troughs in velocity changes lagging about a few hours behind tidal peaks (highlighted by the green arrows in Fig. 2b). This lag is consistently visible in daily observations throughout the monitoring period and particularly marked with a 2-hour delay when averaging the two-week daily changes of the seismic response and the water level (Fig. 2d). This delay likely indicates the hydraulic diffusion rate in the medium, providing valuable insights into the hydrogeologi-

cal processes of fluid movement and pressure transmission in porous media.

The preliminary results demonstrate that using telecommunication cables equipped with DAS technology effectively monitors coastal dynamics. By utilizing telecommunication cables, we successfully monitored changes in seismic velocity in shallow subsurface structures along the banks of the Qiantang River, shedding light on the impacts of tidal changes and weather events. This provides new insights into the mechanisms of river erosion and weather-induced disasters, aiding in the assessment of coastal resilience in the context of climate change and coastal hazards. DAS showcases its potential by transforming extensive cable lengths into seismic sensor arrays, achieving meter-scale spatial resolution across tens to hundreds of kilometers and enabling hourly tracking of coastal dynamics for cost-effective long-term monitoring. Furthermore, the high-resolution capabilities of this technology show great promise for urban geological safety monitoring, particularly in tracking landslide and levee break hazards, which have recently become pressing concerns in southern China. The reliability of using ambient urban noise as continuous seismic sources, as opposed to less consistent man-made sources, establishes DAS as a dependable tool for early warning systems and disaster prevention. Adapting this method to different settings necessitates careful consideration of local geological and environmental conditions. In regions like mountainous areas, particularly for cryosphere studies, it is essential to account for variables such as air temperature and pressure. Furthermore, specific installation parameters for fiber cables, such as those found in urban utility tunnels where cables are suspended on stands, must be thoroughly evaluated to ensure effective deployment.

However, the distributed nature of DAS poses challenges in managing large data volumes and necessitates high accuracy in rapid hazard response. Future efforts could integrate artificial intelligence and cloud computing to analyze huge DAS data in conjunction with hydrological, meteorological, and geotechnical information. With the development of smart cities and the progression of urban digital transformation, the application scope of telecommunication cables is expected to expand across entire coastal urban areas, enhancing the scope for comprehensive coastal monitoring with advanced fiber optic sensing technology.

Conflict of interest

The authors declare that they have no conflict of interest.

Acknowledgments

This work was supported by scientific grants from Zhejiang University (2023QZJH07). We thank Hangzhou Junyun Technology Co., Ltd. for providing the optical cables and their longstanding support. We are also grateful to Hefei Zhidi Sensing Technology Co., Ltd. for supplying the demodulation equipment and their continued support. We thank the Zhejiang Hydrological Monitoring Center for providing hydrological data. We appreciate the support and assistance during the experimental process from Associate Professor Chaoqiang Xi from Anhui University of Science and Technology, Dr. Benxin Chi from Innovation Academy for Precision Measurement Science and Technology, Chinese Academy of Sciences, as well as Nanjing Jianqiao Energy Technology Co., Ltd. and Wuhan Optical Valley Interlink Technology Co., Ltd.

Author contributions

Feng Cheng, Jianghai Xia, and Chi Zhang contributed to experiments design, data analysis and manuscript writing. Feng Cheng and Baoshan Wang conducted the experiments. Feng Cheng, Jianghai Xia, Baoshan Wang, and Chi Zhang contributed to writing reviewing and editing.

Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.scib.2025.02.023.

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文献 9/9

Thaw-driven permeability evolution of frozen soil with segregated ice: insights from low-field nuclear magnetic resonance and LBM simulations

Zheng Wang; Zongwei Gan; Yaning Zhang; Bingxi Li; Zixuan Huo; Chi Zhang

Acta Geotechnica (2026); DOI: 10.1007/s11440-026-02925-7

证据用途：其他合作论文背景；张弛老师角色：末位合作者；证据等级：C

Acta Geotechnica
https://doi.org/10.1007/s11440-026-02925-7

RESEARCH PAPER

Thaw-driven permeability evolution of frozen soil with segregated ice: insights from low-field nuclear magnetic resonance and LBM simulations

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Received: 20 May 2025 / Accepted: 24 December 2025
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Abstract
In this work, we propose a numerical thawing model, based on the lattice Boltzmann method (LBM), to predict the permeability evolution of the thawing frozen soil with segregated ice. Our numerical model considers the effects of soil pore structures, pore-scale water melting temperature phase change and water migration. We also measured the low-field nuclear magnetic resonance (NMR) responses of the thawing process within different porous media and developed a quantitative relationship between the permeability coefficient and melting temperature considering the heterogeneous feature, adhesive force and temperature-driven force. Our numerical model is first validated at the macroscopic scale using a freezing-thawing experimental setup for temperature distribution and at the microscopic scale using NMR measurements for permeability characterization. Numerical results indicate that the thawing-induced permeability changes in unsaturated frozen soil are closely related to its upper surface temperature and porosity. We find that the greatest liquid mass fraction change at the segregated ice layer is the primary driver of thaw-induced soil settlement. This work will not only bridge the knowledge gap between mesoscopic features and macroscopic hydrological properties of unsaturated frozen soil, but also be beneficial for the large-scale hydrological changes towards climate warming.

Keywords Lattice Boltzmann method · Frozen soil · Permeability · Segregated ice · Thawing process

1 Introduction

Seasonally frozen soils are widely distributed in high-latitude and mountainous regions, where their thawing processes exert profound impacts on ecosystems, infrastructure, and sustainable engineering practices [8, 16]. In response to growing environmental concerns, sustainable geomaterials including fly ash-based composites have been increasingly adopted in roadbed construction across these cold regions [1, 17]. With the intensification of climate change, particularly global warming, not only is the degradation of permafrost accelerated, but the spatial extent of seasonally frozen soils is also expanding. These near-surface changes can trigger ground deformation, alter groundwater dynamics, and heighten geohazard risks, which in turn challenge the long-term stability and performance of geomaterial-based infrastructure [31, 44]. During the thawing process, the melting of segregated ice within soil microstructures induces substantial changes in permeability and pore water pressure, disrupting the natural hydrogeological balance and affecting subsurface transport dynamics, such as flow through buried pipelines [34, 47, 48]. In cryospheric regions, repeated freeze-thaw cycles further trigger mountainous geohazards including rockslides and thaw settlement, which can compromise infrastructure stability and lead to costly maintenance or failure events [46, 50, 51]. Therefore, a better understanding of the permeability characteristics of frozen soil with segregated ice plays a significant role in building a thawing numerical model for predicting frozen soil settlement within sustainable roadbed area.

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Published online: 24 February 2026

In recent decades, experimental and numerical studies have investigated the permeability characteristics of frozen soils containing segregated ice. For laboratory experiments, various technologies have been developed and employed to obtain dynamic permeability measurements in frozen soils during thaw [6, 35, 39, 47]. For example, Watanabe and Osada [43] measured the soil freezing and thawing curves and hydraulic conductivity of an arable soil under different freeze-thaw conditions ranging from − 0.5 °C to 0 °C. Laboratory data indicate that when water flows through frozen soil, it exhibits a higher unfrozen water content than under static conditions. They also observed that, as temperature rose from − 0.5 °C to − 0.2 °C, hydraulic conductivity increased by more than four orders of magnitude in parallel with the increase in unfrozen water content. Li et al. [27] focused on the meltwater infiltration curves and the complex hydrological phenomena in seasonal frozen soil area by field-scale measurements and theoretical analysis. They found that the thicker snow cover during the thaw period leads to a smaller proportion of meltwater infiltrating the soil in the seasonally frozen farmland. Dai et al. [10] used random forests analysis to examine seasonal deep-soil water infiltration on the Qinghai-Tibet Plateau, finding that deep infiltration virtually ceased once the lower soil layer froze to about 29 cm and then surged dramatically during thaw. Their results also show that rainfall intensity had little impact, where most precipitation was lost to evapotranspiration, and that deep-layer volumetric moisture exerted far greater control over infiltration dynamics than shallow moisture content.

For numerical simulations, the permeability numerical models aim to capture the fluid flow in porous media at different scales, which can be classified into empirical models, pore-scale direct simulations, and continuum-scale models [9]. As a special porous medium, frozen soil is typically simplified as either a saturated system (i.e., consisting of solid particles, water, and ice) or an unsaturated system (which also contains air/gas) [62, 63]. The heterogeneous features of porous structures significantly affect the energy and solute transport, which raises extreme challenges in numerical simulations, especially for frozen soils with segregated ice [14]. For empirical models, the Kozeny-Carman equation is widely applied for calculating hydraulic conductivity when frozen soil is modeled as a spherical particle [5, 40]. For the continuum-scale models, Xiao et al. [54] applied finite-element method (FEM) with Darcy's law to develop a soil permeability model to predict water migration during thaw. Similarly, Chen and Zhang [7] predicted the hydraulic conductivity of unsaturated frozen soil. Zeng et al. [61] applied the Richards equation to overcome Darcy's law's shortcomings in modeling transient porous media flows, thereby enhancing its numerical stability. Wang et al. [51] developed a finite-element consolidation-thawing model to predict settlement in soils containing segregated ice, enabling global permeability coefficients to be calculated as temperature-dependent functions. For the pore-scale direct simulations, the lattice Boltzmann method is a mesoscopic computational technique that simulates fluid flow by evolving particle distribution functions on a discrete lattice through simple collision and streaming steps. By recovering the Navier-Stokes equations in the low Mach limit, it offers excellent parallel efficiency and easy handling of complex boundaries [13, 15, 25, 45].

In the past decades, LBM has been extensively developed to simulate the complex fluid flow behaviors in porous media [2, 28]. For example, Eshghinejadfard et al. [11] used LBM to analyze the effects of face-centered cubic (FCC) and body-centered cubic (BCC) particle arrangements on permeability, highlighting how geometric resolution influences simulation results. To address some complex fluid flow problems, LBM can be coupled with other numerical techniques. Li [28] combined LBM with the discrete simplified Boltzmann Gas-Kinetic (DSBGK) model to solve gas permeability. Xu et al. [58] simulated seepage by approximating soil particles as spheres, simplifying boundary conditions and enhancing computational efficiency. Wang and Sun [45] integrated microscale permeability and traction-separation outputs from LBM into a macroscopic finite-element model via homogenization, yielding a dual-permeability framework that incorporates discontinuities. The effectiveness of LBM for permeability analysis has been extensively validated, confirming its suitability for simulating frozen soil thaw scenarios. Beyond LBM, Yang et al. [52] employed fractal theory to develop a local apparent liquid permeability (LALP) model to investigate nonlinear seepage behavior at the REV scale in low-permeability sandstone samples.

The application of LBM in frozen-soil applications stems from its exceptional ability to simulate multiphase fluid flow and phase-change processes. Phase change is the primary challenge in frozen soils, as it triggers the freeze-thaw cycle. Li et al. [29] employed an enthalpy-based model to simulate the water-ice phase transition in frozen soils, and Tian et al. [41] applied the same method to investigate larger-scale freezing processes. Hu et al. [20] validated this approach against analytical solutions. Xia et al. [56] coupled the Frozen Soil Quartet Structure Generation Set (FSQSGS) with LBM to study thermal conductivity at the representative elementary volume (REV) scale, while Gan et al. [13] examined moisture transport during the freezing of saturated soils. Yang et al. [53] analyzed seepage behavior in low-permeability carbonate rocks under various conditions. Because soil is often unsaturated, with pore ice forming at gas-liquid

interfaces under cohesive and adhesive forces, where multiphase flow modeling should be incorporated into frozen-soil simulations.

LBM also provides several models for studying permeability in thawing frozen soils, including the Shan-Chen, color-gradient, free-energy, and phase-field approaches. The Shan-Chen model employs a pseudo-potential method to simulate interphase interactions (e.g., water-gas) and has been widely applied to freezing-thawing processes in unsaturated soils [26, 65]. The color-gradient model, adapted from lattice gas automata for immiscible fluids, and the free-energy model, based on the thermodynamic pressure tensor of a free-energy functional, both simulate multiphase flow in specific soil environments [2]. The phase-field model, which excels at capturing mesoscale capillary forces, is particularly well suited for tracking phase-change interfaces around spherical soil particles [30, 55]. To further improve stability and accuracy, especially in complex pore geometries. Many recent studies have adopted the multi-relaxation-time (MRT) scheme in place of the traditional single-relaxation-time (SRT) approach [37].

In this work, we aim to study the permeability characteristics of unsaturated frozen soil with segregated ice during the thawing process. To better understand the mechanisms behind this special phenomenon, we develop a thawing numerical model to capture the permeability evolution of the frozen soil segregated ice based on MRT LBM. In our model, we consider the effects of phase changes, water transport, soil pore structure, pore water melting temperature, and other mesoscopic mechanical properties on its permeability evolution in the calculation of permeability during the thawing processes. In our numerical model, we employ the Shan-Chen model to obtain the distribution of temperature and water migration. To validate the correctness and effectiveness of our numerical model, we compare the numerical results with our laboratory experimental data obtained from both a freezing-thawing experiment and a nuclear magnetic resonance (NMR) permeability measurement. Furthermore, numerical simulations are also helpful to better understand the effects of upper surface temperature state and porosity distribution on the evolution of the full-field temperature, thawing fronts, and liquid mass fraction, which help us to better understand the thaw-driven permeability evolution mechanisms.

The remainder of this work is organized as follows. The mathematical formula for our numerical model and the validation method are stated in Sect. 2. Validation results and further discussion are described in Sect. 3. Conclusions are drawn in Sect. 4.

2 Model development

As shown in Fig. 1, the computational process of the model contains three steps: (a) generating the heterogeneous structure, (b) calculating the freezing process, and (c) simulating the thawing process. In the first step, to generate the heterogeneous structure, a stochastic growth method was employed to construct solid matrices with a predefined porosity. In this study, the D2Q9 growth model was employed to generate the physical structure, which denotes a two-dimensional lattice (D2) with nine discrete velocity directions (Q9). Cores were randomly initialized within the domain, and solid regions were expanded from these cores in nine discrete directions. The growth process was terminated once the global porosity of the system reached the target value ε. The white and brown regions in Fig. 1a represent the pores and the solid matrix, respectively. The second step was designed to establish a heterogeneous temperature distribution for thawing process. Freezing boundary conditions were applied at the top and bottom of the domain until the freezing front reaches a height of 80 lattice units. The green and blue regions in Fig. 1b represent the ice and water phases, respectively. In the third step, a higher upper surface temperature was applied to simulate the thawing process, which continues until the temperature variation within the lattice approaches 0. The blue regions in Fig. 1c represent the thawing process completed throughout the entire domain.

2.1 Physical structure

To describe the heterogeneous structure of porous soil media, a stochastic growth method is employed to generate the pore structure. As shown in Fig. 2, a single growth nucleus produces approximately spherical particles, while two nuclei result in ellipsoidal formations. The locations of the growth cores are randomly initialized within the computational domain according to the target porosity to ensure a statistically heterogeneous microstructure. With increasing numbers of growth nuclei, the particle units develop progressively more irregular morphologies. Subsequently, as these particle units continue to develop, the porous structure gradually takes shape with a corresponding reduction in porosity. The growth process stops once the target porosity is reached, resulting in the desired porous medium structure. Compared with conventional shrinking methods, the proposed stochastic growth algorithm is capable of generating particle morphologies that are more irregular and realistic. In comparison to the sequential trilateration method, it achieves comparable pore-space statistics with substantially higher computational efficiency. In future work, we will consider adopting the fully

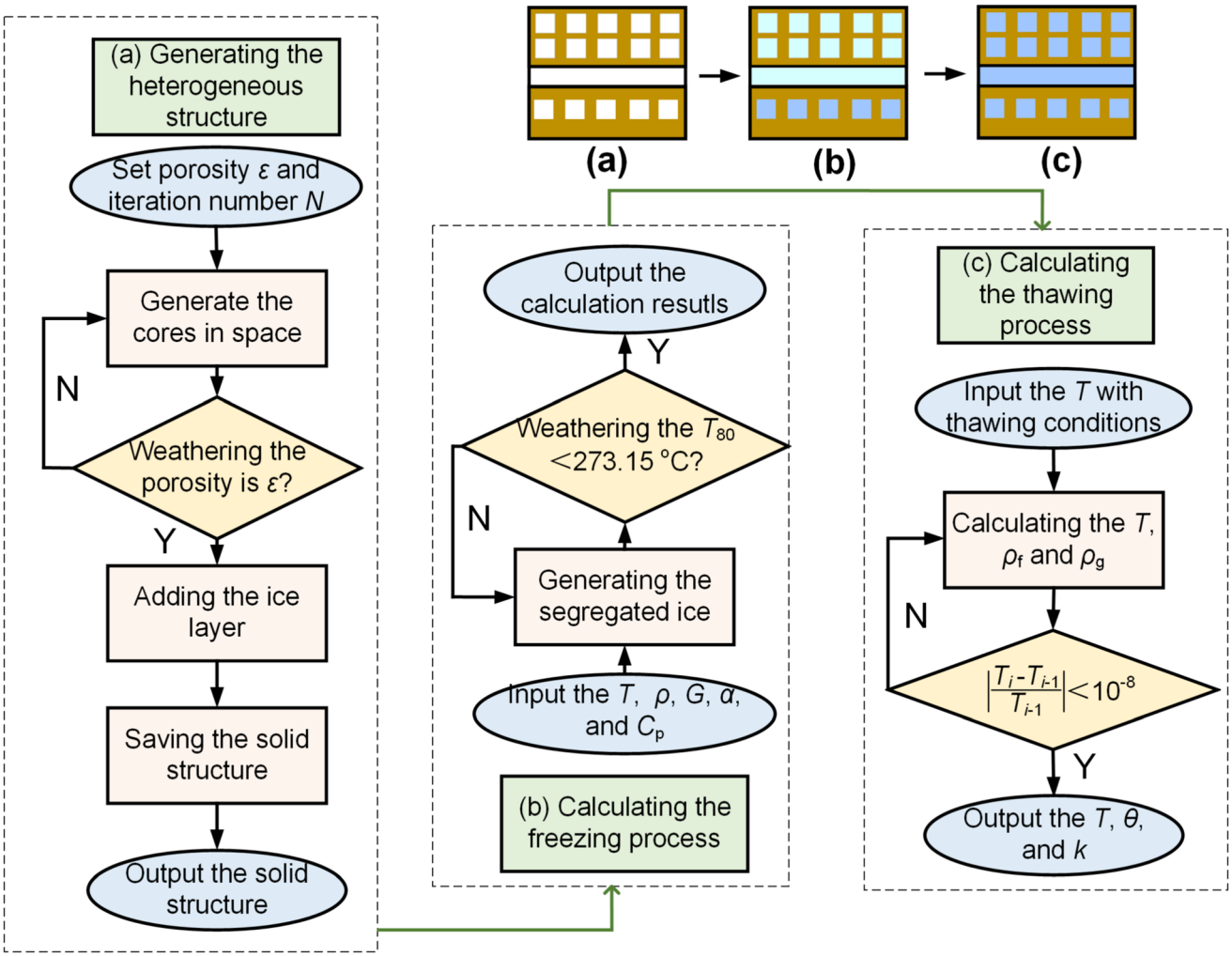

Fig. 1 Schematic diagram of the construction process for the segregated ice melting permeability model: **a** the generation of the heterogeneous structure of frozen soil; **b** the simulation of the freezing process, which serves as the initial condition for **c** the simulation of the thawing process

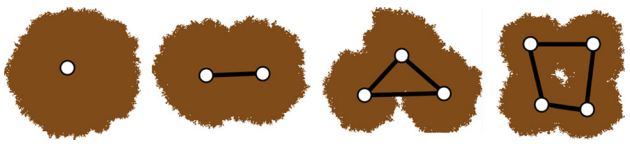

Fig. 2 Shapes of soil particles generated with different numbers of growth cores (1, 2, 3, and 4) using stochastic growth method

improved stochastic growth algorithm to systematically assess how particle-size distribution, seed clustering, localized growth-inhibition zones, and spatially varying growth probabilities influence the resulting pore structure.

As shown in Fig. 3, the generated soil structure exhibits strong morphological similarity to natural soil samples. In Fig. 3a, the image was obtained through in-situ CT scanning using a GE Vtomex system, with the sample of 500-mesh quartz soil in a saturated state. Figure 3b presents the stochastic growth structure generated according to the methodology illustrated in Fig. 2. As demonstrated in Fig. 3c, d, quantitative comparison indicates that the porosity of the numerical structure (0.30) is consistent with that of the CT-scanned natural soil (0.31), confirming the high structural similarity between them. Thus, the numerical structure provides a solid matrix for constructing the pseudo-potential model. It should be noted that different structure generation methods have different capabilities and limitations. Sequential trilateration-based approaches, which explicitly place spherical particles with controlled positions, provide strong control over particle segregation and are well suited for constructing highly ordered or deliberately segregated soil structures [42]. In contrast, the stochastic growth method adopted in this study does not explicitly prescribe particle positions. When used in its original form, it tends to generate statistically uniform soil structures, as illustrated by the uniform-wise structure shown in Fig. 3(d). This limitation is explicitly acknowledged. To overcome this shortcoming, the stochastic growth framework was extended by introducing PSD-

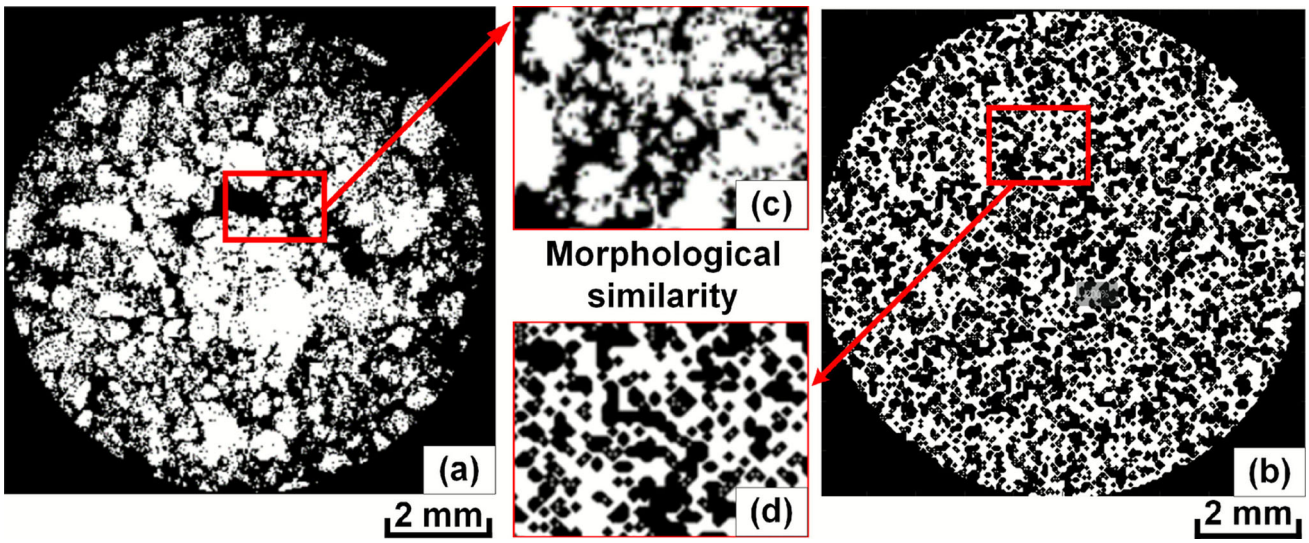

Fig. 3 Comparison between the CT-scanned structure and the structure generated using the stochastic growth method: **a** CT-scanned image; **b** full structure generated by the stochastic growth method; **c** local magnification of the CT image; **d** local magnification of the structure generated by the stochastic growth method

controlled multi-scale seeding, clustered core distributions, and localized growth-inhibition mechanisms, enabling the generation of particle agglomerations and large pore spaces that are more representative of coarse-grained soils.

2.2 Pseudo-potential model

In describing the mixing flow problem of multiphase flow, this study employs a modified multi-relaxation-time pseudo-potential model to simulate the thawing process of unsaturated frozen soil. The basic heat and mass transfer behavior of different phases is expressed by the density distribution function Eq. (1) and the temperature distribution function Eq. (2).

$$f_i^{\sigma}(x+\boldsymbol{e}_i\Delta t, t+\Delta t) = f_i(x,t) - \left(\boldsymbol{M}^{-1}\boldsymbol{S}\boldsymbol{M}\right)_{\sigma}[f_i(x,t) - f_i^{\mathrm{eq}}(x,t)] \tag{1}$$

$$g_i(x+\boldsymbol{e}_i\Delta t, t+\Delta t) = g_i(x,t) - \frac{1}{\tau_{\mathrm{T}}}[g_i(x,t) - g_i^{\mathrm{eq}}(x,t)] + \omega_i\frac{L}{C}[fr(t+\Delta t) - fr(t)] \tag{2}$$

where the $\boldsymbol{e}_i$ is the discrete velocity, $\boldsymbol{M}$ is the transformation matrix, $\boldsymbol{S}$ is the diagonal relaxation matrix, and τ_{T} is the relaxation time. The ω_i, L, C, and fr are the weighting coefficient corresponding to the nine discrete directions, latent heat, specific heat, and liquid mass fraction of the fluids, respectively. The equations of density equilibrium function f_i^{eq} and temperature equilibrium function g_i^{eq} are given as follows:

$$f_i^{\mathrm{eq}}(x,t) = \omega_i\rho_{\sigma}[1 + \frac{(\boldsymbol{e}_i\boldsymbol{u}_{\sigma}^{\mathrm{eq}})}{\boldsymbol{c}_{\mathrm{s}}^2} + \frac{(\boldsymbol{e}_i\boldsymbol{u}_{\sigma}^{\mathrm{eq}})^2}{2\boldsymbol{c}_{\mathrm{s}}^4} - \frac{(\boldsymbol{u}_{\sigma}^{\mathrm{eq}})^2}{2\boldsymbol{c}_{\mathrm{s}}^2}] \tag{3}$$

$$g_i^{\mathrm{eq}}(x,t) = \omega_i T[1 + \frac{(\boldsymbol{e}_i\boldsymbol{u}_{\sigma}^{\mathrm{eq}})}{\boldsymbol{c}_{\mathrm{s}}^2} + \frac{(\boldsymbol{e}_i\boldsymbol{u}_{\sigma}^{\mathrm{eq}})^2}{2\boldsymbol{c}_{\mathrm{s}}^4} - \frac{(\boldsymbol{u}_{\sigma}^{\mathrm{eq}})^2}{2\boldsymbol{c}_{\mathrm{s}}^2}] \tag{4}$$

where ω_i is the weighting coefficient corresponding to the nine discrete directions, ρ_{σ} and T represent the fluids' density and temperature, and $\boldsymbol{u}^{\mathrm{eq}}$ is the macroscopic equilibrium velocity. In the Porter-type force term, the $\boldsymbol{u}^{\mathrm{eq}}$ is given by Eq. (5) [33]:

$$\boldsymbol{u}^{\mathrm{eq}} = \frac{1}{\rho_{\sigma}}\sum_{i=0}^{m-1}\left(\boldsymbol{e}_i f_i + \frac{\delta_t}{2}\boldsymbol{F}_{\sigma}^{\mathrm{tot}}\right) \tag{5}$$

The force terms proposed by Porter are utilized in the equilibrium functions, which extend the He-Shan-Doolen force scheme to a multiphase pseudo-potential model [33]. $F\sigma^{\mathrm{tot}}$ is composed of mechanical terms at the mesoscopic scale. For ice-water and gas terms, these are expressed as shown in Eq. (6):

$$\boldsymbol{F}_{\sigma}^{\mathrm{tot,liquid}} = \boldsymbol{F}_{\sigma}^{\mathrm{adh}} + \boldsymbol{F}_{\sigma}^{\mathrm{coh}} + \boldsymbol{F}_{\sigma}^{\mathrm{T}} \tag{6}$$

where the $F\sigma^{\mathrm{adh}}$, $F\sigma^{\mathrm{coh}}$, and $F\sigma^{\mathrm{T}}$ are the adhesive force, cohesive force, and temperature-driven force, respectively. And the expressions for different forces are given as follows:

$$\boldsymbol{F}_{\sigma}^{\mathrm{adh}}(x,t) = -\rho_{\sigma}(x,t)G_{\sigma}^{\mathrm{adh}}\sum_i \omega_i s_i(x+\boldsymbol{e}_i\Delta t)\boldsymbol{e}_i \tag{7}$$

$$\boldsymbol{F}_{\sigma}^{\mathrm{coh}}(x,t) = -\psi_{\sigma}(x,t)G^{\mathrm{coh}}\sum_i \omega_i\psi_{\sigma 1}(x+\boldsymbol{e}_i\Delta t)\boldsymbol{e}_i \tag{8}$$

$$\boldsymbol{F}_{\mathrm{T}} = G^{\mathrm{T}}\cdot\nabla\varphi\cdot\Delta x \tag{9}$$

where the $G\sigma^{\mathrm{adh}}$, $G\sigma^{\mathrm{coh}}$, and $G\sigma^{\mathrm{T}}$ are the coefficient of fluid wettability between solid and fluid, coefficient controlling the strength between varied fluids, and coefficient controlled by temperature gradients. s_i is the floating variable that is used to distinguish the types of fluids, and $\boldsymbol{e}_i$ is the discrete velocity. Ψ is the pseudopotentials of the particles of fluid component σ and σ_1, which is usually considered as fluid density. φ is the total water potential.

For the gas term, it is expressed as shown in Eq. (10):

$$\boldsymbol{F}_{\sigma}^{\mathrm{tot,gas}} = \boldsymbol{F}_{\sigma}^{\mathrm{adh}} + \boldsymbol{F}_{\sigma}^{\mathrm{coh}} \tag{10}$$

The $\boldsymbol{M}$ in Eq. (1) is the transformation matrix which represents the relaxation time type of distribution function is MRT, and the expression is given as follows:

$$\boldsymbol{M} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ -4 & -1 & -1 & -1 & -1 & 2 & 2 & 2 & 2 \\ 4 & -2 & -2 & -2 & -2 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & -1 & 0 & 1 & -1 & -1 & 1 \\ 0 & -2 & 0 & 2 & 0 & 1 & -1 & -1 & 1 \\ 0 & 0 & 1 & 0 & -1 & 1 & 1 & -1 & -1 \\ 0 & 0 & -2 & 0 & 2 & 1 & 1 & -1 & -1 \\ 0 & 1 & -1 & 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & -1 & 1 & -1 \end{bmatrix} \tag{11}$$

The expression of relaxation matrix $\boldsymbol{S}$ is given as:

$$\boldsymbol{S} = \mathrm{diag}(1, \tau_v, \tau_v, 1, \tau_{\mathrm{T}}, 1, \tau_{\mathrm{T}}, \tau_v, \tau_v) \tag{12}$$

where the τ_{T} and τ_{v} are the relaxation times of thermal diffusivity and kinematic viscosity, and their expressions are given as follows:

$$\tau_{\mathrm{T}} = \frac{\alpha_{\mathrm{e}}}{c_{\mathrm{s}}^2 \Delta t} + 0.5 \tag{13}$$

$$\alpha_{\mathrm{e}} = \frac{\lambda}{\rho_{\sigma} C_{\mathrm{p}}} \tag{14}$$

$$\tau_{\mathrm{v}} = \frac{v_{\mathrm{e}}}{c_{\mathrm{s}}^2 \Delta t} + 0.5 \tag{15}$$

where the α_{e} is the effective thermal diffusivity, λ is the thermal conductivity, v_{e} is the kinematic viscosity, and the c_{s} is the speed of sound.

The expression of the discrete velocity $\boldsymbol{e}_i$ is Eq. (16):

$$\boldsymbol{e}_i = \begin{cases} (0,0)^{\mathrm{T}} \; i = 0 \\ c(\cos[\frac{\pi}{2}(x-1)], \sin[\frac{\pi}{2}(x-1)])^{\mathrm{T}} \; i = 1,2,3,4 \\ \sqrt{2}c(\cos[\frac{\pi}{2}(x-5)+\frac{\pi}{4}], \sin[\frac{\pi}{2}(x-5)+\frac{\pi}{4}])^{\mathrm{T}} \; i = 5,6,7,8 \end{cases} \tag{16}$$

Enthalpy H_i is primarily used to determine the phase state of a fluid, and its expression is given as Eq. (17):

$$H_i = C_{\mathrm{p}} T_i + fr_{\mathrm{f},i-1} L \tag{17}$$

The liquid mass fraction $fr_{\mathrm{f},i}$ is expressed as:

$$fr_{\mathrm{f},i} = \begin{cases} 0 & H \le H_{\mathrm{s}} \\ \dfrac{H_i - H_{\mathrm{s}}}{H_{\mathrm{f}} - H_{\mathrm{s}}} & H_{\mathrm{s}} \le H \le H_{\mathrm{f}} \\ 1 & H > H_{\mathrm{f}} \end{cases} \tag{18}$$

Finally, the simulation results of density ρ_{σ} and temperature T are extended to the macroscopic scale through Eqs. (19) and (20).

$$\boldsymbol{\rho}_{\sigma} = \sum_i f_i^{\sigma} \tag{19}$$

$$\mathrm{T} = \sum_i g_i \tag{20}$$

2.3 Permeability characteristic

In order to bridge the correlation between mesoscopic and macroscopic permeability characteristics, this study introduces a cross-scale modeling approach for permeability. Figure 4 illustrates the frozen soil before the thawing process. As shown in Fig. 4a, the total height of the two-dimensional soil column is 14 cm × 10 cm, consisting of three layers from bottom to top: (b) thawing layer, (c) segregated ice layer, and (d) unfrozen layer. The segregated ice refers to the ice layer that forms near the 0 °C isothermal line, and its melting has a significant impact on the hydrology of the soil [12]. The initial frozen temperature distribution ranges from − 10 °C to 10 °C, and due to the presence of segregated ice and soil particles, the temperature distribution is heterogeneous. When the thawing layer and segregated ice layer have not yet fully melted, the permeability is strongly correlated with temperature.

The permeability coefficient determines the volumetric mass flow, which depends on the inherent properties of the soil. To refine the macro-scale empirical equation for k in Eq. (21), it is necessary to first calculate the volumetric mass Q_{v} based on the developed model, which is derived from Darcy's law [64]:

$$k = -\frac{Q_{\mathrm{v}} A \Delta S}{\mu L} \tag{21}$$

where A is the cross section area of the soil, ΔS is the hydraulic head difference, μ is the dynamic viscosity, and L is the flow path length. The volumetric mass flow is obtained by the inlet and outlet velocities of soil Eq. (22).

$$Q_{\mathrm{v}} = (v_{\mathrm{out}} - v_{\mathrm{in}}) A \tag{22}$$

where v_{in} and v_{out} are the inlet and outlet velocities of the flow.

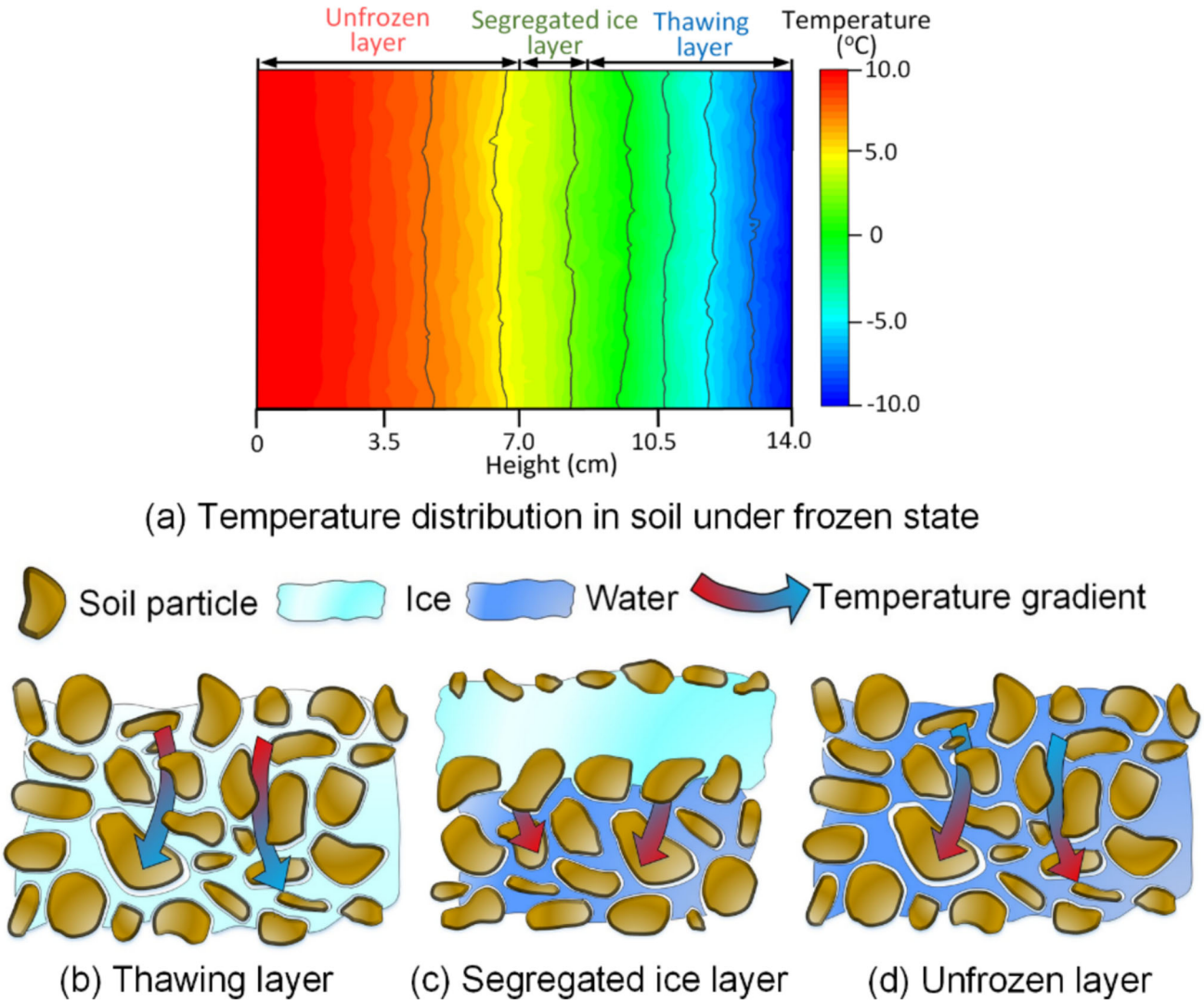

Fig. 4 Temperature distribution and schematic diagrams of soil-water-ice phase structures in different zones during the frozen soil thawing process: **a** the temperature distribution in soil under frozen state with the size of 14 cm × 10 cm; **b**, **c**, and **d** the thawing layer, segregated ice layer, and unfrozen layer at different height of frozen soil

The hydraulic head difference is expressed with the pressure difference between top and bottom Eq. (23):

$$\Delta S = \frac{\Delta P}{\rho_{\text{water}} g} \tag{23}$$

where ρ_{water} and g represent the density of water and gravitational acceleration, respectively.

The velocity difference Δv is also obtained by pressure difference ΔP, which is get from the difference value of macroscopic density between the inlet and outlet cross sections.

$$\Delta v = \frac{\Delta P}{\mu k L} \tag{24}$$

$$\Delta P = c_s^2 \rho_{\text{in}} - c_s^2 \rho_{\text{out}} \tag{25}$$

The pore size influences the curvature of the pore walls, which in turn affects the surface free energy of the grid points inside the pores, leading to variations in freezing temperature across different locations. In this study, the Gibbs-Thomson effect is applied to quantify these variations in freezing temperature T_{f} at different points [4, 49].

$$\Delta T_{\text{f}} = T_0 - T_{\text{f}} = \frac{T_0 \sigma_{\text{sl}}}{r \Delta H \rho_s} \tag{26}$$

where the ΔT_{f} is freezing point depression, the T_0 is freezing point of pure water, the σ_{sl} is ice–water interfacial free energy, and the r is pore diameter, which is obtained directly from the heterogeneous pore structure generated by the stochastic growth method. The ΔH is latent heat of solidification, and the ρ_{s} is density of ice. The model was developed based on the structure of 500-mesh soil, so the

minimum spacing in the matrix is also set to 0.074 mm, which corresponds to the mesh size of the particles.

The unfrozen water content in the pores depends on the interfacial curvature radius of the pores and the ambient temperature, which affects the permeability during the thawing process. In this study, the Gibbs-Thomson equation was used to calculate the freezing temperature in the pores, as given in Eq. (27) [36].

$$\frac{T_0 - T_m}{T_m} = \frac{2\Delta\gamma}{r_{ave}\Delta H \rho_i} \tag{27}$$

where the T_m is the melting temperature, $\Delta\gamma$ is the surface tension between the solid wall and the liquid wall, and r_{ave} is size of the pore, which is calculated by statistically averaging all pore radii within the representative elementary volume of the simulated soil matrix, weighted by the local pore volume fraction. ρ is the density of ice. ΔH is the melting latent heat.

Figure 5 presents the distribution of lattice numbers within the matrix corresponding to various melting temperatures. The freezing temperatures are primarily concentrated within the range of − 0.0035 °C to − 0.0015 °C, indicating that the melting temperature of 500-mesh soil is minimally influenced by the pore structure.

The initial setting conditions are shown in Table 1. The porosity value depends on the type of soil. When the sand content is higher, the porosity increases, whereas higher silt-clay content results in lower porosity. According to the experimental data supported by field measurements, the porosity ranges approximately from 0.1 to 0.5 [21, 32]. The unsaturated saturation reaches its peak value of 0.9, as reported in the work of Broms [3]. In addition, the model adopts Dirichlet boundary conditions at the top and bottom boundaries, while periodic boundary conditions are applied to the lateral sides.

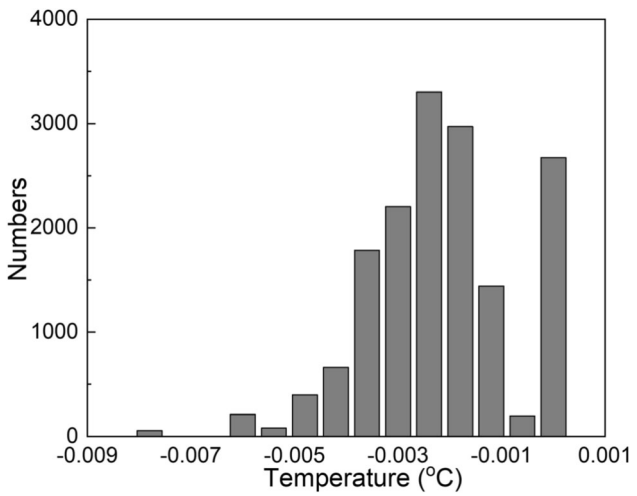

Fig. 5 Number of lattice nodes in the frozen soil matrix corresponding to different melting temperatures, as determined by the Gibbs–Thomson equation

Table 1 The initial conditions for thawing frozen soil

Parameters	Value
Upper surface temperature	− 10 °C
Lower surface temperature	10 °C
Temperature distribution of soil	− 10 °C ~ 10 °C
Height of segregated ice layer	75 mm
Height of ice lens layer	5 mm
Saturation	0.9
Porosity	0.3

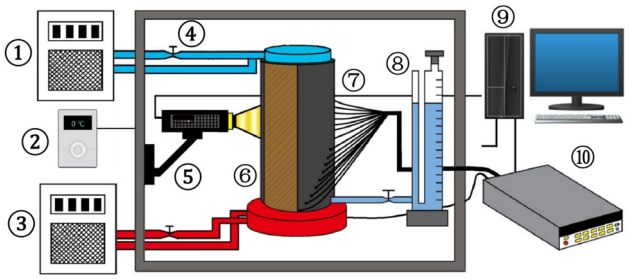

Fig. 6 Schematic diagram of the freezing–thawing experimental setup. The temperature system consist of 1—temperature controller (top), 2—temperature controller, 3—temperature controller (bottom), 4—thermostatic chamber, and 7—thermocouple, 5—camera is used to record the morphology of 6—sample, 8 is the Mariotte bottle for the water supply in freezing process, 9—computer and 10—data acquisition system are for the data collection and recording

2.4 Validation method

In this study, a freezing–thawing experimental setup was established to validate the macroscopic accuracy of the model. As shown in Fig. 6, 500-mesh quartz sand was adopted as soil samples to simulate the thawing process [51]. The axial temperature of the samples was controlled by temperature controllers, and the temperature distribution was recorded using thermocouples. Initially, the top and bottom temperatures were set to − 10 °C and 10 °C, respectively, to simulate the freezing process of the samples. Subsequently, both the top and bottom temperatures were set to 10 °C to simulate the melting process of the samples. The results of the temperature variations were collected and compared with the LBM results.

To validate the accuracy of the numerical model, the one-dimensional analytical solution was provided for comparison with the results obtained from the LBM, which was developed based on heat transfer theory [11, 18]. The temperature field and the position of the phase interface

during the solid–liquid phase change process can be described using the error function and complementary error function, as shown in Eq. (28).

The position of the phase interface X as a function of time t can be expressed as:

$$X(t) = 2z\sqrt{\alpha t} \tag{28}$$

where the z is the root of the transcendental equation and its expression is given as Eq. (29), and α is the thermal diffusivity.

$$\frac{C_{\mathrm{p}}(T_{\mathrm{b}} - T_{\mathrm{m}})}{Le^{\mathrm{k}}\mathrm{erf}(z)} - \frac{C_{\mathrm{p}}(T_{\mathrm{m}} - T_0)}{Le^{\mathrm{k}}\mathrm{erfc}(z)} = z_f\sqrt{\pi} \tag{29}$$

where the e is the constant value equal to about 2.72. The temperature distribution expressions for the solid phase and liquid phase are shown in Eq. (30):

$$T(x,t) = \begin{cases} T_{\mathrm{b}} - \dfrac{(T_{\mathrm{b}} - T_{\mathrm{m}})\mathrm{erf}\left(x/2\sqrt{t}\right)}{\mathrm{erf}(z)} & 0 < x < X(t) \\ T_0 - \dfrac{(T_{\mathrm{m}} - T_0)\mathrm{erf}\left(x/2\sqrt{t}\right)}{\mathrm{erf}(z)} & x > X(t) \end{cases} \tag{30}$$

where the T_{b} and T_{m} are the constant heat source and melting temperatures, respectively.

The permeability coefficient can be determined using the SDR model based on NMR technology. NMR enables accurate quantification of unfrozen water content, and it reflects the dynamic distribution of water within the pore space through the geometric mean of the T_2 relaxation time as well. In this study, the CPMG sequence of NMR was employed assisted with the SDR model to invert the melting and permeability processes of pore water, investigating the temperature-dependent permeability characteristics of frozen soil samples with varying porosities. The results obtained provide a validation reference for the permeability coefficients derived from the LBM model.

The T_2 relaxation time in NMR is correlated with the ratio of pore volume to specific surface area (V/S) [23]. Consequently, the Schlumberger-Doll Research (SDR) equation (Eq. (31)) has become a key physical model linking pore geometry to permeability [22]. Studies by Straley et al. and Kleinberg et al. demonstrated that the SDR equation effectively predicts permeability by comparing measured permeability of reservoir sandstones with SDR-based estimations [24, 38]. In this study, the SDR expression is employed to invert the pore water permeability of frozen soil, aiming to achieve a more accurate characterization and numerical validation of its permeability.

$$k = A\phi^4 T_{2\mathrm{LM}}^2 \tag{31}$$

where the k denotes the intrinsic permeability and A is a constant related to the structure characteristics. Φ denotes porosity, and $T_{2\mathrm{LM}}$ represents the logarithmic mean of the T_2 relaxation time, which is calculated according to Eq. (32). As temperature varies, the solid–liquid phase transitions of pore water alter the relaxation signals in different pores, making $T_{2\mathrm{LM}}$ a comprehensive indicator of the water distribution state within the pore structure.

$$T_{2\mathrm{LM}} = \exp\left(\frac{\sum y_i \ln x_i}{y_i}\right) \tag{32}$$

As shown in Fig. 7, the NMR-based permeability measurement setup consists of the following main components: A—constant-temperature water bath, B—NMR instrument, C—NMR control system, and D—temperature-controlled chamber. The water bath is connected to the NMR instrument to maintain the permanent magnet under stable thermal conditions, preventing magnetization loss due to elevated temperatures. The NMR control system is responsible for calibrating the center frequency of the sample, estimating the signal amplitude range, and acquiring the T_2 relaxation signals during the experiment.

The freeze–thaw permeability measurement procedure is as follows: The saturated soil sample is placed in the temperature-controlled chamber for testing. During the freezing phase, the sample temperature is sequentially decreased to 15 °C, 1 °C, 0 °C, − 1 °C, − 2 °C, − 5 °C, − 10 °C, and − 20 °C. At each target temperature, the sample is held in the chamber for 4 h to ensure thermal equilibrium. Once equilibrium is reached, the sample is transferred to the NMR instrument to measure the distribution of T_2 relaxation times. In the subsequent thawing phase, the temperature is gradually increased in reverse order to − 10 °C, − 5 °C, − 2 °C, − 1 °C, 0 °C, 1 °C, and finally 15 °C, with the same sequence repeated as in the freezing phase.

The porosity of the soil sample should be determined before using the SDR equation to calculate permeability. In this study, porosity is estimated using the T_2 relaxation time distribution obtained from NMR measurements of saturated samples. Figure 8 presents the porosity comparison between saturated soil samples with particle sizes of 0.150 mm, 0.025 mm, and 0.013 mm, corresponding to typical sand, silt, and clay particle sizes, respectively. As shown in the figure, the relative peak areas of the saturated samples are 12,605, 13,346, and 14,200, respectively, and values correspond to the peak areas of calibration samples with water contents of 0.31, 0.33, and 0.35, respectively. Therefore, the porosities of the three soil samples are inferred to be 0.31, 0.33, and 0.35, respectively.

The maximum $T_{2\mathrm{LM}}$ value should be determined before fitting the parameter A in SDR permeability equation Eq. (31). Figure 9 illustrates the variation of $T_{2\mathrm{LM}}$ during

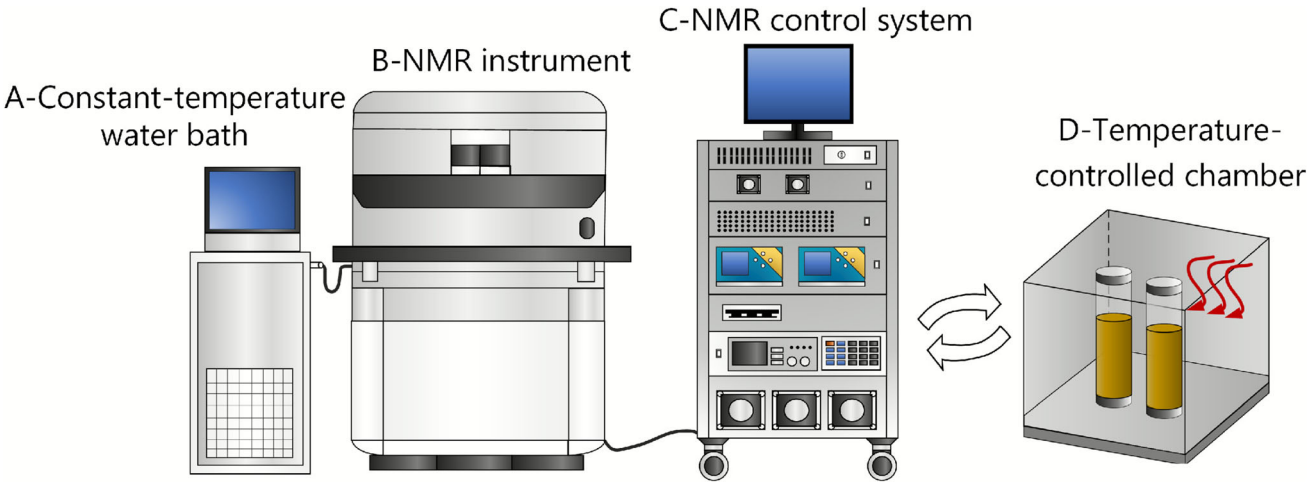

Fig. 7 Schematic diagram of the NMR permeability measurement setup. The A—constant-temperature water bath is used to keep the B—NMR instrument at stable thermal conditions, C—NMR control system is the operation platform for the data collection from samples in D—temperature-controlled chamber

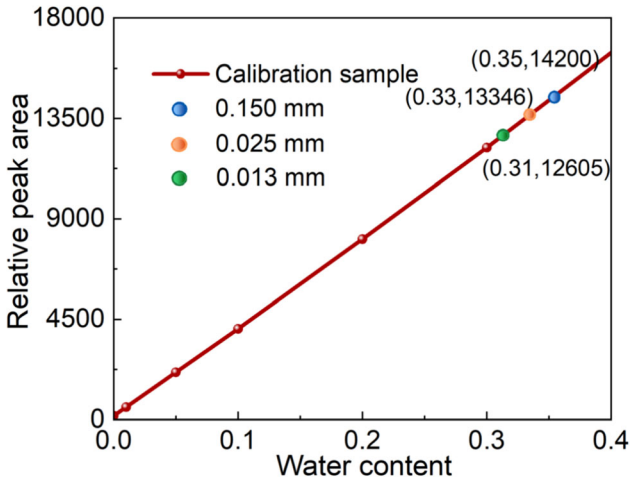

Fig. 8 Comparison of porosity between soils samples with different particle sizes (0.150 mm, 0.025 mm, 0.013 mm) and the calibration samples

the freeze–thaw cycles for soil samples with particle sizes of 0.150 mm, 0.025 mm, and 0.013 mm. All samples exhibit pronounced hysteresis behavior during both freezing and thawing processes, indicating that permeability estimation must consider both the temperature value and the direction of temperature change rather than relying on a single temperature point. The maximum T_{2LM} values during the melting phase for the three particle sizes are 221.9 ms, 23.6 ms, and 13.1 ms, respectively. Larger particle sizes correspond to larger pore structures, which result in longer relaxation times.

3 Results and discussion

3.1 Validation of model

Figure 10 compares the analytical, experimental, and LBM results are to validate the model's reliability. The thawing process of frozen soil under boundary conditions of − 10 °C and 10 °C lasts for 3 h, with the thawing front disappearing around the 10 cm position at approximately 1 h. The relative error between the LBM results and analytical data is 0.15%, consistent with the validation approaches used by Hu et al. [20] and Huang et al. [18] to verify LBM results. For further validation, measurements obtained from the freezing–thawing experimental setup are used for comparison, showing a relative error of 0.32% between the LBM and experimental results. This demonstrates that the LBM aligns well with both the experimental data and the analytical model. A slight numerical fluctuation can be observed between 5 and 7 cm in the LBM results at 1 h, which originates from local heterogeneity near the segregated ice layer where solid, liquid, and gas phases coexist. This fluctuation reflects the physical non-uniformity of the pore structure. The slight deviation observed between the LBM and analytical results during the temperature transition stage (around 1 h) arises from the heterogeneous soil–ice–water distribution considered in the LBM, which is not included in the analytical formulation. This difference highlights the advantage of the LBM in reproducing non-uniform temperature evolution in frozen soils.

The physical parameters of the 0.150 mm, 0.025 mm, and 0.013 mm soil samples derived from the above experimental results are summarized in Table 2. Based on Eq. (31), the corresponding A values for particle sizes of

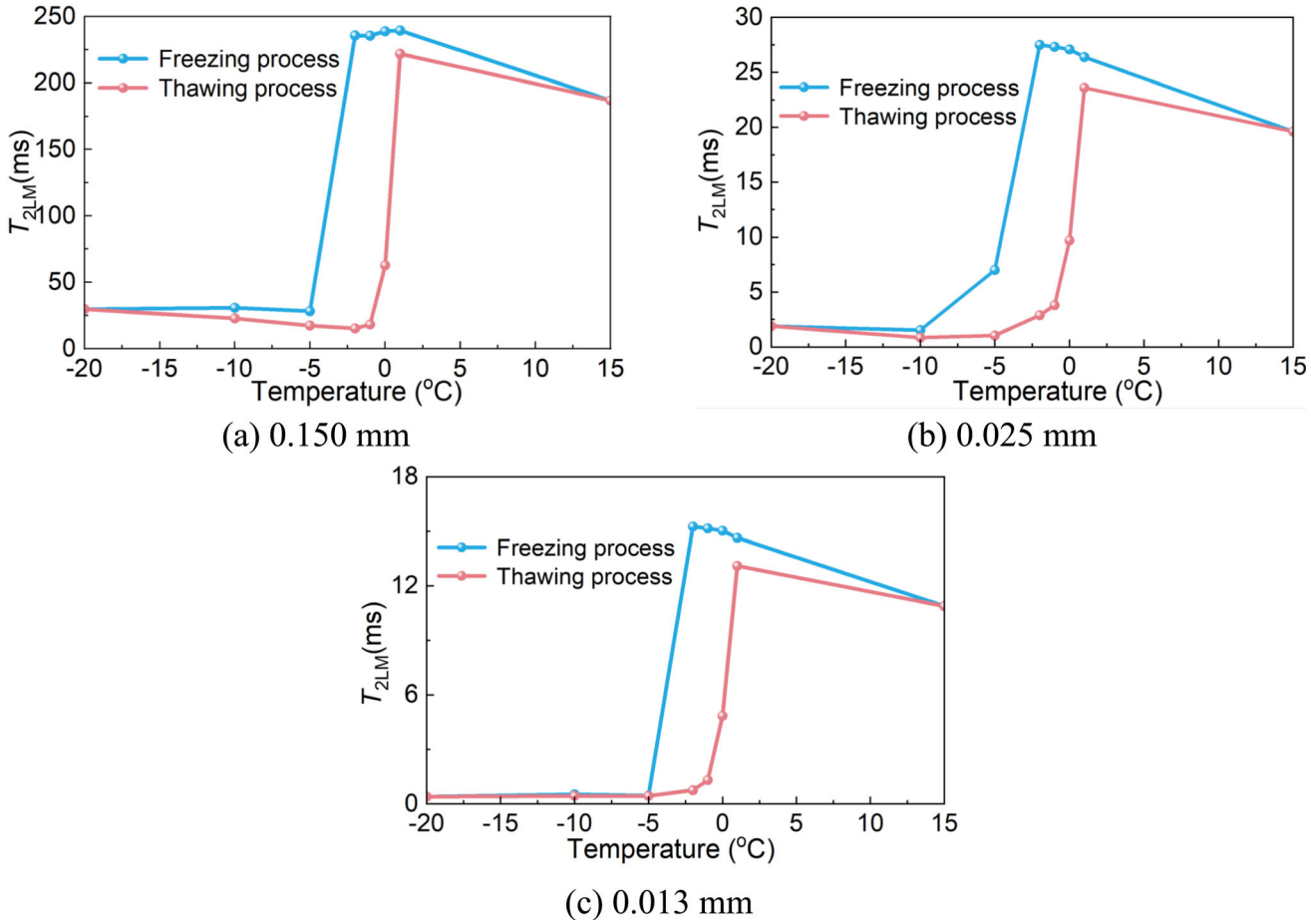

Fig. 9 Variation of the T_{2LM} during the freeze–thaw process with different particle sizes **a** 0.150 mm, **b** 0.025 mm, **c** 0.013 mm

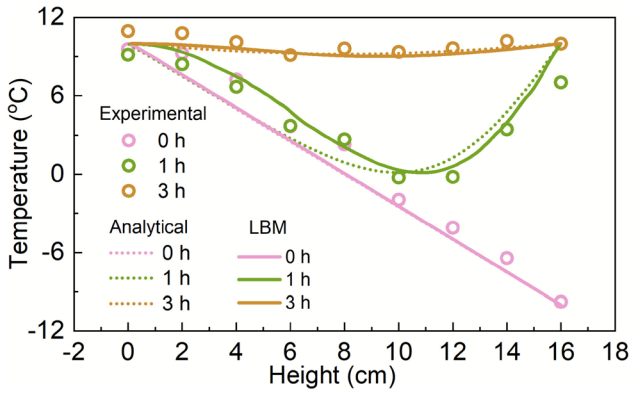

Fig. 10 Temperatures distribution comparisons along the soil column between experimental results, analytical results, and LBM results at different thawing times (0 h, 1 h, 3 h)

0.150 mm, 0.025 mm, and 0.013 mm are calculated as 1.38×10^{-2} m^2/ms^2, 4.78×10^{-16} m^2/ms^2, and 6.73×10^{-17} m^2/ms^2, respectively.

The relationship between permeability and temperature derived from the SDR equation and NMR experimental results is illustrated in Fig. 11. As expected, the permeability increases with particle size. At a melting temperature of 1 °C, the maximum permeability values for the three samples are 2.54×10^{-8} m/s, 9.57×10^{-9} m/s, and 3.39×10^{-9} m/s, respectively. As the temperature further rises to 15 °C, a slight decrease in permeability is observed, which may be attributed to localized pore contraction caused by thaw settlement under gravitational effects. Taking the 0.150 mm sample as an example, the permeability ranges from 4.6×10^{-10} m/s to 1.5×10^{-10} m/s as the temperature increases from − 20 °C to − 5 °C. When the temperature further increases from − 5 °C to 1 °C, the permeability increases

Table 2 Physical properties of soil samples

Porosity	Permeability (m/s)	T_{2LM} (ms)	A (m^2/ms^2)
0.35	2.54×10^{-8}	221.9	3.25×10^{-18}
0.33	9.57×10^{-9}	23.6	1.47×10^{-16}
0.31	3.39×10^{-9}	13.1	2.18×10^{-16}

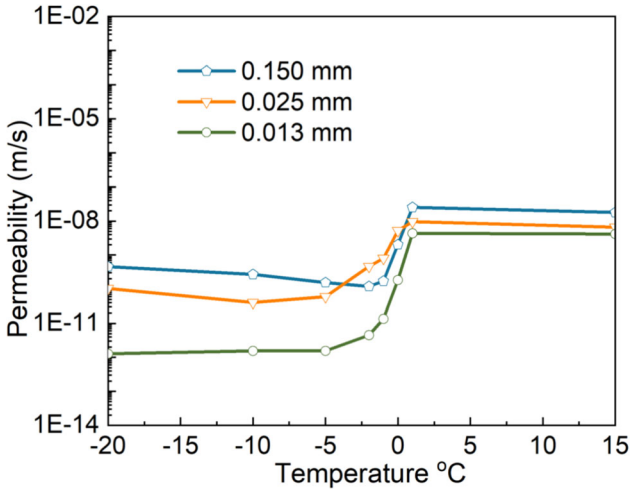

Fig. 11 Variation of permeability for thawing frozen soil samples with different particle sizes (0.150 mm, 0.025 mm, 0.013 mm)

significantly from 1.5×10^{-10} m/s to 2.5×10^{-8} m/s. This indicates that although the soil is predominantly in a frozen state, unfrozen water still flows through finer pores. The associated seepage, on the order of 10^{-10} m/s to 10^{-9} m/s, is minimal and thus exerts a limited impact on the overall frost heave and thaw settlement processes.

Based on the porosity and intrinsic permeability data provided in Table 2, the expression can be fitted derived from the Kozeny–Carman equation, which is given as follows [59]:

$$k = -2.9^{-11}\left(1 - e^{19.5x}\right) \tag{33}$$

As shown in Fig. 12, the permeability coefficients obtained from LBM simulations and those predicted by the SDR fitting curve are compared for soil samples with

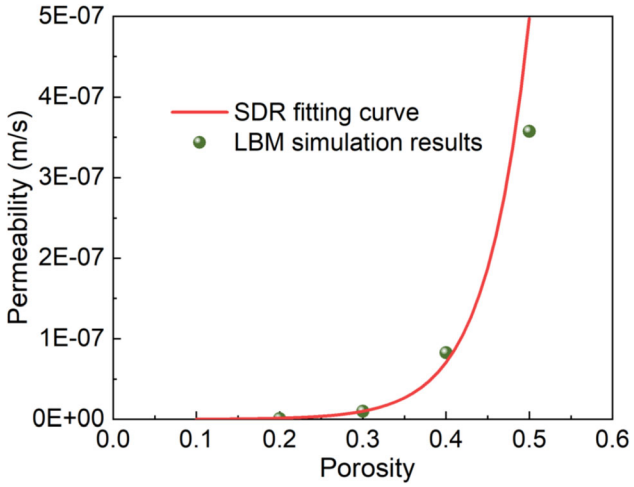

Fig. 12 Permeability comparison between LBM simulation results and SDR fitting curves for soil samples with different porosities (0.2, 0.3, 0.4, 0.5)

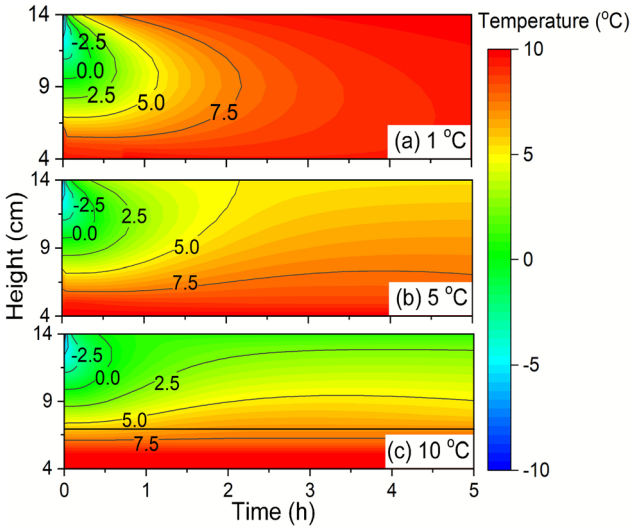

Fig. 13 Variation of temperature distribution under different upper surface temperatures. The soil with an upper surface temperature of **a** 1 °C shows the longest time to reach thermal equilibrium, whereas those with **b** 5 °C and **c** 10 °C reach a steady state faster

porosities of 0.2, 0.3, 0.4, and 0.5. The results indicate that permeability increases exponentially with porosity. The overall trend of the LBM simulation results exhibits good agreement with the experimental data.

3.2 Upper surface temperature

As shown in Fig. 13, the upper surface temperatures are set at 1 °C, 5 °C, and 10 °C to predict the duration of the melting process under different thermal conditions. The melting state at 1 °C requires approximately 3 h, while the upper boundary temperatures of 5 °C and 10 °C result in durations of 2.6 h and 2.2 h, respectively. Compared to the duration of segregated ice formation [19], the melting process is significantly faster than the freezing process. This difference arises from the varying temperature gradient requirements for the formation and melting of segregated ice. During the melting process, water migration is primarily influenced by the initial condition of external load, whereas the key factor is the distance between the freezing front and the water-rich region in the freezing process. The simulation results show good agreement with the macroscopic FEM results [51], indicating that the macroscopic results obtained by integrating the temperature distribution equation g are reasonable. The progress of the thawing process primarily depends on the temperature gradients indicating that the dimensionless values used in this part are reasonable.

As shown in Fig. 14, the liquid mass fraction distribution is presented under different upper surface temperatures at 1 h. The thawing front refers to the 0 °C interface, which marks the transition where ice begins to melt into liquid

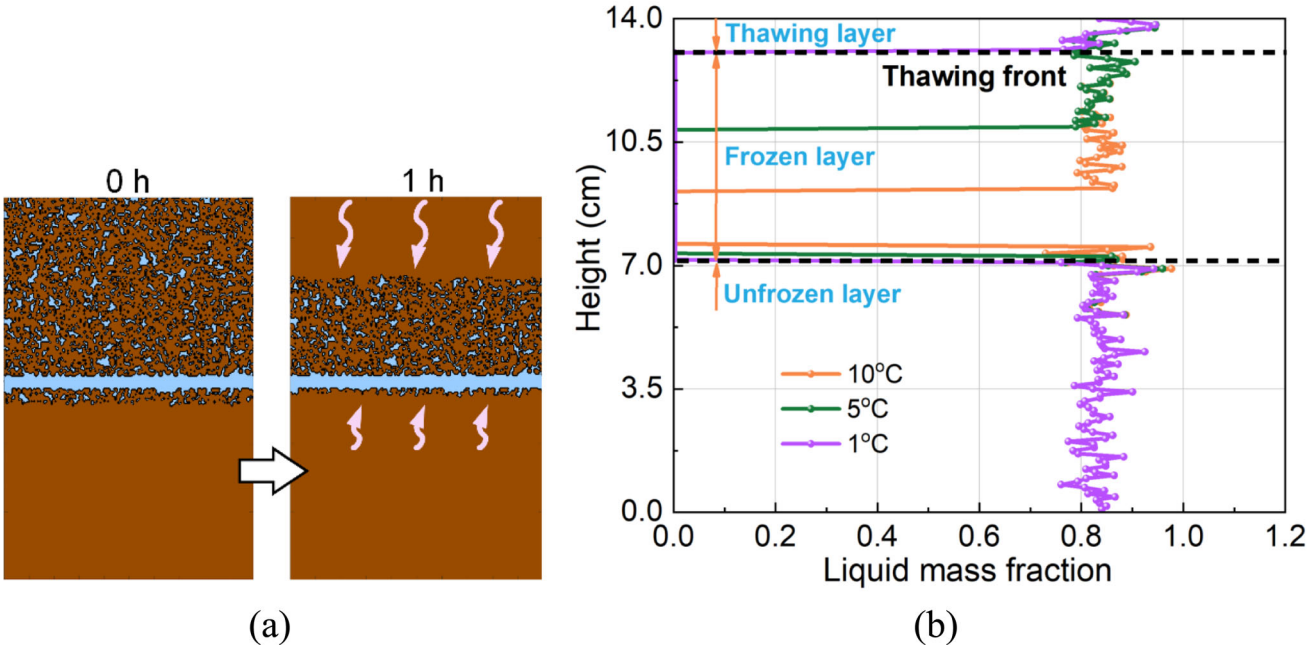

Fig. 14 Influence of upper surface temperature on liquid mass fraction: **a** ice distribution at 0 h and 1 h under an upper surface temperature of 10 °C; **b** liquid mass fraction with different upper temperatures at 1 h(1 °C, 5 °C, 10 °C)

due to an increase in temperature [60]. The values of the liquid mass fraction were normalized to ensure a consistent comparison of results. Figure 14a compares the liquid mass fraction at 0 h and 1 h. The melting process mainly happened at the top of soil while the lower melting front moved slowly. As shown in Fig. 14b, the macroscopic simulation results show that the region with the highest liquid mass fraction (at 7 cm) also corresponds to the water-rich segregated ice zone [57]. When the upper surface temperature is 1 °C, the upper thawing front is located at 13.1 cm, while it shifts to 9.2 cm when the upper surface temperature is 10 °C. Meanwhile, the lower thawing front remains around 7.1 cm to 7.5 cm under upper surface temperatures of 1 °C, 5 °C, and 10 °C. Additionally, the upward speed of lower thawing fronts remains similar across different temperatures, as the variation of the upper surface temperature cannot drive the temperature distribution of soil below 7 cm significantly.

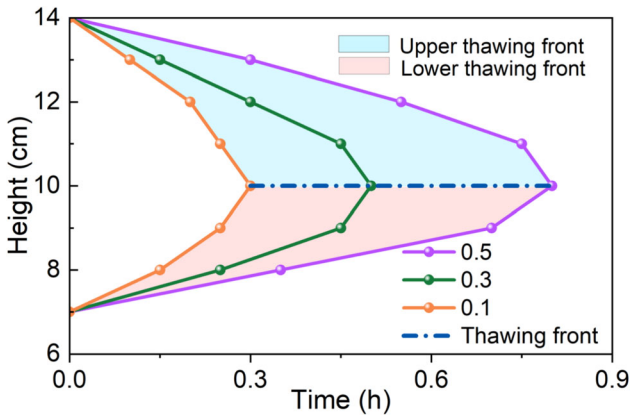

Fig. 15 Thawing fronts of frozen soil with different porosities (0.1, 0.3, 0.5) at thawing duration

3.3 Porosity

In this section, all the results in which porosity is used as a variable are calculated under the upper surface temperature of 10 °C. Figure 15 illustrates the thawing front for soils with porosities ranging from 0.1 to 0.5. As the porosity increases from 0.1 to 0.5, the intersection time of the upper and lower thawing fronts increases from 0.3 h to 0.8 h. This phenomenon occurs because the thermal conductivity of water is lower than that of solid particles. Meanwhile, the thawing front consistently stabilizes at a height of 10 cm, indicating that porosity has no impact on the final temperature distribution when the thawing process is sufficiently prolonged. Additionally, the gradient of change in the upper and lower thawing fronts gradually decreased over time, which was caused by the diminishing temperature gradient. A similar conclusion was also observed in the macroscopic simulation results [51]. Compared to the upper thawing front, the lower thawing front exhibits a smaller gradient of change, as the variation in temperature distribution within the domain is primarily influenced by the surface rather than the subsurface.

Figure 16 illustrates the liquid mass fraction in samples with varying porosities ranging from 0.1 to 0.5. In Fig. 16a, the water migrates through the pores in frozen soil at 1 h.

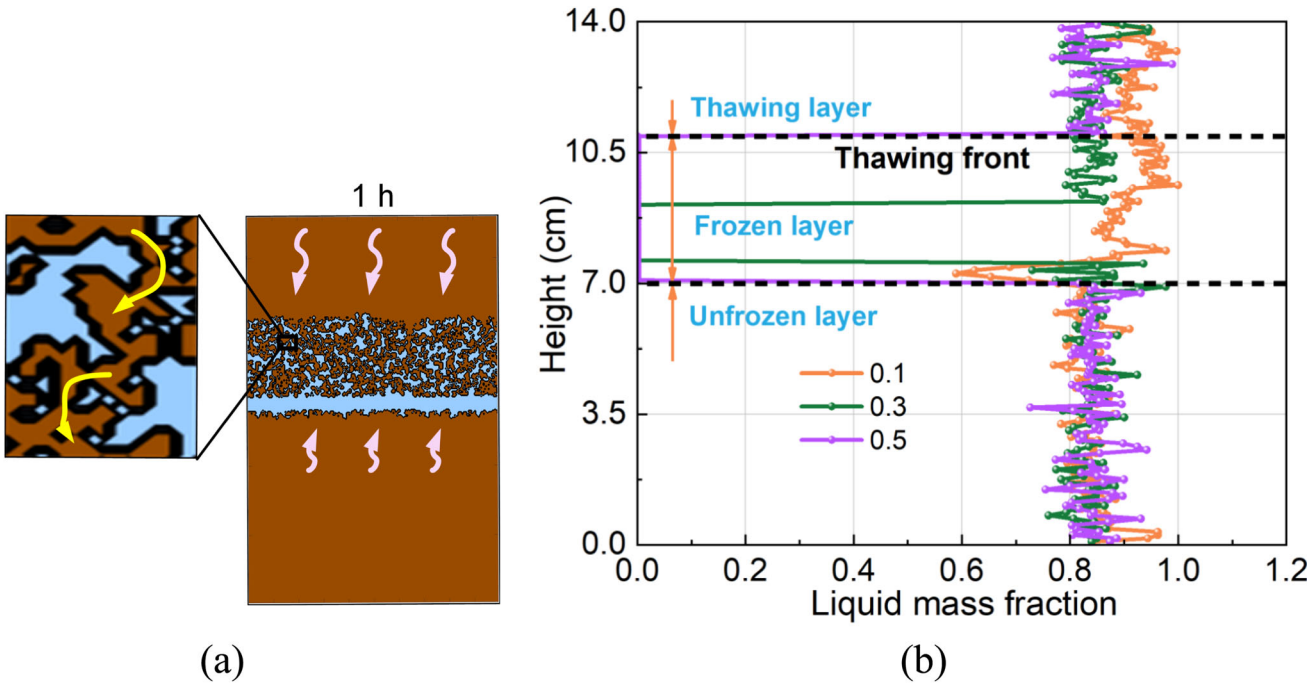

Fig. 16 Influence of porosity on liquid mass fraction: **a** local magnification of pores in ice distribution; **b** liquid mass fraction with different porosities at 1 h (0.1, 0.3, 0.5)

As shown in Fig. 16b, for example with 0.1 porosity, the ice has completely melted into water within 1 h, which is faster compared to samples with larger porosities (0.3 and 0.5). This is attributed to the fact that solid soil particles have a thermal diffusivity approximately twice that of water, while the presence of more ice and air delays the downward progression of the thawing front. The largest difference in liquid mass fraction is observed at a height of 7.3 cm to 7.9 cm, corresponding to the location of melting segregated ice. This phenomenon occurs because this region initially absorbed a significant amount of water from the bottom of the soil during the freezing process, aligning well with the experimental observations [60]. A similar phenomenon was also observed in the freezing model proposed by [57], where locations with larger temperature differences are more likely to exhibit significant fluctuations in water potential. This is attributed to the temperature-driven force at the mesoscopic scale, which is proposed to describe the influence of temperature gradients on equilibrium distribution velocity, thereby causing variations in the liquid mass fraction. During the melting state, the segregated ice layer becomes the most dynamic region in the soil and is particularly susceptible to thaw settlement under external loads.

4 Conclusion

This study developed a thawing model for frozen soil based on the lattice Boltzmann method. The heterogeneous structure of frozen soil was generated with the stochastic growth method according to the natural soil obtained by in situ CT scanning. The conclusions are as follows:

In terms of model validation, the relative error of temperature between the LBM and experimental results is 0.32%, while that is 0.15% compared with analytical results. Also, the overall trend of the LBM simulation results exhibits good agreement with the experimental data because the heterogeneity of the frozen soil was taken into account.

In terms of upper surface temperature, the duration of thawing for soils decreases from 3 h to 2.2 h with an increases of upper surface temperature from 1 °C to 10 °C. The water flows generally downward with the top have a higher temperature gradient due to the influence of the mesoscopic temperature-driven force. According to the liquid mass fraction results, the upper thawing front is more significantly influenced by the upper surface temperature compared to the lower thawing front.

In terms of porosity, as porosity increases from 0.1 to 0.5, the time for the upper and lower thawing fronts to intersect increases from 0.3 h to 0.8 h, while the thawing front remains positioned at a height of 10 cm. The middle height of soil (7.3 cm to 7.9 cm) is always observed with a

large liquid mass fraction corresponding to the location of segregated ice.

Acknowledgements This study was supported by the National Natural Science Foundation of China (52276056) and Hochschuljubiläumsfonds der Stadt Wien (H-523078/2024). The authors extend their gratitude to Ms. Liu Hong from Shiyanjia Lab (www.shiyanjia.com) for providing invaluable assistance with CT imaging.

Funding Open access funding provided by University of Vienna.

Declarations

Conflict of interest The authors declare no competing interests.

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